

UNREAL ENGINE 5 PLUGIN

Runtime File Importer

Blueprint node reference. Import, play, edit and export audio, 3D models, images, video, text, data, documents, subtitles, archives and fonts while the game runs.

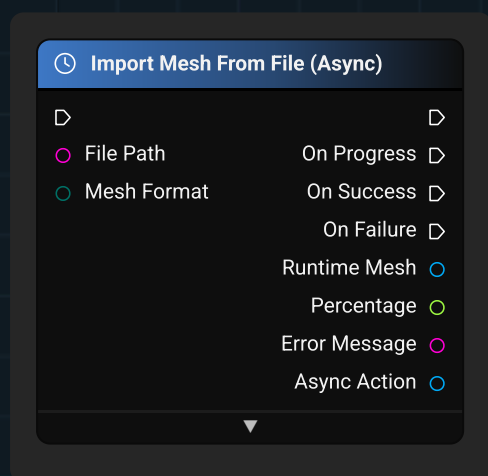
585 nodes

135 async

23 events

47 structs

40 enums



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Overview

This reference lists every Blueprint node of the plugin with its input and output pins, followed by the plugin's objects, widgets, structs and enums.

Getting started

1. Enable the plugin: **Edit > Plugins**, search for **Runtime File Importer**, tick it and restart the editor.
2. In any Blueprint, right-click and search for a node, for example **Import Audio from File**. Most nodes belong to the **Runtime File Importer Subsystem**: connect a **Get Runtime File Importer Subsystem** node to their **Target** pin.
3. Check the result: nodes that can fail have an **Error Message** output (empty on success) and return **false** or **None**. Print the message while developing.
4. For large files and downloads, use the **(Async)** version of the node so the game keeps running while the work happens on another thread.

Example: import a sound and play it

Import Audio From File (Async) → **On Success** → engine node **Play Sound 2D** with the **Sound Wave** output. Connect **On Failure** to **Print String** with **Error Message**.

Example: let the player pick a model

Open File Dialog (Async) with **Get File Dialog Filter** (File Type: Mesh) → **On Success** → **Import Mesh From File (Async)** with the first of **File Paths** → **On Success** → **Spawn Runtime Mesh Actor**.

How the nodes work

Functions

Blue nodes run when executed and finish before the next node runs. Green (pure) nodes have no execution pins and only read values. A function that reads a large file or downloads something blocks the game until it is done, so use its async version in gameplay.

Async nodes

Async nodes (clock icon) continue from the plain output pin right away and do the work in the background. They then fire **On Success** or **On Failure**, and many fire **On Progress** with a **Percentage** from 0 to 100 while they work. All events of one node share the same data pins; pins that do not apply to an event are empty. The **Async Action** pin is the running operation: call the engine's **Cancel** node on it to stop it, after which no more events fire.

Events

The subsystem has events such as **On Audio Imported** and **On Mesh Import Failed**. They fire for every import and export, whichever node started it, which makes them a good place for logging or a loading screen. Bind them with **Bind Event to ...** on the subsystem.

Objects and variables

Imported results are objects with Blueprint variables: a **Runtime Sound Wave** has Volume, Pitch, Looping and Playback Time; a **Runtime Video** has Texture, Time, Rate and Volume. Each of these can also be read and changed with subsystem functions that take the object as an input.

Advanced and optional pins

Pins marked **ADVANCED** are hidden until you expand the node with the arrow at its bottom. Pins marked **OPTIONAL** can stay unconnected; their default values are used. Options structs (for example **Runtime Mesh Import Options**) can be split

or made with a Make node.

Supported formats

Formats are detected from the file contents (and the extension) when a format pin is left on **Auto Detect**.

Kind	Read	Write
Audio	MP3, WAV, FLAC, OGG Vorbis, Uncompressed (Raw PCM), AIFF, OGG Opus	WAV (16-bit PCM), OGG Vorbis, FLAC (Lossless), Uncompressed (Raw PCM)
3D models	FBX, OBJ, glTF (.gltf), glTF Binary (.glb), STL, PLY, USD (.usd, .usda, .usdc), USDZ, Alembic (.abc), Collada (.dae)	FBX (binary), OBJ, glTF (.gltf), glTF Binary (.glb), STL, PLY, USD (.usda), USDZ, Alembic (.abc), Collada (.dae)
Images	PNG, JPEG, BMP, TGA, EXR, HDR, TIFF, ICO, ICNS, GIF	PNG, JPEG, BMP, TGA, EXR, HDR, GIF
Video	MP4, MOV, AVI, MKV, WebM, WMV, GIF, MPEG-TS, MPEG	MP4, AVI, GIF, Image Sequence
Text & documents	Plain Text, JSON, XML, CSV, TSV, INI, Markdown, HTML, RTF, SubRip Subtitles (SRT), WebVTT Subtitles (VTT), SubStation Alpha Subtitles (ASS/SSA), Word Document (DOCX), OpenDocument Text (ODT), Excel Spreadsheet (XLSX), OpenDocument Spreadsheet (ODS), YAML, PDF Document. Google Docs, Sheets, Slides and Drive links.	Text in any encoding (UTF-8, UTF-16, ...), Markdown and HTML from documents
Data	JSON, XML, CSV, TSV, INI, Excel Spreadsheet (XLSX), OpenDocument Spreadsheet (ODS), YAML	JSON, XML, CSV, TSV, INI, YAML
Subtitles	SubRip (SRT), WebVTT (VTT), SubStation Alpha (ASS/SSA)	SubRip (SRT), WebVTT (VTT), SubStation Alpha (ASS)
Archives	ZIP (including ZIP64), GZip, Zlib, raw Deflate	ZIP, GZip, Zlib, raw Deflate
Fonts	TTF, OTF, TTC, OTC, WOFF	–

Platform notes

- Supported platform: Windows 64-bit (Win64), in the editor and in packaged games (Development and Shipping).
- The Open, Save and Folder dialogs use the Windows file dialogs. Call [Are File Dialogs Supported](#) before showing a dialog in cross-platform code.
- In packaged games, exporting an engine Static or Skeletal Mesh needs **Allow CPU Access** on the mesh. Meshes imported by the plugin always work.
- Downloads use HTTP or HTTPS. Google Docs, Sheets, Slides and Drive share links are turned into their download links automatically.

Pin colors

The node drawings use the editor's pin colors. A grid icon is an array of that type.

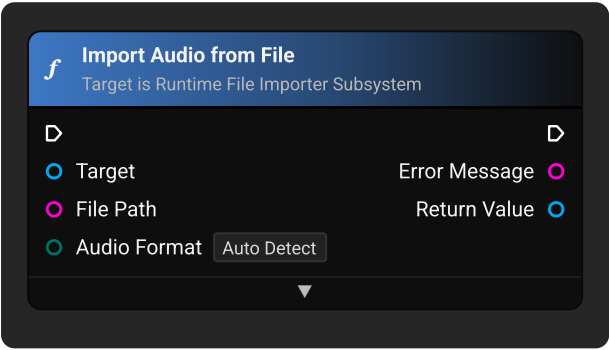
 Execution	 Boolean	 Byte / enum
 Integer	 Float	 Name
 String	 Text	 Object reference
 Struct	 Vector	 Rotator
 Transform	 Any struct	 Array (here of String)

Audio

Import sounds from files, memory or URLs as Runtime Sound Waves, play and control them, capture the microphone, stream audio in, read audio info and tags, and export or convert audio files.

Import 6 nodes

Import Audio from File FUNCTION



Decodes an audio file (MP3, WAV, FLAC, OGG Vorbis, OGG Opus, AIFF or raw PCM) into a playable Sound Wave. Blocks until finished, which can hitch on large files; use the Async version to avoid that. Sample Rate / Num Channels override the file's own values when > 0; they are required for raw PCM.

Target: Runtime File Importer Subsystem
 Async version: [Import Audio From File \(Async\)](#)
 C++: `URuntimeFileImporterSubsystem::ImportAudioFromFile`

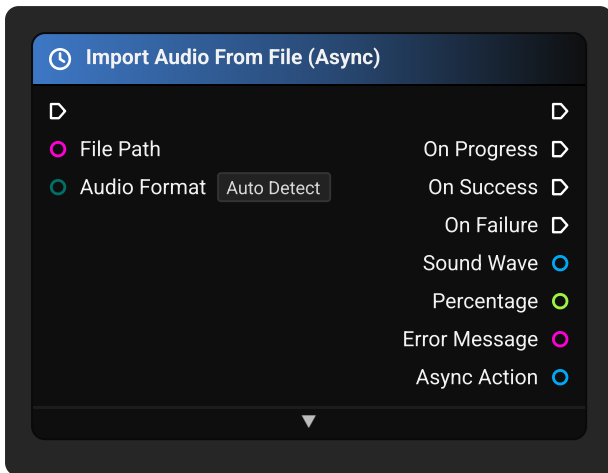
INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
File Path	String	Full path of the file on disk.
Audio Format	EAudio Format	Format of the audio. Auto Detect reads it from the contents. Default: <code>Auto Detect</code>
Sample Rate ADVANCED	Integer	Sample rate of the result in Hz. -1 (or 0): keep the source's rate. Default: <code>-1</code>
Num Channels ADVANCED	Integer	Channels of the result (1 mono, 2 stereo). -1 (or 0): keep the source's. Default: <code>-1</code>

OUTPUTS

Pin	Type	Description
Error Message	String	Empty on success; otherwise why it failed.
Return Value	Runtime Sound Wave Object Reference	The result, or None/empty if it failed (Error Message says why).

Import Audio From File (Async) ASYNC



Decodes an audio file (MP3, WAV, FLAC, OGG Vorbis, OGG Opus, AIFF or raw PCM) into a playable Sound Wave without blocking the game thread. Sample Rate / Num Channels override the file's own values when > 0; they are required for raw PCM.

Blocking version: [Import Audio from File](#)

C++:

UImportAudioFromFileAsyncAction::ImportAudioFromFileAsync

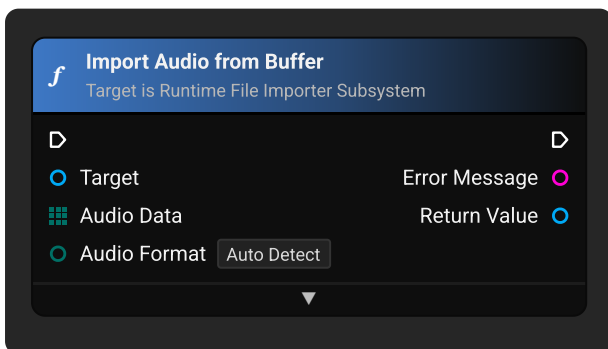
INPUTS

Pin	Type	Description
File Path	String	Full path of the file on disk.
Audio Format	EAudio Format	Format of the audio. Auto Detect reads it from the contents. Default: Auto Detect
Sample Rate ADVANCED	Integer	Sample rate of the result in Hz. -1 (or 0): keep the source's rate. Default: -1
Num Channels ADVANCED	Integer	Channels of the result (1 mono, 2 stereo). -1 (or 0): keep the source's. Default: -1

OUTPUTS

Pin	Type	Description
On Progress	Exec	Fires as the file is read and decoded, with Percentage from 0 to 100.
On Success	Exec	Fires when the operation has finished successfully.
On Failure	Exec	Fires if the operation failed or was cancelled; Error Message says why.
Sound Wave	Runtime Sound Wave Object Reference	The Runtime Sound Wave.
Percentage	Float	Progress from 0 to 100.
Error Message	String	Empty on success; otherwise why it failed.
Async Action	Import Audio from File Async Action Object Reference	This node's operation. Call Cancel on it to stop the operation (no more events fire).

Import Audio from Buffer FUNCTION



Decodes encoded audio bytes already in memory (e.g. received over the network) into a playable Sound Wave. Sample Rate / Num Channels override the data's own values when > 0; they are required for raw PCM.

Target: Runtime File Importer Subsystem

Async version: [Import Audio From Buffer \(Async\)](#)

C++: URuntimeFileImporterSubsystem::ImportAudioFromBuffer

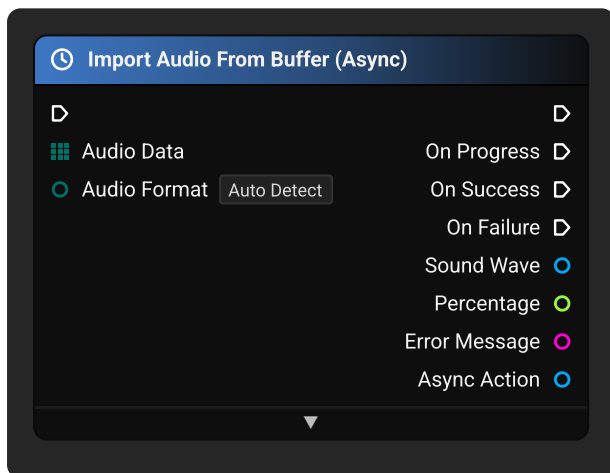
INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Audio Data	Array of Byte	Bytes of an audio file (MP3, WAV, FLAC, OGG, Opus, AIFF, ...).
Audio Format	EAudio Format	Format of the audio. Auto Detect reads it from the contents. Default: Auto Detect
Sample Rate ADVANCED	Integer	Sample rate of the result in Hz. -1 (or 0): keep the source's rate. Default: -1
Num Channels ADVANCED	Integer	Channels of the result (1 mono, 2 stereo). -1 (or 0): keep the source's. Default: -1

OUTPUTS

Pin	Type	Description
Error Message	String	Empty on success; otherwise why it failed.
Return Value	Runtime Sound Wave Object Reference	The result, or None/empty if it failed (Error Message says why).

Import Audio From Buffer (Async) ASYNC



Decodes encoded audio bytes already in memory (e.g. received over the network) into a playable Sound Wave without blocking the game thread. Sample Rate / Num Channels override the data's own values when > 0; they are required for raw PCM.

Blocking version: [Import Audio from Buffer](#)

C++:

```
UImportAudioFromBufferAsyncAction::ImportAudioFromBufferAsync
```

INPUTS

Pin	Type	Description
Audio Data	Array of Byte	Bytes of an audio file (MP3, WAV, FLAC, OGG, Opus, AIFF, ...).
Audio Format	EAudio Format	Format of the audio. Auto Detect reads it from the contents. Default: Auto Detect
Sample Rate ADVANCED	Integer	Sample rate of the result in Hz. -1 (or 0): keep the source's rate. Default: -1
Num Channels ADVANCED	Integer	Channels of the result (1 mono, 2 stereo). -1 (or 0): keep the source's. Default: -1

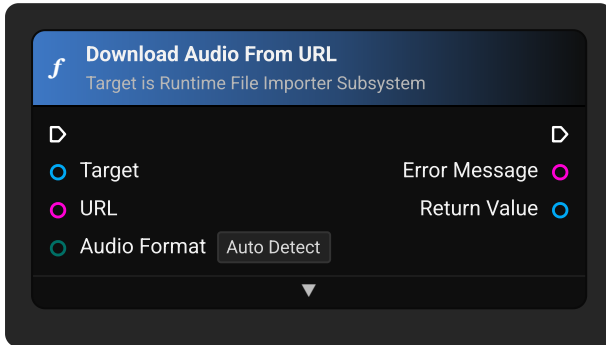
OUTPUTS

Pin	Type	Description
On Progress	Exec	Fires as the data is decoded, with Percentage from 0 to 100.
On Success	Exec	Fires when the operation has finished successfully.
On Failure	Exec	Fires if the operation failed or was cancelled; Error Message says why.
Sound Wave	Runtime Sound Wave Object Reference	The Runtime Sound Wave.
Percentage	Float	Progress from 0 to 100.

Pin	Type	Description
Error Message	String	Empty on success; otherwise why it failed.
Async Action	Import Audio from Buffer Async Action Object Reference	This node's operation. Call Cancel on it to stop the operation (no more events fire).

Download Audio From URL

FUNCTION



Downloads an audio file over HTTP(S) and decodes it into a playable Sound Wave. This FREEZES the game until the download finishes or times out; prefer the Async version. If Save To File Path is set, the downloaded file is also written there, unmodified.

Target: Runtime File Importer Subsystem

Async version: [Download Audio From URL \(Async\)](#)

C++: URuntimeFileImporterSubsystem::DownloadAudioFromURL

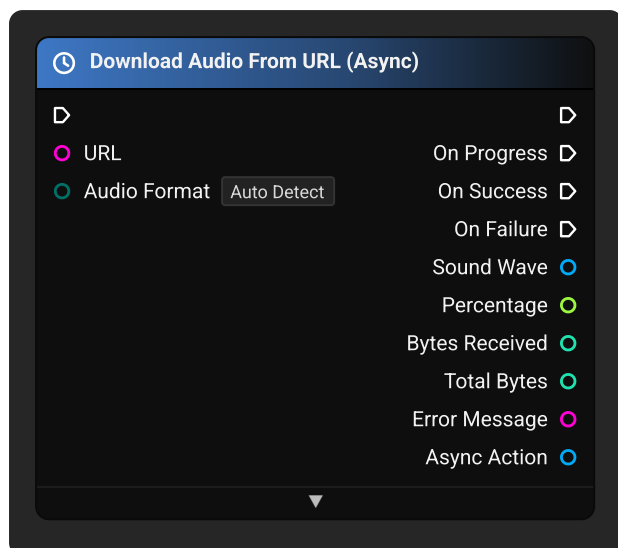
INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
URL	String	The http(s) address to download. Google Docs, Sheets, Slides and Drive share links are converted automatically.
Audio Format	EAudio Format	Format of the audio. Auto Detect reads it from the contents. Default: Auto Detect
Save to File Path	String	Optional. Also writes the downloaded file there, unmodified. Default: (empty)
Timeout Seconds	Float	Gives up (and fails) if the download takes longer than this. Default: 30.0

OUTPUTS

Pin	Type	Description
Error Message	String	Empty on success; otherwise why it failed.
Return Value	Runtime Sound Wave Object Reference	The result, or None/empty if it failed (Error Message says why).

Download Audio From URL (Async) ASYNC



Downloads an audio file over HTTP(S) and decodes it into a playable Sound Wave. If Save To File Path is set, the downloaded file is also written there, unmodified. Timeout Seconds <= 0 uses the engine's default HTTP timeout.

Blocking version: [Download Audio From URL](#)

C++:

```
UDownloadAudioFromURLAsyncAction::DownloadAudioFromURLAsync
```

INPUTS

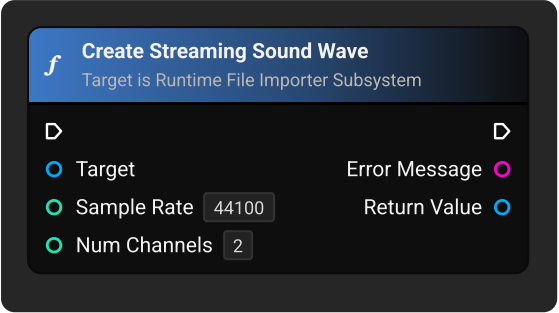
Pin	Type	Description
URL	String	The http(s) address to download. Google Docs, Sheets, Slides and Drive share links are converted automatically.
Audio Format	EAudio Format	Format of the audio. Auto Detect reads it from the contents. Default: <code>Auto Detect</code>
Save to File Path ADVANCED	String	Optional. Also writes the downloaded file there, unmodified. Default: <code>(empty)</code>
Timeout Seconds ADVANCED	Float	Gives up (and fails) if the download takes longer than this. Default: <code>60.0</code>

OUTPUTS

Pin	Type	Description
On Progress	Exec	Fires repeatedly while downloading. Percentage (0-100) and Total Bytes are 0 if the server doesn't report the file size.
On Success	Exec	Fires when the operation has finished successfully.
On Failure	Exec	Fires if the operation failed or was cancelled; Error Message says why.
Sound Wave	Runtime Sound Wave Object Reference	The Runtime Sound Wave.
Percentage	Float	Progress from 0 to 100.
Bytes Received	Integer	Bytes downloaded so far.
Total Bytes	Integer	Size of the download in bytes, or 0 while the server hasn't said.
Error Message	String	Empty on success; otherwise why it failed.
Async Action	Download Audio from URL Async Action Object Reference	This node's operation. Call Cancel on it to stop the operation (no more events fire).

Create Streaming Sound Wave

FUNCTION

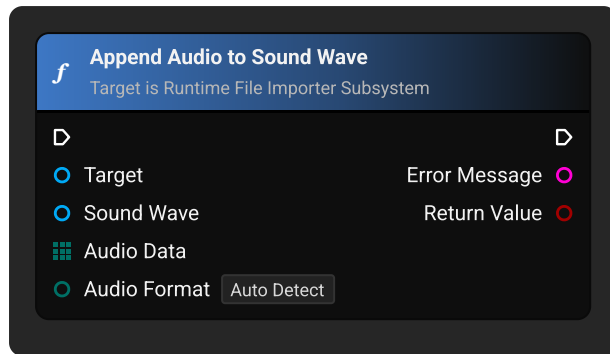


Creates an empty Runtime Sound Wave that audio can be appended to while it plays (e.g. text-to-speech or voice chat arriving in pieces). Playback waits for more audio instead of ending, until Finish Streaming.

Target: Runtime File Importer Subsystem
C++: URuntimeFileImporterSubsystem::CreateStreamingSoundWave

INPUTS		
Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Sample Rate	Integer	Samples per second (Hz), e.g. 44100 or 48000. Default: 44100
Num Channels	Integer	Number of channels (1 mono, 2 stereo). Default: 2
OUTPUTS		
Pin	Type	Description
Error Message	String	Empty on success; otherwise why it failed.
Return Value	Runtime Sound Wave Object Reference	The result, or None/empty if it failed (Error Message says why).

Append Audio to Sound Wave FUNCTION



Decodes audio (any supported format; raw PCM must already match the sound wave) and appends it to a Runtime Sound Wave, converting it to the sound wave's sample rate and channel count. Appends are applied in the order they were started, including async ones still in progress.

Target: Runtime File Importer Subsystem

Async version: [Append Audio To Sound Wave \(Async\)](#)

C++: `URuntimeFileImporterSubsystem::AppendAudioToSoundWave`

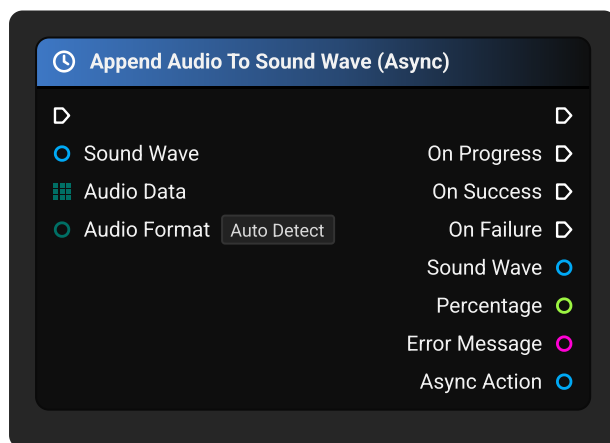
INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Sound Wave	Runtime Sound Wave Object Reference	The Runtime Sound Wave.
Audio Data	Array of Byte	Bytes of an audio file (MP3, WAV, FLAC, OGG, Opus, AIFF, ...).
Audio Format	EAudio Format	Format of the audio. Auto Detect reads it from the contents. Default: Auto Detect

OUTPUTS

Pin	Type	Description
Error Message	String	Empty on success; otherwise why it failed.
Return Value	Boolean	True on success. False if it failed (Error Message says why).

Append Audio To Sound Wave (Async) ASYNC



Decodes audio (any supported format; raw PCM must already match the sound wave) on a background thread and appends it to a Runtime Sound Wave, converted to its sample rate and channel count. Appends are applied in the order they were started, even when a later one finishes decoding first.

Blocking version: [Append Audio to Sound Wave](#)

C++:

`UAppendAudioToSoundWaveAsyncAction::AppendAudioToSoundWaveAsync`

INPUTS

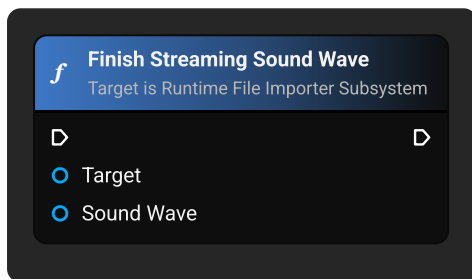
Pin	Type	Description
Sound Wave	Runtime Sound Wave Object Reference	The Runtime Sound Wave.
Audio Data	Array of Byte	Bytes of an audio file (MP3, WAV, FLAC, OGG, Opus, AIFF, ...).
Audio Format	EAudio Format	Format of the audio. Auto Detect reads it from the contents. Default: Auto Detect

OUTPUTS

Pin	Type	Description
On Progress	 Exec	Fires as the audio is decoded and converted, with Percentage from 0 to 100.
On Success	 Exec	Fires when the operation has finished successfully.
On Failure	 Exec	Fires if the operation failed or was cancelled; Error Message says why.
Sound Wave	 Runtime Sound Wave Object Reference	The Runtime Sound Wave.
Percentage	 Float	Progress from 0 to 100.
Error Message	 String	Empty on success; otherwise why it failed.
Async Action	 Append Audio to Sound Wave Async Action Object Reference	This node's operation. Call Cancel on it to stop the operation (no more events fire).

Finish Streaming Sound Wave

FUNCTION



Marks a streaming sound wave as complete (after any appends still in progress), so playback finishes at its end.

Target: Runtime File Importer Subsystem

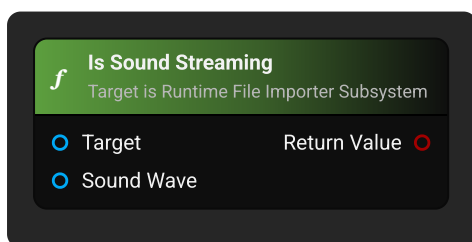
C++: `URuntimeFileImporterSubsystem::FinishStreamingSoundWave`

INPUTS

Pin	Type	Description
Target	 Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Sound Wave	 Runtime Sound Wave Object Reference	The Runtime Sound Wave.

Is Sound Streaming

PURE



True while more audio can still be appended to a sound (until Finish Streaming Sound Wave). Same as its Is Streaming variable.

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::IsSoundStreaming`

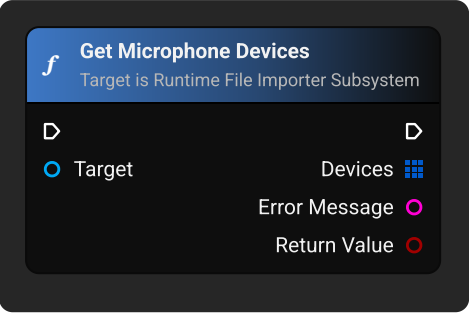
INPUTS

Pin	Type	Description
Target	 Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Sound Wave	 Runtime Sound Wave Object Reference	The Runtime Sound Wave.

OUTPUTS

Pin	Type	Description
Return Value	 Boolean	True while more audio can still be appended to a sound (until Finish Streaming Sound Wave).

Get Microphone Devices FUNCTION



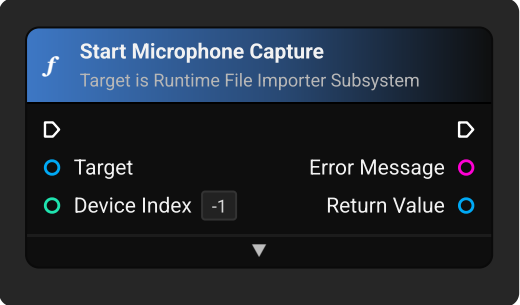
Lists the microphones and other audio inputs that can be recorded from. Returns false if there are none.

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::GetMicrophoneDevices`

INPUTS		
Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
OUTPUTS		
Pin	Type	Description
Devices	Array of Runtime Microphone Device	The audio input devices found.
Error Message	String	Empty on success; otherwise why it failed.
Return Value	Boolean	False if there are none.

Start Microphone Capture FUNCTION



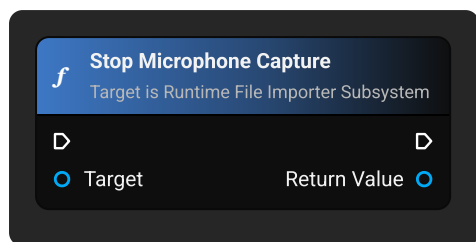
Starts recording from a microphone (Device Index -1 = the default one) into a streaming Runtime Sound Wave, which can be played live, exported, or kept after Stop Microphone Capture. One capture runs at a time. Android needs the RECORD_AUDIO permission (and a runtime request); iOS needs a microphone usage description.

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::StartMicrophoneCapture`

INPUTS		
Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Device Index	Integer	Index in Get Microphone Devices. -1: the system's default microphone. Default: -1
Mono	ADVANCED Boolean	Record one channel instead of the device's own channel count. Default: false
OUTPUTS		
Pin	Type	Description
Error Message	String	Empty on success; otherwise why it failed.
Return Value	Runtime Sound Wave Object Reference	The result, or None/empty if it failed (Error Message says why).

Stop Microphone Capture FUNCTION



Stops recording and returns the finished recording (null if nothing was being recorded).

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::StopMicrophoneCapture`

INPUTS

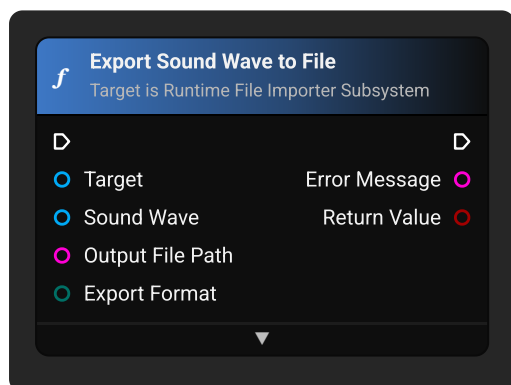
Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.

OUTPUTS

Pin	Type	Description
Return Value	Runtime Sound Wave Object Reference	Stops recording and returns the finished recording (null if nothing was being recorded).

Export 6 nodes

Export Sound Wave to File FUNCTION



Encodes a Sound Wave imported by this plugin and writes it to disk in the chosen format. Quality (1-100) only applies to OGG Vorbis. Blocks until finished; use the Async version to avoid hitches. Sample Rate / Num Channels resample / remix the exported audio (0 keeps the sound's own), e.g. 16000 and 1 for speech services.

Target: Runtime File Importer Subsystem

Async version: [Export Sound Wave To File \(Async\)](#)

C++: `URuntimeFileImporterSubsystem::ExportSoundWaveToFile`

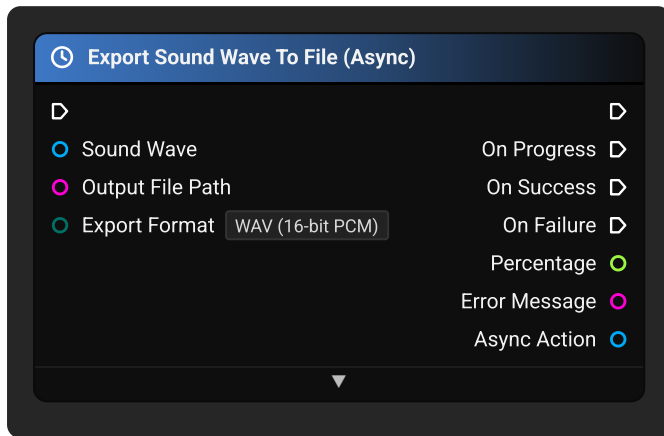
INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Sound Wave	Sound Wave Object Reference	The Runtime Sound Wave.
Output File Path	String	Full path of the file to write. Missing folders are created; an existing file is replaced.
Export Format	EAudio Export Format	Format to write.
Quality ADVANCED	Integer	Quality of lossy formats, 0 to 100 (higher: better and larger). Default: 70
Sample Rate ADVANCED	Integer	Sample rate of the result in Hz. 0: keep the source's rate. Default: 0
Num Channels ADVANCED	Integer	Channels of the result (1 mono, 2 stereo). 0: keep the source's. Default: 0

OUTPUTS

Pin	Type	Description
Error Message	 String	Empty on success; otherwise why it failed.
Return Value	 Boolean	True on success. False if it failed (Error Message says why).

Export Sound Wave To File (Async) ASYNC



Encodes a Sound Wave imported by this plugin and writes it to disk in the chosen format. Quality (1-100) only applies to OGG Vorbis. Sample Rate / Num Channels resample / remix the audio (0 keeps the sound's own).

Blocking version: [Export Sound Wave to File](#)

C++:

```
UExportSoundWaveToFileAsyncAction::ExportSoundWaveToFileAsync
```

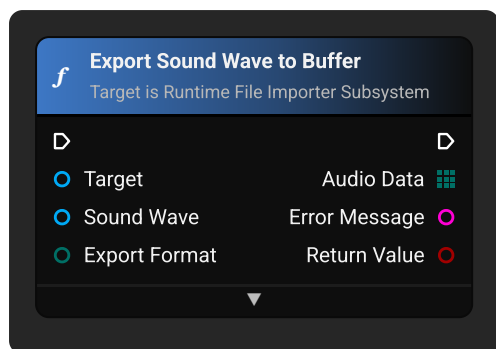
INPUTS

Pin	Type	Description
Sound Wave	 Sound Wave Object Reference	The Runtime Sound Wave.
Output File Path	 String	Full path of the file to write. Missing folders are created; an existing file is replaced.
Export Format	 EAudio Export Format	Format to write. Default: WAV (16-bit PCM)
Quality ADVANCED	 Integer	Quality of lossy formats, 0 to 100 (higher: better and larger). Default: 70
Sample Rate ADVANCED	 Integer	Sample rate of the result in Hz. 0: keep the source's rate. Default: 0
Num Channels ADVANCED	 Integer	Channels of the result (1 mono, 2 stereo). 0: keep the source's. Default: 0

OUTPUTS

Pin	Type	Description
On Progress	 Exec	Fires as the audio is encoded and written, with Percentage from 0 to 100.
On Success	 Exec	Fires when the operation has finished successfully.
On Failure	 Exec	Fires if the operation failed or was cancelled; Error Message says why.
Percentage	 Float	Progress from 0 to 100.
Error Message	 String	Empty on success; otherwise why it failed.
Async Action	 Export Sound Wave to File Async Action Object Reference	This node's operation. Call Cancel on it to stop the operation (no more events fire).

Export Sound Wave to Buffer FUNCTION



Encodes a Sound Wave imported by this plugin into bytes in the chosen format (e.g. to send over the network). Quality (1-100) only applies to OGG Vorbis. Sample Rate / Num Channels resample / remix the audio (0 keeps the sound's own).

Target: Runtime File Importer Subsystem

Async version: [Export Sound Wave To Buffer \(Async\)](#)

C++: `URuntimeFileImporterSubsystem::ExportSoundWaveToBuffer`

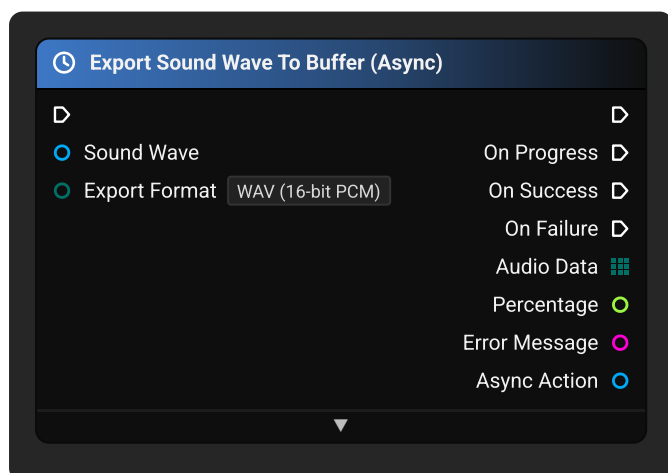
INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Sound Wave	Sound Wave Object Reference	The Runtime Sound Wave.
Export Format	EAudio Export Format	Format to write.
Quality ADVANCED	Integer	Quality of lossy formats, 0 to 100 (higher: better and larger). Default: 70
Sample Rate ADVANCED	Integer	Sample rate of the result in Hz. 0: keep the source's rate. Default: 0
Num Channels ADVANCED	Integer	Channels of the result (1 mono, 2 stereo). 0: keep the source's. Default: 0

OUTPUTS

Pin	Type	Description
Audio Data	Array of Byte	Bytes of an audio file (MP3, WAV, FLAC, OGG, Opus, AIFF, ...).
Error Message	String	Empty on success; otherwise why it failed.
Return Value	Boolean	True on success. False if it failed (Error Message says why).

Export Sound Wave To Buffer (Async) ASYNC



Encodes a Sound Wave imported by this plugin into bytes in the chosen format (e.g. to send over the network). Quality (1-100) only applies to OGG Vorbis. Sample Rate / Num Channels resample / remix the audio (0 keeps the sound's own).

Blocking version: [Export Sound Wave to Buffer](#)

C++:

`UExportSoundWaveToBufferAsyncAction::ExportSoundWaveToBufferAsync`

INPUTS

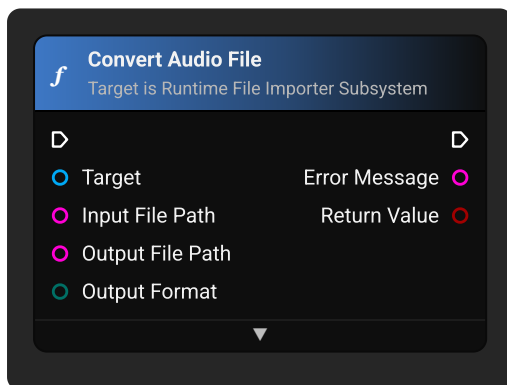
Pin	Type	Description
Sound Wave	Sound Wave Object Reference	The Runtime Sound Wave.
Export Format	EAudio Export Format	Format to write. Default: WAV (16-bit PCM)
Quality ADVANCED	Integer	Quality of lossy formats, 0 to 100 (higher: better and larger). Default: 70
Sample Rate ADVANCED	Integer	Sample rate of the result in Hz. 0: keep the source's rate. Default: 0
Num Channels ADVANCED	Integer	Channels of the result (1 mono, 2 stereo). 0: keep the source's. Default: 0

OUTPUTS

Pin	Type	Description
On Progress	Exec	Fires as the audio is encoded, with Percentage from 0 to 100.
On Success	Exec	Fires when the operation has finished successfully.
On Failure	Exec	Fires if the operation failed or was cancelled; Error Message says why.
Audio Data	Array of Byte	Bytes of an audio file (MP3, WAV, FLAC, OGG, Opus, AIFF, ...).
Percentage	Float	Progress from 0 to 100.
Error Message	String	Empty on success; otherwise why it failed.
Async Action	Export Sound Wave to Buffer Async Action Object Reference	This node's operation. Call Cancel on it to stop the operation (no more events fire).

Convert Audio File

FUNCTION



Converts an audio file to another format (e.g. MP3 to OGG). Raw PCM input is assumed to be 44100 Hz stereo. Quality (1-100) only applies to OGG Vorbis output. Sample Rate / Num Channels resample / remix the audio (0 keeps the file's own).

Target: Runtime File Importer Subsystem

Async version: [Convert Audio File \(Async\)](#)

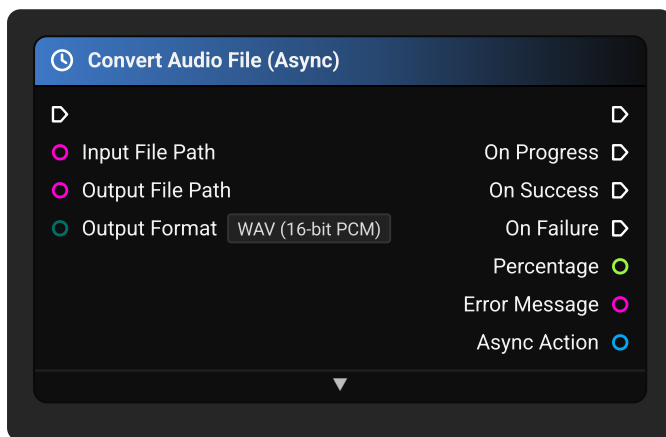
C++: `URuntimeFileImporterSubsystem::ConvertAudioFile`

INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Input File Path	String	Full path of the file to convert.
Output File Path	String	Full path of the file to write. Missing folders are created; an existing file is replaced.
Output Format	EAudio Export Format	Format to write.
Input Format ADVANCED	EAudio Format	Format of the input. Auto Detect reads it from the contents. Default: Auto Detect
Quality ADVANCED	Integer	Quality of lossy formats, 0 to 100 (higher: better and larger). Default: 70

Pin	Type	Description
Sample Rate ADVANCED	Integer	Sample rate of the result in Hz. 0: keep the source's rate. Default: 0
Num Channels ADVANCED	Integer	Channels of the result (1 mono, 2 stereo). 0: keep the source's. Default: 0
OUTPUTS		
Pin	Type	Description
Error Message	String	Empty on success; otherwise why it failed.
Return Value	Boolean	True on success. False if it failed (Error Message says why).

Convert Audio File (Async) ASYNC



Converts an audio file to another format (e.g. MP3 to OGG) without blocking the game thread. Raw PCM input is assumed to be 44100 Hz stereo. Quality (1-100) only applies to OGG Vorbis output. Sample Rate / Num Channels resample / remix the audio (0 keeps the file's own).

Blocking version: [Convert Audio File](#)

C++:

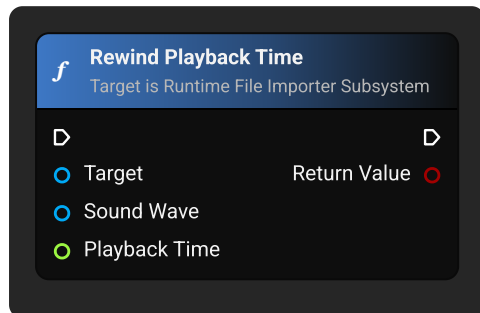
```
UConvertAudioFileAsyncAction::ConvertAudioFileAsync
```

INPUTS		
Pin	Type	Description
Input File Path	String	Full path of the file to convert.
Output File Path	String	Full path of the file to write. Missing folders are created; an existing file is replaced.
Output Format	EAudio Export Format	Format to write. Default: WAV (16-bit PCM)
Input Format ADVANCED	EAudio Format	Format of the input. Auto Detect reads it from the contents. Default: Auto Detect
Quality ADVANCED	Integer	Quality of lossy formats, 0 to 100 (higher: better and larger). Default: 70
Sample Rate ADVANCED	Integer	Sample rate of the result in Hz. 0: keep the source's rate. Default: 0
Num Channels ADVANCED	Integer	Channels of the result (1 mono, 2 stereo). 0: keep the source's. Default: 0
OUTPUTS		
Pin	Type	Description
On Progress	Exec	Fires as the file is decoded and re-encoded, with Percentage from 0 to 100.
On Success	Exec	Fires when the operation has finished successfully.
On Failure	Exec	Fires if the operation failed or was cancelled; Error Message says why.
Percentage	Float	Progress from 0 to 100.
Error Message	String	Empty on success; otherwise why it failed.

Pin	Type	Description
Async Action	 Convert Audio File Async Action Object Reference	This node's operation. Call Cancel on it to stop the operation (no more events fire).

Playback 13 nodes

Rewind Playback Time FUNCTION



Jumps playback to Playback Time (in seconds). Works while playing or stopped; a stopped sound resumes from here when played again. Returns false if the time is outside the sound's duration.

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::RewindPlaybackTime`

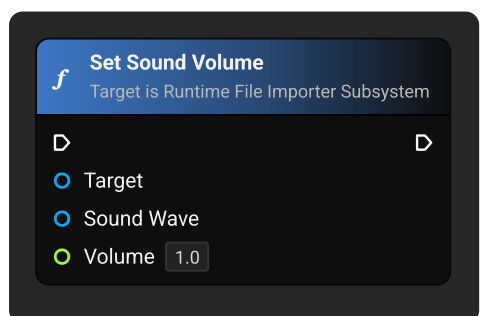
INPUTS

Pin	Type	Description
Target	 Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Sound Wave	 Runtime Sound Wave Object Reference	The Runtime Sound Wave.
Playback Time	 Float	Where to continue playing from, in seconds.

OUTPUTS

Pin	Type	Description
Return Value	 Boolean	False if the time is outside the sound's duration.

Set Sound Volume FUNCTION



Sets a sound's volume multiplier (0 or higher). Takes effect immediately, even while playing. Same as setting its Volume variable.

Target: Runtime File Importer Subsystem

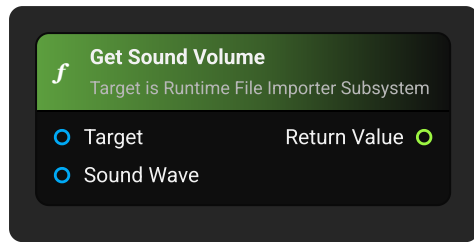
C++: `URuntimeFileImporterSubsystem::SetSoundVolume`

INPUTS

Pin	Type	Description
Target	 Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Sound Wave	 Runtime Sound Wave Object Reference	The Runtime Sound Wave.
Volume	 Float	Volume multiplier (1: unchanged, 0: silent). Default: 1.0

Get Sound Volume

PURE



A sound's volume multiplier. Same as its Volume variable.

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::GetSoundVolume`

INPUTS

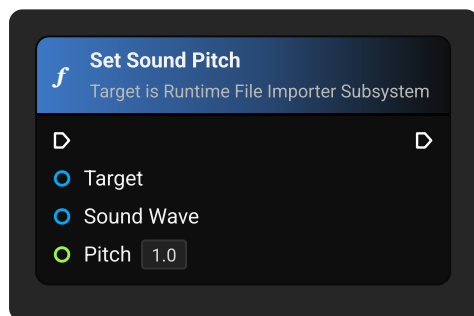
Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Sound Wave	Runtime Sound Wave Object Reference	The Runtime Sound Wave.

OUTPUTS

Pin	Type	Description
Return Value	Float	A sound's volume multiplier.

Set Sound Pitch

FUNCTION



Sets a sound's pitch multiplier (clamped to 0.125 - 4; also changes playback speed). Takes effect immediately. Same as setting its Pitch variable.

Target: Runtime File Importer Subsystem

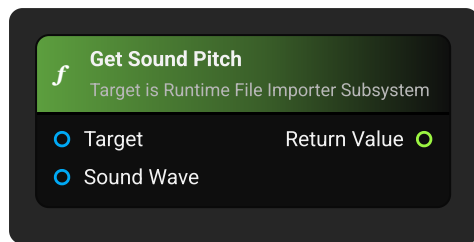
C++: `URuntimeFileImporterSubsystem::SetSoundPitch`

INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Sound Wave	Runtime Sound Wave Object Reference	The Runtime Sound Wave.
Pitch	Float	Pitch multiplier (1: unchanged, 2: an octave higher). Default: 1.0

Get Sound Pitch

PURE



A sound's pitch multiplier. Same as its Pitch variable.

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::GetSoundPitch`

INPUTS

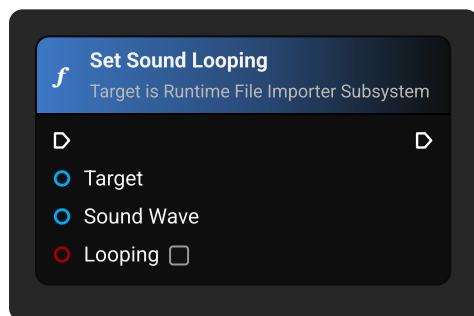
Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Sound Wave	Runtime Sound Wave Object Reference	The Runtime Sound Wave.

OUTPUTS

Pin	Type	Description
Return Value	Float	A sound's pitch multiplier.

Set Sound Looping

FUNCTION



Sets whether a sound starts over when it reaches its end. Takes effect immediately; ignored while streaming. Same as setting its Looping variable.

Target: Runtime File Importer Subsystem

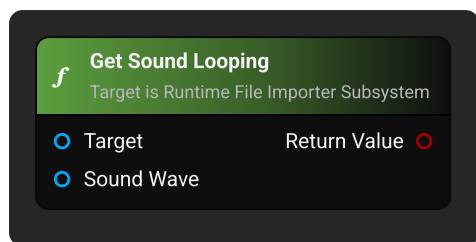
C++: `URuntimeFileImporterSubsystem::SetSoundLooping`

INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Sound Wave	Runtime Sound Wave Object Reference	The Runtime Sound Wave.
Looping	Boolean	Start again from the beginning when the end is reached.

Get Sound Looping

PURE



Whether a sound loops. Same as its Looping variable.

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::GetSoundLooping`

INPUTS

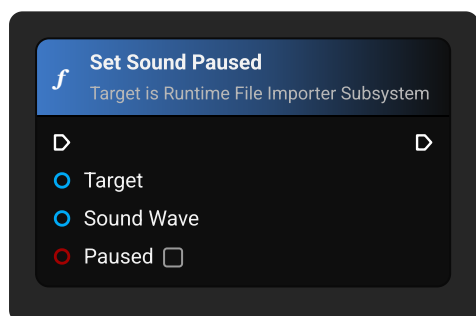
Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Sound Wave	Runtime Sound Wave Object Reference	The Runtime Sound Wave.

OUTPUTS

Pin	Type	Description
Return Value	Boolean	Whether a sound loops.

Set Sound Paused

FUNCTION



Pauses or resumes a sound and fires its On Playback Paused / Resumed events. Same as setting its Paused variable.

Target: Runtime File Importer Subsystem

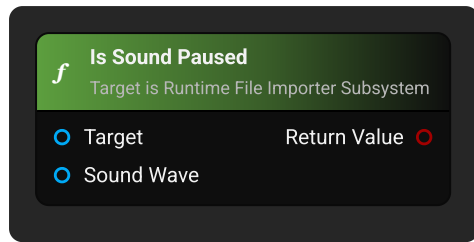
C++: `URuntimeFileImporterSubsystem::SetSoundPaused`

INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Sound Wave	Runtime Sound Wave Object Reference	The Runtime Sound Wave.
Paused	Boolean	True to pause, false to resume.

Is Sound Paused

PURE



Whether a sound is paused. Same as its Paused variable.

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::IsSoundPaused`

INPUTS

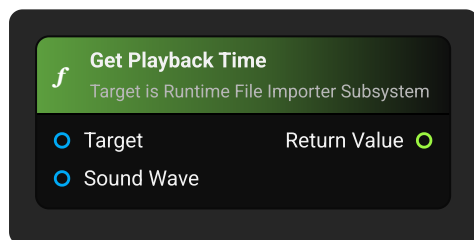
Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Sound Wave	Runtime Sound Wave Object Reference	The Runtime Sound Wave.

OUTPUTS

Pin	Type	Description
Return Value	Boolean	Whether a sound is paused.

Get Playback Time

PURE



A sound's current playback position, in seconds. Same as its Playback Time variable.

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::GetPlaybackTime`

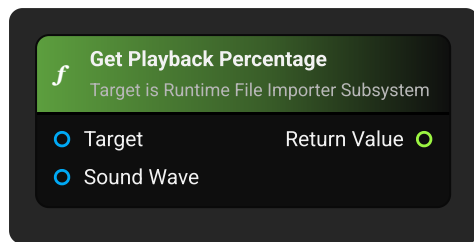
INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Sound Wave	Runtime Sound Wave Object Reference	The Runtime Sound Wave.

OUTPUTS

Pin	Type	Description
Return Value	Float	A sound's current playback position, in seconds.

Get Playback Percentage PURE



A sound's playback progress from 0 to 100. Same as its Playback Percentage variable.

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::GetPlaybackPercentage`

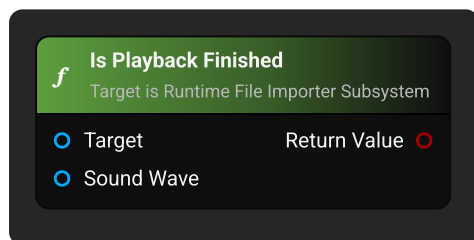
INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Sound Wave	Runtime Sound Wave Object Reference	The Runtime Sound Wave.

OUTPUTS

Pin	Type	Description
Return Value	Float	A sound's playback progress from 0 to 100.

Is Playback Finished PURE



True once a non-looping sound has played to the end. Same as its Is Playback Finished variable.

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::IsPlaybackFinished`

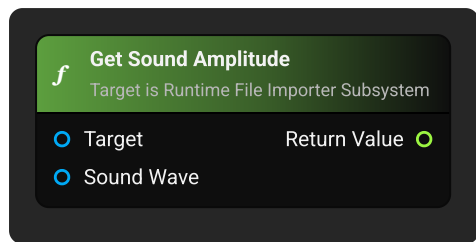
INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Sound Wave	Runtime Sound Wave Object Reference	The Runtime Sound Wave.

OUTPUTS

Pin	Type	Description
Return Value	Boolean	True once a non-looping sound has played to the end.

Get Sound Amplitude PURE



Loudness (RMS, 0-1) of the audio a sound just played. Same as its Current Amplitude variable.

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::GetSoundAmplitude`

INPUTS

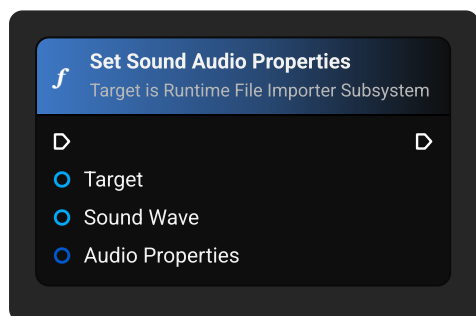
Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Sound Wave	Runtime Sound Wave Object Reference	The Runtime Sound Wave.

OUTPUTS

Pin	Type	Description
Return Value	Float	Loudness (RMS, 0-1) of the audio a sound just played.

Properties 2 nodes

Set Sound Audio Properties FUNCTION



Sets every runtime-changeable property of a sound (volume, pitch, attenuation, sound class, submixes, ...). Volume, pitch and looping apply immediately; the rest the next time it starts playing. Same as setting its Audio Properties variable.

Target: Runtime File Importer Subsystem

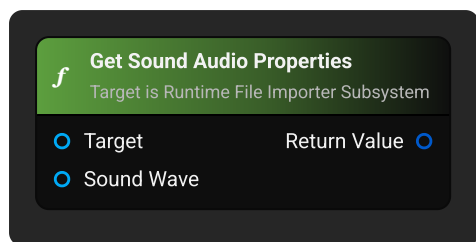
C++: `URuntimeFileImporterSubsystem::SetSoundAudioProperties`

INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Sound Wave	Runtime Sound Wave Object Reference	The Runtime Sound Wave.
Audio Properties	Runtime Audio Properties	Every property of the sound wave that can be changed at runtime.

Get Sound Audio Properties

PURE



Every runtime-changeable property of a sound. Same as its Audio Properties variable.

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::GetSoundAudioProperties`

INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Sound Wave	Runtime Sound Wave Object Reference	The Runtime Sound Wave.

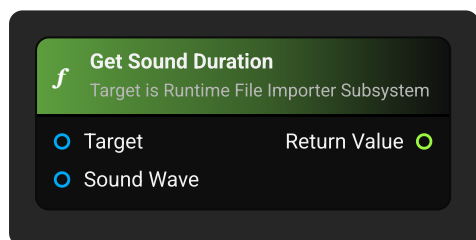
OUTPUTS

Pin	Type	Description
Return Value	Runtime Audio Properties	Every runtime-changeable property of a sound.

Info 3 nodes

Get Sound Duration

PURE



Length of a sound in seconds (for streaming sounds, what has been received so far). Same as its Duration variable.

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::GetSoundDuration`

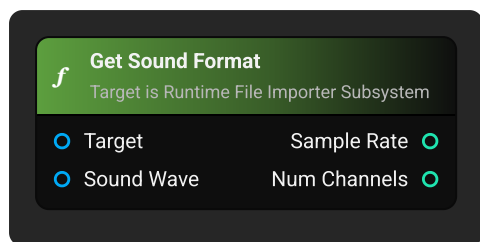
INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Sound Wave	Runtime Sound Wave Object Reference	The Runtime Sound Wave.

OUTPUTS

Pin	Type	Description
Return Value	Float	Length of a sound in seconds (for streaming sounds, what has been received so far).

Get Sound Format PURE



A sound's sample rate and channel count. Same as its Sample Rate and Num Channels variables.

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::GetSoundFormat`

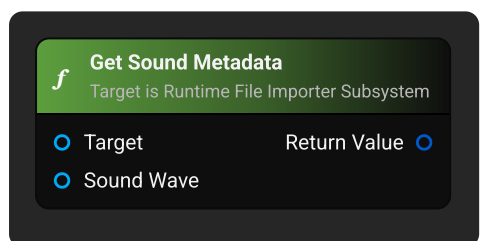
INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Sound Wave	Runtime Sound Wave Object Reference	The Runtime Sound Wave.

OUTPUTS

Pin	Type	Description
Sample Rate	Integer	Samples per second (Hz), e.g. 44100 or 48000.
Num Channels	Integer	Number of channels (1 mono, 2 stereo).

Get Sound Metadata PURE



Tags read from a sound's file (title, artist, album, ...). Same as its Metadata variable.

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::GetSoundMetadata`

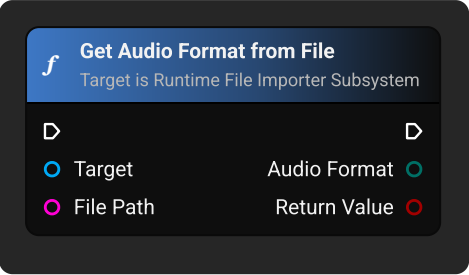
INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Sound Wave	Runtime Sound Wave Object Reference	The Runtime Sound Wave.

OUTPUTS

Pin	Type	Description
Return Value	Runtime Audio Metadata	Tags read from a sound's file (title, artist, album, ...).

Get Audio Format from File FUNCTION

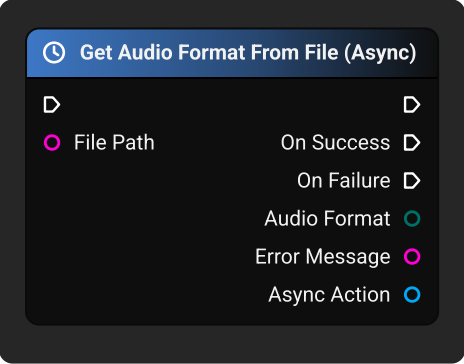


Detects a file's audio format from its contents (falling back to its extension), so it can be stored and passed to the import nodes. Returns false if the format isn't recognized.

Target: Runtime File Importer Subsystem
Async version: [Get Audio Format From File \(Async\)](#)
C++: `URuntimeFileImporterSubsystem::GetAudioFormatFromFile`

INPUTS		
Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
File Path	String	Full path of the file on disk.
OUTPUTS		
Pin	Type	Description
Audio Format	EAudio Format	Format of the audio. Auto Detect reads it from the contents.
Return Value	Boolean	False if the format isn't recognized.

Get Audio Format From File (Async) ASYNC



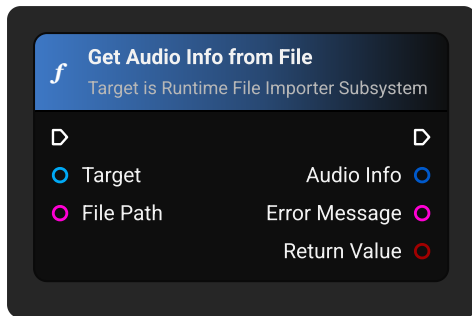
Detects a file's audio format from its contents (falling back to its extension), for use with the import nodes.

Blocking version: [Get Audio Format from File](#)
C++: `UGetAudioFormatFromFileAsyncAction::GetAudioFormatFromFileAsync`

INPUTS		
Pin	Type	Description
File Path	String	Full path of the file on disk.
OUTPUTS		
Pin	Type	Description
On Success	Exec	Fires when the operation has finished successfully.
On Failure	Exec	Fires if the operation failed or was cancelled; Error Message says why.
Audio Format	EAudio Format	Format of the audio. Auto Detect reads it from the contents.
Error Message	String	Empty on success; otherwise why it failed.
Async Action	Get Audio Format from File Async Action Object Reference	This node's operation. Call Cancel on it to stop the operation (no more events fire).

Get Audio Info from File

FUNCTION



Reads an audio file's format, duration, sample rate, channels, bitrate and tags (title, artist, album, ...) without importing it. Returns false if the file can't be read or isn't a supported format.

Target: Runtime File Importer Subsystem

Async version: [Get Audio Info From File \(Async\)](#)

C++: `URuntimeFileImporterSubsystem::GetAudioInfoFromFile`

INPUTS

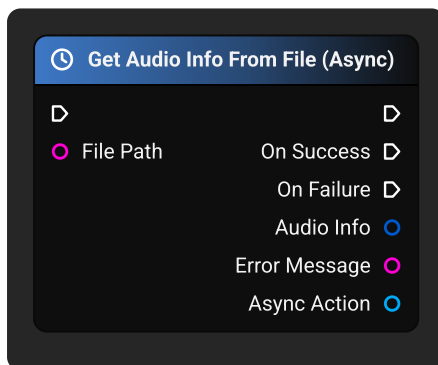
Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
File Path	String	Full path of the file on disk.

OUTPUTS

Pin	Type	Description
Audio Info	Runtime Audio Info	Format, duration, sample rate, channels, bit rate and tags of the audio.
Error Message	String	Empty on success; otherwise why it failed.
Return Value	Boolean	False if the file can't be read or isn't a supported format.

Get Audio Info From File (Async)

ASYNC



Reads an audio file's format, duration, sample rate, channels, bitrate and tags (title, artist, album, ...) without importing it.

Blocking version: [Get Audio Info from File](#)

C++: `UGetAudioInfoFromFileAsyncAction::GetAudioInfoFromFileAsync`

INPUTS

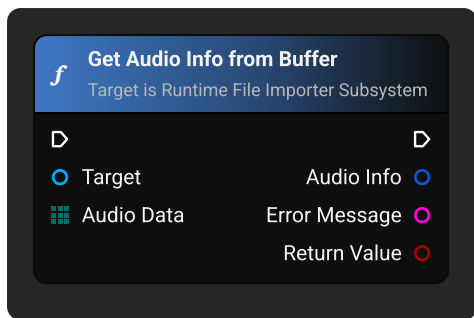
Pin	Type	Description
File Path	String	Full path of the file on disk.

OUTPUTS

Pin	Type	Description
On Success	Exec	Fires when the operation has finished successfully.
On Failure	Exec	Fires if the operation failed or was cancelled; Error Message says why.
Audio Info	Runtime Audio Info	Format, duration, sample rate, channels, bit rate and tags of the audio.
Error Message	String	Empty on success; otherwise why it failed.
Async Action	Get Audio Info from File Async Action Object Reference	This node's operation. Call Cancel on it to stop the operation (no more events fire).

Get Audio Info from Buffer

FUNCTION



Reads the format, duration, sample rate, channels, bitrate and tags of encoded audio in memory without importing it.

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::GetAudioInfoFromBuffer`

INPUTS

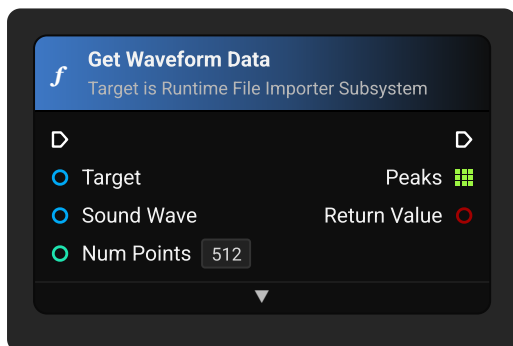
Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Audio Data	Array of Byte	Bytes of an audio file (MP3, WAV, FLAC, OGG, Opus, AIFF, ...).

OUTPUTS

Pin	Type	Description
Audio Info	Runtime Audio Info	Format, duration, sample rate, channels, bit rate and tags of the audio.
Error Message	String	Empty on success; otherwise why it failed.
Return Value	Boolean	True on success. False if it failed (Error Message says why).

Get Waveform Data

FUNCTION



Peak levels (0-1) of Num Points equal slices of a Runtime Sound Wave, for drawing its waveform. Start Time / End Time (seconds) select a part of it; an End Time of 0 means the end.

Target: Runtime File Importer Subsystem

Async version: [Get Waveform Data \(Async\)](#)

C++: `URuntimeFileImporterSubsystem::GetWaveformData`

INPUTS

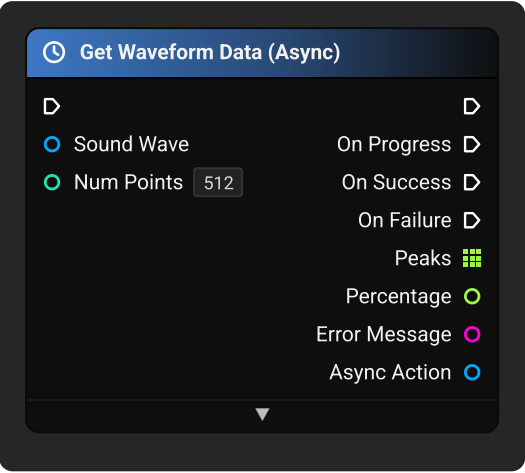
Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Sound Wave	Runtime Sound Wave Object Reference	The Runtime Sound Wave.
Num Points	Integer	How many values to compute across the range (e.g. one per pixel of a waveform widget). Default: 512
Start Time	Float	Start of the range, in seconds. Default: 0.0
End Time	Float	End of the range, in seconds. 0: the end. Default: 0.0

OUTPUTS

Pin	Type	Description
Peaks	Array of Float	Peak amplitude (0 to 1) of each point.

Pin	Type	Description
Return Value	 Boolean	Peak levels (0-1) of Num Points equal slices of a Runtime Sound Wave, for drawing its waveform.

Get Waveform Data (Async) ASYNC



Peak levels (0-1) of Num Points equal slices of a Runtime Sound Wave, for drawing its waveform. Start Time / End Time (seconds) select a part of it; an End Time of 0 means the end.

Blocking version: [Get Waveform Data](#)

C++: `UGetWaveformDataAsyncAction::GetWaveformDataAsync`

INPUTS

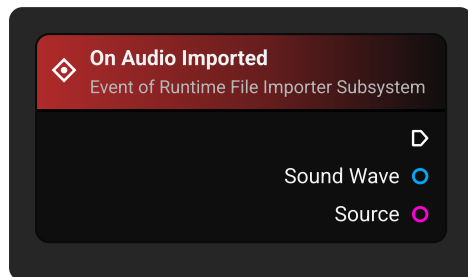
Pin	Type	Description
Sound Wave	 Runtime Sound Wave Object Reference	The Runtime Sound Wave.
Num Points	 Integer	How many values to compute across the range (e.g. one per pixel of a waveform widget). Default: 512
Start Time ADVANCED	 Float	Start of the range, in seconds. Default: 0.0
End Time ADVANCED	 Float	End of the range, in seconds. 0: the end. Default: 0.0

OUTPUTS

Pin	Type	Description
On Progress	 Exec	Fires as the operation advances (Percentage from 0 to 100). Other outputs are not set yet.
On Success	 Exec	Fires when the operation has finished successfully.
On Failure	 Exec	Fires if the operation failed or was cancelled; Error Message says why.
Peaks	 Array of Float	Peak amplitude (0 to 1) of each point.
Percentage	 Float	Progress from 0 to 100.
Error Message	 String	Empty on success; otherwise why it failed.
Async Action	 Get Waveform Data Async Action Object Reference	This node's operation. Call Cancel on it to stop the operation (no more events fire).

On Audio Imported

EVENT



Any audio import succeeded. Source is the file path or URL (empty for buffers).

Bind it with "Bind Event to On Audio Imported" (or "Assign On Audio Imported") on the Runtime File Importer Subsystem.

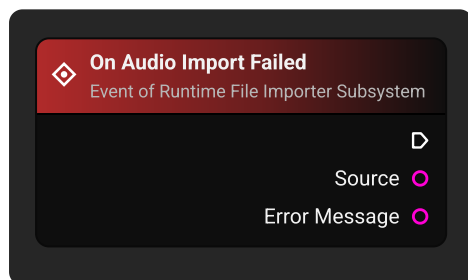
C++: `URuntimeFileImporterSubsystem::OnAudioImported`

OUTPUTS

Pin	Type	Description
Sound Wave	Runtime Sound Wave Object Reference	The Runtime Sound Wave.
Source	String	File path or URL the import came from (empty for buffers and strings).

On Audio Import Failed

EVENT



Any audio import (from a file, buffer or URL) failed. Source is the file path or URL (empty for buffers).

Bind it with "Bind Event to On Audio Import Failed" (or "Assign On Audio Import Failed") on the Runtime File Importer Subsystem.

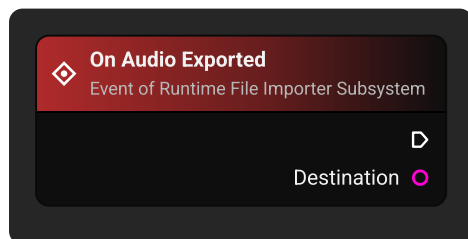
C++: `URuntimeFileImporterSubsystem::OnAudioImportFailed`

OUTPUTS

Pin	Type	Description
Source	String	File path or URL the import came from (empty for buffers and strings).
Error Message	String	Empty on success; otherwise why it failed.

On Audio Exported

EVENT



Any audio export or conversion succeeded. Destination is the output file path (empty for buffers).

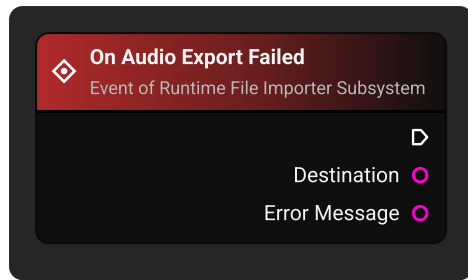
Bind it with "Bind Event to On Audio Exported" (or "Assign On Audio Exported") on the Runtime File Importer Subsystem.

C++: `URuntimeFileImporterSubsystem::OnAudioExported`

OUTPUTS

Pin	Type	Description
Destination	String	Output file path (empty when writing to a buffer).

On Audio Export Failed EVENT



Any audio export or conversion failed. Destination is the output file path (empty for buffers).

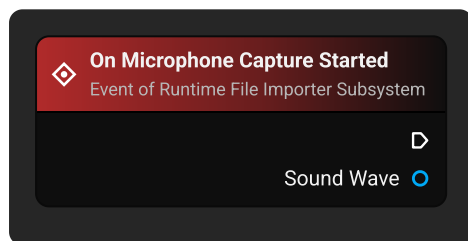
Bind it with "Bind Event to On Audio Export Failed" (or "Assign On Audio Export Failed") on the Runtime File Importer Subsystem.

C++: `URuntimeFileImporterSubsystem::OnAudioExportFailed`

OUTPUTS

Pin	Type	Description
Destination	String	Output file path (empty when writing to a buffer).
Error Message	String	Empty on success; otherwise why it failed.

On Microphone Capture Started EVENT



Microphone capture started, recording into Sound Wave.

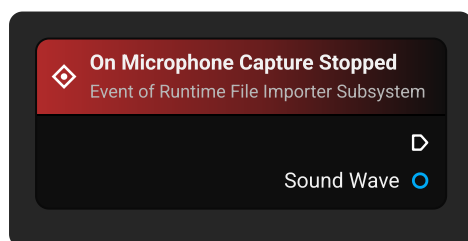
Bind it with "Bind Event to On Microphone Capture Started" (or "Assign On Microphone Capture Started") on the Runtime File Importer Subsystem.

C++: `URuntimeFileImporterSubsystem::OnMicrophoneCaptureStarted`

OUTPUTS

Pin	Type	Description
Sound Wave	Runtime Sound Wave Object Reference	The Runtime Sound Wave.

On Microphone Capture Stopped EVENT



Microphone capture stopped; Sound Wave now holds the complete recording.

Bind it with "Bind Event to On Microphone Capture Stopped" (or "Assign On Microphone Capture Stopped") on the Runtime File Importer Subsystem.

C++: `URuntimeFileImporterSubsystem::OnMicrophoneCaptureStopped`

OUTPUTS

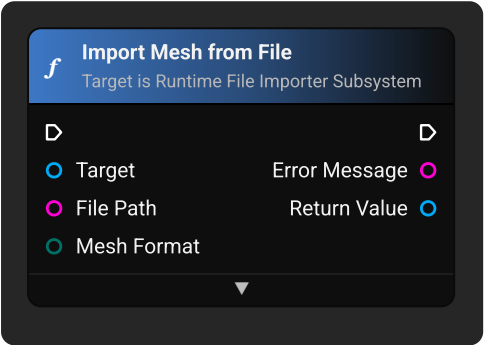
Pin	Type	Description
Sound Wave	Runtime Sound Wave Object Reference	The Runtime Sound Wave.

Mesh

Import 3D models as Runtime Meshes (static or skeletal, with materials, skeletons, animations and physics), spawn them in the level, export any mesh asset, animation or actor to model files, convert between mesh types, inspect and edit mesh assets, and edit geometry with Mesh Descriptions.

Import 10 nodes

Import Mesh from File FUNCTION



Imports a 3D model (FBX, OBJ, glTF/GLB, STL, PLY, USD/USDA/USDC/USDZ, Alembic, Collada) as a Runtime Mesh, with its materials and textures (files next to the model are found automatically). Models with bones, morph targets or animations get a Skeletal Mesh, Skeleton, Animations and Physics Asset (see Options' Mesh Type); others a Static Mesh. Blocks until finished, which can hitch on large files; use the Async version to avoid that.

Target: Runtime File Importer Subsystem
Async version: [Import Mesh From File \(Async\)](#)
C++: `URuntimeFileImporterSubsystem::ImportMeshFromFile`

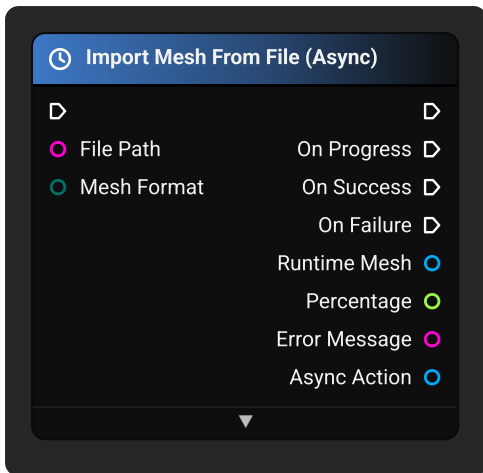
INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
File Path	String	Full path of the file on disk.
Mesh Format	EMesh Format	Format of the file. Auto Detect reads it from the contents and extension.
Options	ADVANCED OPTIONAL Runtime Mesh Import Options	How a 3D model file is read and turned into a Runtime Mesh.

OUTPUTS

Pin	Type	Description
Error Message	String	Empty on success; otherwise why it failed.
Return Value	Runtime Mesh Object Reference	The result, or None/empty if it failed (Error Message says why).

Import Mesh From File (Async) ASYNC



Imports a 3D model (FBX, OBJ, glTF/GLB, STL, PLY, USD/USDA/USDC/USDZ, Alembic, Collada) as a Runtime Mesh with its materials and textures, without blocking the game thread. Models with bones, morph targets or animations get a Skeletal Mesh, Skeleton, Animations and Physics Asset (see Options' Mesh Type); others a Static Mesh.

Blocking version: [Import Mesh from File](#)

C++: `UImportMeshAsyncAction::ImportMeshFromFileAsync`

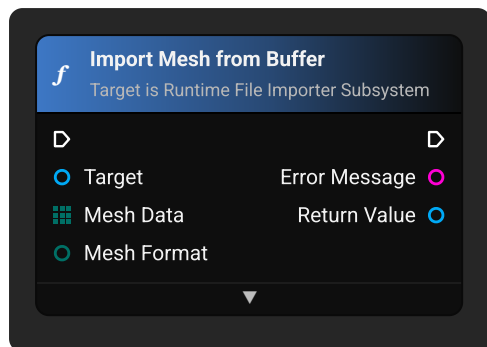
INPUTS

Pin	Type	Description
File Path	String	Full path of the file on disk.
Mesh Format	EMesh Format	Format of the file. Auto Detect reads it from the contents and extension.
Options ADVANCED OPTIONAL	Runtime Mesh Import Options	How a 3D model file is read and turned into a Runtime Mesh.

OUTPUTS

Pin	Type	Description
On Progress	Exec	Fires as the operation advances (Percentage from 0 to 100). Other outputs are not set yet.
On Success	Exec	Fires when the operation has finished successfully.
On Failure	Exec	Fires if the operation failed or was cancelled; Error Message says why.
Runtime Mesh	Runtime Mesh Object Reference	The Runtime Mesh (an imported model).
Percentage	Float	Progress from 0 to 100.
Error Message	String	Empty on success; otherwise why it failed.
Async Action	Import Mesh Async Action Object Reference	This node's operation. Call Cancel on it to stop the operation (no more events fire).

Import Mesh from Buffer FUNCTION



Imports a 3D model from bytes already in memory (e.g. received over the network). Files the model refers to can't be loaded this way: use a self-contained format (GLB, FBX, USDZ, STL, PLY) for textures.

Target: Runtime File Importer Subsystem

Async version: [Import Mesh From Buffer \(Async\)](#)

C++: `URuntimeFileImporterSubsystem::ImportMeshFromBuffer`

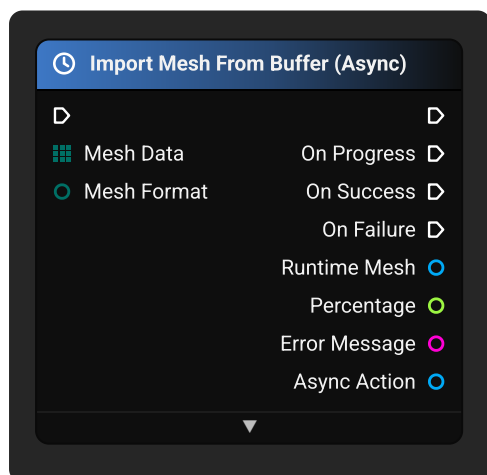
INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Mesh Data	Array of Byte	Bytes of a 3D model file.
Mesh Format	EMesh Format	Format of the file. Auto Detect reads it from the contents and extension.
Options ADVANCED OPTIONAL	Runtime Mesh Import Options	How a 3D model file is read and turned into a Runtime Mesh.

OUTPUTS

Pin	Type	Description
Error Message	String	Empty on success; otherwise why it failed.
Return Value	Runtime Mesh Object Reference	The result, or None/empty if it failed (Error Message says why).

Import Mesh From Buffer (Async) ASYNC



Imports a 3D model from bytes already in memory without blocking the game thread. Use a self-contained format (GLB, FBX, USDZ, ...) for textures.

Blocking version: [Import Mesh from Buffer](#)

C++: `UImportMeshAsyncAction::ImportMeshFromBufferAsync`

INPUTS

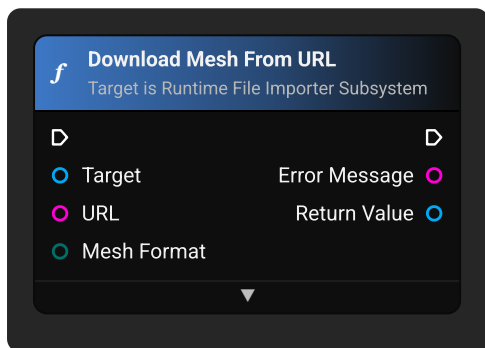
Pin	Type	Description
Mesh Data	Array of Byte	Bytes of a 3D model file.
Mesh Format	EMesh Format	Format of the file. Auto Detect reads it from the contents and extension.
Options ADVANCED OPTIONAL	Runtime Mesh Import Options	How a 3D model file is read and turned into a Runtime Mesh.

OUTPUTS

Pin	Type	Description
On Progress	 Exec	Fires as the operation advances (Percentage from 0 to 100). Other outputs are not set yet.
On Success	 Exec	Fires when the operation has finished successfully.
On Failure	 Exec	Fires if the operation failed or was cancelled; Error Message says why.
Runtime Mesh	 Runtime Mesh Object Reference	The Runtime Mesh (an imported model).
Percentage	 Float	Progress from 0 to 100.
Error Message	 String	Empty on success; otherwise why it failed.
Async Action	 Import Mesh Async Action Object Reference	This node's operation. Call Cancel on it to stop the operation (no more events fire).

Download Mesh From URL

FUNCTION



Downloads a 3D model over HTTP(S) and imports it. This FREEZES the game until the download finishes or times out; prefer the Async version. Use a self-contained format (GLB, FBX, USDZ, ...) for textures. If Save To File Path is set, the downloaded file is also written there, unmodified.

Target: Runtime File Importer Subsystem

Async version: [Download Mesh From URL \(Async\)](#)

C++: `URuntimeFileImporterSubsystem::DownloadMeshFromURL`

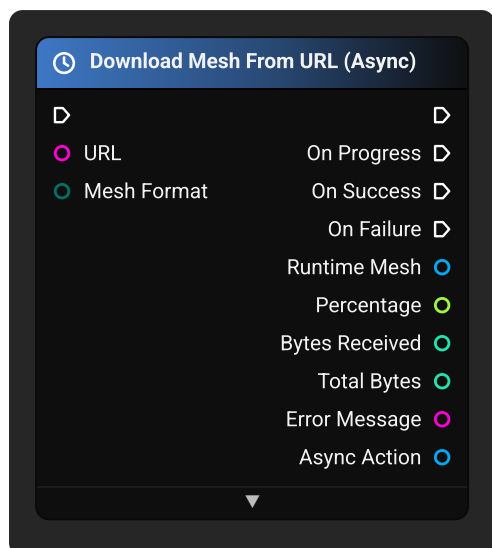
INPUTS

Pin	Type	Description
Target	 Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
URL	 String	The http(s) address to download. Google Docs, Sheets, Slides and Drive share links are converted automatically.
Mesh Format	 EMesh Format	Format of the file. Auto Detect reads it from the contents and extension.
Options	 Runtime Mesh Import Options	How a 3D model file is read and turned into a Runtime Mesh.
Save to File Path	 String	Optional. Also writes the downloaded file there, unmodified. Default: (empty)
Timeout Seconds	 Float	Gives up (and fails) if the download takes longer than this. Default: 60.0

OUTPUTS

Pin	Type	Description
Error Message	 String	Empty on success; otherwise why it failed.
Return Value	 Runtime Mesh Object Reference	The result, or None/empty if it failed (Error Message says why).

Download Mesh From URL (Async) ASYNC



Downloads a 3D model over HTTP(S) and imports it. Use a self-contained format (GLB, FBX, USDZ, ...) for textures. If Save To File Path is set, the downloaded file is also written there, unmodified.

Blocking version: [Download Mesh From URL](#)

C++: `UDownloadMeshFromURLAsyncAction::DownloadMeshFromURLAsync`

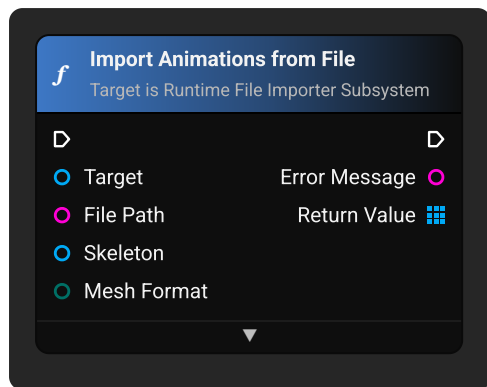
INPUTS

Pin	Type	Description
URL	String	The http(s) address to download. Google Docs, Sheets, Slides and Drive share links are converted automatically.
Mesh Format	EMesh Format	Format of the file. Auto Detect reads it from the contents and extension.
Options ADVANCED OPTIONAL	Runtime Mesh Import Options	How a 3D model file is read and turned into a Runtime Mesh.
Save to File Path ADVANCED	String	Optional. Also writes the downloaded file there, unmodified. Default: (empty)
Timeout Seconds ADVANCED	Float	Gives up (and fails) if the download takes longer than this. Default: 120.0

OUTPUTS

Pin	Type	Description
On Progress	Exec	Fires repeatedly while downloading. Percentage (0-100) and Total Bytes are 0 if the server doesn't report the file size.
On Success	Exec	Fires when the operation has finished successfully.
On Failure	Exec	Fires if the operation failed or was cancelled; Error Message says why.
Runtime Mesh	Runtime Mesh Object Reference	The Runtime Mesh (an imported model).
Percentage	Float	Progress from 0 to 100.
Bytes Received	Integer	Bytes downloaded so far.
Total Bytes	Integer	Size of the download in bytes, or 0 while the server hasn't said.
Error Message	String	Empty on success; otherwise why it failed.
Async Action	Download Mesh from URLAsync Action Object Reference	This node's operation. Call Cancel on it to stop the operation (no more events fire).

Import Animations from File FUNCTION



Imports the animations of a file (FBX takes, glTF animations; with or without a mesh, e.g. animation packs) as Animation Sequences for an existing Skeleton, matching bones by name. Options' axis, unit, scale and frame rate apply.

Target: Runtime File Importer Subsystem

Async version: [Import Animations From File \(Async\)](#)

C++: `URuntimeFileImporterSubsystem::ImportAnimationsFromFile`

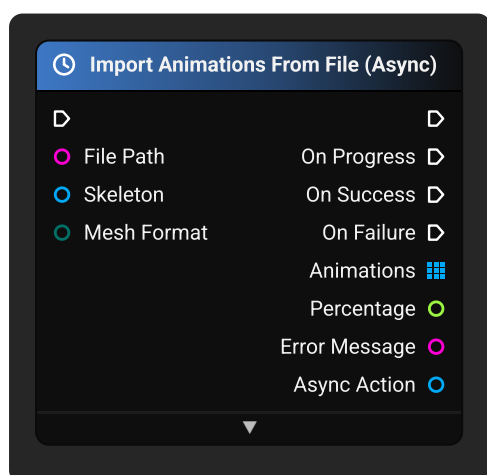
INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
File Path	String	Full path of the file on disk.
Skeleton	Skeleton Object Reference	The Skeleton the animations are made for (bones matched by name).
Mesh Format	EMesh Format	Format of the file. Auto Detect reads it from the contents and extension.
Options	ADVANCED OPTIONAL Runtime Mesh Import Options	How a 3D model file is read and turned into a Runtime Mesh.

OUTPUTS

Pin	Type	Description
Error Message	String	Empty on success; otherwise why it failed.
Return Value	Array of Anim Sequence Object Reference	The result, or None/empty if it failed (Error Message says why).

Import Animations From File (Async) ASYNC



Imports the animations of a file (FBX takes, glTF animations; with or without a mesh) as Animation Sequences for an existing Skeleton, e.g. a character's, matching bones by name. Without blocking the game thread.

Blocking version: [Import Animations from File](#)

C++: `UImportAnimationsAsyncAction::ImportAnimationsFromFileAsync`

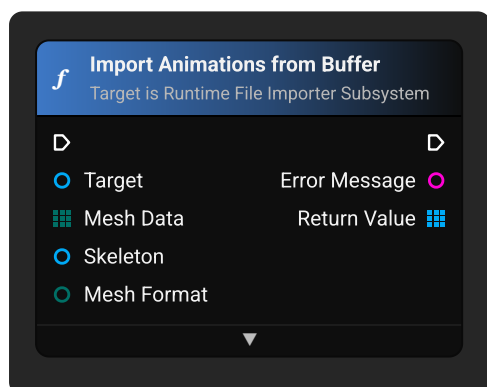
INPUTS

Pin	Type	Description
File Path	String	Full path of the file on disk.
Skeleton	Skeleton Object Reference	The Skeleton the animations are made for (bones matched by name).

Pin	Type	Description
Mesh Format	EMesh Format	Format of the file. Auto Detect reads it from the contents and extension.
Options ADVANCED OPTIONAL Runtime Mesh Import Options		
OUTPUTS		
Pin	Type	Description
On Progress	Exec	Fires as the operation advances (Percentage from 0 to 100). Other outputs are not set yet.
On Success	Exec	Fires when the operation has finished successfully.
On Failure	Exec	Fires if the operation failed or was cancelled; Error Message says why.
Animations	Array of Anim Sequence Object Reference	Animation Sequences to include.
Percentage	Float	Progress from 0 to 100.
Error Message	String	Empty on success; otherwise why it failed.
Async Action	Import Animations Async Action Object Reference	This node's operation. Call Cancel on it to stop the operation (no more events fire).

Import Animations from Buffer

FUNCTION



Imports the animations of a model in memory as Animation Sequences for an existing Skeleton (bones matched by name).

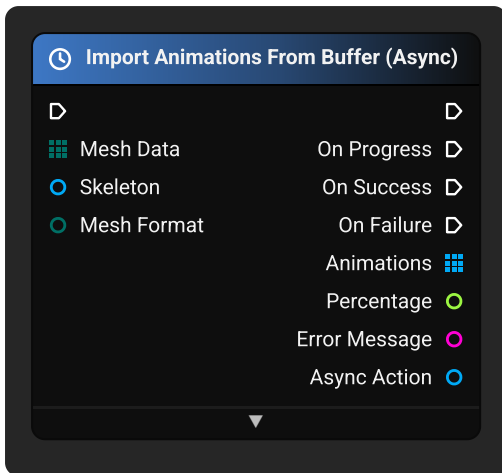
Target: Runtime File Importer Subsystem

Async version: [Import Animations From Buffer \(Async\)](#)

C++: `URuntimeFileImporterSubsystem::ImportAnimationsFromBuffer`

INPUTS		
Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Mesh Data	Array of Byte	Bytes of a 3D model file.
Skeleton	Skeleton Object Reference	The Skeleton the animations are made for (bones matched by name).
Mesh Format	EMesh Format	Format of the file. Auto Detect reads it from the contents and extension.
Options ADVANCED OPTIONAL Runtime Mesh Import Options		
OUTPUTS		
Pin	Type	Description
Error Message	String	Empty on success; otherwise why it failed.
Return Value	Array of Anim Sequence Object Reference	The result, or None/empty if it failed (Error Message says why).

Import Animations From Buffer (Async) ASYNC



Imports the animations of a model in memory as Animation Sequences for an existing Skeleton (bones matched by name).

Blocking version: [Import Animations from Buffer](#)

C++: `UImportAnimationsAsyncAction::ImportAnimationsFromBufferAsync`

INPUTS

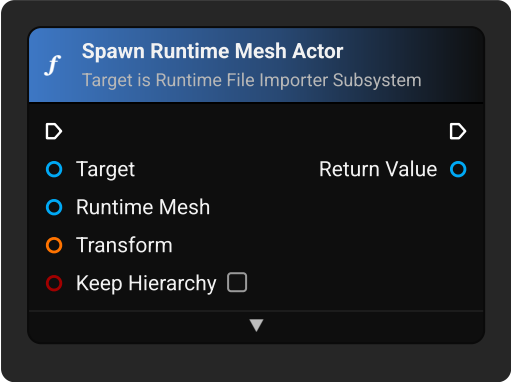
Pin	Type	Description
Mesh Data	Array of Byte	Bytes of a 3D model file.
Skeleton	Skeleton Object Reference	The Skeleton the animations are made for (bones matched by name).
Mesh Format	EMesh Format	Format of the file. Auto Detect reads it from the contents and extension.
Options ADVANCED OPTIONAL	Runtime Mesh Import Options	How a 3D model file is read and turned into a Runtime Mesh.

OUTPUTS

Pin	Type	Description
On Progress	Exec	Fires as the operation advances (Percentage from 0 to 100). Other outputs are not set yet.
On Success	Exec	Fires when the operation has finished successfully.
On Failure	Exec	Fires if the operation failed or was cancelled; Error Message says why.
Animations	Array of Anim Sequence Object Reference	Animation Sequences to include.
Percentage	Float	Progress from 0 to 100.
Error Message	String	Empty on success; otherwise why it failed.
Async Action	Import Animations Async Action Object Reference	This node's operation. Call Cancel on it to stop the operation (no more events fire).

Spawn Runtime Mesh Actor

FUNCTION



Spawns an actor showing a Runtime Mesh. Static models: with Keep Hierarchy, the model's nodes become separate components (named after them) that can be moved, hidden or given other materials; otherwise one component shows the whole model. Skeletal models get a Skeletal Mesh Component, playing Animation To Play (by name) if set.

Target: Runtime File Importer Subsystem

C++: URuntimeFileImporterSubsystem::SpawnRuntimeMeshActor

INPUTS

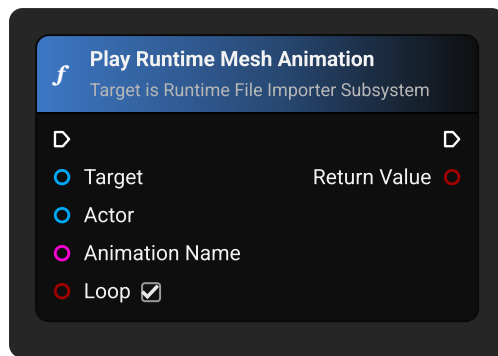
Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Runtime Mesh	Runtime Mesh Object Reference	The Runtime Mesh (an imported model).
Transform	Transform	Location, rotation and scale.
Keep Hierarchy	Boolean	Rebuild the model's node hierarchy as separate components (named after the nodes) that can be moved, hidden or given other materials. Off: one component shows the whole model. Default: <code>false</code>
Animation to Play ADVANCED	String	Name of an animation of the model to play on spawn (skeletal models). Empty: none. Default: <code>(empty)</code>
Loop Animation ADVANCED	Boolean	Loop Animation To Play. Default: <code>true</code>

OUTPUTS

Pin	Type	Description
Return Value	Runtime Mesh Actor Object Reference	Spawns an actor showing a Runtime Mesh.

Play Runtime Mesh Animation

FUNCTION



Plays one of a Runtime Mesh Actor's animations, by its name in the file (empty: the first one). Returns false if there's no such animation.

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::PlayRuntimeMeshAnimation`

INPUTS

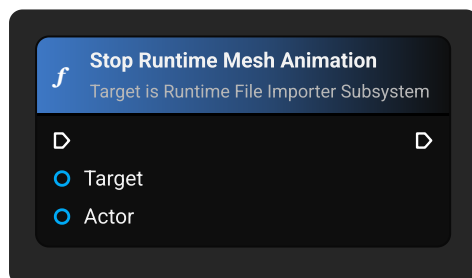
Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Actor	Runtime Mesh Actor Object Reference	The Runtime Mesh Actor.
Animation Name	String	Name of the animation in the model file. Empty: the first one.
Loop	Boolean	Play the animation in a loop. Default: <code>true</code>

OUTPUTS

Pin	Type	Description
Return Value	Boolean	False if there's no such animation.

Stop Runtime Mesh Animation

FUNCTION



Stops the animation a Runtime Mesh Actor is playing.

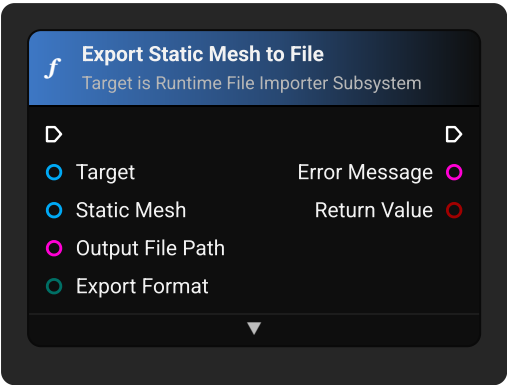
Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::StopRuntimeMeshAnimation`

INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Actor	Runtime Mesh Actor Object Reference	The Runtime Mesh Actor.

Export Static Mesh to File FUNCTION



Writes a Static Mesh to a 3D model file. In packaged games the mesh needs "Allow CPU Access" (meshes from this plugin always work). Material colors are exported for this plugin's materials, names for others. OBJ, glTF, USDA and Collada write their textures (and .mtl/.bin files) next to it.

Target: Runtime File Importer Subsystem
Async version: [Export Static Mesh To File \(Async\)](#)
C++: `URuntimeFileImporterSubsystem::ExportStaticMeshToFile`

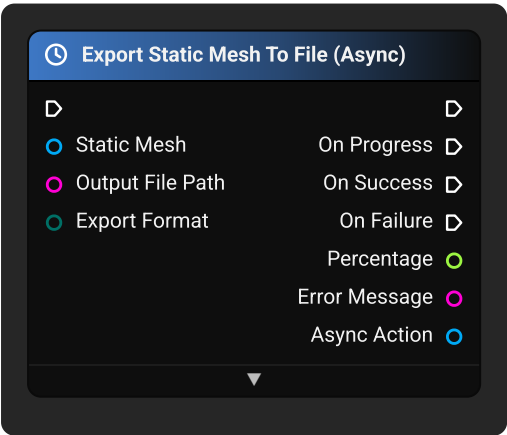
INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Static Mesh	Static Mesh Object Reference	The Static Mesh asset.
Output File Path	String	Full path of the file to write. Missing folders are created; an existing file is replaced.
Export Format	EMesh Export Format	Format to write.
Options	ADVANCED OPTIONAL Runtime Mesh Export Options	How a mesh is written to a 3D model file.

OUTPUTS

Pin	Type	Description
Error Message	String	Empty on success; otherwise why it failed.
Return Value	Boolean	True on success. False if it failed (Error Message says why).

Export Static Mesh To File (Async) ASYNC



Writes a Static Mesh to a 3D model file. In packaged games the mesh needs "Allow CPU Access" (meshes from this plugin always work).

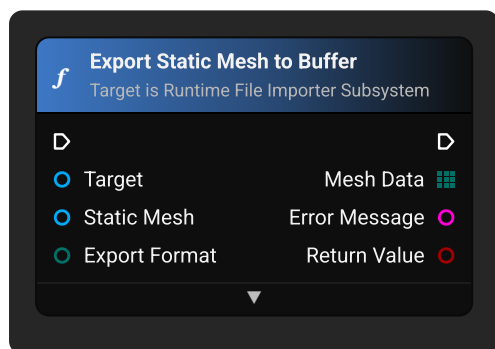
Blocking version: [Export Static Mesh to File](#)
C++: `UExportMeshToFileAsyncAction::ExportStaticMeshToFileAsync`

INPUTS

Pin	Type	Description
Static Mesh	Static Mesh Object Reference	The Static Mesh asset.

Pin	Type	Description
Output File Path	 String	Full path of the file to write. Missing folders are created; an existing file is replaced.
Export Format	 EMesh Export Format	Format to write.
Options ADVANCED OPTIONAL	 Runtime Mesh Export Options	How a mesh is written to a 3D model file.
OUTPUTS		
Pin	Type	Description
On Progress	 Exec	Fires as the operation advances (Percentage from 0 to 100). Other outputs are not set yet.
On Success	 Exec	Fires when the operation has finished successfully.
On Failure	 Exec	Fires if the operation failed or was cancelled; Error Message says why.
Percentage	 Float	Progress from 0 to 100.
Error Message	 String	Empty on success; otherwise why it failed.
Async Action	 Export Mesh to File Async Action Object Reference	This node's operation. Call Cancel on it to stop the operation (no more events fire).

Export Static Mesh to Buffer FUNCTION



Encodes a Static Mesh into 3D file bytes (e.g. to send over the network). Textures are embedded where the format allows it.

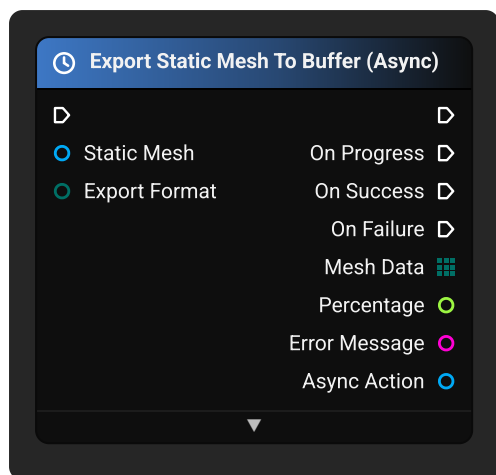
Target: Runtime File Importer Subsystem

Async version: [Export Static Mesh To Buffer \(Async\)](#)

C++: `URuntimeFileImporterSubsystem::ExportStaticMeshToBuffer`

INPUTS		
Pin	Type	Description
Target	 Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Static Mesh	 Static Mesh Object Reference	The Static Mesh asset.
Export Format	 EMesh Export Format	Format to write.
Options ADVANCED OPTIONAL	 Runtime Mesh Export Options	How a mesh is written to a 3D model file.
OUTPUTS		
Pin	Type	Description
Mesh Data	 Array of Byte	Bytes of a 3D model file.
Error Message	 String	Empty on success; otherwise why it failed.
Return Value	 Boolean	True on success. False if it failed (Error Message says why).

Export Static Mesh To Buffer (Async) ASYNC



Encodes a Static Mesh into 3D file bytes (e.g. to send over the network).

Blocking version: [Export Static Mesh to Buffer](#)

C++: `UExportMeshToBufferAsyncAction::ExportStaticMeshToBufferAsync`

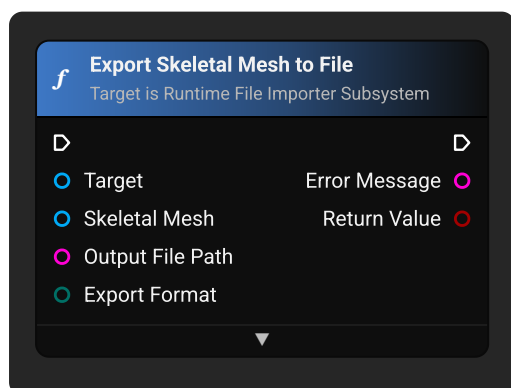
INPUTS

Pin	Type	Description
Static Mesh	Static Mesh Object Reference	The Static Mesh asset.
Export Format	EMesh Export Format	Format to write.
Options ADVANCED OPTIONAL	Runtime Mesh Export Options	How a mesh is written to a 3D model file.

OUTPUTS

Pin	Type	Description
On Progress	Exec	Fires as the operation advances (Percentage from 0 to 100). Other outputs are not set yet.
On Success	Exec	Fires when the operation has finished successfully.
On Failure	Exec	Fires if the operation failed or was cancelled; Error Message says why.
Mesh Data	Array of Byte	Bytes of a 3D model file.
Percentage	Float	Progress from 0 to 100.
Error Message	String	Empty on success; otherwise why it failed.
Async Action	Export Mesh to Buffer Async Action Object Reference	This node's operation. Call Cancel on it to stop the operation (no more events fire).

Export Skeletal Mesh to File FUNCTION



Writes a Skeletal Mesh (bones, skin weights, morph targets, materials) to a 3D model file, optionally with Animations. FBX and glTF keep the skeleton; other formats get the reference pose. In packaged games the mesh's LOD needs "Allow CPU Access" (meshes from this plugin always work).

Target: Runtime File Importer Subsystem

Async version: [Export Skeletal Mesh To File \(Async\)](#)

C++: `URuntimeFileImporterSubsystem::ExportSkeletalMeshToFile`

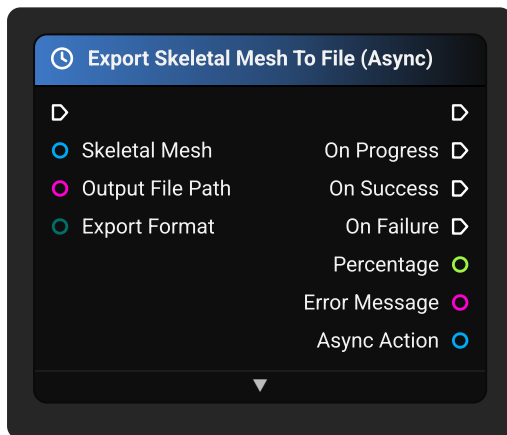
INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Skeletal Mesh	Skeletal Mesh Object Reference	The Skeletal Mesh asset.
Output File Path	String	Full path of the file to write. Missing folders are created; an existing file is replaced.
Export Format	EMesh Export Format	Format to write.
Options ADVANCED OPTIONAL	Runtime Mesh Export Options	How a mesh is written to a 3D model file.
Animations ADVANCED OPTIONAL	Array of Anim Sequence Object Reference	Animation Sequences to include.

OUTPUTS

Pin	Type	Description
Error Message	String	Empty on success; otherwise why it failed.
Return Value	Boolean	True on success. False if it failed (Error Message says why).

Export Skeletal Mesh To File (Async) ASYNC



Writes a Skeletal Mesh (bones, skin weights, morph targets, materials) to a 3D model file, optionally with Animations. FBX and glTF keep the skeleton; other formats get the reference pose. In packaged games the mesh's LOD needs "Allow CPU Access" (meshes from this plugin always work).

Blocking version: [Export Skeletal Mesh to File](#)

C++: `UEExportMeshToFileAsyncAction::ExportSkeletalMeshToFileAsync`

INPUTS

Pin	Type	Description
Skeletal Mesh	Skeletal Mesh Object Reference	The Skeletal Mesh asset.
Output File Path	String	Full path of the file to write. Missing folders are created; an existing file is replaced.
Export Format	EMesh Export Format	Format to write.
Options ADVANCED OPTIONAL	Runtime Mesh Export Options	How a mesh is written to a 3D model file.
Animations ADVANCED OPTIONAL	Array of Anim Sequence Object Reference	Animation Sequences to include.

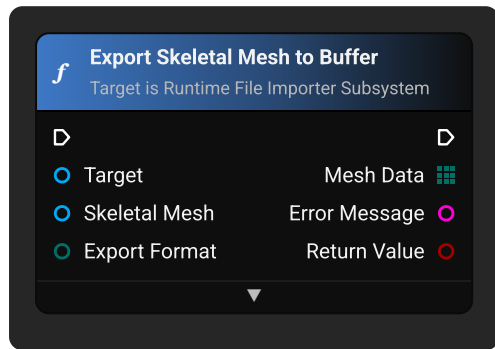
OUTPUTS

Pin	Type	Description
On Progress	Exec	Fires as the operation advances (Percentage from 0 to 100). Other outputs are not set yet.
On Success	Exec	Fires when the operation has finished successfully.

Pin	Type	Description
On Failure	 Exec	Fires if the operation failed or was cancelled; Error Message says why.
Percentage	 Float	Progress from 0 to 100.
Error Message	 String	Empty on success; otherwise why it failed.
Async Action	 Export Mesh to File Async Action Object Reference	This node's operation. Call Cancel on it to stop the operation (no more events fire).

Export Skeletal Mesh to Buffer

FUNCTION



Encodes a Skeletal Mesh (and optionally Animations) into 3D file bytes.

Target: Runtime File Importer Subsystem

Async version: [Export Skeletal Mesh To Buffer \(Async\)](#)

C++: `URuntimeFileImporterSubsystem::ExportSkeletalMeshToBuffer`

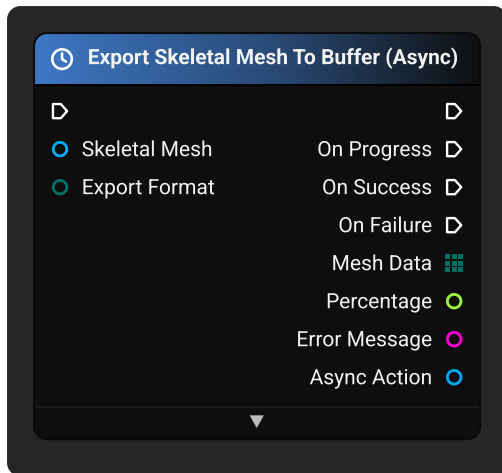
INPUTS

Pin	Type	Description
Target	 Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Skeletal Mesh	 Skeletal Mesh Object Reference	The Skeletal Mesh asset.
Export Format	 EMesh Export Format	Format to write.
Options ADVANCED OPTIONAL	 Runtime Mesh Export Options	How a mesh is written to a 3D model file.
Animations ADVANCED OPTIONAL	 Array of Anim Sequence Object Reference	Animation Sequences to include.

OUTPUTS

Pin	Type	Description
Mesh Data	 Array of Byte	Bytes of a 3D model file.
Error Message	 String	Empty on success; otherwise why it failed.
Return Value	 Boolean	True on success. False if it failed (Error Message says why).

Export Skeletal Mesh To Buffer (Async) ASYNC



Encodes a Skeletal Mesh (and optionally Animations) into 3D file bytes.

Blocking version: [Export Skeletal Mesh to Buffer](#)

C++: `UExportMeshToBufferAsyncAction::ExportSkeletalMeshToBufferAsync`

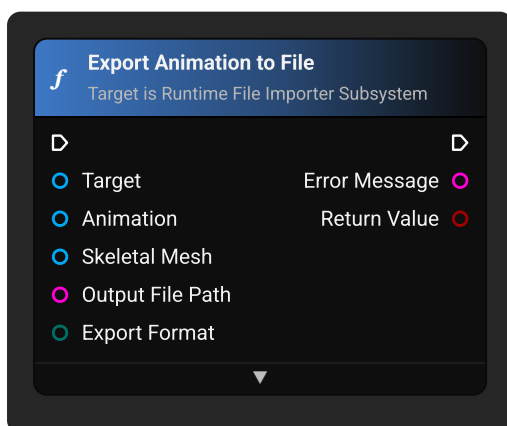
INPUTS

Pin	Type	Description
Skeletal Mesh	Skeletal Mesh Object Reference	The Skeletal Mesh asset.
Export Format	EMesh Export Format	Format to write.
Options ADVANCED OPTIONAL	Runtime Mesh Export Options	How a mesh is written to a 3D model file.
Animations ADVANCED OPTIONAL	Array of Anim Sequence Object Reference	Animation Sequences to include.

OUTPUTS

Pin	Type	Description
On Progress	Exec	Fires as the operation advances (Percentage from 0 to 100). Other outputs are not set yet.
On Success	Exec	Fires when the operation has finished successfully.
On Failure	Exec	Fires if the operation failed or was cancelled; Error Message says why.
Mesh Data	Array of Byte	Bytes of a 3D model file.
Percentage	Float	Progress from 0 to 100.
Error Message	String	Empty on success; otherwise why it failed.
Async Action	Export Mesh to Buffer Async Action Object Reference	This node's operation. Call Cancel on it to stop the operation (no more events fire).

Export Animation to File FUNCTION



Writes an Animation Sequence to an FBX or glTF file: on Skeletal Mesh (a skinned, animated model) or, without one, as an animated skeleton. Options' Animation Frame Rate sets the sampling (0: the animation's own rate).

Target: Runtime File Importer Subsystem

Async version: [Export Animation To File \(Async\)](#)

C++: `URuntimeFileImporterSubsystem::ExportAnimationToFile`

INPUTS

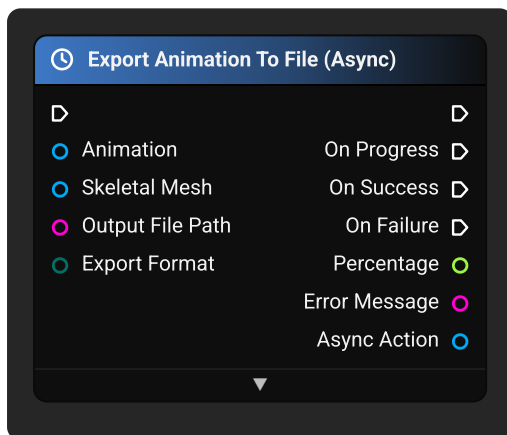
Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Animation	Anim Sequence Object Reference	The Animation Sequence.
Skeletal Mesh	Skeletal Mesh Object Reference	The Skeletal Mesh asset.
Output File Path	String	Full path of the file to write. Missing folders are created; an existing file is replaced.
Export Format	EMesh Export Format	Format to write.
Options	ADVANCED OPTIONAL Runtime Mesh Export Options	How a mesh is written to a 3D model file.

OUTPUTS

Pin	Type	Description
Error Message	String	Empty on success; otherwise why it failed.
Return Value	Boolean	True on success. False if it failed (Error Message says why).

Export Animation To File (Async)

ASYNC



Writes an Animation Sequence to an FBX or glTF file: on Skeletal Mesh (a skinned, animated model) or, without one, as an animated skeleton.

Blocking version: [Export Animation to File](#)

C++: `UExportMeshToFileAsyncAction::ExportAnimationToFileAsync`

INPUTS

Pin	Type	Description
Animation	Anim Sequence Object Reference	The Animation Sequence.
Skeletal Mesh	Skeletal Mesh Object Reference	The Skeletal Mesh asset.
Output File Path	String	Full path of the file to write. Missing folders are created; an existing file is replaced.
Export Format	EMesh Export Format	Format to write.
Options	ADVANCED OPTIONAL Runtime Mesh Export Options	How a mesh is written to a 3D model file.

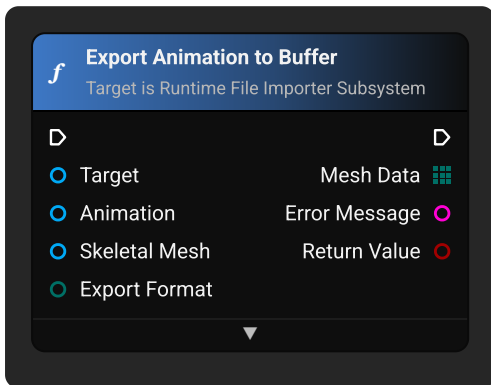
OUTPUTS

Pin	Type	Description
On Progress	Exec	Fires as the operation advances (Percentage from 0 to 100). Other outputs are not set yet.
On Success	Exec	Fires when the operation has finished successfully.

Pin	Type	Description
On Failure	Exec	Fires if the operation failed or was cancelled; Error Message says why.
Percentage	Float	Progress from 0 to 100.
Error Message	String	Empty on success; otherwise why it failed.
Async Action	Export Mesh to File Async Action Object Reference	This node's operation. Call Cancel on it to stop the operation (no more events fire).

Export Animation to Buffer

FUNCTION



Encodes an Animation Sequence (on Skeletal Mesh, or as an animated skeleton) into FBX or glTF bytes.

Target: Runtime File Importer Subsystem

Async version: [Export Animation To Buffer \(Async\)](#)

C++: `URuntimeFileImporterSubsystem::ExportAnimationToBuffer`

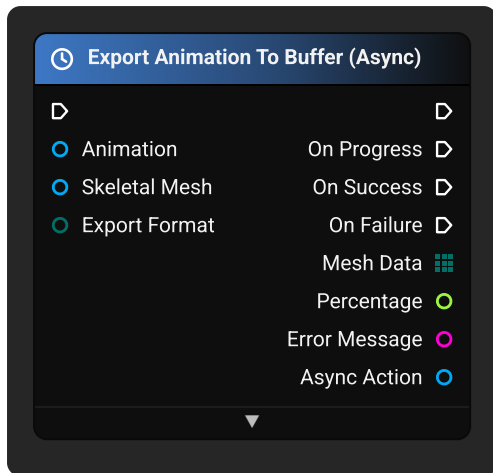
INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Animation	Anim Sequence Object Reference	The Animation Sequence.
Skeletal Mesh	Skeletal Mesh Object Reference	The Skeletal Mesh asset.
Export Format	EMesh Export Format	Format to write.
Options	Runtime Mesh Export Options	How a mesh is written to a 3D model file.

OUTPUTS

Pin	Type	Description
Mesh Data	Array of Byte	Bytes of a 3D model file.
Error Message	String	Empty on success; otherwise why it failed.
Return Value	Boolean	True on success. False if it failed (Error Message says why).

Export Animation To Buffer (Async) ASYNC



Encodes an Animation Sequence (on Skeletal Mesh, or as an animated skeleton) into FBX or glTF bytes.

Blocking version: [Export Animation to Buffer](#)

C++: `UExportMeshToBufferAsyncAction::ExportAnimationToBufferAsync`

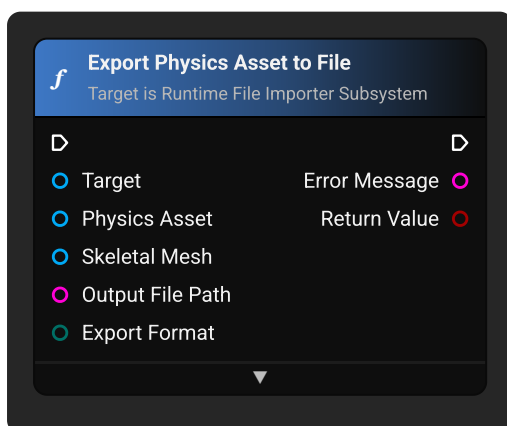
INPUTS

Pin	Type	Description
Animation	Anim Sequence Object Reference	The Animation Sequence.
Skeletal Mesh	Skeletal Mesh Object Reference	The Skeletal Mesh asset.
Export Format	EMesh Export Format	Format to write.
Options ADVANCED OPTIONAL	Runtime Mesh Export Options	How a mesh is written to a 3D model file.

OUTPUTS

Pin	Type	Description
On Progress	Exec	Fires as the operation advances (Percentage from 0 to 100). Other outputs are not set yet.
On Success	Exec	Fires when the operation has finished successfully.
On Failure	Exec	Fires if the operation failed or was cancelled; Error Message says why.
Mesh Data	Array of Byte	Bytes of a 3D model file.
Percentage	Float	Progress from 0 to 100.
Error Message	String	Empty on success; otherwise why it failed.
Async Action	Export Mesh to Buffer Async Action Object Reference	This node's operation. Call Cancel on it to stop the operation (no more events fire).

Export Physics Asset to File FUNCTION



Writes a Physics Asset's bodies (spheres, boxes, capsules, convex hulls) as meshes on the bones of Skeletal Mesh (empty: a loaded Skeletal Mesh using the asset).

Target: Runtime File Importer Subsystem

Async version: [Export Physics Asset To File \(Async\)](#)

C++: `URuntimeFileImporterSubsystem::ExportPhysicsAssetToFile`

INPUTS

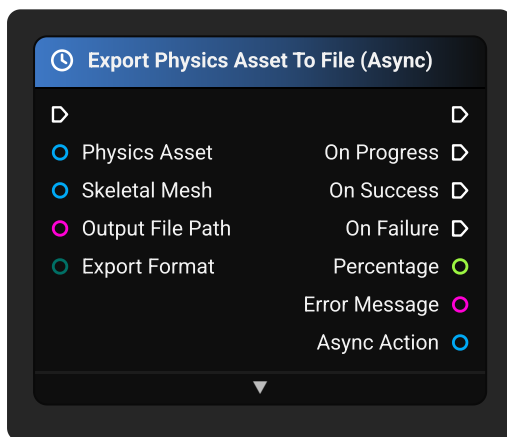
Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Physics Asset	Physics Asset Object Reference	The Physics Asset.
Skeletal Mesh	Skeletal Mesh Object Reference	The Skeletal Mesh asset.
Output File Path	String	Full path of the file to write. Missing folders are created; an existing file is replaced.
Export Format	EMesh Export Format	Format to write.
Options	ADVANCED OPTIONAL Runtime Mesh Export Options	How a mesh is written to a 3D model file.

OUTPUTS

Pin	Type	Description
Error Message	String	Empty on success; otherwise why it failed.
Return Value	Boolean	True on success. False if it failed (Error Message says why).

Export Physics Asset To File (Async)

ASYNC



Writes a Physics Asset's bodies as meshes on the bones of Skeletal Mesh (empty; a loaded Skeletal Mesh using the asset).

Blocking version: [Export Physics Asset to File](#)

C++: `UExportMeshToFileAsyncAction::ExportPhysicsAssetToFileAsync`

INPUTS

Pin	Type	Description
Physics Asset	Physics Asset Object Reference	The Physics Asset.
Skeletal Mesh	Skeletal Mesh Object Reference	The Skeletal Mesh asset.
Output File Path	String	Full path of the file to write. Missing folders are created; an existing file is replaced.
Export Format	EMesh Export Format	Format to write.
Options	ADVANCED OPTIONAL Runtime Mesh Export Options	How a mesh is written to a 3D model file.

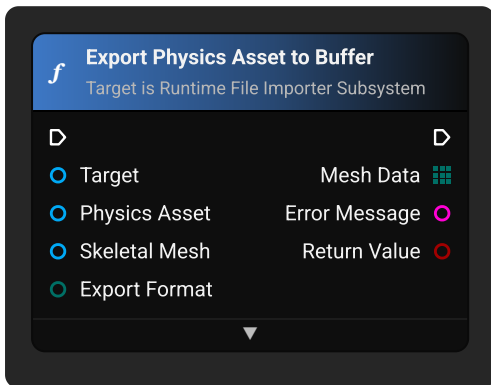
OUTPUTS

Pin	Type	Description
On Progress	Exec	Fires as the operation advances (Percentage from 0 to 100). Other outputs are not set yet.
On Success	Exec	Fires when the operation has finished successfully.

Pin	Type	Description
On Failure	 Exec	Fires if the operation failed or was cancelled; Error Message says why.
Percentage	 Float	Progress from 0 to 100.
Error Message	 String	Empty on success; otherwise why it failed.
Async Action	 Export Mesh to File Async Action Object Reference	This node's operation. Call Cancel on it to stop the operation (no more events fire).

Export Physics Asset to Buffer

FUNCTION



Encodes a Physics Asset's bodies into 3D file bytes.

Target: Runtime File Importer Subsystem

Async version: [Export Physics Asset To Buffer \(Async\)](#)

C++: `URuntimeFileImporterSubsystem::ExportPhysicsAssetToBuffer`

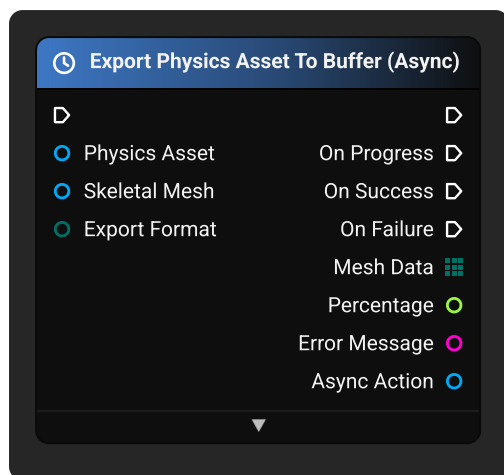
INPUTS

Pin	Type	Description
Target	 Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Physics Asset	 Physics Asset Object Reference	The Physics Asset.
Skeletal Mesh	 Skeletal Mesh Object Reference	The Skeletal Mesh asset.
Export Format	 EMesh Export Format	Format to write.
Options	ADVANCED OPTIONAL  Runtime Mesh Export Options	How a mesh is written to a 3D model file.

OUTPUTS

Pin	Type	Description
Mesh Data	 Array of Byte	Bytes of a 3D model file.
Error Message	 String	Empty on success; otherwise why it failed.
Return Value	 Boolean	True on success. False if it failed (Error Message says why).

Export Physics Asset To Buffer (Async) ASYNC



Encodes a Physics Asset's bodies (on Skeletal Mesh's bones) into 3D file bytes.

Blocking version: [Export Physics Asset to Buffer](#)

C++: `UExportMeshToBufferAsyncAction::ExportPhysicsAssetToBufferAsync`

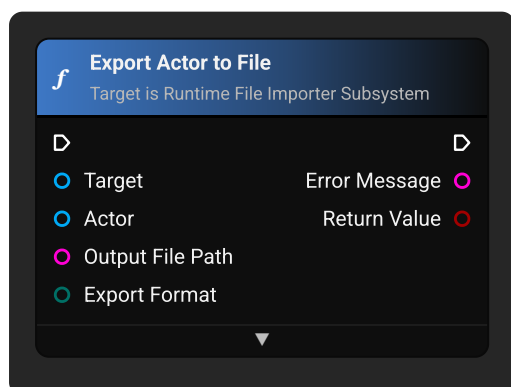
INPUTS

Pin	Type	Description
Physics Asset	Physics Asset Object Reference	The Physics Asset.
Skeletal Mesh	Skeletal Mesh Object Reference	The Skeletal Mesh asset.
Export Format	EMesh Export Format	Format to write.
Options ADVANCED OPTIONAL	Runtime Mesh Export Options	How a mesh is written to a 3D model file.

OUTPUTS

Pin	Type	Description
On Progress	Exec	Fires as the operation advances (Percentage from 0 to 100). Other outputs are not set yet.
On Success	Exec	Fires when the operation has finished successfully.
On Failure	Exec	Fires if the operation failed or was cancelled; Error Message says why.
Mesh Data	Array of Byte	Bytes of a 3D model file.
Percentage	Float	Progress from 0 to 100.
Error Message	String	Empty on success; otherwise why it failed.
Async Action	Export Mesh to Buffer Async Action Object Reference	This node's operation. Call Cancel on it to stop the operation (no more events fire).

Export Actor to File FUNCTION



Writes every visible static mesh (components and instances) and skeletal mesh (as posed now) of an actor, relative to the actor, e.g. to export a level piece or a player creation.

Target: Runtime File Importer Subsystem

Async version: [Export Actor To File \(Async\)](#)

C++: `URuntimeFileImporterSubsystem::ExportActorToFile`

INPUTS

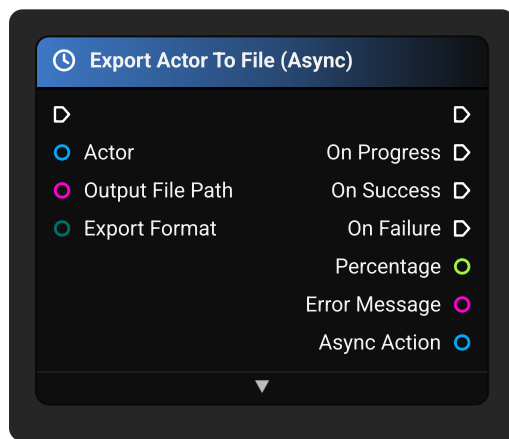
Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Actor	Actor Object Reference	The actor. Its visible static and skeletal mesh components are used.
Output File Path	String	Full path of the file to write. Missing folders are created; an existing file is replaced.
Export Format	EMesh Export Format	Format to write.
Options	ADVANCED OPTIONAL Runtime Mesh Export Options	How a mesh is written to a 3D model file.

OUTPUTS

Pin	Type	Description
Error Message	String	Empty on success; otherwise why it failed.
Return Value	Boolean	True on success. False if it failed (Error Message says why).

Export Actor To File (Async)

ASYNC



Writes every visible static mesh (and instance) and skeletal mesh (as posed now) of an actor, relative to the actor.

Blocking version: [Export Actor to File](#)

C++: `UExportMeshToFileAsyncAction::ExportActorToFileAsync`

INPUTS

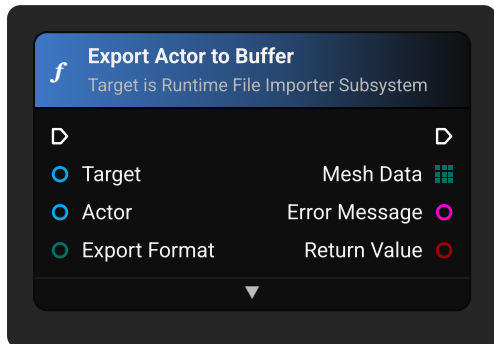
Pin	Type	Description
Actor	Actor Object Reference	The actor. Its visible static and skeletal mesh components are used.
Output File Path	String	Full path of the file to write. Missing folders are created; an existing file is replaced.
Export Format	EMesh Export Format	Format to write.
Options	ADVANCED OPTIONAL Runtime Mesh Export Options	How a mesh is written to a 3D model file.

OUTPUTS

Pin	Type	Description
On Progress	Exec	Fires as the operation advances (Percentage from 0 to 100). Other outputs are not set yet.
On Success	Exec	Fires when the operation has finished successfully.
On Failure	Exec	Fires if the operation failed or was cancelled; Error Message says why.
Percentage	Float	Progress from 0 to 100.
Error Message	String	Empty on success; otherwise why it failed.

Pin	Type	Description
Async Action	 Export Mesh to File Async Action Object Reference	This node's operation. Call Cancel on it to stop the operation (no more events fire).

Export Actor to Buffer FUNCTION



Encodes an actor's visible static and skeletal meshes into 3D file bytes.

Target: Runtime File Importer Subsystem

Async version: [Export Actor To Buffer \(Async\)](#)

C++: `URuntimeFileImporterSubsystem::ExportActorToBuffer`

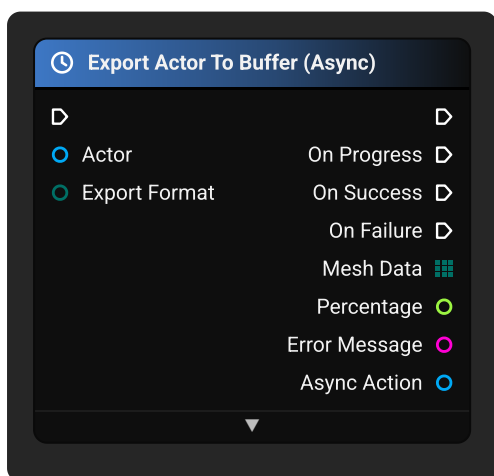
INPUTS

Pin	Type	Description
Target	 Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Actor	 Actor Object Reference	The actor. Its visible static and skeletal mesh components are used.
Export Format	 EMesh Export Format	Format to write.
Options ADVANCED OPTIONAL	 Runtime Mesh Export Options	How a mesh is written to a 3D model file.

OUTPUTS

Pin	Type	Description
Mesh Data	 Array of Byte	Bytes of a 3D model file.
Error Message	 String	Empty on success; otherwise why it failed.
Return Value	 Boolean	True on success. False if it failed (Error Message says why).

Export Actor To Buffer (Async) ASYNC



Encodes an actor's visible static and skeletal meshes into 3D file bytes.

Blocking version: [Export Actor to Buffer](#)

C++: `UExportMeshToBufferAsyncAction::ExportActorToBufferAsync`

INPUTS

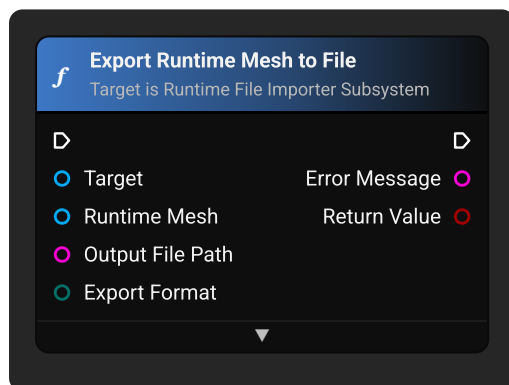
Pin	Type	Description
Actor	Actor Object Reference	The actor. Its visible static and skeletal mesh components are used.
Export Format	EMesh Export Format	Format to write.
Options	ADVANCED OPTIONAL Runtime Mesh Export Options	How a mesh is written to a 3D model file.

OUTPUTS

Pin	Type	Description
On Progress	Exec	Fires as the operation advances (Percentage from 0 to 100). Other outputs are not set yet.
On Success	Exec	Fires when the operation has finished successfully.
On Failure	Exec	Fires if the operation failed or was cancelled; Error Message says why.
Mesh Data	Array of Byte	Bytes of a 3D model file.
Percentage	Float	Progress from 0 to 100.
Error Message	String	Empty on success; otherwise why it failed.
Async Action	Export Mesh to Buffer Async Action Object Reference	This node's operation. Call Cancel on it to stop the operation (no more events fire).

Export Runtime Mesh to File

FUNCTION



Writes a Runtime Mesh (hierarchy, skeleton, animations, materials and textures included) to a 3D model file.

Target: Runtime File Importer Subsystem

Async version: [Export Runtime Mesh To File \(Async\)](#)

C++: `URuntimeFileImporterSubsystem::ExportRuntimeMeshToFile`

INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Runtime Mesh	Runtime Mesh Object Reference	The Runtime Mesh (an imported model).
Output File Path	String	Full path of the file to write. Missing folders are created; an existing file is replaced.
Export Format	EMesh Export Format	Format to write.
Options	ADVANCED OPTIONAL Runtime Mesh Export Options	How a mesh is written to a 3D model file.

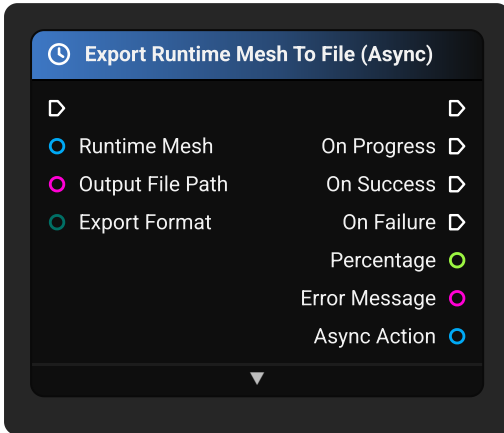
OUTPUTS

Pin	Type	Description
Error Message	String	Empty on success; otherwise why it failed.

Pin	Type	Description
Return Value	 Boolean	True on success. False if it failed (Error Message says why).

Export Runtime Mesh To File (Async)

ASYNC



Writes a Runtime Mesh (hierarchy, skeleton, animations, materials and textures included) to a 3D model file.

Blocking version: [Export Runtime Mesh to File](#)

C++: `UExportMeshToFileAsyncAction::ExportRuntimeMeshToFileAsync`

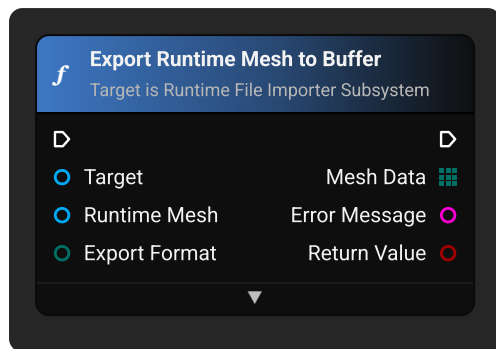
INPUTS

Pin	Type	Description
Runtime Mesh	 Runtime Mesh Object Reference	The Runtime Mesh (an imported model).
Output File Path	 String	Full path of the file to write. Missing folders are created; an existing file is replaced.
Export Format	 EMesh Export Format	Format to write.
Options	 Runtime Mesh Export Options	How a mesh is written to a 3D model file.

OUTPUTS

Pin	Type	Description
On Progress	 Exec	Fires as the operation advances (Percentage from 0 to 100). Other outputs are not set yet.
On Success	 Exec	Fires when the operation has finished successfully.
On Failure	 Exec	Fires if the operation failed or was cancelled; Error Message says why.
Percentage	 Float	Progress from 0 to 100.
Error Message	 String	Empty on success; otherwise why it failed.
Async Action	 Export Mesh to File Async Action Object Reference	This node's operation. Call Cancel on it to stop the operation (no more events fire).

Export Runtime Mesh to Buffer FUNCTION



Encodes a Runtime Mesh into bytes (e.g. to send over the network). Textures are embedded where the format allows it (GLB, FBX, USDZ, glTF).

Target: Runtime File Importer Subsystem

Async version: [Export Runtime Mesh To Buffer \(Async\)](#)

C++: `URuntimeFileImporterSubsystem::ExportRuntimeMeshToBuffer`

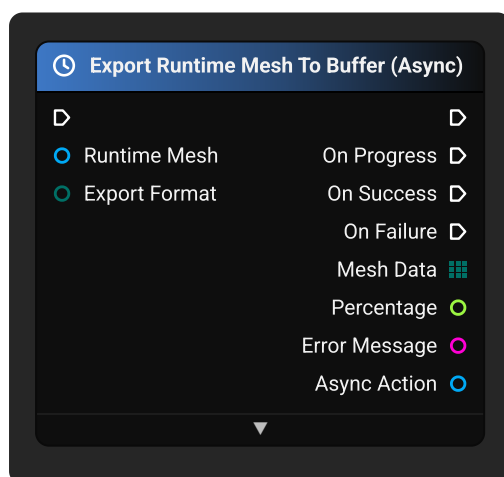
INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Runtime Mesh	Runtime Mesh Object Reference	The Runtime Mesh (an imported model).
Export Format	EMesh Export Format	Format to write.
Options	ADVANCED OPTIONAL Runtime Mesh Export Options	How a mesh is written to a 3D model file.

OUTPUTS

Pin	Type	Description
Mesh Data	Array of Byte	Bytes of a 3D model file.
Error Message	String	Empty on success; otherwise why it failed.
Return Value	Boolean	True on success. False if it failed (Error Message says why).

Export Runtime Mesh To Buffer (Async) ASYNC



Encodes a Runtime Mesh into 3D file bytes (e.g. to send over the network).

Blocking version: [Export Runtime Mesh to Buffer](#)

C++: `UExportMeshToBufferAsyncAction::ExportRuntimeMeshToBufferAsync`

INPUTS

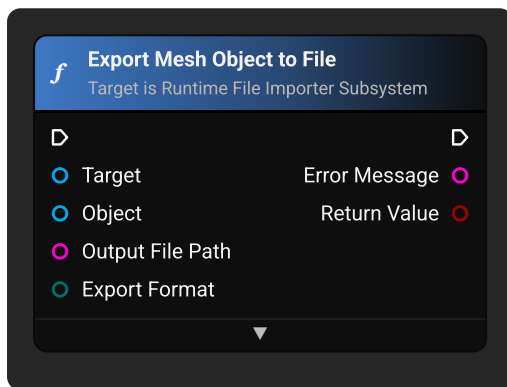
Pin	Type	Description
Runtime Mesh	Runtime Mesh Object Reference	The Runtime Mesh (an imported model).
Export Format	EMesh Export Format	Format to write.
Options	ADVANCED OPTIONAL Runtime Mesh Export Options	How a mesh is written to a 3D model file.

OUTPUTS

Pin	Type	Description
On Progress	 Exec	Fires as the operation advances (Percentage from 0 to 100). Other outputs are not set yet.
On Success	 Exec	Fires when the operation has finished successfully.
On Failure	 Exec	Fires if the operation failed or was cancelled; Error Message says why.
Mesh Data	 Array of Byte	Bytes of a 3D model file.
Percentage	 Float	Progress from 0 to 100.
Error Message	 String	Empty on success; otherwise why it failed.
Async Action	 Export Mesh to Buffer Async Action Object Reference	This node's operation. Call Cancel on it to stop the operation (no more events fire).

Export Mesh Object to File

FUNCTION



Writes any mesh object to a 3D model file, detecting what it is: Static or Skeletal Mesh, Skeleton, Animation Sequence, Physics Asset, Runtime Mesh, actor, or Static/Skeletal Mesh Component. Source Options give what some sources need (Skeletal Mesh for animations and physics assets, extra Animations, a pose).

Target: Runtime File Importer Subsystem

Async version: [Export Mesh Object To File \(Async\)](#)

C++: `URuntimeFileImporterSubsystem::ExportMeshObjectToFile`

INPUTS

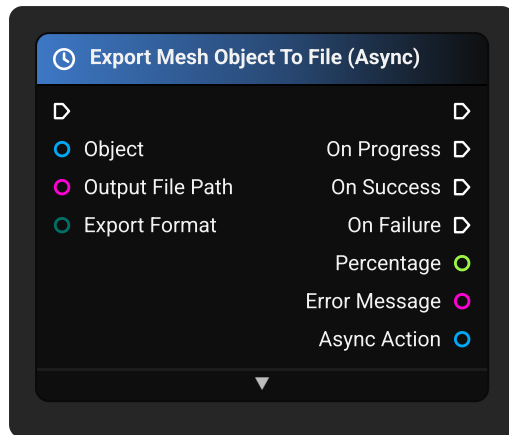
Pin	Type	Description
Target	 Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Object	 Object Object Reference	Any supported object (detected automatically).
Output File Path	 String	Full path of the file to write. Missing folders are created; an existing file is replaced.
Export Format	 EMesh Export Format	Format to write.
Options	 Runtime Mesh Export Options	How a mesh is written to a 3D model file.
Source Options	 Runtime Mesh Convert Options	What some sources need: a Skeletal Mesh for animations and physics assets, extra Animations, a pose.

OUTPUTS

Pin	Type	Description
Error Message	 String	Empty on success; otherwise why it failed.
Return Value	 Boolean	True on success. False if it failed (Error Message says why).

Export Mesh Object To File (Async)

ASYNC



Writes any mesh object to a 3D model file, detecting what it is: Static or Skeletal Mesh, Skeleton, Animation Sequence, Physics Asset, Runtime Mesh, actor, or Static/Skeletal Mesh Component. Source Options give what some sources need (Skeletal Mesh for animations and physics assets, extra Animations, a pose).

Blocking version: [Export Mesh Object to File](#)

C++: `UExportMeshToFileAsyncAction::ExportMeshObjectToFileAsync`

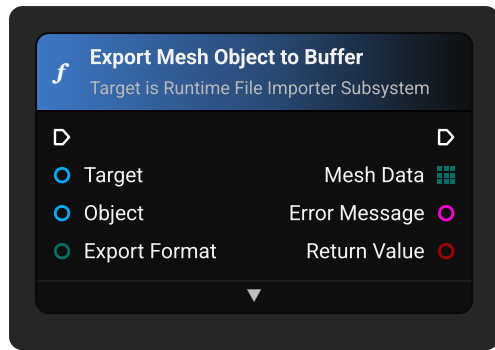
INPUTS

Pin	Type	Description
Object	Object Object Reference	Any supported object (detected automatically).
Output File Path	String	Full path of the file to write. Missing folders are created; an existing file is replaced.
Export Format	EMesh Export Format	Format to write.
Options ADVANCED OPTIONAL	Runtime Mesh Export Options	How a mesh is written to a 3D model file.
Source Options ADVANCED OPTIONAL	Runtime Mesh Convert Options	What some sources need: a Skeletal Mesh for animations and physics assets, extra Animations, a pose.

OUTPUTS

Pin	Type	Description
On Progress	Exec	Fires as the operation advances (Percentage from 0 to 100). Other outputs are not set yet.
On Success	Exec	Fires when the operation has finished successfully.
On Failure	Exec	Fires if the operation failed or was cancelled; Error Message says why.
Percentage	Float	Progress from 0 to 100.
Error Message	String	Empty on success; otherwise why it failed.
Async Action	Export Mesh to File Async Action Object Reference	This node's operation. Call Cancel on it to stop the operation (no more events fire).

Export Mesh Object to Buffer FUNCTION



Encodes any mesh object (detected automatically, see Export Mesh Object To File) into 3D file bytes.

Target: Runtime File Importer Subsystem

Async version: [Export Mesh Object To Buffer \(Async\)](#)

C++: `URuntimeFileImporterSubsystem::ExportMeshObjectToBuffer`

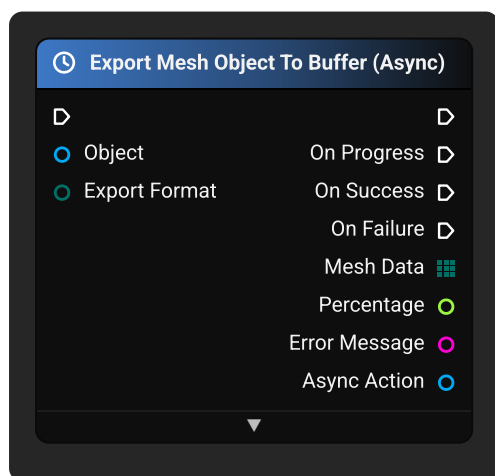
INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Object	Object Object Reference	Any supported object (detected automatically).
Export Format	EMesh Export Format	Format to write.
Options ADVANCED OPTIONAL	Runtime Mesh Export Options	How a mesh is written to a 3D model file.
Source Options ADVANCED OPTIONAL	Runtime Mesh Convert Options	What some sources need: a Skeletal Mesh for animations and physics assets, extra Animations, a pose.

OUTPUTS

Pin	Type	Description
Mesh Data	Array of Byte	Bytes of a 3D model file.
Error Message	String	Empty on success; otherwise why it failed.
Return Value	Boolean	True on success. False if it failed (Error Message says why).

Export Mesh Object To Buffer (Async) ASYNC



Encodes any mesh object (detected automatically, see Export Mesh Object To File) into 3D file bytes.

Blocking version: [Export Mesh Object To Buffer](#)

C++: `UExportMeshToBufferAsyncAction::ExportMeshObjectToBufferAsync`

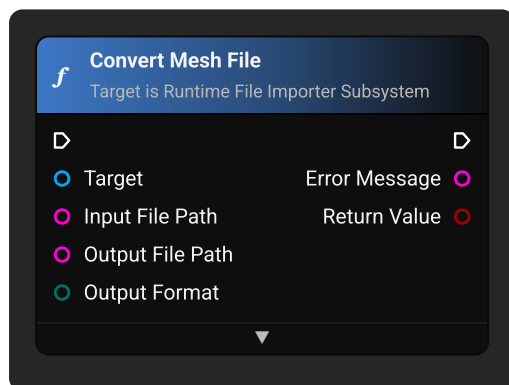
INPUTS

Pin	Type	Description
Object	Object Object Reference	Any supported object (detected automatically).

Pin	Type	Description
Export Format	EMesh Export Format	Format to write.
Options ADVANCED OPTIONAL	Runtime Mesh Export Options	How a mesh is written to a 3D model file.
Source Options ADVANCED OPTIONAL	Runtime Mesh Convert Options	What some sources need: a Skeletal Mesh for animations and physics assets, extra Animations, a pose.
OUTPUTS		
Pin	Type	Description
On Progress	Exec	Fires as the operation advances (Percentage from 0 to 100). Other outputs are not set yet.
On Success	Exec	Fires when the operation has finished successfully.
On Failure	Exec	Fires if the operation failed or was cancelled; Error Message says why.
Mesh Data	Array of Byte	Bytes of a 3D model file.
Percentage	Float	Progress from 0 to 100.
Error Message	String	Empty on success; otherwise why it failed.
Async Action	Export Mesh to Buffer Async Action Object Reference	This node's operation. Call Cancel on it to stop the operation (no more events fire).

Convert Mesh File

FUNCTION



Converts a 3D model file to another format (e.g. FBX to GLB), materials, textures, skeletons and animations included. Import Options' axis, unit, scale and transform settings are applied to the result.

Target: Runtime File Importer Subsystem

Async version: [Convert Mesh File \(Async\)](#)

C++: `URuntimeFileImporterSubsystem::ConvertMeshFile`

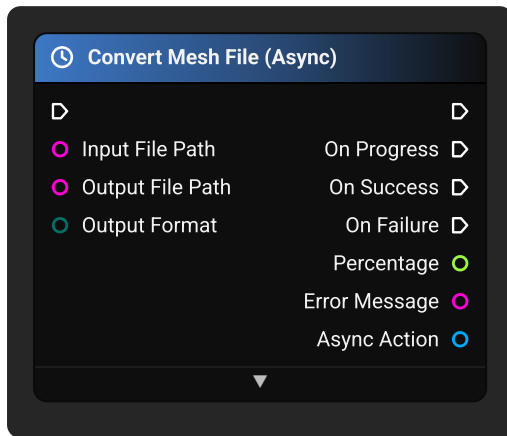
INPUTS		
Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Input File Path	String	Full path of the file to convert.
Output File Path	String	Full path of the file to write. Missing folders are created; an existing file is replaced.
Output Format	EMesh Export Format	Format to write.
Input Format ADVANCED	EMesh Format	Format of the input. Auto Detect reads it from the contents.
Import Options ADVANCED OPTIONAL	Runtime Mesh Import Options	How a 3D model file is read and turned into a Runtime Mesh.
Export Options ADVANCED OPTIONAL	Runtime Mesh Export Options	How a mesh is written to a 3D model file.

OUTPUTS

Pin	Type	Description
Error Message	 String	Empty on success; otherwise why it failed.
Return Value	 Boolean	True on success. False if it failed (Error Message says why).

Convert Mesh File (Async)

ASYNC



Converts a 3D model file to another format (e.g. FBX to GLB), materials, textures, skeletons and animations included.

Blocking version: [Convert Mesh File](#)

C++: `UExportMeshToFileAsyncAction::ConvertMeshFileAsync`

INPUTS

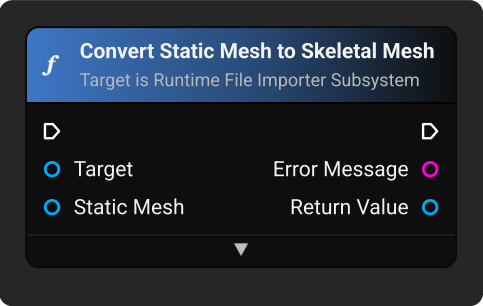
Pin	Type	Description
Input File Path	 String	Full path of the file to convert.
Output File Path	 String	Full path of the file to write. Missing folders are created; an existing file is replaced.
Output Format	 EMesh Export Format	Format to write.
Input Format ADVANCED	 EMesh Format	Format of the input. Auto Detect reads it from the contents.
Import Options ADVANCED OPTIONAL	 Runtime Mesh Import Options	How a 3D model file is read and turned into a Runtime Mesh.
Export Options ADVANCED OPTIONAL	 Runtime Mesh Export Options	How a mesh is written to a 3D model file.

OUTPUTS

Pin	Type	Description
On Progress	 Exec	Fires as the operation advances (Percentage from 0 to 100). Other outputs are not set yet.
On Success	 Exec	Fires when the operation has finished successfully.
On Failure	 Exec	Fires if the operation failed or was cancelled; Error Message says why.
Percentage	 Float	Progress from 0 to 100.
Error Message	 String	Empty on success; otherwise why it failed.
Async Action	 Export Mesh to File Async Action Object Reference	This node's operation. Call Cancel on it to stop the operation (no more events fire).

Convert Static Mesh to Skeletal Mesh

FUNCTION



Makes a Skeletal Mesh (with a new Skeleton, or Options' Skeleton, and optionally a Physics Asset) from a Static Mesh, bound to a single root bone. Materials are kept. For a bone per node of a model, use Convert Mesh Object To Skeletal Mesh on its Runtime Mesh.

Target: Runtime File Importer Subsystem

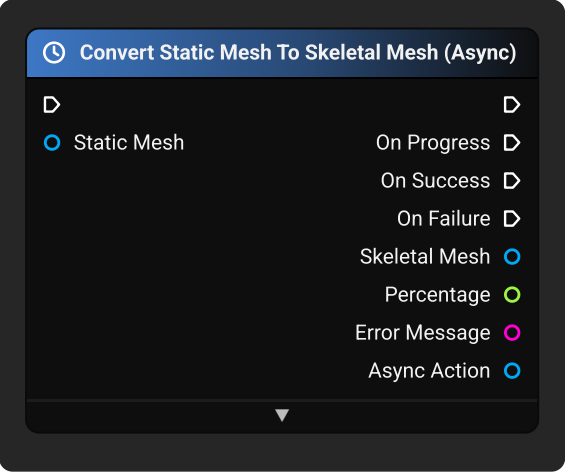
Async version: [Convert Static Mesh To Skeletal Mesh \(Async\)](#)

C++: `URuntimeFileImporterSubsystem::ConvertStaticMeshToSkeletalMesh`

INPUTS		
Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Static Mesh	Static Mesh Object Reference	The Static Mesh asset.
Options	ADVANCEDOPTIONAL Runtime Mesh Convert Options	How Convert nodes turn one kind of mesh object into another.
OUTPUTS		
Pin	Type	Description
Error Message	String	Empty on success; otherwise why it failed.
Return Value	Skeletal Mesh Object Reference	The result, or None/empty if it failed (Error Message says why).

Convert Static Mesh To Skeletal Mesh (Async)

ASYNC



Makes a Skeletal Mesh (with a Skeleton and, optionally, a Physics Asset) from a Static Mesh, bound to a single root bone.

Blocking version: [Convert Static Mesh to Skeletal Mesh](#)

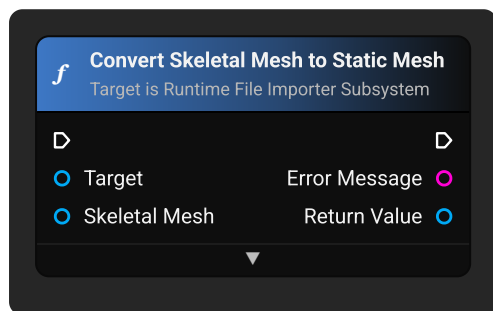
C++:

`UConvertToSkeletalMeshAsyncAction::ConvertStaticMeshToSkeletalMeshAsync`

INPUTS		
Pin	Type	Description
Static Mesh	Static Mesh Object Reference	The Static Mesh asset.
Options	ADVANCEDOPTIONAL Runtime Mesh Convert Options	How Convert nodes turn one kind of mesh object into another.
OUTPUTS		
Pin	Type	Description
On Progress	Exec	Fires as the operation advances (Percentage from 0 to 100). Other outputs are not set yet.

Pin	Type	Description
On Success	Exec	Fires when the operation has finished successfully.
On Failure	Exec	Fires if the operation failed or was cancelled; Error Message says why.
Skeletal Mesh	Skeletal Mesh Object Reference	The Skeletal Mesh asset.
Percentage	Float	Progress from 0 to 100.
Error Message	String	Empty on success; otherwise why it failed.
Async Action	Convert to Skeletal Mesh Async Action Object Reference	This node's operation. Call Cancel on it to stop the operation (no more events fire).

Convert Skeletal Mesh to Static Mesh FUNCTION



Makes a Static Mesh from a Skeletal Mesh, in its reference pose or posed by Options' Pose Animation at Pose Time. Materials are kept.

Target: Runtime File Importer Subsystem

Async version: [Convert Skeletal Mesh To Static Mesh \(Async\)](#)

C++: `URuntimeFileImporterSubsystem::ConvertSkeletalMeshToStaticMesh`

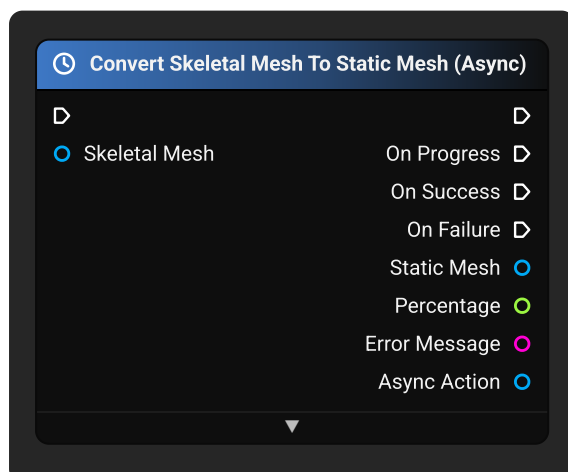
INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Skeletal Mesh	Skeletal Mesh Object Reference	The Skeletal Mesh asset.
Options	ADVANCED OPTIONAL Runtime Mesh Convert Options	How Convert nodes turn one kind of mesh object into another.

OUTPUTS

Pin	Type	Description
Error Message	String	Empty on success; otherwise why it failed.
Return Value	Static Mesh Object Reference	The result, or None/empty if it failed (Error Message says why).

Convert Skeletal Mesh To Static Mesh (Async) ASYNC



Makes a Static Mesh from a Skeletal Mesh, in its reference pose or posed by Options' Pose Animation at Pose Time.

Blocking version: [Convert Skeletal Mesh to Static Mesh](#)

C++:

`UConvertToStaticMeshAsyncAction::ConvertSkeletalMeshToStaticMeshAsync`

INPUTS

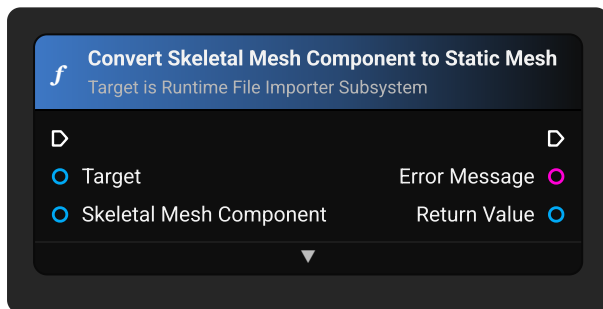
Pin	Type	Description
Skeletal Mesh	Skeletal Mesh Object Reference	The Skeletal Mesh asset.
Options	ADVANCED OPTIONAL Runtime Mesh Convert Options	How Convert nodes turn one kind of mesh object into another.

OUTPUTS

Pin	Type	Description
On Progress	Exec	Fires as the operation advances (Percentage from 0 to 100). Other outputs are not set yet.
On Success	Exec	Fires when the operation has finished successfully.
On Failure	Exec	Fires if the operation failed or was cancelled; Error Message says why.
Static Mesh	Static Mesh Object Reference	The Static Mesh asset.
Percentage	Float	Progress from 0 to 100.
Error Message	String	Empty on success; otherwise why it failed.
Async Action	Convert to Static Mesh Async Action Object Reference	This node's operation. Call Cancel on it to stop the operation (no more events fire).

Convert Skeletal Mesh Component to Static Mesh

FUNCTION



Makes a Static Mesh of a Skeletal Mesh Component as it is posed right now (animation, morph targets, materials), e.g. to freeze a character.

Target: Runtime File Importer Subsystem

Async version: [Convert Skeletal Mesh Component To Static Mesh \(Async\)](#)

C++:

```
URuntimeFileImporterSubsystem::ConvertSkeletalMeshComponentToStaticMesh
```

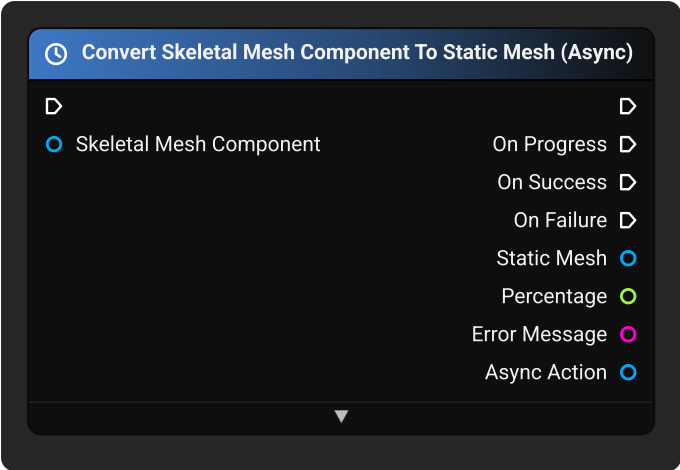
INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Skeletal Mesh Component	Skeletal Mesh Component Object Reference	The component, as posed right now.
Options	ADVANCED OPTIONAL Runtime Mesh Convert Options	How Convert nodes turn one kind of mesh object into another.

OUTPUTS

Pin	Type	Description
Error Message	String	Empty on success; otherwise why it failed.
Return Value	Static Mesh Object Reference	The result, or None/empty if it failed (Error Message says why).

Convert Skeletal Mesh Component To Static Mesh (Async) ASYNC



Makes a Static Mesh of a Skeletal Mesh Component as it is posed right now (animation, morph targets, materials).

Blocking version: [Convert Skeletal Mesh Component to Static Mesh](#)

C++:

```
UConvertToStaticMeshAsyncAction::ConvertSkeletalMeshComponentToStaticMeshAsync
```

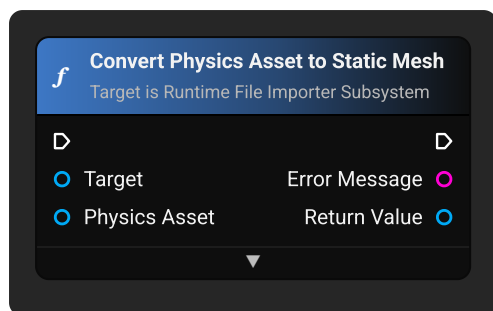
INPUTS

Pin	Type	Description
Skeletal Mesh Component	Skeletal Mesh Component Object Reference	The component, as posed right now.
Options ADVANCED OPTIONAL	Runtime Mesh Convert Options	How Convert nodes turn one kind of mesh object into another.

OUTPUTS

Pin	Type	Description
On Progress	Exec	Fires as the operation advances (Percentage from 0 to 100). Other outputs are not set yet.
On Success	Exec	Fires when the operation has finished successfully.
On Failure	Exec	Fires if the operation failed or was cancelled; Error Message says why.
Static Mesh	Static Mesh Object Reference	The Static Mesh asset.
Percentage	Float	Progress from 0 to 100.
Error Message	String	Empty on success; otherwise why it failed.
Async Action	Convert to Static Mesh Async Action Object Reference	This node's operation. Call Cancel on it to stop the operation (no more events fire).

Convert Physics Asset to Static Mesh FUNCTION



Makes a Static Mesh of a Physics Asset's bodies, placed on the bones of Options' Skeletal Mesh (empty: a loaded Skeletal Mesh using the asset).

Target: Runtime File Importer Subsystem

Async version: [Convert Physics Asset To Static Mesh \(Async\)](#)

C++: `URuntimeFileImporterSubsystem::ConvertPhysicsAssetToStaticMesh`

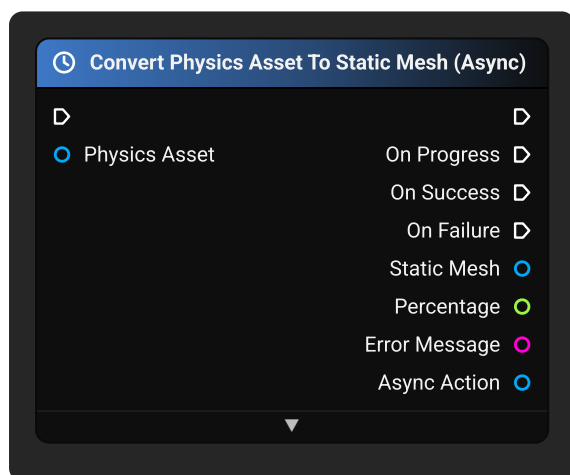
INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Physics Asset	Physics Asset Object Reference	The Physics Asset.
Options ADVANCED OPTIONAL	Runtime Mesh Convert Options	How Convert nodes turn one kind of mesh object into another.

OUTPUTS

Pin	Type	Description
Error Message	String	Empty on success; otherwise why it failed.
Return Value	Static Mesh Object Reference	The result, or None/empty if it failed (Error Message says why).

Convert Physics Asset To Static Mesh (Async) ASYNC



Makes a Static Mesh of a Physics Asset's bodies, placed on the bones of Options' Skeletal Mesh (empty: a loaded Skeletal Mesh using the asset).

Blocking version: [Convert Physics Asset to Static Mesh](#)

C++:

`UConvertToStaticMeshAsyncAction::ConvertPhysicsAssetToStaticMeshAsync`

INPUTS

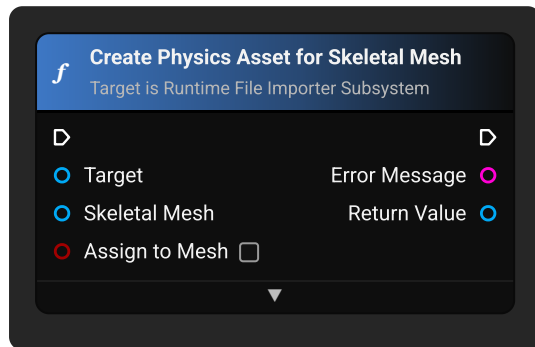
Pin	Type	Description
Physics Asset	Physics Asset Object Reference	The Physics Asset.
Options ADVANCED OPTIONAL	Runtime Mesh Convert Options	How Convert nodes turn one kind of mesh object into another.

OUTPUTS

Pin	Type	Description
On Progress	Exec	Fires as the operation advances (Percentage from 0 to 100). Other outputs are not set yet.
On Success	Exec	Fires when the operation has finished successfully.
On Failure	Exec	Fires if the operation failed or was cancelled; Error Message says why.

Pin	Type	Description
Static Mesh	 Static Mesh Object Reference	The Static Mesh asset.
Percentage	 Float	Progress from 0 to 100.
Error Message	 String	Empty on success; otherwise why it failed.
Async Action	 Convert to Static Mesh Async Action Object Reference	This node's operation. Call Cancel on it to stop the operation (no more events fire).

Create Physics Asset for Skeletal Mesh FUNCTION



Generates a Physics Asset for a Skeletal Mesh: a body fitted to the vertices of each bone (Physics Asset Options choose the shape and size limit), joined by constraints for ragdolls. With Assign To Mesh, the mesh uses it right away.

Target: Runtime File Importer Subsystem

Async version: [Create Physics Asset For Skeletal Mesh \(Async\)](#)

C++:

```
URuntimeFileImporterSubsystem::CreatePhysicsAssetForSkeletalMesh
```

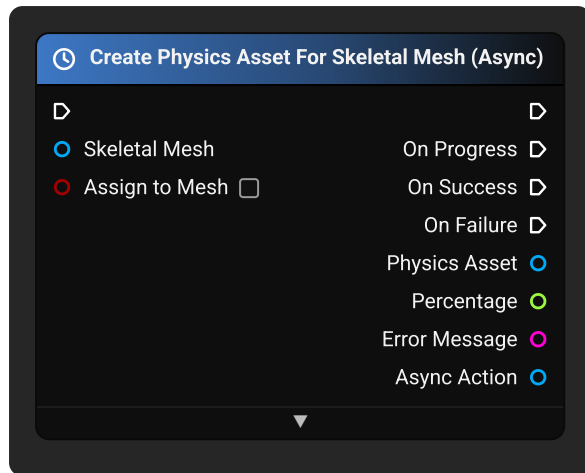
INPUTS

Pin	Type	Description
Target	 Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Skeletal Mesh	 Skeletal Mesh Object Reference	The Skeletal Mesh asset.
Physics Asset Options	ADVANCED OPTIONAL  Runtime Physics Asset Options	Shape and size limit of the generated bodies.
Assign to Mesh	 Boolean	Make the Skeletal Mesh use the new Physics Asset right away. Default: <code>false</code>

OUTPUTS

Pin	Type	Description
Error Message	 String	Empty on success; otherwise why it failed.
Return Value	 Physics Asset Object Reference	The result, or None/empty if it failed (Error Message says why).

Create Physics Asset For Skeletal Mesh (Async) ASYNC



Generates a Physics Asset for a Skeletal Mesh: a body fitted to the vertices of each bone, joined by constraints. With Assign To Mesh, the mesh uses it right away (components showing it get the new collision).

Blocking version: [Create Physics Asset for Skeletal Mesh](#)

C++:

```
UCreatePhysicsAssetAsyncAction::CreatePhysicsAssetForSkeletalMeshAsync
```

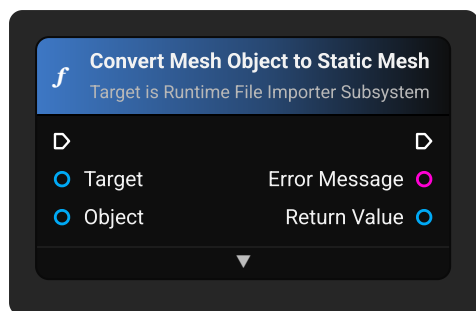
INPUTS

Pin	Type	Description
Skeletal Mesh	Skeletal Mesh Object Reference	The Skeletal Mesh asset.
Physics Asset Options ADVANCED OPTIONAL	Runtime Physics Asset Options	Shape and size limit of the generated bodies.
Assign to Mesh	Boolean	Make the Skeletal Mesh use the new Physics Asset right away. Default: <code>false</code>

OUTPUTS

Pin	Type	Description
On Progress	Exec	Fires as the operation advances (Percentage from 0 to 100). Other outputs are not set yet.
On Success	Exec	Fires when the operation has finished successfully.
On Failure	Exec	Fires if the operation failed or was cancelled; Error Message says why.
Physics Asset	Physics Asset Object Reference	The Physics Asset.
Percentage	Float	Progress from 0 to 100.
Error Message	String	Empty on success; otherwise why it failed.
Async Action	Create Physics Asset Async Action Object Reference	This node's operation. Call Cancel on it to stop the operation (no more events fire).

Convert Mesh Object to Static Mesh FUNCTION



Makes a Static Mesh from any mesh object, detecting what it is (skeletal sources are posed; actors and hierarchies are merged into one mesh).

Target: Runtime File Importer Subsystem

Async version: [Convert Mesh Object To Static Mesh \(Async\)](#)

C++: `URuntimeFileImporterSubsystem::ConvertMeshObjectToStaticMesh`

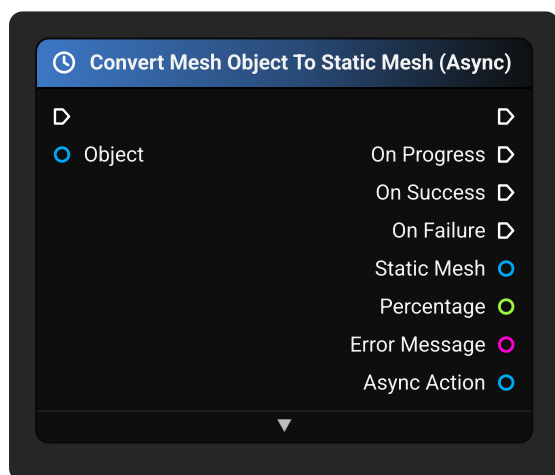
INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Object	Object Object Reference	Any supported object (detected automatically).
Options	ADVANCED OPTIONAL Runtime Mesh Convert Options	How Convert nodes turn one kind of mesh object into another.

OUTPUTS

Pin	Type	Description
Error Message	String	Empty on success; otherwise why it failed.
Return Value	Static Mesh Object Reference	The result, or None/empty if it failed (Error Message says why).

Convert Mesh Object To Static Mesh (Async) ASYNC



Makes a Static Mesh from any mesh object, detecting what it is (skeletal sources are posed; actors are merged into one mesh).

Blocking version: [Convert Mesh Object to Static Mesh](#)

C++:

`UConvertToStaticMeshAsyncAction::ConvertMeshObjectToStaticMeshAsync`

INPUTS

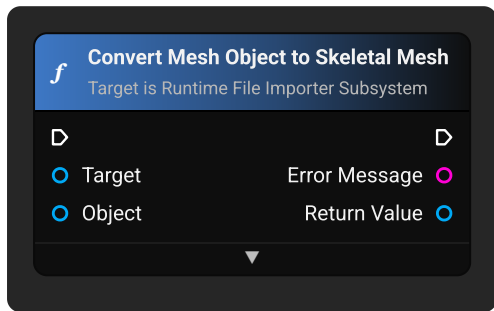
Pin	Type	Description
Object	Object Object Reference	Any supported object (detected automatically).
Options	ADVANCED OPTIONAL Runtime Mesh Convert Options	How Convert nodes turn one kind of mesh object into another.

OUTPUTS

Pin	Type	Description
On Progress	Exec	Fires as the operation advances (Percentage from 0 to 100). Other outputs are not set yet.
On Success	Exec	Fires when the operation has finished successfully.
On Failure	Exec	Fires if the operation failed or was cancelled; Error Message says why.

Pin	Type	Description
Static Mesh	Static Mesh Object Reference	The Static Mesh asset.
Percentage	Float	Progress from 0 to 100.
Error Message	String	Empty on success; otherwise why it failed.
Async Action	Convert to Static Mesh Async Action Object Reference	This node's operation. Call Cancel on it to stop the operation (no more events fire).

Convert Mesh Object to Skeletal Mesh FUNCTION



Makes a Skeletal Mesh from any mesh object, detecting what it is (hierarchies get a bone per node with Bone Per Node).

Target: Runtime File Importer Subsystem

Async version: [Convert Mesh Object To Skeletal Mesh \(Async\)](#)

C++: `URuntimeFileImporterSubsystem::ConvertMeshObjectToSkeletalMesh`

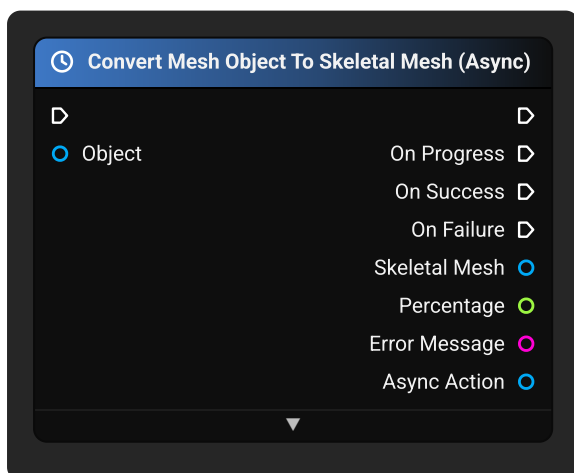
INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Object	Object Object Reference	Any supported object (detected automatically).
Options ADVANCED OPTIONAL	Runtime Mesh Convert Options	How Convert nodes turn one kind of mesh object into another.

OUTPUTS

Pin	Type	Description
Error Message	String	Empty on success; otherwise why it failed.
Return Value	Skeletal Mesh Object Reference	The result, or None/empty if it failed (Error Message says why).

Convert Mesh Object To Skeletal Mesh (Async) ASYNC



Makes a Skeletal Mesh from any mesh object, detecting what it is (hierarchies get a bone per node with Bone Per Node).

Blocking version: [Convert Mesh Object to Skeletal Mesh](#)

C++:

`UConvertToSkeletalMeshAsyncAction::ConvertMeshObjectToSkeletalMeshAsync`

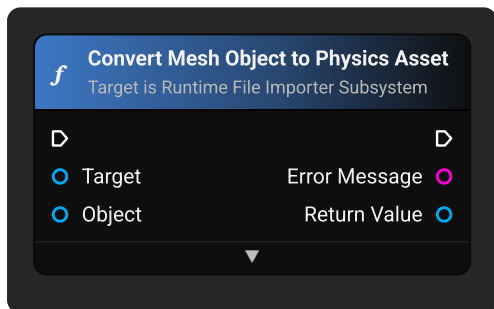
INPUTS

Pin	Type	Description
Object	 Object Object Reference	Any supported object (detected automatically).
Options	ADVANCED OPTIONAL  Runtime Mesh Convert Options	How Convert nodes turn one kind of mesh object into another.

OUTPUTS

Pin	Type	Description
On Progress	 Exec	Fires as the operation advances (Percentage from 0 to 100). Other outputs are not set yet.
On Success	 Exec	Fires when the operation has finished successfully.
On Failure	 Exec	Fires if the operation failed or was cancelled; Error Message says why.
Skeletal Mesh	 Skeletal Mesh Object Reference	The Skeletal Mesh asset.
Percentage	 Float	Progress from 0 to 100.
Error Message	 String	Empty on success; otherwise why it failed.
Async Action	 Convert to Skeletal Mesh Async Action Object Reference	This node's operation. Call Cancel on it to stop the operation (no more events fire).

Convert Mesh Object to Physics Asset FUNCTION



Makes a Physics Asset from any mesh object (non-skeletal sources first become a Skeletal Mesh, which the asset is made for).

Target: Runtime File Importer Subsystem

Async version: [Convert Mesh Object To Physics Asset \(Async\)](#)

C++: `URuntimeFileImporterSubsystem::ConvertMeshObjectToPhysicsAsset`

INPUTS

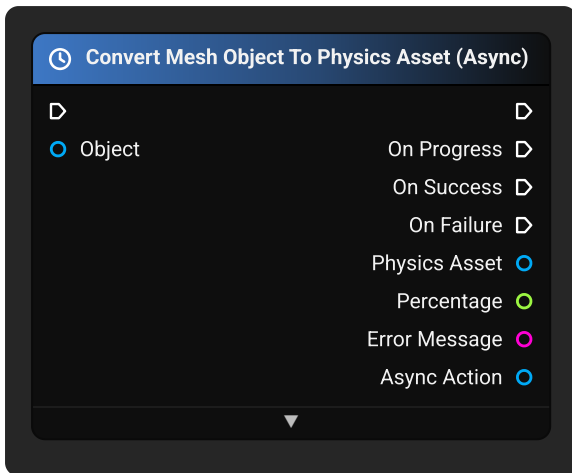
Pin	Type	Description
Target	 Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Object	 Object Object Reference	Any supported object (detected automatically).
Options	ADVANCED OPTIONAL  Runtime Mesh Convert Options	How Convert nodes turn one kind of mesh object into another.

OUTPUTS

Pin	Type	Description
Error Message	 String	Empty on success; otherwise why it failed.
Return Value	 Physics Asset Object Reference	The result, or None/empty if it failed (Error Message says why).

Convert Mesh Object To Physics Asset (Async)

ASYNC



Makes a Physics Asset from any mesh object (non-skeletal sources first become a Skeletal Mesh, which the asset is made for).

Blocking version: [Convert Mesh Object to Physics Asset](#)

C++:

```
UCreatePhysicsAssetAsyncAction::ConvertMeshObjectToPhysicsAssetAsync
```

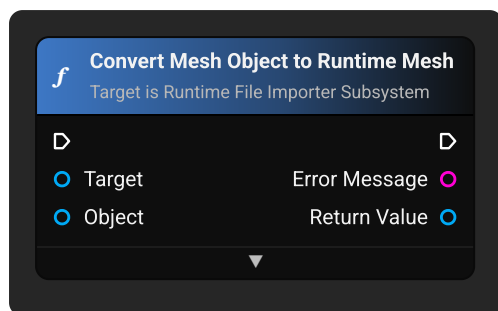
INPUTS

Pin	Type	Description
Object	Object Object Reference	Any supported object (detected automatically).
Options	Runtime Mesh Convert Options	How Convert nodes turn one kind of mesh object into another.

OUTPUTS

Pin	Type	Description
On Progress	Exec	Fires as the operation advances (Percentage from 0 to 100). Other outputs are not set yet.
On Success	Exec	Fires when the operation has finished successfully.
On Failure	Exec	Fires if the operation failed or was cancelled; Error Message says why.
Physics Asset	Physics Asset Object Reference	The Physics Asset.
Percentage	Float	Progress from 0 to 100.
Error Message	String	Empty on success; otherwise why it failed.
Async Action	Create Physics Asset Async Action Object Reference	This node's operation. Call Cancel on it to stop the operation (no more events fire).

Convert Mesh Object to Runtime Mesh FUNCTION



Makes a Runtime Mesh from any mesh object, detecting what it is: it keeps everything this plugin works with (hierarchy, skeleton, Options' Animations, materials), for Spawn Runtime Mesh Actor or export.

Target: Runtime File Importer Subsystem

Async version: [Convert Mesh Object To Runtime Mesh \(Async\)](#)

C++: `URuntimeFileImporterSubsystem::ConvertMeshObjectToRuntimeMesh`

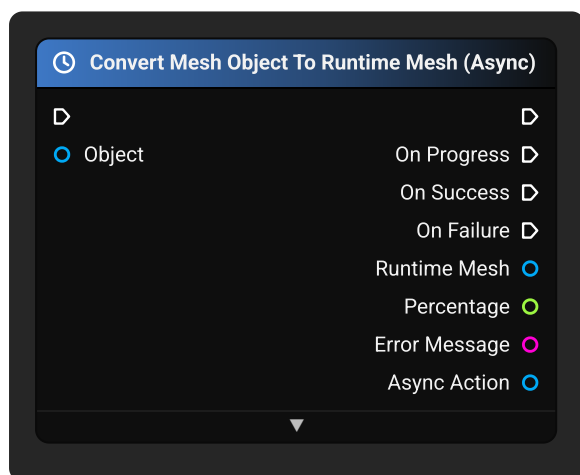
INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Object	Object Object Reference	Any supported object (detected automatically).
Options	Runtime Mesh Convert Options	How Convert nodes turn one kind of mesh object into another.

OUTPUTS

Pin	Type	Description
Error Message	String	Empty on success; otherwise why it failed.
Return Value	Runtime Mesh Object Reference	The result, or None/empty if it failed (Error Message says why).

Convert Mesh Object To Runtime Mesh (Async) ASYNC



Makes a Runtime Mesh from any mesh object, detecting what it is: it keeps everything this plugin works with (hierarchy, skeleton, Options' Animations, materials), for Spawn Runtime Mesh Actor or export.

Blocking version: [Convert Mesh Object to Runtime Mesh](#)

C++:

`UConvertToRuntimeMeshAsyncAction::ConvertMeshObjectToRuntimeMeshAsync`

INPUTS

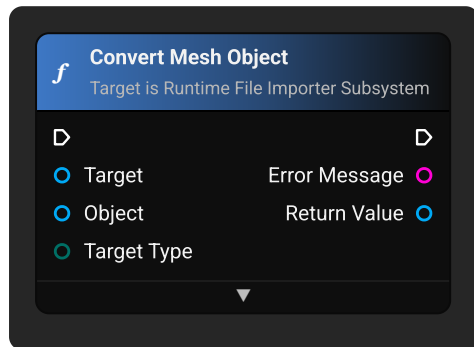
Pin	Type	Description
Object	Object Object Reference	Any supported object (detected automatically).
Options	Runtime Mesh Convert Options	How Convert nodes turn one kind of mesh object into another.

OUTPUTS

Pin	Type	Description
On Progress	Exec	Fires as the operation advances (Percentage from 0 to 100). Other outputs are not set yet.
On Success	Exec	Fires when the operation has finished successfully.
On Failure	Exec	Fires if the operation failed or was cancelled; Error Message says why.

Pin	Type	Description
Runtime Mesh	Runtime Mesh Object Reference	The Runtime Mesh (an imported model).
Percentage	Float	Progress from 0 to 100.
Error Message	String	Empty on success; otherwise why it failed.
Async Action	Convert to Runtime Mesh Async Action Object Reference	This node's operation. Call Cancel on it to stop the operation (no more events fire).

Convert Mesh Object FUNCTION



Converts any mesh object (Static or Skeletal Mesh, Skeleton, Animation Sequence, Physics Asset, Runtime Mesh, Mesh Description, actor, mesh component; detected automatically, see Get Mesh Object Type) into Target Type. Cast the result to that type.

Target: Runtime File Importer Subsystem

Async version: [Convert Mesh Object \(Async\)](#)

C++: `URuntimeFileImporterSubsystem::ConvertMeshObject`

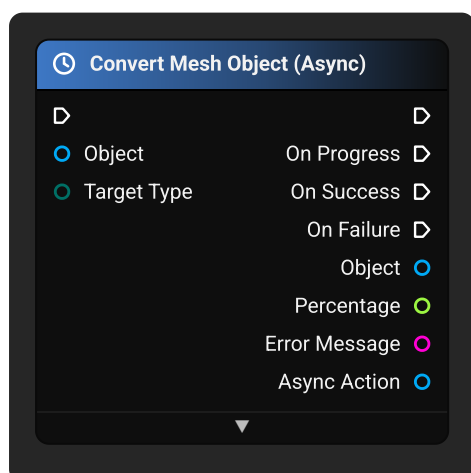
INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Object	Object Object Reference	Any supported object (detected automatically).
Target Type	ERuntime Mesh Convert Target	What to convert into.
Options	ADVANCED OPTIONAL Runtime Mesh Convert Options	How Convert nodes turn one kind of mesh object into another.

OUTPUTS

Pin	Type	Description
Error Message	String	Empty on success; otherwise why it failed.
Return Value	Object Object Reference	The result, or None/empty if it failed (Error Message says why).

Convert Mesh Object (Async) ASYNC



Converts any mesh object (Static or Skeletal Mesh, Skeleton, Animation Sequence, Physics Asset, Runtime Mesh, Mesh Description, actor, mesh component; detected automatically) into the Target Type. Cast the result to the type you asked for.

Blocking version: [Convert Mesh Object](#)

C++: `UConvertMeshObjectAsyncAction::ConvertMeshObjectAsync`

INPUTS

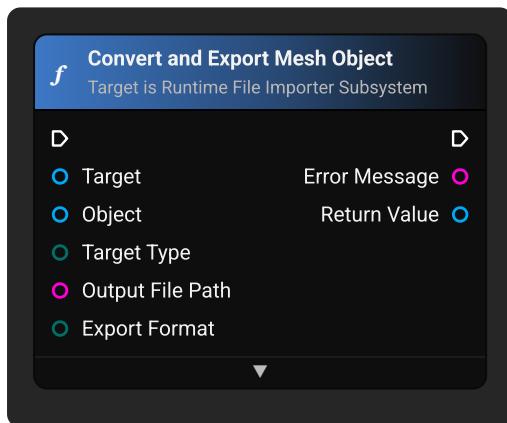
Pin	Type	Description
Object	Object Object Reference	Any supported object (detected automatically).
Target Type	ERuntime Mesh Convert Target	What to convert into.
Options	Runtime Mesh Convert Options	How Convert nodes turn one kind of mesh object into another.

OUTPUTS

Pin	Type	Description
On Progress	Exec	Fires as the operation advances (Percentage from 0 to 100). Other outputs are not set yet.
On Success	Exec	Fires when the operation has finished successfully.
On Failure	Exec	Fires if the operation failed or was cancelled; Error Message says why.
Object	Object Object Reference	Any supported object (detected automatically).
Percentage	Float	Progress from 0 to 100.
Error Message	String	Empty on success; otherwise why it failed.
Async Action	Convert Mesh Object Async Action Object Reference	This node's operation. Call Cancel on it to stop the operation (no more events fire).

Convert and Export Mesh Object

FUNCTION



Converts any mesh object into Target Type, then writes the result to a 3D model file, e.g. a Static Mesh as a Skeletal Mesh in an FBX file, or an actor merged into one mesh. Returns the converted object (null on failure).

Target: Runtime File Importer Subsystem

Async version: [Convert And Export Mesh Object \(Async\)](#)

C++: `URuntimeFileImporterSubsystem::ConvertAndExportMeshObject`

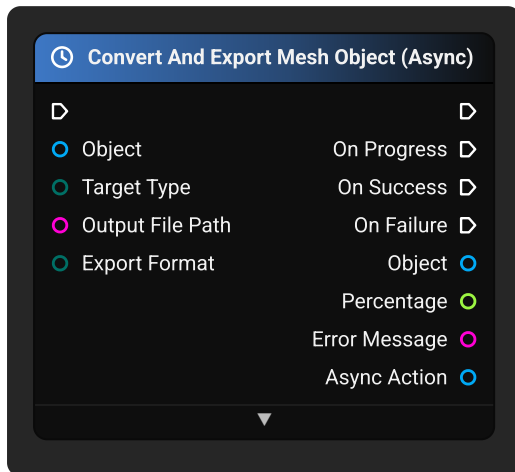
INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Object	Object Object Reference	Any supported object (detected automatically).
Target Type	ERuntime Mesh Convert Target	What to convert into.
Output File Path	String	Full path of the file to write. Missing folders are created; an existing file is replaced.
Export Format	EMesh Export Format	Format to write.
Options	Runtime Mesh Convert Options	How Convert nodes turn one kind of mesh object into another.
Export Options	Runtime Mesh Export Options	How a mesh is written to a 3D model file.

OUTPUTS

Pin	Type	Description
Error Message	 String	Empty on success; otherwise why it failed.
Return Value	 Object Object Reference	The converted object (null on failure).

Convert And Export Mesh Object (Async) ASYNC



Converts any mesh object into the Target Type, then writes the result to a 3D model file. Outputs the converted object too. E.g. a Static Mesh as a Skeletal Mesh in an FBX file, or an actor merged into one mesh.

Blocking version: [Convert and Export Mesh Object](#)

C++: `UConvertMeshObjectAsyncAction::ConvertAndExportMeshObjectAsync`

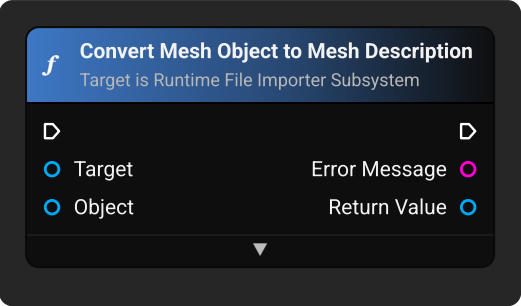
INPUTS

Pin	Type	Description
Object	 Object Object Reference	Any supported object (detected automatically).
Target Type	 ERuntime Mesh Convert Target	What to convert into.
Output File Path	 String	Full path of the file to write. Missing folders are created; an existing file is replaced.
Export Format	 EMesh Export Format	Format to write.
Options ADVANCED OPTIONAL	 Runtime Mesh Convert Options	How Convert nodes turn one kind of mesh object into another.
Export Options ADVANCED OPTIONAL	 Runtime Mesh Export Options	How a mesh is written to a 3D model file.

OUTPUTS

Pin	Type	Description
On Progress	 Exec	Fires as the operation advances (Percentage from 0 to 100). Other outputs are not set yet.
On Success	 Exec	Fires when the operation has finished successfully.
On Failure	 Exec	Fires if the operation failed or was cancelled; Error Message says why.
Object	 Object Object Reference	Any supported object (detected automatically).
Percentage	 Float	Progress from 0 to 100.
Error Message	 String	Empty on success; otherwise why it failed.
Async Action	 Convert Mesh Object Async Action Object Reference	This node's operation. Call Cancel on it to stop the operation (no more events fire).

Convert Mesh Object to Mesh Description FUNCTION



Makes editable geometry from any mesh object (Static/Skeletal Mesh, Runtime Mesh, actor, component, ...; skeletal sources are posed). Edit it with the Mesh Description nodes (Get/Set Vertex Position, Set Vertex Instance UV, Create Triangle, Generate UVs, Recompute Normals And Tangents, ...), then make a mesh of it again with Convert Mesh Object To Static Mesh / Skeletal Mesh, export it, or rebuild an existing Static Mesh with Build From Static Mesh Descriptions.

Target: Runtime File Importer Subsystem
Async version: [Convert Mesh Object To Mesh Description \(Async\)](#)
C++:
URuntimeFileImporterSubsystem::ConvertMeshObjectToMeshDescription

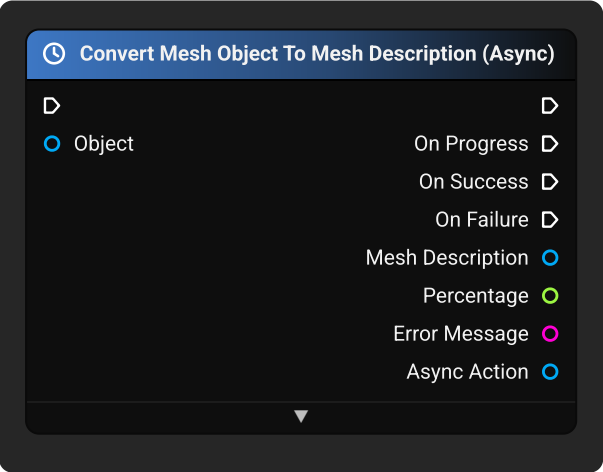
INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Object	Object Object Reference	Any supported object (detected automatically).
Options	ADVANCED OPTIONAL Runtime Mesh Convert Options	How Convert nodes turn one kind of mesh object into another.

OUTPUTS

Pin	Type	Description
Error Message	String	Empty on success; otherwise why it failed.
Return Value	Runtime Mesh Description Object Reference	The result, or None/empty if it failed (Error Message says why).

Convert Mesh Object To Mesh Description (Async) ASYNC



Makes editable geometry from any mesh object (skeletal sources are posed): edit it with the Mesh Description nodes (vertices, triangles, UVs, normals, colors), then convert it back with Convert Mesh Object To Static Mesh.

Blocking version: [Convert Mesh Object to Mesh Description](#)
C++:
UConvertToMeshDescriptionAsyncAction::ConvertMeshObjectToMeshDescriptionAsync

INPUTS

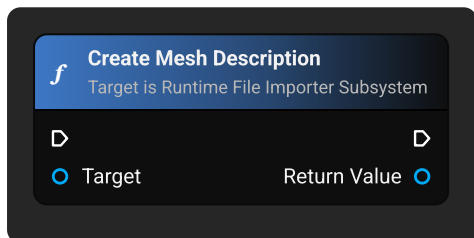
Pin	Type	Description
Object	Object Object Reference	Any supported object (detected automatically).
Options	ADVANCED OPTIONAL Runtime Mesh Convert Options	How Convert nodes turn one kind of mesh object into another.

OUTPUTS

Pin	Type	Description
On Progress	 Exec	Fires as the operation advances (Percentage from 0 to 100). Other outputs are not set yet.
On Success	 Exec	Fires when the operation has finished successfully.
On Failure	 Exec	Fires if the operation failed or was cancelled; Error Message says why.
Mesh Description	 Runtime Mesh Description Object Reference	The Runtime Mesh Description made (editable geometry).
Percentage	 Float	Progress from 0 to 100.
Error Message	 String	Empty on success; otherwise why it failed.
Async Action	 Convert to Mesh Description Async Action Object Reference	This node's operation. Call Cancel on it to stop the operation (no more events fire).

Create Mesh Description

FUNCTION



An empty Runtime Mesh Description to build geometry from scratch (Create Vertex, Create Triangle, ...), with one UV channel.

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::CreateMeshDescription`

INPUTS

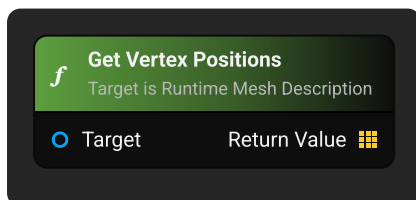
Pin	Type	Description
Target	 Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.

OUTPUTS

Pin	Type	Description
Return Value	 Runtime Mesh Description Object Reference	An empty Runtime Mesh Description to build geometry from scratch (Create Vertex, Create Triangle, ...), with one UV channel.

Get Vertex Positions

PURE



Position of every vertex, indexed by vertex ID.

Target: Runtime Mesh Description

C++: `URuntimeMeshDescription::GetVertexPositions`

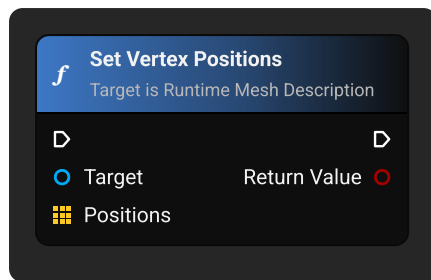
INPUTS

Pin	Type	Description
Target	 Runtime Mesh Description Object Reference	The Runtime Mesh Description to use.

OUTPUTS

Pin	Type	Description
Return Value	 Array of Vector	Position of every vertex, indexed by vertex ID.

Set Vertex Positions FUNCTION



Moves every vertex at once (array indexed by vertex ID). Returns false if the array size doesn't match.

Target: Runtime Mesh Description

C++: `URuntimeMeshDescription::SetVertexPositions`

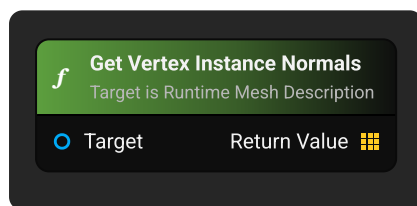
INPUTS

Pin	Type	Description
Target	Runtime Mesh Description Object Reference	The Runtime Mesh Description to use.
Positions	Array of Vector	One position per vertex.

OUTPUTS

Pin	Type	Description
Return Value	Boolean	False if the array size doesn't match.

Get Vertex Instance Normals PURE



Normal of every vertex instance, indexed by vertex instance ID.

Target: Runtime Mesh Description

C++: `URuntimeMeshDescription::GetVertexInstanceNormals`

INPUTS

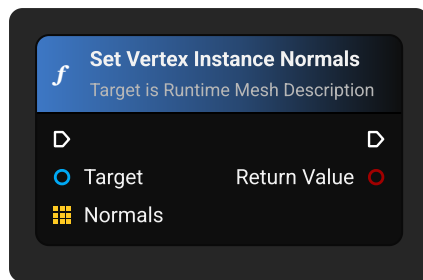
Pin	Type	Description
Target	Runtime Mesh Description Object Reference	The Runtime Mesh Description to use.

OUTPUTS

Pin	Type	Description
Return Value	Array of Vector	Normal of every vertex instance, indexed by vertex instance ID.

Set Vertex Instance Normals

FUNCTION



Sets every vertex instance's normal (array indexed by vertex instance ID). Returns false if the array size doesn't match.

Target: Runtime Mesh Description

C++: `URuntimeMeshDescription::SetVertexInstanceNormals`

INPUTS

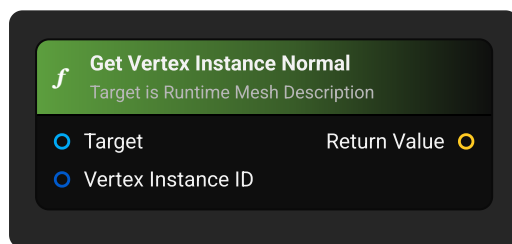
Pin	Type	Description
Target	Runtime Mesh Description Object Reference	The Runtime Mesh Description to use.
Normals	Array of Vector	One normal per vertex instance.

OUTPUTS

Pin	Type	Description
Return Value	Boolean	False if the array size doesn't match.

Get Vertex Instance Normal

PURE



Normal of one vertex instance.

Target: Runtime Mesh Description

C++: `URuntimeMeshDescription::GetVertexInstanceNormal`

INPUTS

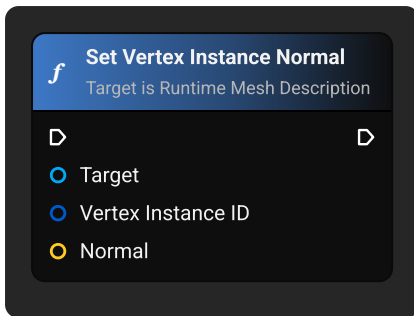
Pin	Type	Description
Target	Runtime Mesh Description Object Reference	The Runtime Mesh Description to use.
Vertex Instance ID	Vertex Instance ID	The vertex instance.

OUTPUTS

Pin	Type	Description
Return Value	Vector	Normal of one vertex instance.

Set Vertex Instance Normal

FUNCTION



Sets the normal of one vertex instance.

Target: Runtime Mesh Description

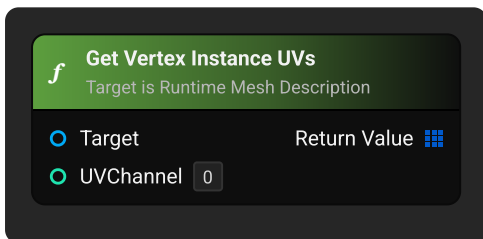
C++: `URuntimeMeshDescription::SetVertexInstanceNormal`

INPUTS

Pin	Type	Description
Target	Runtime Mesh Description Object Reference	The Runtime Mesh Description to use.
Vertex Instance ID	Vertex Instance ID	The vertex instance.
Normal	Vector	The normal.

Get Vertex Instance UVs

PURE



UVs of one channel for every vertex instance, indexed by vertex instance ID.

Target: Runtime Mesh Description

C++: `URuntimeMeshDescription::GetVertexInstanceUVs`

INPUTS

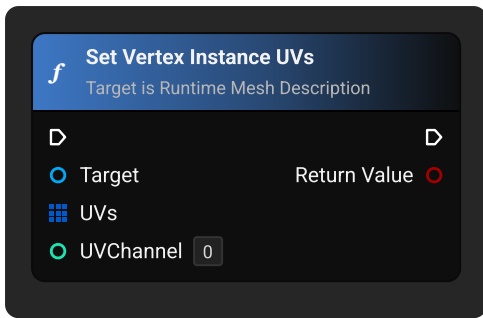
Pin	Type	Description
Target	Runtime Mesh Description Object Reference	The Runtime Mesh Description to use.
UVChannel	Integer	UV channel (0: the first). Default: 0

OUTPUTS

Pin	Type	Description
Return Value	Array of Vector 2D	UVs of one channel for every vertex instance, indexed by vertex instance ID.

Set Vertex Instance UVs

FUNCTION



Sets one UV channel of every vertex instance (array indexed by vertex instance ID). Returns false if the channel doesn't exist or the size doesn't match.

Target: Runtime Mesh Description

C++: `URuntimeMeshDescription::SetVertexInstanceUVs`

INPUTS

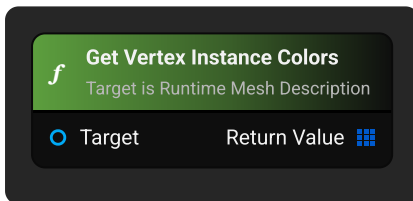
Pin	Type	Description
Target	Runtime Mesh Description Object Reference	The Runtime Mesh Description to use.
UVs	Array of Vector 2D	One UV per vertex instance.
UVChannel	Integer	UV channel (0: the first). Default: 0

OUTPUTS

Pin	Type	Description
Return Value	Boolean	False if the channel doesn't exist or the size doesn't match.

Get Vertex Instance Colors

PURE



Linear color of every vertex instance, indexed by vertex instance ID.

Target: Runtime Mesh Description

C++: `URuntimeMeshDescription::GetVertexInstanceColors`

INPUTS

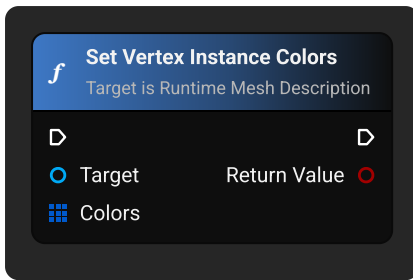
Pin	Type	Description
Target	Runtime Mesh Description Object Reference	The Runtime Mesh Description to use.

OUTPUTS

Pin	Type	Description
Return Value	Array of Linear Color	Linear color of every vertex instance, indexed by vertex instance ID.

Set Vertex Instance Colors

FUNCTION



Sets every vertex instance's color (array indexed by vertex instance ID). Returns false if the array size doesn't match.

Target: Runtime Mesh Description

C++: `URuntimeMeshDescription::SetVertexInstanceColors`

INPUTS

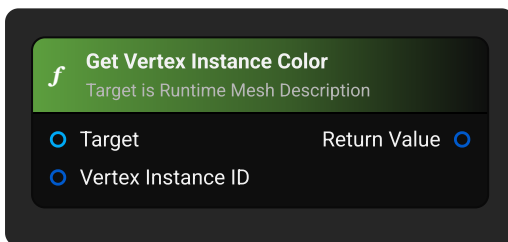
Pin	Type	Description
Target	Runtime Mesh Description Object Reference	The Runtime Mesh Description to use.
Colors	Array of Linear Color	One color per vertex instance.

OUTPUTS

Pin	Type	Description
Return Value	Boolean	False if the array size doesn't match.

Get Vertex Instance Color

PURE



Vertex color of one vertex instance.

Target: Runtime Mesh Description

C++: `URuntimeMeshDescription::GetVertexInstanceColor`

INPUTS

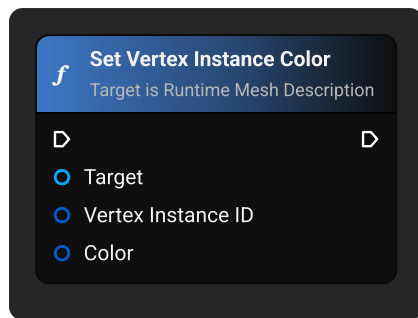
Pin	Type	Description
Target	Runtime Mesh Description Object Reference	The Runtime Mesh Description to use.
Vertex Instance ID	Vertex Instance ID	The vertex instance.

OUTPUTS

Pin	Type	Description
Return Value	Linear Color	Vertex color of one vertex instance.

Set Vertex Instance Color

FUNCTION



Sets the vertex color of one vertex instance.

Target: Runtime Mesh Description

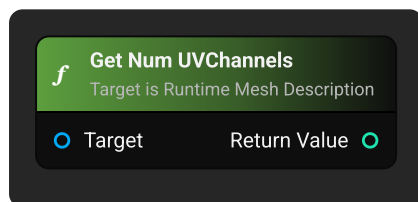
C++: `URuntimeMeshDescription::SetVertexInstanceColor`

INPUTS

Pin	Type	Description
Target	Runtime Mesh Description Object Reference	The Runtime Mesh Description to use.
Vertex Instance ID	Vertex Instance ID	The vertex instance.
Color	Linear Color	The color.

Get Num UVChannels

PURE



Number of UV channels (1 to 8).

Target: Runtime Mesh Description

C++: `URuntimeMeshDescription::GetNumUVChannels`

INPUTS

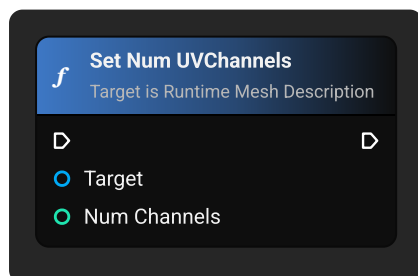
Pin	Type	Description
Target	Runtime Mesh Description Object Reference	The Runtime Mesh Description to use.

OUTPUTS

Pin	Type	Description
Return Value	Integer	Number of UV channels (1 to 8).

Set Num UVChannels

FUNCTION



Adds or removes UV channels (1 to 8); new channels start at (0, 0).

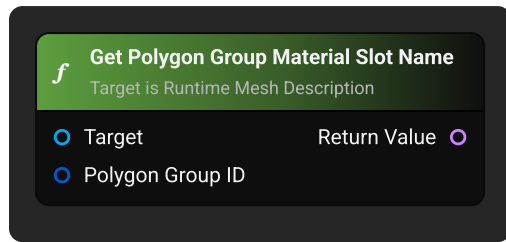
Target: Runtime Mesh Description

C++: `URuntimeMeshDescription::SetNumUVChannels`

INPUTS

Pin	Type	Description
Target	Runtime Mesh Description Object Reference	The Runtime Mesh Description to use.
Num Channels	Integer	Number of channels (1 mono, 2 stereo).

Get Polygon Group Material Slot Name PURE



Material slot name of a polygon group.

Target: Runtime Mesh Description

C++: `URuntimeMeshDescription::GetPolygonGroupMaterialSlotName`

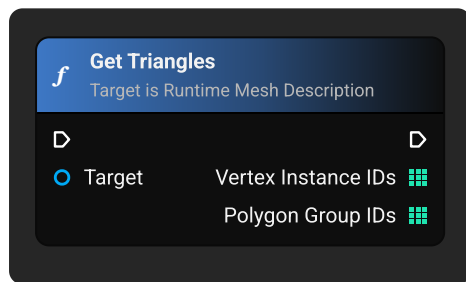
INPUTS

Pin	Type	Description
Target	Runtime Mesh Description Object Reference	The Runtime Mesh Description to use.
Polygon Group ID	Polygon Group ID	The polygon group (one per material section).

OUTPUTS

Pin	Type	Description
Return Value	Name	Material slot name of a polygon group.

Get Triangles FUNCTION



The vertex instance IDs of every triangle, three per triangle, and each triangle's polygon group.

Target: Runtime Mesh Description

C++: `URuntimeMeshDescription::GetTriangles`

INPUTS

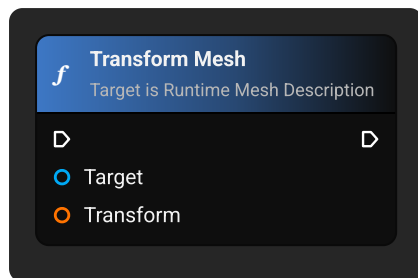
Pin	Type	Description
Target	Runtime Mesh Description Object Reference	The Runtime Mesh Description to use.

OUTPUTS

Pin	Type	Description
Vertex Instance IDs	Array of Integer	Three vertex instance IDs per triangle.
Polygon Group IDs	Array of Integer	The polygon group (material section) of each triangle.

Transform Mesh

FUNCTION



Moves, turns and scales the whole mesh (normals and tangents follow).

Target: Runtime Mesh Description

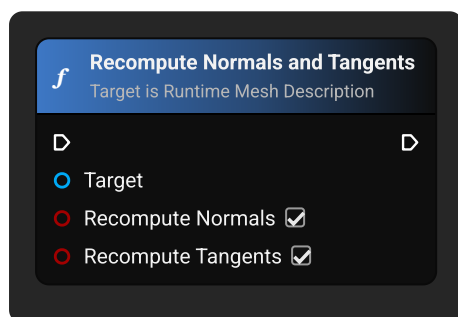
C++: `URuntimeMeshDescription::TransformMesh`

INPUTS

Pin	Type	Description
Target	Runtime Mesh Description Object Reference	The Runtime Mesh Description to use.
Transform	Transform	Location, rotation and scale.

Recompute Normals and Tangents

FUNCTION



Recomputes the normals (smooth where triangles share vertices) and tangents, e.g. after moving vertices.

Target: Runtime Mesh Description

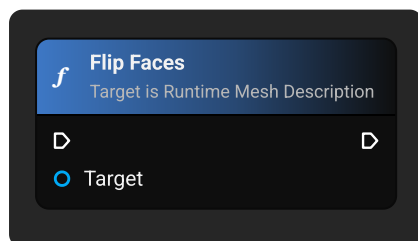
C++: `URuntimeMeshDescription::RecomputeNormalsAndTangents`

INPUTS

Pin	Type	Description
Target	Runtime Mesh Description Object Reference	The Runtime Mesh Description to use.
Recompute Normals	Boolean	Compute smooth normals from the triangles. Default: <code>true</code>
Recompute Tangents	Boolean	Compute tangents from the normals and UVs. Default: <code>true</code>

Flip Faces

FUNCTION



Turns every triangle around (front faces become back faces), flipping the normals.

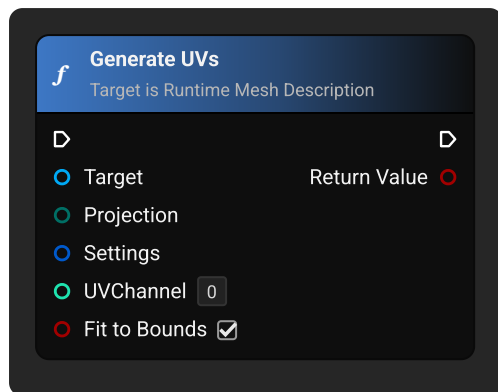
Target: Runtime Mesh Description

C++: `URuntimeMeshDescription::FlipFaces`

INPUTS

Pin	Type	Description
Target	Runtime Mesh Description Object Reference	The Runtime Mesh Description to use.

Generate UVs FUNCTION



Projects the mesh onto a UV channel (added if missing, up to 8). Settings place the projection gizmo; with Fit To Bounds, its Position and Size come from the mesh's bounds (its Rotation, Scale and UV Tile are still used).

Target: Runtime Mesh Description

C++: `URuntimeMeshDescription::GenerateUVs`

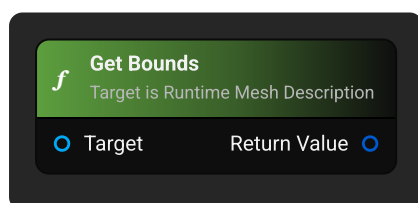
INPUTS

Pin	Type	Description
Target	Runtime Mesh Description Object Reference	The Runtime Mesh Description to use.
Projection	ERuntime UVProjection	How the UVs are projected.
Settings	UV Map Settings	Position, size and rotation of the projection gizmo.
UVChannel	Integer	UV channel (0: the first). Default: 0
Fit to Bounds	Boolean	Size the projection to the mesh's bounds (ignores the gizmo's size). Default: <code>true</code>

OUTPUTS

Pin	Type	Description
Return Value	Boolean	Projects the mesh onto a UV channel (added if missing, up to 8).

Get Bounds PURE



Bounds of the vertices.

Target: Runtime Mesh Description

C++: `URuntimeMeshDescription::GetBounds`

INPUTS

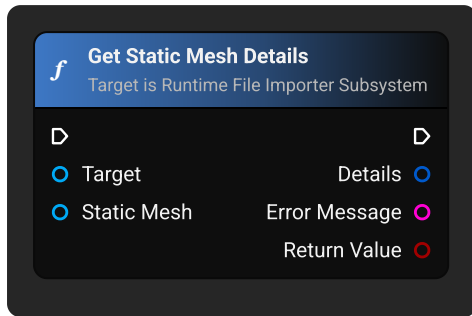
Pin	Type	Description
Target	Runtime Mesh Description Object Reference	The Runtime Mesh Description to use.

OUTPUTS

Pin	Type	Description
Return Value	Box	Bounds of the vertices.

Get Static Mesh Details

FUNCTION



Everything about a Static Mesh: materials (index, slot name, material, UV densities), LODs with their sections, vertex/triangle counts, UV channels and their ranges, Nanite, light map, bounds, collision and sockets. Change them with the Editing nodes (Set Static Mesh Material, Set Static Mesh Section Material, ...).

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::GetStaticMeshDetails`

INPUTS

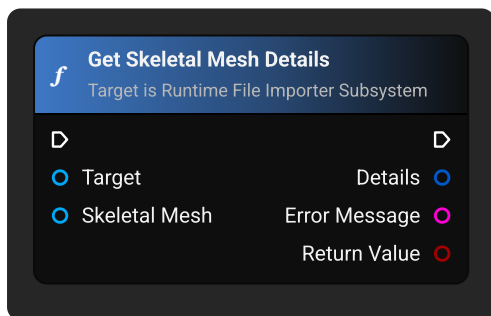
Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Static Mesh	Static Mesh Object Reference	The Static Mesh asset.

OUTPUTS

Pin	Type	Description
Details	Runtime Static Mesh Details	Everything about a Static Mesh asset (Get Static Mesh Details).
Error Message	String	Empty on success; otherwise why it failed.
Return Value	Boolean	True on success. False if it failed (Error Message says why).

Get Skeletal Mesh Details

FUNCTION



Everything about a Skeletal Mesh: materials, LODs with their sections, UV channels, bones, morph targets, sockets, skeleton and physics asset.

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::GetSkeletalMeshDetails`

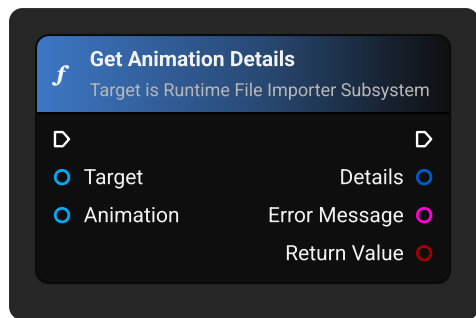
INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Skeletal Mesh	Skeletal Mesh Object Reference	The Skeletal Mesh asset.

OUTPUTS

Pin	Type	Description
Details	Runtime Skeletal Mesh Details	Everything about a Skeletal Mesh asset (Get Skeletal Mesh Details).
Error Message	String	Empty on success; otherwise why it failed.
Return Value	Boolean	True on success. False if it failed (Error Message says why).

Get Animation Details FUNCTION



Everything about an Animation Sequence: length, frame rate, animated bones, curves, notifies, root motion.

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::GetAnimationDetails`

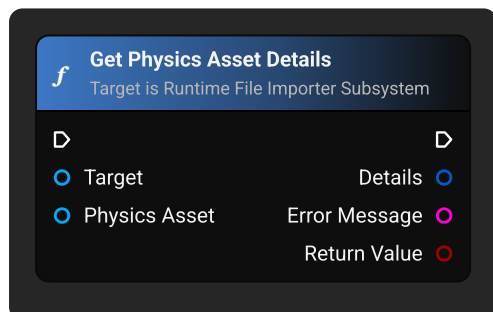
INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Animation	Anim Sequence Object Reference	The Animation Sequence.

OUTPUTS

Pin	Type	Description
Details	Runtime Animation Details	Everything about an Animation Sequence (Get Animation Details).
Error Message	String	Empty on success; otherwise why it failed.
Return Value	Boolean	True on success. False if it failed (Error Message says why).

Get Physics Asset Details FUNCTION



Everything about a Physics Asset: its bodies (shapes, mass, simulation) and constraints (bones, limits).

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::GetPhysicsAssetDetails`

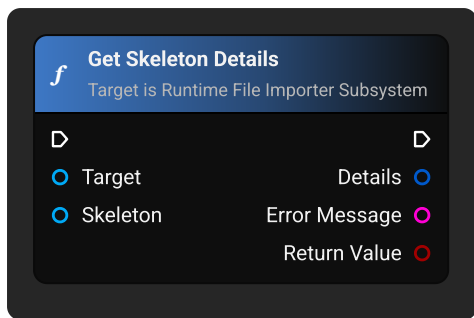
INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Physics Asset	Physics Asset Object Reference	The Physics Asset.

OUTPUTS

Pin	Type	Description
Details	Runtime Physics Asset Details	Everything about a Physics Asset (Get Physics Asset Details).
Error Message	String	Empty on success; otherwise why it failed.
Return Value	Boolean	True on success. False if it failed (Error Message says why).

Get Skeleton Details FUNCTION



Everything about a Skeleton: bones (with reference pose), sockets, virtual bones, curve names.

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::GetSkeletonDetails`

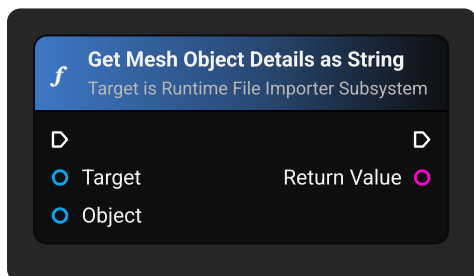
INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Skeleton	Skeleton Object Reference	The Skeleton the animations are made for (bones matched by name).

OUTPUTS

Pin	Type	Description
Details	Runtime Skeleton Details	Everything about a Skeleton (Get Skeleton Details).
Error Message	String	Empty on success; otherwise why it failed.
Return Value	Boolean	True on success. False if it failed (Error Message says why).

Get Mesh Object Details as String FUNCTION



Readable, multi-line details of any mesh object (meshes, animations, physics assets, skeletons, Runtime Meshes, components, actors, mesh descriptions), e.g. to print.

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::GetMeshObjectDetailsAsString`

INPUTS

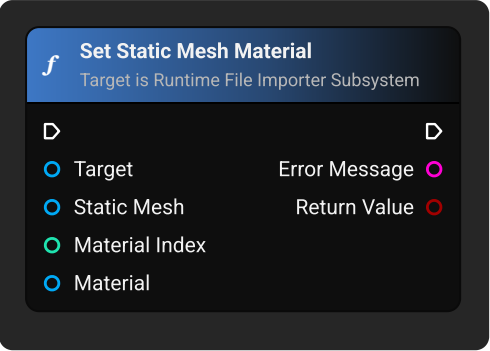
Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Object	Object Object Reference	Any supported object (detected automatically).

OUTPUTS

Pin	Type	Description
Return Value	String	Readable, multi-line details of any mesh object (meshes, animations, physics assets, skeletons, Runtime Meshes, components, actors, mesh descriptions), e.g.

Set Static Mesh Material

FUNCTION



Sets the material of a Static Mesh's slot, for every component showing it (works in packaged games, unlike the mesh's own Set Material). A component's Set Material overrides it for that component only.

Target: Runtime File Importer Subsystem
C++: URuntimeFileImporterSubsystem::SetStaticMeshMaterial

INPUTS

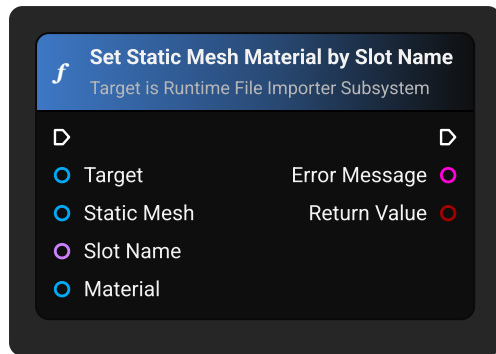
Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Static Mesh	Static Mesh Object Reference	The Static Mesh asset.
Material Index	Integer	Index of the material slot.
Material	Material Interface Object Reference	The material.

OUTPUTS

Pin	Type	Description
Error Message	String	Empty on success; otherwise why it failed.
Return Value	Boolean	True on success. False if it failed (Error Message says why).

Set Static Mesh Material by Slot Name

FUNCTION



Sets the material of a Static Mesh's slot found by its name (or the name it had in the imported file).

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::SetStaticMeshMaterialBySlotName`

INPUTS

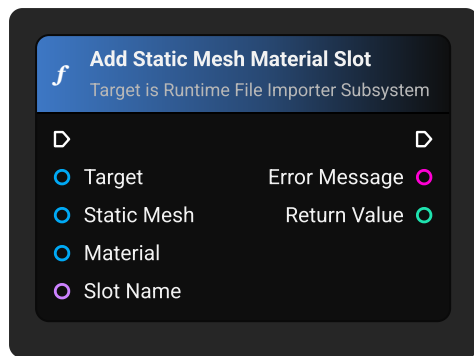
Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Static Mesh	Static Mesh Object Reference	The Static Mesh asset.
Slot Name	Name	Name of the material slot.
Material	Material Interface Object Reference	The material.

OUTPUTS

Pin	Type	Description
Error Message	String	Empty on success; otherwise why it failed.
Return Value	Boolean	True on success. False if it failed (Error Message says why).

Add Static Mesh Material Slot

FUNCTION



Adds a material slot (names are made unique; empty: the material's name). Returns its index (-1 on failure); point sections at it with Set Static Mesh Section Material.

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::AddStaticMeshMaterialSlot`

INPUTS

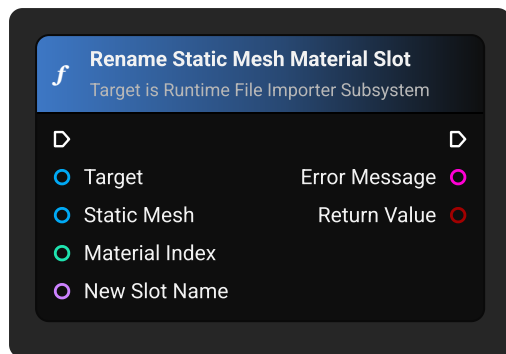
Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Static Mesh	Static Mesh Object Reference	The Static Mesh asset.
Material	Material Interface Object Reference	The material.
Slot Name	Name	Name of the material slot.

OUTPUTS

Pin	Type	Description
Error Message	String	Empty on success; otherwise why it failed.
Return Value	Integer	Its index (-1 on failure); point sections at it with Set Static Mesh Section Material.

Rename Static Mesh Material Slot

FUNCTION



Renames a material slot of a Static Mesh (Set Static Mesh Material By Slot Name uses the name).

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::RenameStaticMeshMaterialSlot`

INPUTS

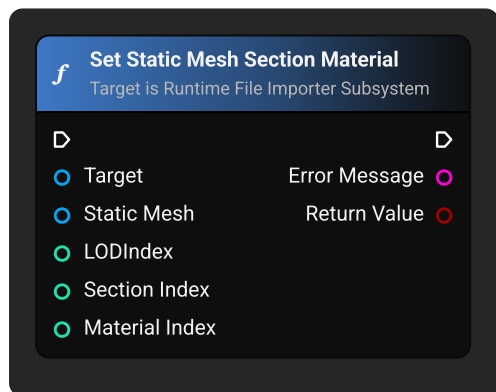
Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Static Mesh	Static Mesh Object Reference	The Static Mesh asset.
Material Index	Integer	Index of the material slot.
New Slot Name	Name	The new name of the slot.

OUTPUTS

Pin	Type	Description
Error Message	String	Empty on success; otherwise why it failed.
Return Value	Boolean	True on success. False if it failed (Error Message says why).

Set Static Mesh Section Material

FUNCTION



Makes a section of a LOD draw with another material slot.

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::SetStaticMeshSectionMaterial`

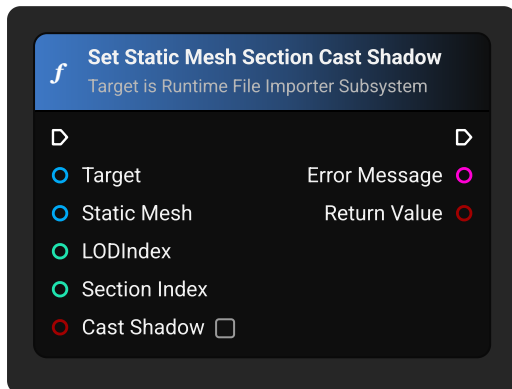
INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Static Mesh	Static Mesh Object Reference	The Static Mesh asset.
LODIndex	Integer	Level of detail (0: the most detailed).
Section Index	Integer	Index of the section in the LOD.
Material Index	Integer	Index of the material slot.

OUTPUTS

Pin	Type	Description
Error Message	 String	Empty on success; otherwise why it failed.
Return Value	 Boolean	True on success. False if it failed (Error Message says why).

Set Static Mesh Section Cast Shadow FUNCTION



Turns shadow casting on or off for one section (material part) of a Static Mesh LOD.

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::SetStaticMeshSectionCastShadow`

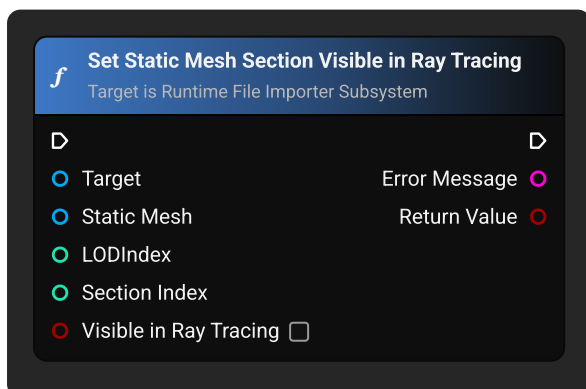
INPUTS

Pin	Type	Description
Target	 Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Static Mesh	 Static Mesh Object Reference	The Static Mesh asset.
LODIndex	 Integer	Level of detail (0: the most detailed).
Section Index	 Integer	Index of the section in the LOD.
Cast Shadow	 Boolean	Whether the section casts shadows.

OUTPUTS

Pin	Type	Description
Error Message	 String	Empty on success; otherwise why it failed.
Return Value	 Boolean	True on success. False if it failed (Error Message says why).

Set Static Mesh Section Visible in Ray Tracing FUNCTION



Shows or hides one section of a Static Mesh LOD in ray tracing (reflections, shadows, GI).

Target: Runtime File Importer Subsystem

C++:

`URuntimeFileImporterSubsystem::SetStaticMeshSectionVisibleInRayTracing`

INPUTS

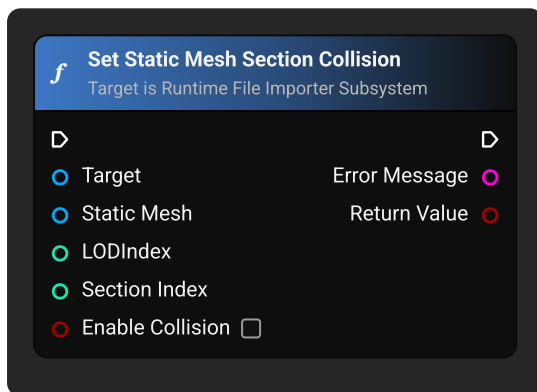
Pin	Type	Description
Target	 Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Static Mesh	 Static Mesh Object Reference	The Static Mesh asset.
LODIndex	 Integer	Level of detail (0: the most detailed).
Section Index	 Integer	Index of the section in the LOD.
Visible in Ray Tracing	 Boolean	Whether the section is visible to ray tracing.

OUTPUTS

Pin	Type	Description
Error Message	 String	Empty on success; otherwise why it failed.
Return Value	 Boolean	True on success. False if it failed (Error Message says why).

Set Static Mesh Section Collision

FUNCTION



Includes or excludes a section from the per-triangle (complex) collision, and rebuilds it.

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::SetStaticMeshSectionCollision`

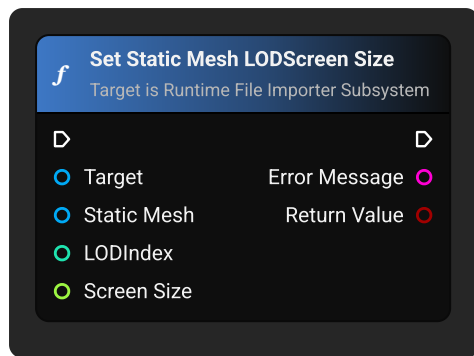
INPUTS

Pin	Type	Description
Target	 Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Static Mesh	 Static Mesh Object Reference	The Static Mesh asset.
LODIndex	 Integer	Level of detail (0: the most detailed).
Section Index	 Integer	Index of the section in the LOD.
Enable Collision	 Boolean	Whether the section takes part in complex (per-triangle) collision.

OUTPUTS

Pin	Type	Description
Error Message	 String	Empty on success; otherwise why it failed.
Return Value	 Boolean	True on success. False if it failed (Error Message says why).

Set Static Mesh LODScreen Size FUNCTION



Screen size (fraction of the screen height) below which the next LOD is drawn.

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::SetStaticMeshLODScreenSize`

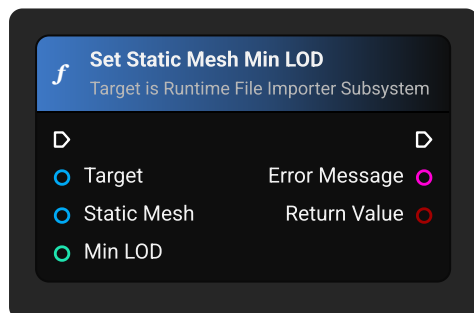
INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Static Mesh	Static Mesh Object Reference	The Static Mesh asset.
LODIndex	Integer	Level of detail (0: the most detailed).
Screen Size	Float	Screen size (fraction of the screen height) below which the next LOD is used.

OUTPUTS

Pin	Type	Description
Error Message	String	Empty on success; otherwise why it failed.
Return Value	Boolean	True on success. False if it failed (Error Message says why).

Set Static Mesh Min LOD FUNCTION



Lowest LOD drawn (more detailed LODs are skipped), e.g. to save memory on low-end devices.

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::SetStaticMeshMinLOD`

INPUTS

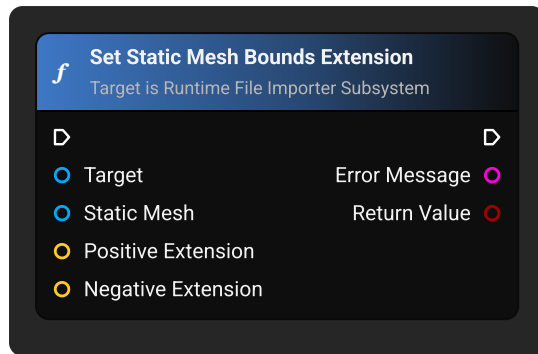
Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Static Mesh	Static Mesh Object Reference	The Static Mesh asset.
Min LOD	Integer	The most detailed LOD that may be drawn.

OUTPUTS

Pin	Type	Description
Error Message	String	Empty on success; otherwise why it failed.
Return Value	Boolean	True on success. False if it failed (Error Message says why).

Set Static Mesh Bounds Extension

FUNCTION



Grows the bounds (in centimeters, per axis), e.g. for materials that move vertices (World Position Offset).

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::SetStaticMeshBoundsExtension`

INPUTS

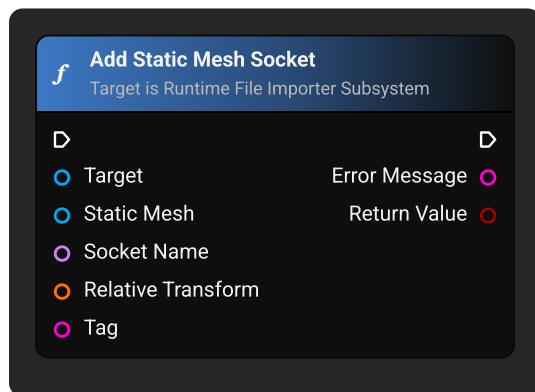
Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Static Mesh	Static Mesh Object Reference	The Static Mesh asset.
Positive Extension	Vector	How far the bounds are extended on the positive side of each axis.
Negative Extension	Vector	How far the bounds are extended on the negative side of each axis.

OUTPUTS

Pin	Type	Description
Error Message	String	Empty on success; otherwise why it failed.
Return Value	Boolean	True on success. False if it failed (Error Message says why).

Add Static Mesh Socket

FUNCTION



Adds a socket (a named point to attach things to). Find it with the mesh's Find Socket, or use its name when attaching.

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::AddStaticMeshSocket`

INPUTS

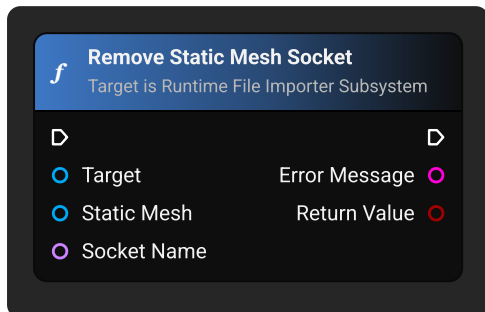
Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Static Mesh	Static Mesh Object Reference	The Static Mesh asset.
Socket Name	Name	Name of the socket.
Relative Transform	Transform	Transform of the socket relative to the mesh (or bone).
Tag	String	Optional tag of the socket.

OUTPUTS

Pin	Type	Description
Error Message	 String	Empty on success; otherwise why it failed.
Return Value	 Boolean	True on success. False if it failed (Error Message says why).

Remove Static Mesh Socket

FUNCTION



Removes a socket from a Static Mesh. Returns false if it has no socket of that name.

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::RemoveStaticMeshSocket`

INPUTS

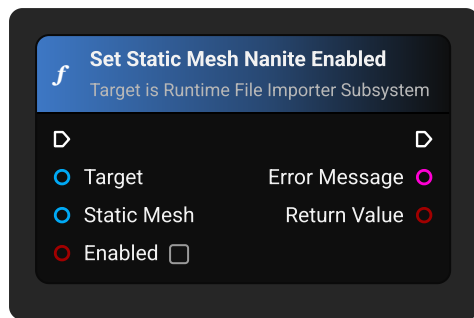
Pin	Type	Description
Target	 Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Static Mesh	 Static Mesh Object Reference	The Static Mesh asset.
Socket Name	 Name	Name of the socket.

OUTPUTS

Pin	Type	Description
Error Message	 String	Empty on success; otherwise why it failed.
Return Value	 Boolean	False if it has no socket of that name.

Set Static Mesh Nanite Enabled

FUNCTION



Turns Nanite on or off for a Static Mesh and rebuilds it. Editor only (e.g. in Editor Utility Widgets): Nanite data is built in the editor, so this fails in packaged games and for meshes made at runtime.

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::SetStaticMeshNaniteEnabled`

INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Static Mesh	Static Mesh Object Reference	The Static Mesh asset.
Enabled	Boolean	True to enable Nanite.

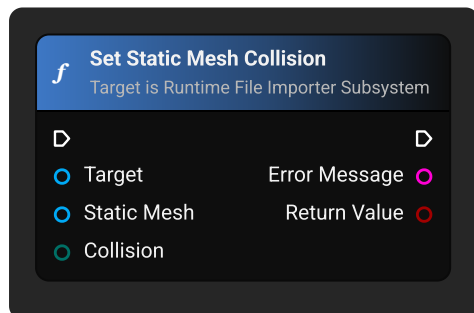
OUTPUTS

Pin	Type	Description
Error Message	String	Empty on success; otherwise why it failed.
Return Value	Boolean	True on success. False if it failed (Error Message says why).

Editing / Static Mesh / Collision 7 nodes

Set Static Mesh Collision

FUNCTION



Replaces a Static Mesh's collision: None, a Box or Convex hull around it, or its triangles (Complex). Components get it right away. Convex and Complex read the triangles (meshes need "Allow CPU Access" in packaged games, except this plugin's).

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::SetStaticMeshCollision`

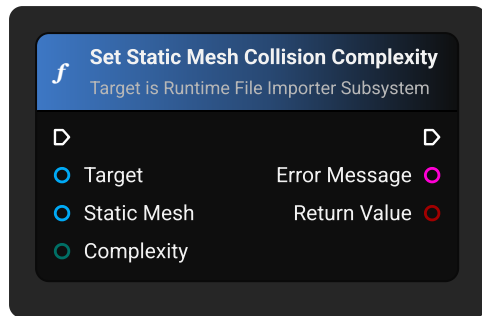
INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Static Mesh	Static Mesh Object Reference	The Static Mesh asset.
Collision	ERuntime Mesh Collision	The collision to generate.

OUTPUTS

Pin	Type	Description
Error Message	String	Empty on success; otherwise why it failed.
Return Value	Boolean	True on success. False if it failed (Error Message says why).

Set Static Mesh Collision Complexity FUNCTION



How simple shapes and triangles are used for physics and traces (the mesh's Collision Complexity).

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::SetStaticMeshCollisionComplexity`

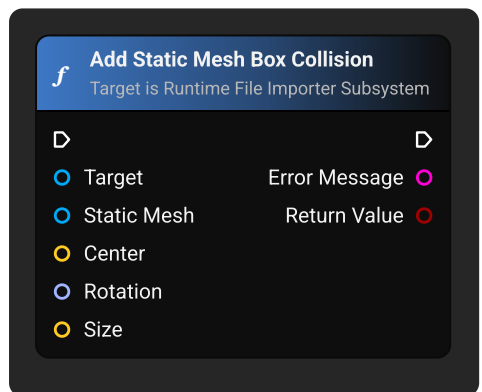
INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Static Mesh	Static Mesh Object Reference	The Static Mesh asset.
Complexity	ERuntime Collision Complexity	Which collision traces and physics use.

OUTPUTS

Pin	Type	Description
Error Message	String	Empty on success; otherwise why it failed.
Return Value	Boolean	True on success. False if it failed (Error Message says why).

Add Static Mesh Box Collision FUNCTION



Adds a box to the simple collision (Size in centimeters, relative to the mesh).

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::AddStaticMeshBoxCollision`

INPUTS

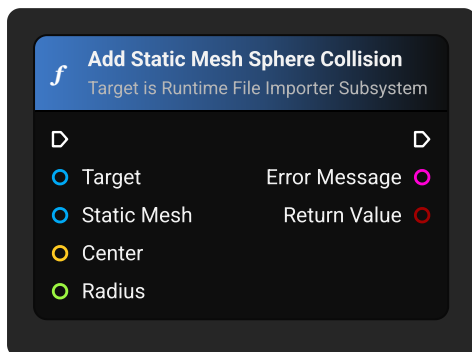
Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Static Mesh	Static Mesh Object Reference	The Static Mesh asset.
Center	Vector	Center of the shape, relative to the mesh.
Rotation	Rotator	Rotation of the shape.
Size	Vector	Full size of the box on each axis.

OUTPUTS

Pin	Type	Description
Error Message	String	Empty on success; otherwise why it failed.

Pin	Type	Description
Return Value	 Boolean	True on success. False if it failed (Error Message says why).

Add Static Mesh Sphere Collision FUNCTION



Adds a sphere to the simple collision of a Static Mesh (Center relative to the mesh).

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::AddStaticMeshSphereCollision`

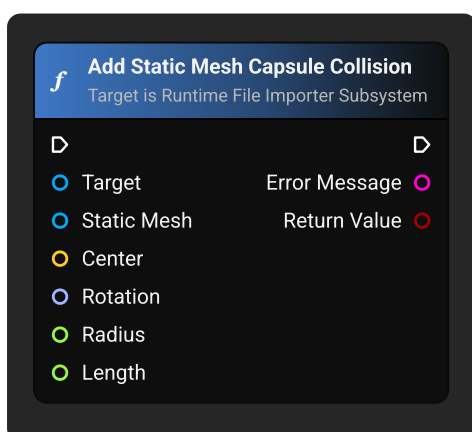
INPUTS

Pin	Type	Description
Target	 Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Static Mesh	 Static Mesh Object Reference	The Static Mesh asset.
Center	 Vector	Center of the shape, relative to the mesh.
Radius	 Float	Radius of the shape.

OUTPUTS

Pin	Type	Description
Error Message	 String	Empty on success; otherwise why it failed.
Return Value	 Boolean	True on success. False if it failed (Error Message says why).

Add Static Mesh Capsule Collision FUNCTION



Adds a capsule (along its Z axis; Length excludes the round ends) to the simple collision.

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::AddStaticMeshCapsuleCollision`

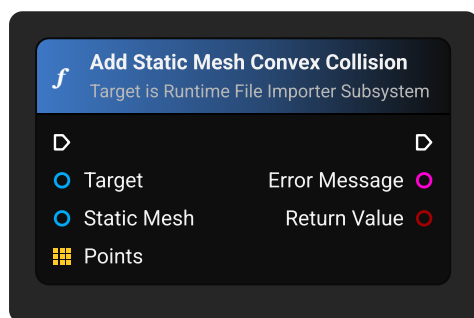
INPUTS

Pin	Type	Description
Target	 Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Static Mesh	 Static Mesh Object Reference	The Static Mesh asset.

Pin	Type	Description
Center	 Vector	Center of the shape, relative to the mesh.
Rotation	 Rotator	Rotation of the shape.
Radius	 Float	Radius of the shape.
Length	 Float	Length of the capsule's straight part.
OUTPUTS		
Pin	Type	Description
Error Message	 String	Empty on success; otherwise why it failed.
Return Value	 Boolean	True on success. False if it failed (Error Message says why).

Add Static Mesh Convex Collision

FUNCTION



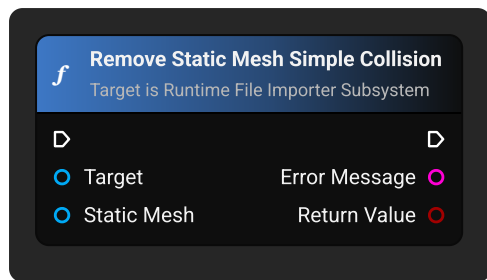
Adds the convex hull of points (relative to the mesh) to the simple collision.

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::AddStaticMeshConvexCollision`

INPUTS		
Pin	Type	Description
Target	 Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Static Mesh	 Static Mesh Object Reference	The Static Mesh asset.
Points	 Array of Vector	Points the convex hull is built around (relative to the mesh).
OUTPUTS		
Pin	Type	Description
Error Message	 String	Empty on success; otherwise why it failed.
Return Value	 Boolean	True on success. False if it failed (Error Message says why).

Remove Static Mesh Simple Collision FUNCTION



Removes every simple collision shape (the triangles still collide if the complexity uses them).

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::RemoveStaticMeshSimpleCollision`

INPUTS

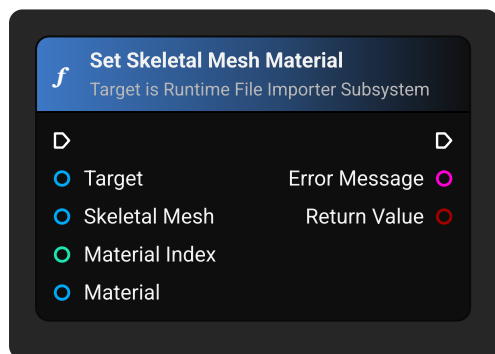
Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Static Mesh	Static Mesh Object Reference	The Static Mesh asset.

OUTPUTS

Pin	Type	Description
Error Message	String	Empty on success; otherwise why it failed.
Return Value	Boolean	True on success. False if it failed (Error Message says why).

Editing / Skeletal Mesh 14 nodes

Set Skeletal Mesh Material FUNCTION



Sets the material of a Skeletal Mesh's slot, for every component showing it.

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::SetSkeletalMeshMaterial`

INPUTS

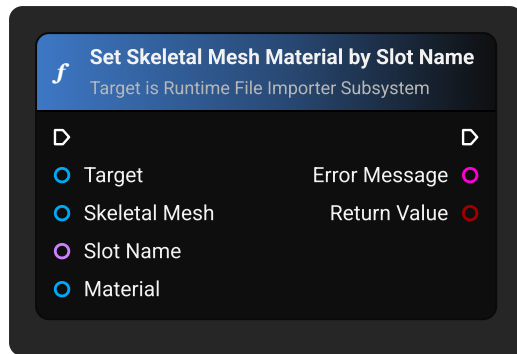
Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Skeletal Mesh	Skeletal Mesh Object Reference	The Skeletal Mesh asset.
Material Index	Integer	Index of the material slot.
Material	Material Interface Object Reference	The material.

OUTPUTS

Pin	Type	Description
Error Message	String	Empty on success; otherwise why it failed.
Return Value	Boolean	True on success. False if it failed (Error Message says why).

Set Skeletal Mesh Material by Slot Name

FUNCTION



Sets the material of a Skeletal Mesh slot, found by its name.

Target: Runtime File Importer Subsystem

C++:

```
URuntimeFileImporterSubsystem::SetSkeletalMeshMaterialBySlotName
```

INPUTS

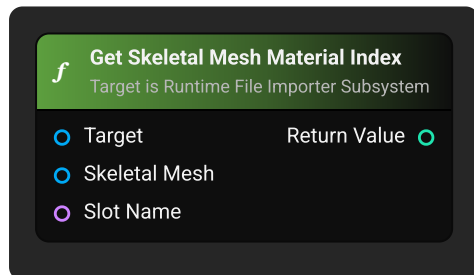
Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Skeletal Mesh	Skeletal Mesh Object Reference	The Skeletal Mesh asset.
Slot Name	Name	Name of the material slot.
Material	Material Interface Object Reference	The material.

OUTPUTS

Pin	Type	Description
Error Message	String	Empty on success; otherwise why it failed.
Return Value	Boolean	True on success. False if it failed (Error Message says why).

Get Skeletal Mesh Material Index

PURE



Index of a Skeletal Mesh's material slot by its name (or the name it had in the imported file), or -1.

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::GetSkeletalMeshMaterialIndex`

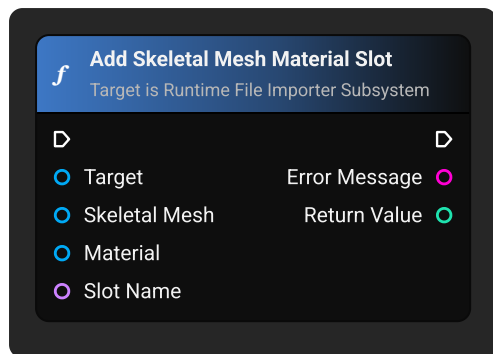
INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Skeletal Mesh	Skeletal Mesh Object Reference	The Skeletal Mesh asset.
Slot Name	Name	Name of the material slot.

OUTPUTS

Pin	Type	Description
Return Value	Integer	Index of a Skeletal Mesh's material slot by its name (or the name it had in the imported file), or -1.

Add Skeletal Mesh Material Slot FUNCTION



Adds a material slot (names are made unique). Returns its index (-1 on failure); point sections at it with Set Skeletal Mesh Section Material.

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::AddSkeletalMeshMaterialSlot`

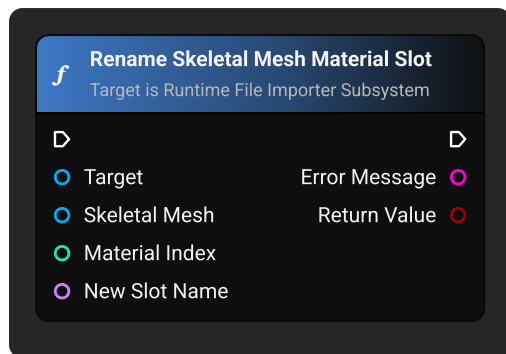
INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Skeletal Mesh	Skeletal Mesh Object Reference	The Skeletal Mesh asset.
Material	Material Interface Object Reference	The material.
Slot Name	Name	Name of the material slot.

OUTPUTS

Pin	Type	Description
Error Message	String	Empty on success; otherwise why it failed.
Return Value	Integer	Its index (-1 on failure); point sections at it with Set Skeletal Mesh Section Material.

Rename Skeletal Mesh Material Slot FUNCTION



Renames a material slot of a Skeletal Mesh.

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::RenameSkeletalMeshMaterialSlot`

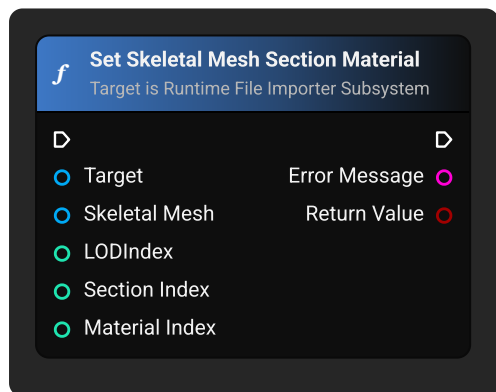
INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Skeletal Mesh	Skeletal Mesh Object Reference	The Skeletal Mesh asset.
Material Index	Integer	Index of the material slot.
New Slot Name	Name	The new name of the slot.

OUTPUTS

Pin	Type	Description
Error Message	String	Empty on success; otherwise why it failed.
Return Value	Boolean	True on success. False if it failed (Error Message says why).

Set Skeletal Mesh Section Material FUNCTION



Makes a section of a LOD draw with another material slot.

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::SetSkeletalMeshSectionMaterial`

INPUTS

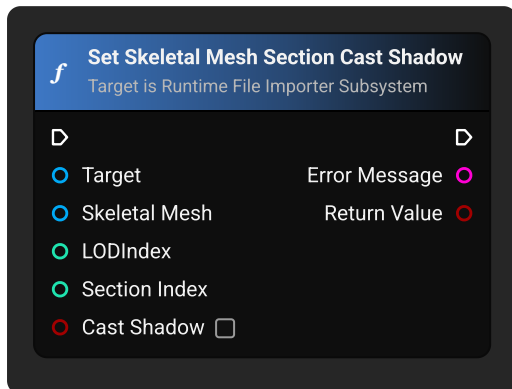
Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Skeletal Mesh	Skeletal Mesh Object Reference	The Skeletal Mesh asset.
LODIndex	Integer	Level of detail (0: the most detailed).
Section Index	Integer	Index of the section in the LOD.
Material Index	Integer	Index of the material slot.

OUTPUTS

Pin	Type	Description
Error Message	 String	Empty on success; otherwise why it failed.
Return Value	 Boolean	True on success. False if it failed (Error Message says why).

Set Skeletal Mesh Section Cast Shadow

FUNCTION



Turns shadow casting on or off for one section (material part) of a Skeletal Mesh LOD.

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::SetSkeletalMeshSectionCastShadow`

INPUTS

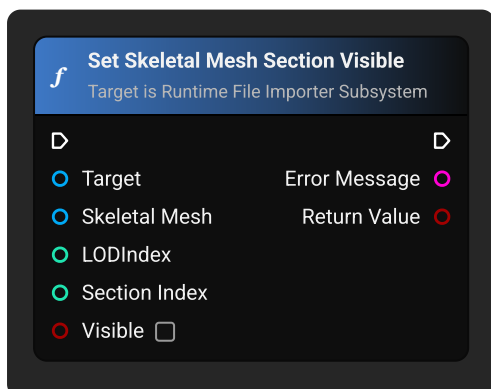
Pin	Type	Description
Target	 Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Skeletal Mesh	 Skeletal Mesh Object Reference	The Skeletal Mesh asset.
LODIndex	 Integer	Level of detail (0: the most detailed).
Section Index	 Integer	Index of the section in the LOD.
Cast Shadow	 Boolean	Whether the section casts shadows.

OUTPUTS

Pin	Type	Description
Error Message	 String	Empty on success; otherwise why it failed.
Return Value	 Boolean	True on success. False if it failed (Error Message says why).

Set Skeletal Mesh Section Visible

FUNCTION



Shows or hides a section on every component showing the mesh (a component's Show Material Section does it for one component).

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::SetSkeletalMeshSectionVisible`

INPUTS

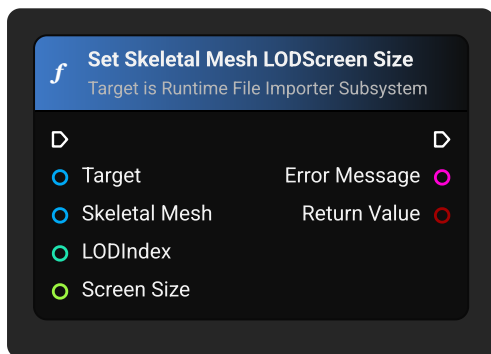
Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Skeletal Mesh	Skeletal Mesh Object Reference	The Skeletal Mesh asset.
LODIndex	Integer	Level of detail (0: the most detailed).
Section Index	Integer	Index of the section in the LOD.
Visible	Boolean	Whether the section is drawn.

OUTPUTS

Pin	Type	Description
Error Message	String	Empty on success; otherwise why it failed.
Return Value	Boolean	True on success. False if it failed (Error Message says why).

Set Skeletal Mesh LODScreen Size

FUNCTION



Sets the screen size below which a Skeletal Mesh switches from this LOD to the next one.

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::SetSkeletalMeshLODScreenSize`

INPUTS

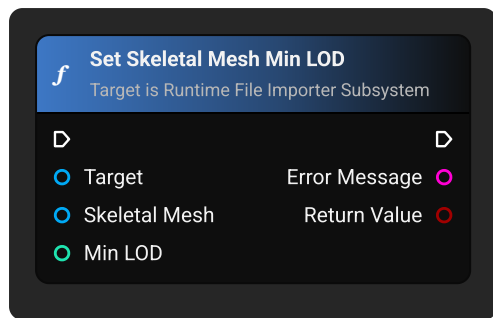
Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Skeletal Mesh	Skeletal Mesh Object Reference	The Skeletal Mesh asset.
LODIndex	Integer	Level of detail (0: the most detailed).
Screen Size	Float	Screen size (fraction of the screen height) below which the next LOD is used.

OUTPUTS

Pin	Type	Description
Error Message	String	Empty on success; otherwise why it failed.
Return Value	Boolean	True on success. False if it failed (Error Message says why).

Set Skeletal Mesh Min LOD

FUNCTION



Sets the most detailed LOD a Skeletal Mesh may draw (higher: cheaper, less detailed).

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::SetSkeletalMeshMinLOD`

INPUTS

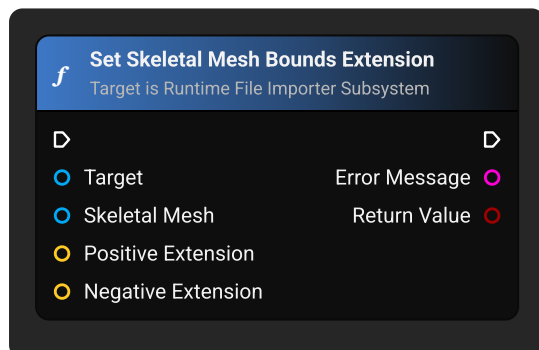
Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Skeletal Mesh	Skeletal Mesh Object Reference	The Skeletal Mesh asset.
Min LOD	Integer	The most detailed LOD that may be drawn.

OUTPUTS

Pin	Type	Description
Error Message	String	Empty on success; otherwise why it failed.
Return Value	Boolean	True on success. False if it failed (Error Message says why).

Set Skeletal Mesh Bounds Extension

FUNCTION



Grows the bounds of a Skeletal Mesh (e.g. so animations that move far from the reference pose are not culled).

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::SetSkeletalMeshBoundsExtension`

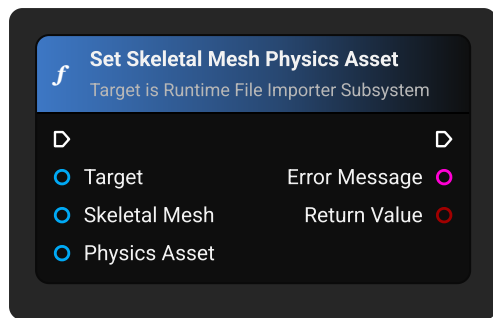
INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Skeletal Mesh	Skeletal Mesh Object Reference	The Skeletal Mesh asset.
Positive Extension	Vector	How far the bounds are extended on the positive side of each axis.
Negative Extension	Vector	How far the bounds are extended on the negative side of each axis.

OUTPUTS

Pin	Type	Description
Error Message	String	Empty on success; otherwise why it failed.
Return Value	Boolean	True on success. False if it failed (Error Message says why).

Set Skeletal Mesh Physics Asset FUNCTION



Makes a Skeletal Mesh use a Physics Asset (collision and ragdoll); components showing it get it right away.

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::SetSkeletalMeshPhysicsAsset`

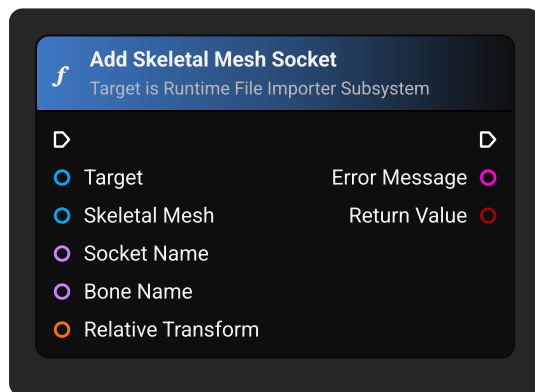
INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Skeletal Mesh	Skeletal Mesh Object Reference	The Skeletal Mesh asset.
Physics Asset	Physics Asset Object Reference	The Physics Asset.

OUTPUTS

Pin	Type	Description
Error Message	String	Empty on success; otherwise why it failed.
Return Value	Boolean	True on success. False if it failed (Error Message says why).

Add Skeletal Mesh Socket FUNCTION



Adds a socket on a bone (Relative Transform is relative to the bone).

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::AddSkeletalMeshSocket`

INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Skeletal Mesh	Skeletal Mesh Object Reference	The Skeletal Mesh asset.
Socket Name	Name	Name of the socket.
Bone Name	Name	Bone the socket is attached to.
Relative Transform OPTIONAL	Transform	Transform of the socket relative to the mesh (or bone).

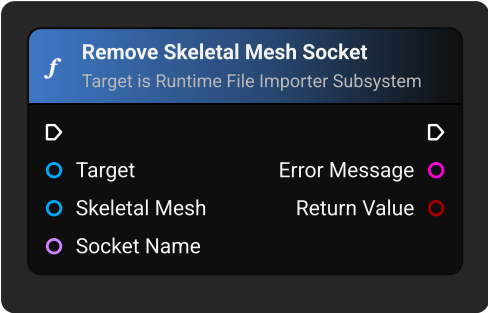
OUTPUTS

Pin	Type	Description
Error Message	String	Empty on success; otherwise why it failed.

Pin	Type	Description
Return Value	 Boolean	True on success. False if it failed (Error Message says why).

Remove Skeletal Mesh Socket

FUNCTION



Removes a socket the mesh itself has (not one of its skeleton's).

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::RemoveSkeletalMeshSocket`

INPUTS

Pin	Type	Description
Target	 Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Skeletal Mesh	 Skeletal Mesh Object Reference	The Skeletal Mesh asset.
Socket Name	 Name	Name of the socket.

OUTPUTS

Pin	Type	Description
Error Message	 String	Empty on success; otherwise why it failed.
Return Value	 Boolean	True on success. False if it failed (Error Message says why).

Set Animation Rate Scale

FUNCTION

f

Set Animation Rate Scale

Target is Runtime File Importer Subsystem

Target

Animation

Rate Scale

Error Message

Return Value

Playback speed of an animation (1: normal, negative: backwards), wherever it plays.

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::SetAnimationRateScale`

INPUTS		
Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Animation	Anim Sequence Object Reference	The Animation Sequence.
Rate Scale	Float	Playback speed multiplier (1: normal).
OUTPUTS		
Pin	Type	Description
Error Message	String	Empty on success; otherwise why it failed.
Return Value	Boolean	True on success. False if it failed (Error Message says why).

Set Animation Root Motion

FUNCTION

f

Set Animation Root Motion

Target is Runtime File Importer Subsystem

Target

Animation

Enable Root Motion

Error Message

Return Value

Root motion: the root bone's movement moves the actor (with Root Motion enabled in its Anim Blueprint or montage).

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::SetAnimationRootMotion`

INPUTS		
Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Animation	Anim Sequence Object Reference	The Animation Sequence.
Enable Root Motion	Boolean	Use the root bone's motion to move the character.
OUTPUTS		
Pin	Type	Description
Error Message	String	Empty on success; otherwise why it failed.
Return Value	Boolean	True on success. False if it failed (Error Message says why).

Get Mesh Format from File

FUNCTION

f

Get Mesh Format from File

Target is Runtime File Importer Subsystem

⌵

○ Target

○ File Path

⌵

○ Mesh Format

○ Return Value

Detects a file's 3D model format from its contents (falling back to its extension). Returns false if it isn't recognized.

Target: Runtime File Importer Subsystem

Async version: [Get Mesh Format From File \(Async\)](#)

C++: `URuntimeFileImporterSubsystem::GetMeshFormatFromFile`

INPUTS		
Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
File Path	String	Full path of the file on disk.
OUTPUTS		
Pin	Type	Description
Mesh Format	EMesh Format	Format of the file. Auto Detect reads it from the contents and extension.
Return Value	Boolean	False if it isn't recognized.

Get Mesh Format From File (Async)

ASYNC

⌚

Get Mesh Format From File (Async)

⌵

○ File Path

⌵

○ On Success

○ On Failure

○ Mesh Format

○ Error Message

○ Async Action

Detects a file's 3D model format from its contents (falling back to its extension).

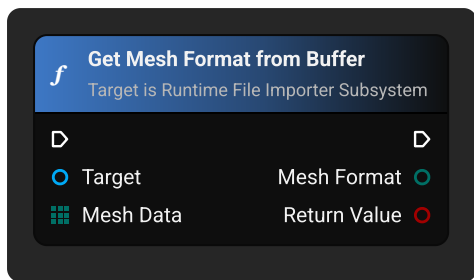
Blocking version: [Get Mesh Format from File](#)

C++: `UGetMeshFormatAsyncAction::GetMeshFormatFromFileAsync`

INPUTS		
Pin	Type	Description
File Path	String	Full path of the file on disk.
OUTPUTS		
Pin	Type	Description
On Success	Exec	Fires when the operation has finished successfully.
On Failure	Exec	Fires if the operation failed or was cancelled; Error Message says why.
Mesh Format	EMesh Format	Format of the file. Auto Detect reads it from the contents and extension.
Error Message	String	Empty on success; otherwise why it failed.
Async Action	Get Mesh Format Async Action Object Reference	This node's operation. Call Cancel on it to stop the operation (no more events fire).

Get Mesh Format from Buffer

FUNCTION



Detects the 3D model format of bytes in memory. Returns false if it isn't recognized.

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::GetMeshFormatFromBuffer`

INPUTS

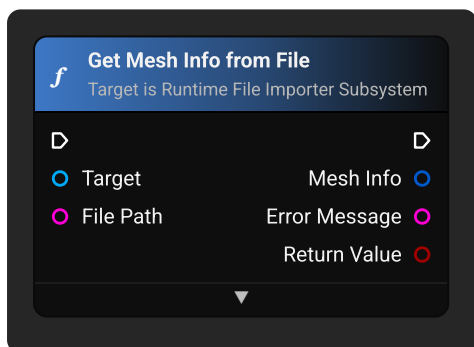
Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Mesh Data	Array of Byte	Bytes of a 3D model file.

OUTPUTS

Pin	Type	Description
Mesh Format	EMesh Format	Format of the file. Auto Detect reads it from the contents and extension.
Return Value	Boolean	False if it isn't recognized.

Get Mesh Info from File

FUNCTION



Reads a 3D model file's counts, bounds, bones, morph targets, animations, materials and metadata without creating anything (the file is still fully parsed). Mesh Info To String turns the result into readable text.

Target: Runtime File Importer Subsystem

Async version: [Get Mesh Info From File \(Async\)](#)

C++: `URuntimeFileImporterSubsystem::GetMeshInfoFromFile`

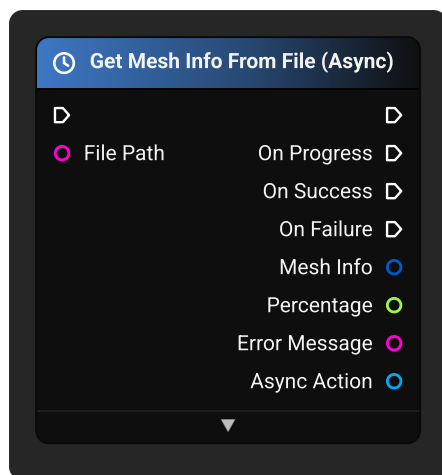
INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
File Path	String	Full path of the file on disk.
Options	ADVANCED OPTIONAL Runtime Mesh Import Options	How a 3D model file is read and turned into a Runtime Mesh.

OUTPUTS

Pin	Type	Description
Mesh Info	Runtime Mesh Info	Mesher, materials, bones, animations and size of the model.
Error Message	String	Empty on success; otherwise why it failed.
Return Value	Boolean	True on success. False if it failed (Error Message says why).

Get Mesh Info From File (Async) ASYNC



Reads a 3D model file's counts, bounds, bones, morph targets, animations, materials and metadata without creating anything, on a background thread. Mesh Info To String turns the result into readable text.

Blocking version: [Get Mesh Info from File](#)

C++: `UGetMeshInfoAsyncAction::GetMeshInfoFromFileAsync`

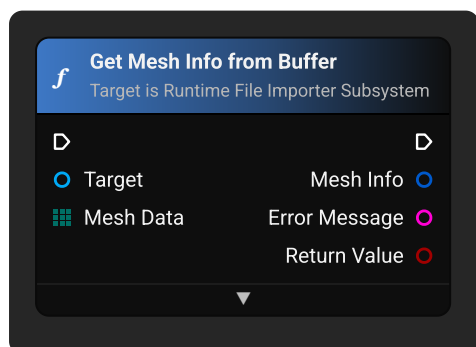
INPUTS

Pin	Type	Description
File Path	String	Full path of the file on disk.
Options ADVANCED OPTIONAL	Runtime Mesh Import Options	How a 3D model file is read and turned into a Runtime Mesh.

OUTPUTS

Pin	Type	Description
On Progress	Exec	Fires as the operation advances (Percentage from 0 to 100). Other outputs are not set yet.
On Success	Exec	Fires when the operation has finished successfully.
On Failure	Exec	Fires if the operation failed or was cancelled; Error Message says why.
Mesh Info	Runtime Mesh Info	Meshes, materials, bones, animations and size of the model.
Percentage	Float	Progress from 0 to 100.
Error Message	String	Empty on success; otherwise why it failed.
Async Action	Get Mesh Info Async Action Object Reference	This node's operation. Call Cancel on it to stop the operation (no more events fire).

Get Mesh Info from Buffer FUNCTION



Reads the info of a 3D model in memory without creating anything.

Target: Runtime File Importer Subsystem

Async version: [Get Mesh Info From Buffer \(Async\)](#)

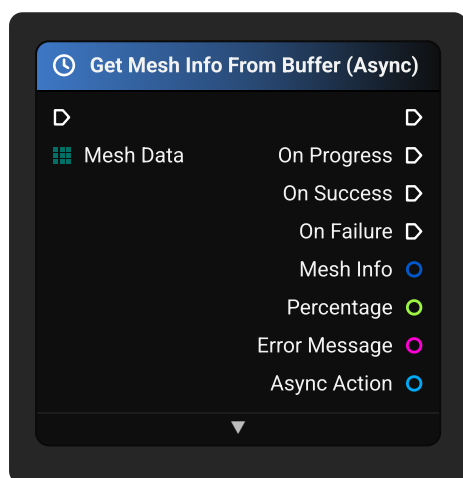
C++: `URuntimeFileImporterSubsystem::GetMeshInfoFromBuffer`

INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.

Pin	Type	Description
Mesh Data	 Array of Byte	Bytes of a 3D model file.
Mesh Format ADVANCED	 EMesh Format	Format of the file. Auto Detect reads it from the contents and extension.
Options ADVANCED OPTIONAL	 Runtime Mesh Import Options	How a 3D model file is read and turned into a Runtime Mesh.
OUTPUTS		
Pin	Type	Description
Mesh Info	 Runtime Mesh Info	Meshes, materials, bones, animations and size of the model.
Error Message	 String	Empty on success; otherwise why it failed.
Return Value	 Boolean	True on success. False if it failed (Error Message says why).

Get Mesh Info From Buffer (Async) ASYNC



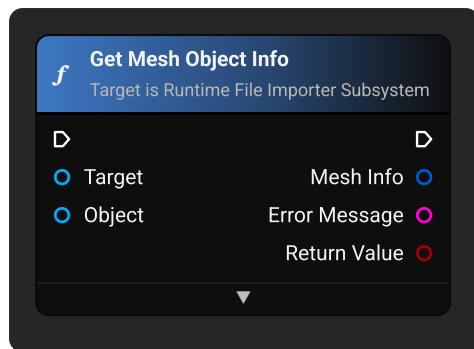
Reads the info of a 3D model in memory, on a background thread.

Blocking version: [Get Mesh Info from Buffer](#)

C++: `UGetMeshInfoAsyncAction::GetMeshInfoFromBufferAsync`

INPUTS		
Pin	Type	Description
Mesh Data	 Array of Byte	Bytes of a 3D model file.
Mesh Format ADVANCED	 EMesh Format	Format of the file. Auto Detect reads it from the contents and extension.
Options ADVANCED OPTIONAL	 Runtime Mesh Import Options	How a 3D model file is read and turned into a Runtime Mesh.
OUTPUTS		
Pin	Type	Description
On Progress	 Exec	Fires as the operation advances (Percentage from 0 to 100). Other outputs are not set yet.
On Success	 Exec	Fires when the operation has finished successfully.
On Failure	 Exec	Fires if the operation failed or was cancelled; Error Message says why.
Mesh Info	 Runtime Mesh Info	Meshes, materials, bones, animations and size of the model.
Percentage	 Float	Progress from 0 to 100.
Error Message	 String	Empty on success; otherwise why it failed.
Async Action	 Get Mesh Info Async Action Object Reference	This node's operation. Call Cancel on it to stop the operation (no more events fire).

Get Mesh Object Info FUNCTION



The info of any mesh object (detected automatically): counts, bounds, bones, morph targets, animations, materials.

Target: Runtime File Importer Subsystem

Async version: [Get Mesh Object Info \(Async\)](#)

C++: `URuntimeFileImporterSubsystem::GetMeshObjectInfo`

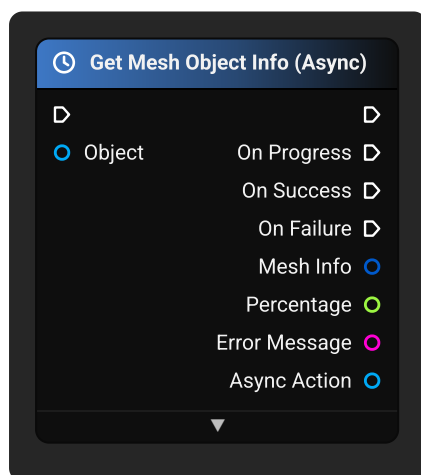
INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Object	Object Object Reference	Any supported object (detected automatically).
Source Options	ADVANCED OPTIONAL Runtime Mesh Convert Options	What some sources need: a Skeletal Mesh for animations and physics assets, extra Animations, a pose.

OUTPUTS

Pin	Type	Description
Mesh Info	Runtime Mesh Info	Meshes, materials, bones, animations and size of the model.
Error Message	String	Empty on success; otherwise why it failed.
Return Value	Boolean	True on success. False if it failed (Error Message says why).

Get Mesh Object Info (Async) ASYNC



The info of any mesh object (detected automatically: meshes, skeletons, animations, physics assets, Runtime Meshes, actors, components).

Blocking version: [Get Mesh Object Info](#)

C++: `UGetMeshInfoAsyncAction::GetMeshObjectInfoAsync`

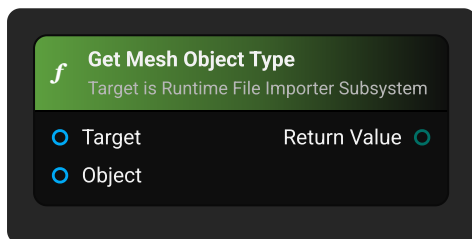
INPUTS

Pin	Type	Description
Object	Object Object Reference	Any supported object (detected automatically).
Source Options	ADVANCED OPTIONAL Runtime Mesh Convert Options	What some sources need: a Skeletal Mesh for animations and physics assets, extra Animations, a pose.

OUTPUTS

Pin	Type	Description
On Progress	 Exec	Fires as the operation advances (Percentage from 0 to 100). Other outputs are not set yet.
On Success	 Exec	Fires when the operation has finished successfully.
On Failure	 Exec	Fires if the operation failed or was cancelled; Error Message says why.
Mesh Info	 Runtime Mesh Info	Meshes, materials, bones, animations and size of the model.
Percentage	 Float	Progress from 0 to 100.
Error Message	 String	Empty on success; otherwise why it failed.
Async Action	 Get Mesh Info Async Action Object Reference	This node's operation. Call Cancel on it to stop the operation (no more events fire).

Get Mesh Object Type PURE



What kind of mesh object something is (Static/Skeletal Mesh, Skeleton, Animation, Physics Asset, Runtime Mesh, actor, component), or None.

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::GetMeshObjectType`

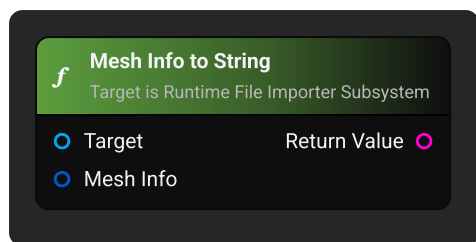
INPUTS

Pin	Type	Description
Target	 Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Object	 Object Object Reference	Any supported object (detected automatically).

OUTPUTS

Pin	Type	Description
Return Value	 ERuntime Mesh Object Type	What kind of mesh object something is (Static/Skeletal Mesh, Skeleton, Animation, Physics Asset, Runtime Mesh, actor, component), or None.

Mesh Info to String PURE



A readable, multi-line description of a model's info (e.g. to print or show in a UI).

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::MeshInfoToString`

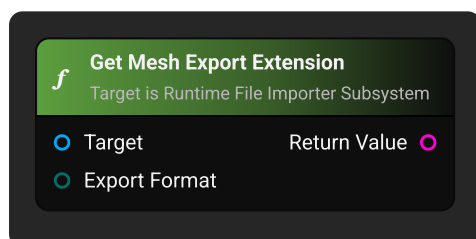
INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Mesh Info	Runtime Mesh Info	Meshes, materials, bones, animations and size of the model.

OUTPUTS

Pin	Type	Description
Return Value	String	A readable, multi-line description of a model's info (e.g.

Get Mesh Export Extension PURE



File extension written for an export format, without the dot (e.g. "glb").

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::GetMeshExportExtension`

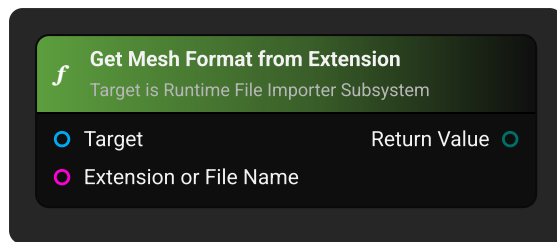
INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Export Format	EMesh Export Format	Format to write.

OUTPUTS

Pin	Type	Description
Return Value	String	File extension written for an export format, without the dot (e.g.

Get Mesh Format from Extension PURE



Import format of a file extension or file name (Auto Detect if unknown).

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::GetMeshFormatFromExtension`

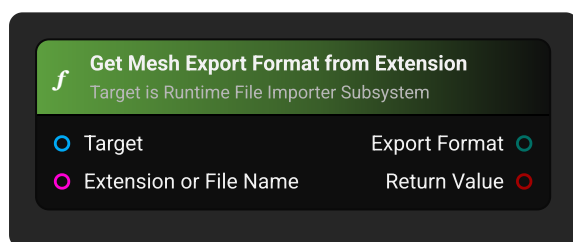
INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Extension or File Name	String	An extension (with or without the dot) or a file name / path / URL ending in one.

OUTPUTS

Pin	Type	Description
Return Value	EMesh Format	Import format of a file extension or file name (Auto Detect if unknown).

Get Mesh Export Format from Extension PURE



Export format matching a file extension or file name, e.g. to export to whatever path a user typed. Returns false if there's none.

Target: Runtime File Importer Subsystem

C++:

`URuntimeFileImporterSubsystem::GetMeshExportFormatFromExtension`

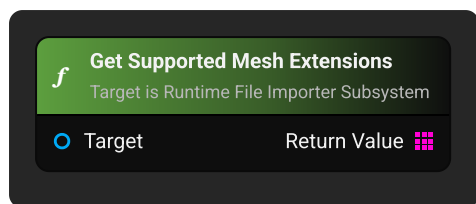
INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Extension or File Name	String	An extension (with or without the dot) or a file name / path / URL ending in one.

OUTPUTS

Pin	Type	Description
Export Format	EMesh Export Format	Format to write.
Return Value	Boolean	False if there's none.

Get Supported Mesh Extensions PURE



File extensions that can be imported (without dots), e.g. for a file picker filter.

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::GetSupportedMeshExtensions`

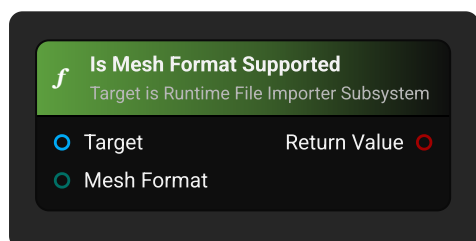
INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.

OUTPUTS

Pin	Type	Description
Return Value	Array of String	File extensions that can be imported (without dots), e.g.

Is Mesh Format Supported PURE



False for formats this platform can't read (Alembic outside Windows, Mac and Linux).

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::IsMeshFormatSupported`

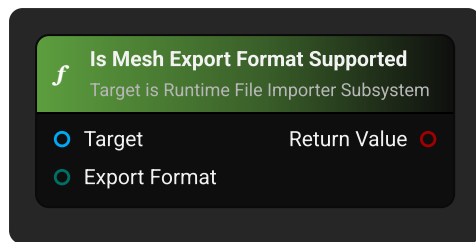
INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Mesh Format	EMesh Format	Format of the file. Auto Detect reads it from the contents and extension.

OUTPUTS

Pin	Type	Description
Return Value	Boolean	False for formats this platform can't read (Alembic outside Windows, Mac and Linux).

Is Mesh Export Format Supported PURE



False for formats this platform can't write (Alembic outside Windows, Mac and Linux).

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::IsMeshExportFormatSupported`

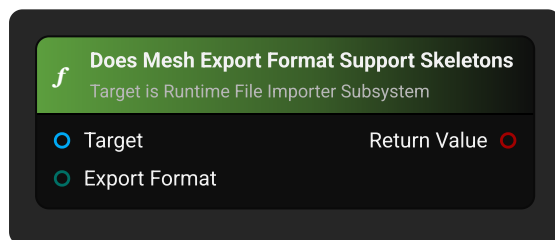
INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Export Format	EMesh Export Format	Format to write.

OUTPUTS

Pin	Type	Description
Return Value	Boolean	False for formats this platform can't write (Alembic outside Windows, Mac and Linux).

Does Mesh Export Format Support Skeletons PURE



True for export formats that keep skeletons, skin weights, morph targets and animations (FBX, glTF).

Target: Runtime File Importer Subsystem

C++:

`URuntimeFileImporterSubsystem::DoesMeshExportFormatSupportSkeletons`

INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Export Format	EMesh Export Format	Format to write.

OUTPUTS

Pin	Type	Description
Return Value	Boolean	True for export formats that keep skeletons, skin weights, morph targets and animations (FBX, glTF).

Get Runtime Mesh Static Mesh PURE

f

Get Runtime Mesh Static Mesh

Target is Runtime File Importer Subsystem

Target

Return Value

Runtime Mesh

The whole model as one Static Mesh (null for skeletal models). Same as its Static Mesh variable.

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::GetRuntimeMeshStaticMesh`

INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Runtime Mesh	Runtime Mesh Object Reference	The Runtime Mesh (an imported model).

OUTPUTS

Pin	Type	Description
Return Value	Static Mesh Object Reference	The whole model as one Static Mesh (null for skeletal models).

Get Runtime Mesh Skeletal Mesh PURE

f

Get Runtime Mesh Skeletal Mesh

Target is Runtime File Importer Subsystem

Target

Return Value

Runtime Mesh

The model as a Skeletal Mesh (null for static models). Same as its Skeletal Mesh variable.

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::GetRuntimeMeshSkeletalMesh`

INPUTS

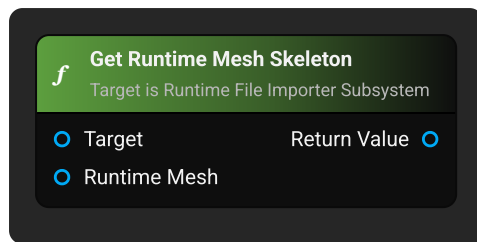
Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Runtime Mesh	Runtime Mesh Object Reference	The Runtime Mesh (an imported model).

OUTPUTS

Pin	Type	Description
Return Value	Skeletal Mesh Object Reference	The model as a Skeletal Mesh (null for static models).

Get Runtime Mesh Skeleton

PURE



The Skeleton of a skeletal model. Same as its Skeleton variable.

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::GetRuntimeMeshSkeleton`

INPUTS

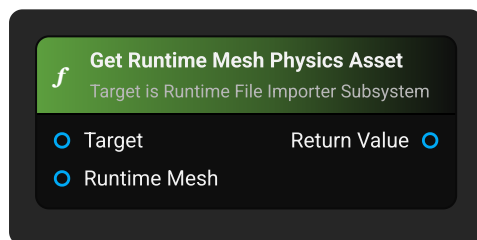
Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Runtime Mesh	Runtime Mesh Object Reference	The Runtime Mesh (an imported model).

OUTPUTS

Pin	Type	Description
Return Value	Skeleton Object Reference	The Skeleton of a skeletal model.

Get Runtime Mesh Physics Asset

PURE



The Physics Asset of a skeletal model. Same as its Physics Asset variable.

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::GetRuntimeMeshPhysicsAsset`

INPUTS

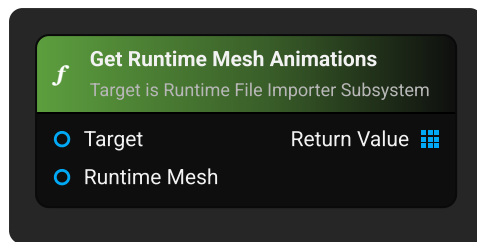
Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Runtime Mesh	Runtime Mesh Object Reference	The Runtime Mesh (an imported model).

OUTPUTS

Pin	Type	Description
Return Value	Physics Asset Object Reference	The Physics Asset of a skeletal model.

Get Runtime Mesh Animations

PURE



The model's animations. Same as its Animations variable.

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::GetRuntimeMeshAnimations`

INPUTS

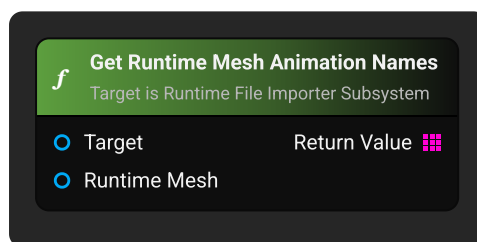
Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Runtime Mesh	Runtime Mesh Object Reference	The Runtime Mesh (an imported model).

OUTPUTS

Pin	Type	Description
Return Value	Array of Anim Sequence Object Reference	The model's animations.

Get Runtime Mesh Animation Names

PURE



Names of the model's animations as the file calls them, in the order of Animations.

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::GetRuntimeMeshAnimationNames`

INPUTS

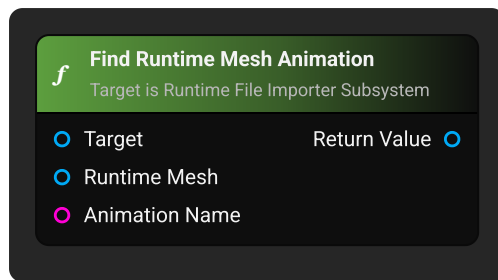
Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Runtime Mesh	Runtime Mesh Object Reference	The Runtime Mesh (an imported model).

OUTPUTS

Pin	Type	Description
Return Value	Array of String	Names of the model's animations as the file calls them, in the order of Animations.

Find Runtime Mesh Animation

PURE



One of the model's animations, by its name in the file, or null.

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::FindRuntimeMeshAnimation`

INPUTS

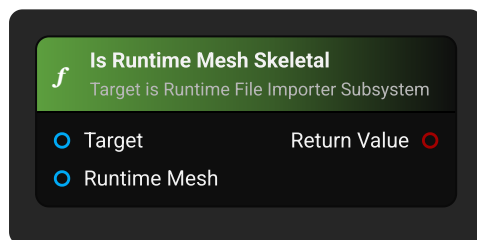
Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Runtime Mesh	Runtime Mesh Object Reference	The Runtime Mesh (an imported model).
Animation Name	String	Name of the animation in the model file. Empty: the first one.

OUTPUTS

Pin	Type	Description
Return Value	Anim Sequence Object Reference	One of the model's animations, by its name in the file, or null.

Is Runtime Mesh Skeletal

PURE



True for models with a Skeletal Mesh.

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::IsRuntimeMeshSkeletal`

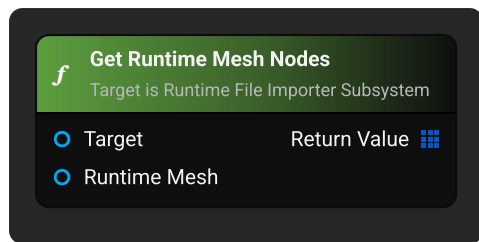
INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Runtime Mesh	Runtime Mesh Object Reference	The Runtime Mesh (an imported model).

OUTPUTS

Pin	Type	Description
Return Value	Boolean	True for models with a Skeletal Mesh.

Get Runtime Mesh Nodes PURE



The model's node hierarchy (names, parents, transforms, meshes, bones).

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::GetRuntimeMeshNodes`

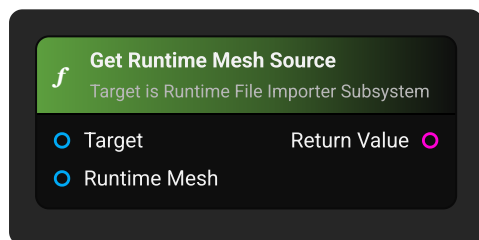
INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Runtime Mesh	Runtime Mesh Object Reference	The Runtime Mesh (an imported model).

OUTPUTS

Pin	Type	Description
Return Value	Array of Runtime Mesh Node Info	The model's node hierarchy (names, parents, transforms, meshes, bones).

Get Runtime Mesh Source PURE



File path or URL the model was imported from (empty for buffers and conversions). Same as its Source variable.

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::GetRuntimeMeshSource`

INPUTS

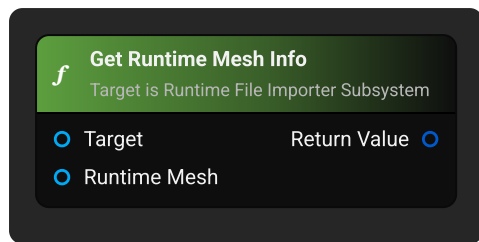
Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Runtime Mesh	Runtime Mesh Object Reference	The Runtime Mesh (an imported model).

OUTPUTS

Pin	Type	Description
Return Value	String	File path or URL the model was imported from (empty for buffers and conversions).

Get Runtime Mesh Info

PURE



Counts, bounds and metadata of a Runtime Mesh. Same as its Mesh Info variable.

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::GetRuntimeMeshInfo`

INPUTS

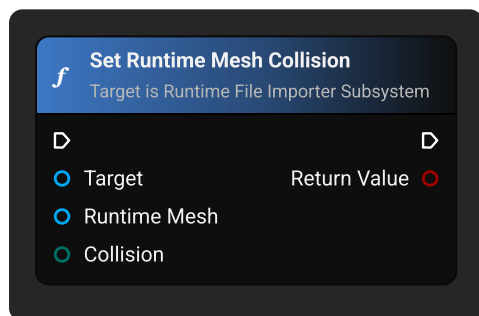
Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Runtime Mesh	Runtime Mesh Object Reference	The Runtime Mesh (an imported model).

OUTPUTS

Pin	Type	Description
Return Value	Runtime Mesh Info	Counts, bounds and metadata of a Runtime Mesh.

Set Runtime Mesh Collision

FUNCTION



Changes the collision of a static model's meshes (None, a Box around them, or the triangles themselves). Components showing them get it right away. Skeletal models collide with their Physics Asset instead.

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::SetRuntimeMeshCollision`

INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Runtime Mesh	Runtime Mesh Object Reference	The Runtime Mesh (an imported model).
Collision	ERuntime Mesh Collision	The collision to generate.

OUTPUTS

Pin	Type	Description
Return Value	Boolean	Changes the collision of a static model's meshes (None, a Box around them, or the triangles themselves).

Get Runtime Mesh Materials

PURE

f

Get Runtime Mesh Materials

Target is Runtime File Importer Subsystem

○ Target

○ Runtime Mesh

Return Value

The material of each slot. Same as its Materials variable.

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::GetRuntimeMeshMaterials`

INPUTS		
Pin	Type	Description
Target	 Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Runtime Mesh	 Runtime Mesh Object Reference	The Runtime Mesh (an imported model).
OUTPUTS		
Pin	Type	Description
Return Value	 Array of Material Interface Object Reference	The material of each slot.

Set Runtime Mesh Materials

FUNCTION

f

Set Runtime Mesh Materials

Target is Runtime File Importer Subsystem

▷

○ Target

○ Runtime Mesh

 Materials

▷

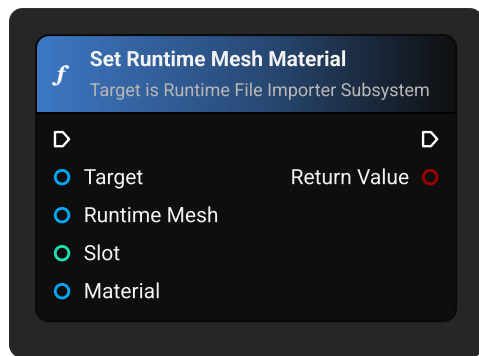
Replaces the materials slot by slot, on every component showing the model. Same as setting its Materials variable.

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::SetRuntimeMeshMaterials`

INPUTS		
Pin	Type	Description
Target	 Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Runtime Mesh	 Runtime Mesh Object Reference	The Runtime Mesh (an imported model).
Materials	 Array of Material Interface Object Reference	One material per slot.

Set Runtime Mesh Material FUNCTION



Replaces one slot's material, on every component showing the model.
Returns false if the slot doesn't exist.

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::SetRuntimeMeshMaterial`

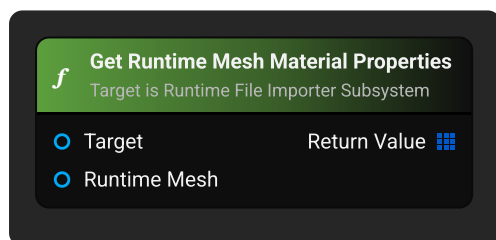
INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Runtime Mesh	Runtime Mesh Object Reference	The Runtime Mesh (an imported model).
Slot	Integer	Index of the material slot.
Material	Material Interface Object Reference	The material.

OUTPUTS

Pin	Type	Description
Return Value	Boolean	False if the slot doesn't exist.

Get Runtime Mesh Material Properties PURE



The model's materials as described in the file (colors, textures, ...). Same as its Material Properties variable.

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::GetRuntimeMeshMaterialProperties`

INPUTS

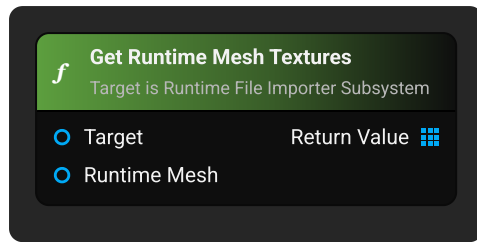
Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Runtime Mesh	Runtime Mesh Object Reference	The Runtime Mesh (an imported model).

OUTPUTS

Pin	Type	Description
Return Value	Array of Runtime Mesh Material Info	The model's materials as described in the file (colors, textures, ...).

Get Runtime Mesh Textures

PURE



Every texture loaded from the model. Same as its Textures variable.

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::GetRuntimeMeshTextures`

INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Runtime Mesh	Runtime Mesh Object Reference	The Runtime Mesh (an imported model).

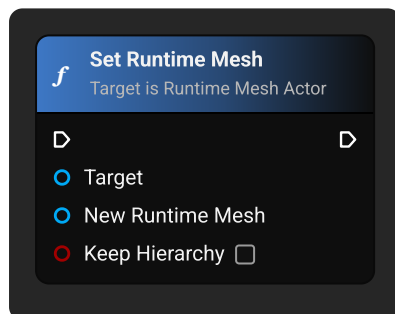
OUTPUTS

Pin	Type	Description
Return Value	Array of Texture 2D Object Reference	Every texture loaded from the model.

General 4 nodes

Set Runtime Mesh

FUNCTION



Shows a Runtime Mesh, replacing what the actor showed before. With Keep Hierarchy, the model's nodes become separate components that can be moved, hidden or given other materials individually.

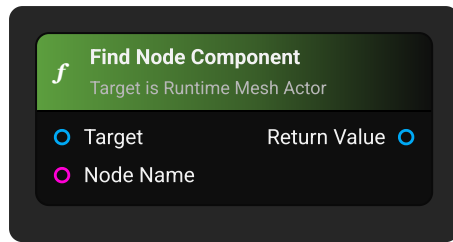
Target: Runtime Mesh Actor

C++: `ARuntimeImportedMeshActor::SetRuntimeMesh`

INPUTS

Pin	Type	Description
Target	Runtime Mesh Actor Object Reference	The Runtime Mesh Actor to use.
New Runtime Mesh	Runtime Mesh Object Reference	The Runtime Mesh to show instead.
Keep Hierarchy	Boolean	Rebuild the model's node hierarchy as separate components (static models only). Default: <code>false</code>

Find Node Component PURE



The node component with this name (the model's node names), or null.

Target: Runtime Mesh Actor

C++: `ARuntimeImportedMeshActor::FindNodeComponent`

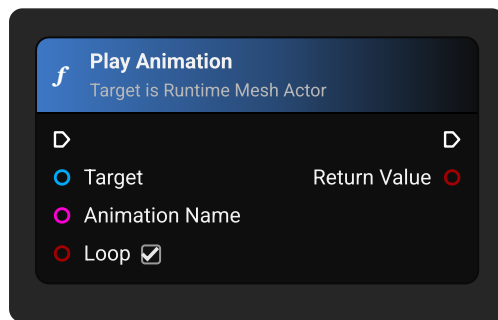
INPUTS

Pin	Type	Description
Target	Runtime Mesh Actor Object Reference	The Runtime Mesh Actor to use.
Node Name	String	Name of the node in the model file.

OUTPUTS

Pin	Type	Description
Return Value	Scene Component Object Reference	The node component with this name (the model's node names), or null.

Play Animation FUNCTION



Plays one of the model's animations, by its name in the file (empty: the first one). Returns false if there's no such animation.

Target: Runtime Mesh Actor

C++: `ARuntimeImportedMeshActor::PlayAnimation`

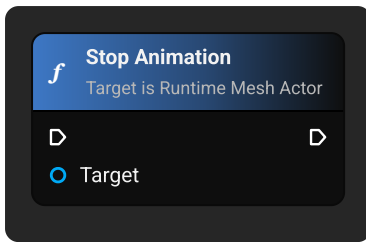
INPUTS

Pin	Type	Description
Target	Runtime Mesh Actor Object Reference	The Runtime Mesh Actor to use.
Animation Name	String	Name of the animation in the model file. Empty: the first one.
Loop	Boolean	Play the animation in a loop. Default: <code>true</code>

OUTPUTS

Pin	Type	Description
Return Value	Boolean	False if there's no such animation.

Stop Animation FUNCTION



Stops the animation being played.

Target: Runtime Mesh Actor

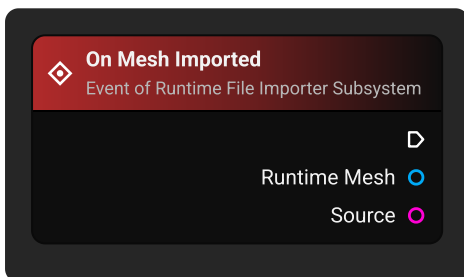
C++: `ARuntimeImportedMeshActor::StopAnimation`

INPUTS

Pin	Type	Description
Target	Runtime Mesh Actor Object Reference	The Runtime Mesh Actor to use.

Events 4 nodes

On Mesh Imported EVENT



Any mesh import succeeded. Source is the file path or URL (empty for buffers).

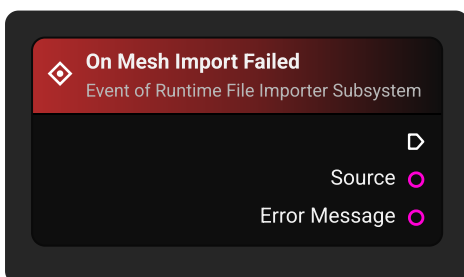
Bind it with "Bind Event to On Mesh Imported" (or "Assign On Mesh Imported") on the Runtime File Importer Subsystem.

C++: `URuntimeFileImporterSubsystem::OnMeshImported`

OUTPUTS

Pin	Type	Description
Runtime Mesh	Runtime Mesh Object Reference	The Runtime Mesh (an imported model).
Source	String	File path or URL the import came from (empty for buffers and strings).

On Mesh Import Failed EVENT



Any mesh import (from a file, buffer or URL) failed. Source is the file path or URL (empty for buffers).

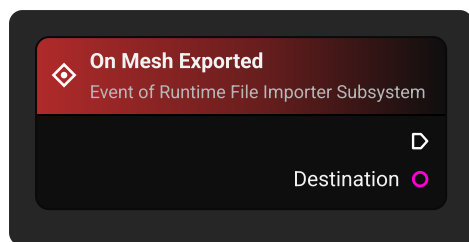
Bind it with "Bind Event to On Mesh Import Failed" (or "Assign On Mesh Import Failed") on the Runtime File Importer Subsystem.

C++: `URuntimeFileImporterSubsystem::OnMeshImportFailed`

OUTPUTS

Pin	Type	Description
Source	String	File path or URL the import came from (empty for buffers and strings).
Error Message	String	Empty on success; otherwise why it failed.

On Mesh Exported EVENT



Any mesh export or conversion succeeded. Destination is the output file path (empty for buffers).

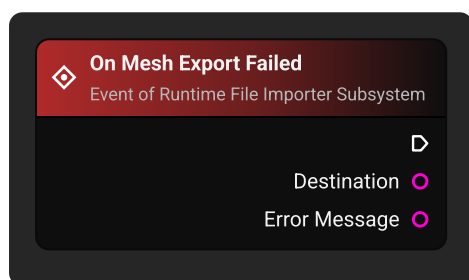
Bind it with "Bind Event to On Mesh Exported" (or "Assign On Mesh Exported") on the Runtime File Importer Subsystem.

```
C++: URuntimeFileImporterSubsystem::OnMeshExported
```

OUTPUTS

Pin	Type	Description
Destination	String	Output file path (empty when writing to a buffer).

On Mesh Export Failed EVENT



Any mesh export or conversion failed. Destination is the output file path (empty for buffers).

Bind it with "Bind Event to On Mesh Export Failed" (or "Assign On Mesh Export Failed") on the Runtime File Importer Subsystem.

```
C++: URuntimeFileImporterSubsystem::OnMeshExportFailed
```

OUTPUTS

Pin	Type	Description
Destination	String	Output file path (empty when writing to a buffer).
Error Message	String	Empty on success; otherwise why it failed.

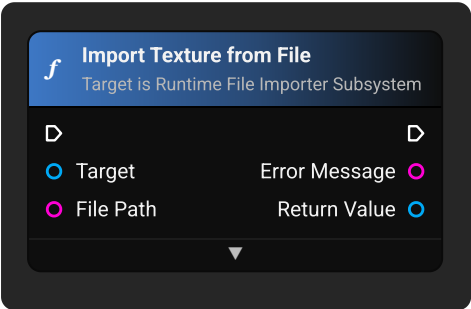
Texture

Import images as textures or as editable Runtime Images, edit pixels (resize, crop, rotate, adjust colors, draw), inspect and change any texture, convert between texture types and export image files.

Import 12 nodes

Import Texture from File

FUNCTION



Imports an image file (PNG, JPEG, BMP, TGA, EXR, HDR, TIFF, ICO, GIF) as a texture, usable anywhere a texture goes (materials, UMG brushes, Create Material From Texture). HDR files become float textures. Blocks until finished; use the Async version for large images.

Target: Runtime File Importer Subsystem

Async version: [Import Texture From File \(Async\)](#)

C++: `URuntimeFileImporterSubsystem::ImportTextureFromFile`

INPUTS

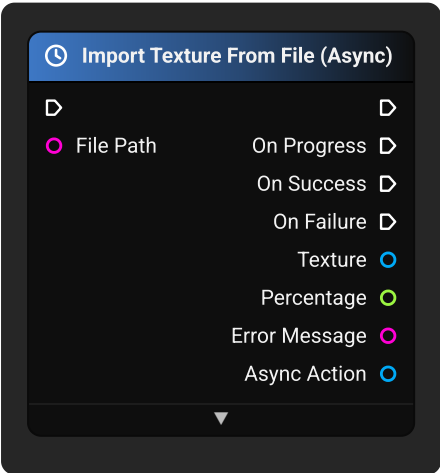
Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
File Path	String	Full path of the file on disk.
Image Format ADVANCED	ERuntime Image Format	Format of the image. Auto Detect reads it from the contents.
Options ADVANCED OPTIONAL	Runtime Texture Import Options	How imported (and converted) images become textures.

OUTPUTS

Pin	Type	Description
Error Message	String	Empty on success; otherwise why it failed.
Return Value	Texture 2D Object Reference	The result, or None/empty if it failed (Error Message says why).

Import Texture From File (Async)

ASYNC



Imports an image file (PNG, JPEG, BMP, TGA, EXR, HDR, TIFF, ICO, GIF) as a texture without blocking the game thread.

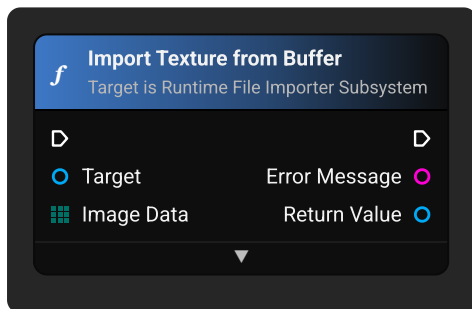
Blocking version: [Import Texture from File](#)

C++: `UImportTextureAsyncAction::ImportTextureFromFileAsync`

INPUTS		
Pin	Type	Description
File Path	 String	Full path of the file on disk.
Image Format ADVANCED	 ERuntime Image Format	Format of the image. Auto Detect reads it from the contents.
Options ADVANCED OPTIONAL	 Runtime Texture Import Options	How imported (and converted) images become textures.
OUTPUTS		
Pin	Type	Description
On Progress	 Exec	Fires as the operation advances (Percentage from 0 to 100). Other outputs are not set yet.
On Success	 Exec	Fires when the operation has finished successfully.
On Failure	 Exec	Fires if the operation failed or was cancelled; Error Message says why.
Texture	 Texture 2D Object Reference	The Texture 2D.
Percentage	 Float	Progress from 0 to 100.
Error Message	 String	Empty on success; otherwise why it failed.
Async Action	 Import Texture Async Action Object Reference	This node's operation. Call Cancel on it to stop the operation (no more events fire).

Import Texture from Buffer

FUNCTION



Imports an image from bytes already in memory (e.g. received over the network) as a texture.

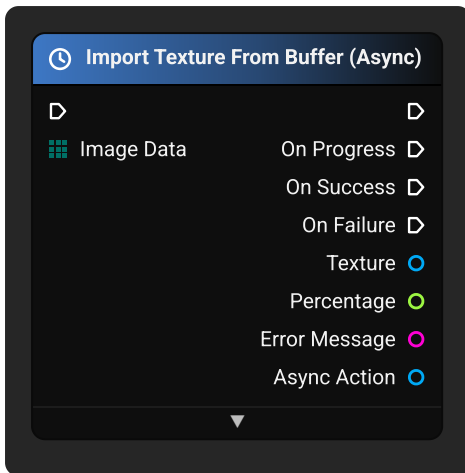
Target: Runtime File Importer Subsystem

Async version: [Import Texture From Buffer \(Async\)](#)

C++: `URuntimeFileImporterSubsystem::ImportTextureFromBuffer`

INPUTS		
Pin	Type	Description
Target	 Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Image Data	 Array of Byte	Bytes of an image file.
Image Format ADVANCED	 ERuntime Image Format	Format of the image. Auto Detect reads it from the contents.
Options ADVANCED OPTIONAL	 Runtime Texture Import Options	How imported (and converted) images become textures.
OUTPUTS		
Pin	Type	Description
Error Message	 String	Empty on success; otherwise why it failed.
Return Value	 Texture 2D Object Reference	The result, or None/empty if it failed (Error Message says why).

Import Texture From Buffer (Async) ASYNC



Imports an image in memory as a texture without blocking the game thread.

Blocking version: [Import Texture from Buffer](#)

C++: `UImportTextureAsyncAction::ImportTextureFromBufferAsync`

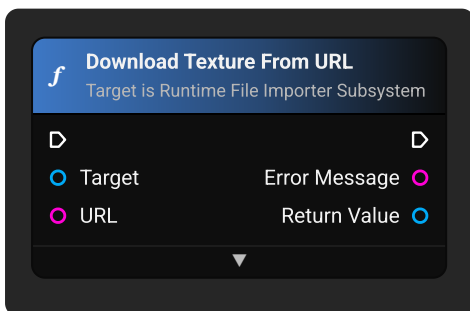
INPUTS

Pin	Type	Description
Image Data	Array of Byte	Bytes of an image file.
Image Format ADVANCED	ERuntime Image Format	Format of the image. Auto Detect reads it from the contents.
Options ADVANCED OPTIONAL	Runtime Texture Import Options	How imported (and converted) images become textures.

OUTPUTS

Pin	Type	Description
On Progress	Exec	Fires as the operation advances (Percentage from 0 to 100). Other outputs are not set yet.
On Success	Exec	Fires when the operation has finished successfully.
On Failure	Exec	Fires if the operation failed or was cancelled; Error Message says why.
Texture	Texture 2D Object Reference	The Texture 2D.
Percentage	Float	Progress from 0 to 100.
Error Message	String	Empty on success; otherwise why it failed.
Async Action	Import Texture Async Action Object Reference	This node's operation. Call Cancel on it to stop the operation (no more events fire).

Download Texture From URL FUNCTION



Downloads an image over HTTP(S) and imports it as a texture. This FREEZES the game until the download finishes or times out; prefer the Async version. If Save To File Path is set, the downloaded file is also written there, unmodified.

Target: Runtime File Importer Subsystem

Async version: [Download Texture From URL \(Async\)](#)

C++: `URuntimeFileImporterSubsystem::DownloadTextureFromURL`

INPUTS

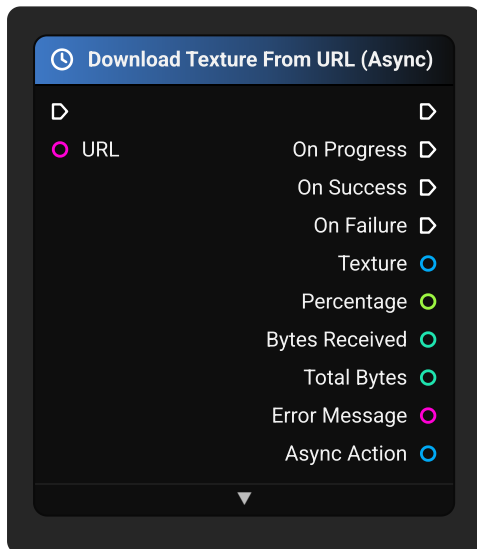
Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
URL	String	The http(s) address to download. Google Docs, Sheets, Slides and Drive share links are converted automatically.
Image Format ADVANCED	ERuntime Image Format	Format of the image. Auto Detect reads it from the contents.
Options ADVANCED OPTIONAL	Runtime Texture Import Options	How imported (and converted) images become textures.
Save to File Path ADVANCED	String	Optional. Also writes the downloaded file there, unmodified. Default: (empty)
Timeout Seconds ADVANCED	Float	Gives up (and fails) if the download takes longer than this. Default: 30.0

OUTPUTS

Pin	Type	Description
Error Message	String	Empty on success; otherwise why it failed.
Return Value	Texture 2D Object Reference	The result, or None/empty if it failed (Error Message says why).

Download Texture From URL (Async)

ASYNC



Downloads an image over HTTP(S) and imports it as a texture. If Save To File Path is set, the downloaded file is also written there.

Blocking version: [Download Texture From URL](#)

C++: `UDownloadTextureFromURLAsyncAction::DownloadTextureFromURLAsync`

INPUTS

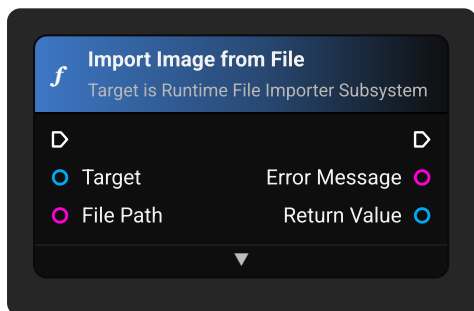
Pin	Type	Description
URL	String	The http(s) address to download. Google Docs, Sheets, Slides and Drive share links are converted automatically.
Image Format ADVANCED	ERuntime Image Format	Format of the image. Auto Detect reads it from the contents.
Options ADVANCED OPTIONAL	Runtime Texture Import Options	How imported (and converted) images become textures.
Save to File Path ADVANCED	String	Optional. Also writes the downloaded file there, unmodified. Default: (empty)
Timeout Seconds ADVANCED	Float	Gives up (and fails) if the download takes longer than this. Default: 60.0

OUTPUTS

Pin	Type	Description
On Progress	Exec	Fires repeatedly while downloading. Percentage (0-100) and Total Bytes are 0 if the server doesn't report the file size.
On Success	Exec	Fires when the operation has finished successfully.
On Failure	Exec	Fires if the operation failed or was cancelled; Error Message says why.
Texture	Texture 2D Object Reference	The Texture 2D.
Percentage	Float	Progress from 0 to 100.
Bytes Received	Integer	Bytes downloaded so far.
Total Bytes	Integer	Size of the download in bytes, or 0 while the server hasn't said.
Error Message	String	Empty on success; otherwise why it failed.
Async Action	Download Texture from URLAsync Action Object Reference	This node's operation. Call Cancel on it to stop the operation (no more events fire).

Import Image from File

FUNCTION



Imports an image file as a Runtime Image: pixels in memory to edit with the Image nodes (no GPU texture yet).

Target: Runtime File Importer Subsystem

Async version: [Import Image From File \(Async\)](#)

C++: `URuntimeFileImporterSubsystem::ImportImageFromFile`

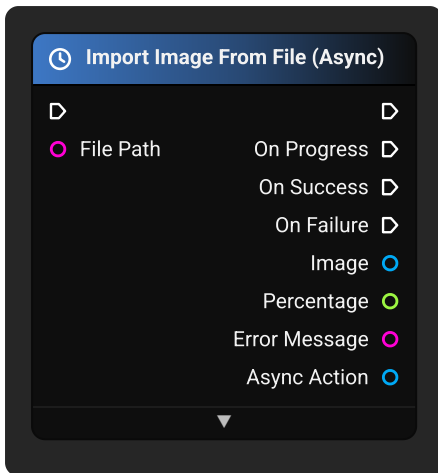
INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
File Path	String	Full path of the file on disk.
Image Format	ADVANCED ERuntime Image Format	Format of the image. Auto Detect reads it from the contents. Default: <code>Auto Detect</code>

OUTPUTS

Pin	Type	Description
Error Message	String	Empty on success; otherwise why it failed.
Return Value	Runtime Image Object Reference	The result, or None/empty if it failed (Error Message says why).

Import Image From File (Async) ASYNC



Imports an image file as a Runtime Image (editable pixels) without blocking the game thread.

Blocking version: [Import Image from File](#)

C++: `UImportImageAsyncAction::ImportImageFromFileAsync`

INPUTS

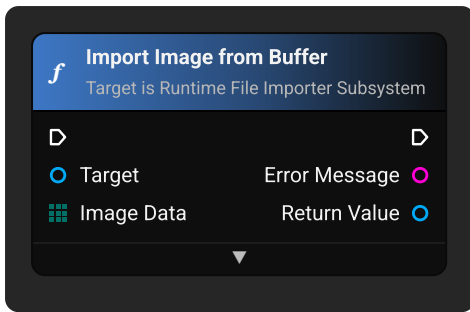
Pin	Type	Description
File Path	 String	Full path of the file on disk.
Image Format ADVANCED	 ERuntime Image Format	Format of the image. Auto Detect reads it from the contents. Default: Auto Detect

OUTPUTS

Pin	Type	Description
On Progress	 Exec	Fires as the operation advances (Percentage from 0 to 100). Other outputs are not set yet.
On Success	 Exec	Fires when the operation has finished successfully.
On Failure	 Exec	Fires if the operation failed or was cancelled; Error Message says why.
Image	 Runtime Image Object Reference	The Runtime Image.
Percentage	 Float	Progress from 0 to 100.
Error Message	 String	Empty on success; otherwise why it failed.
Async Action	 Import Image Async Action Object Reference	This node's operation. Call Cancel on it to stop the operation (no more events fire).

Import Image from Buffer

FUNCTION



Imports an image in memory as a Runtime Image.

Target: Runtime File Importer Subsystem

Async version: [Import Image From Buffer \(Async\)](#)

C++: `URuntimeFileImporterSubsystem::ImportImageFromBuffer`

INPUTS

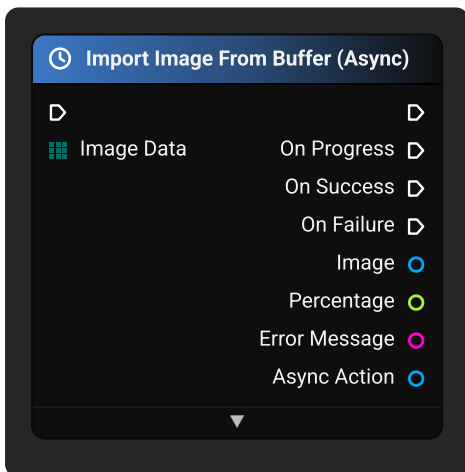
Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Image Data	Array of Byte	Bytes of an image file.
Image Format	ERuntime Image Format	Format of the image. Auto Detect reads it from the contents. Default: <code>Auto Detect</code>

OUTPUTS

Pin	Type	Description
Error Message	String	Empty on success; otherwise why it failed.
Return Value	Runtime Image Object Reference	The result, or None/empty if it failed (Error Message says why).

Import Image From Buffer (Async)

ASYNC



Imports an image in memory as a Runtime Image without blocking the game thread.

Blocking version: [Import Image from Buffer](#)

C++: `UImportImageAsyncAction::ImportImageFromBufferAsync`

INPUTS

Pin	Type	Description
Image Data	Array of Byte	Bytes of an image file.
Image Format	ERuntime Image Format	Format of the image. Auto Detect reads it from the contents. Default: <code>Auto Detect</code>

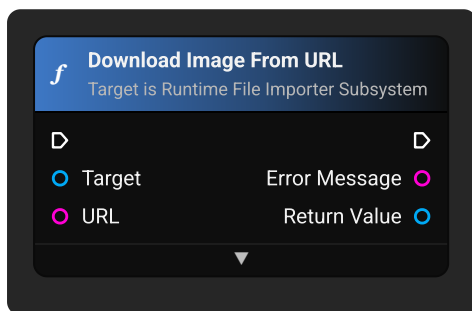
OUTPUTS

Pin	Type	Description
On Progress	Exec	Fires as the operation advances (Percentage from 0 to 100). Other outputs are not set yet.

Pin	Type	Description
On Success	 Exec	Fires when the operation has finished successfully.
On Failure	 Exec	Fires if the operation failed or was cancelled; Error Message says why.
Image	 Runtime Image Object Reference	The Runtime Image.
Percentage	 Float	Progress from 0 to 100.
Error Message	 String	Empty on success; otherwise why it failed.
Async Action	 Import Image Async Action Object Reference	This node's operation. Call Cancel on it to stop the operation (no more events fire).

Download Image From URL

FUNCTION



Downloads an image as a Runtime Image. FREEZES the game until done; prefer the Async version.

Target: Runtime File Importer Subsystem

Async version: [Download Image From URL \(Async\)](#)

C++: `URuntimeFileImporterSubsystem::DownloadImageFromURL`

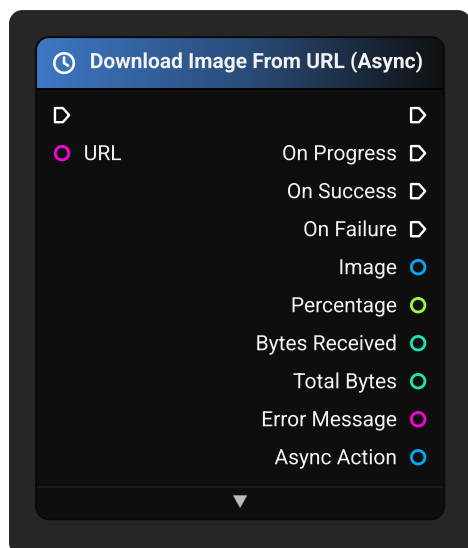
INPUTS

Pin	Type	Description
Target	 Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
URL	 String	The http(s) address to download. Google Docs, Sheets, Slides and Drive share links are converted automatically.
Image Format ADVANCED	 ERuntime Image Format	Format of the image. Auto Detect reads it from the contents. Default: <code>Auto Detect</code>
Save to File Path ADVANCED	 String	Optional. Also writes the downloaded file there, unmodified. Default: <code>(empty)</code>
Timeout Seconds ADVANCED	 Float	Gives up (and fails) if the download takes longer than this. Default: <code>30.0</code>

OUTPUTS

Pin	Type	Description
Error Message	 String	Empty on success; otherwise why it failed.
Return Value	 Runtime Image Object Reference	The result, or None/empty if it failed (Error Message says why).

Download Image From URL (Async) ASYNC



Downloads an image over HTTP(S) as a Runtime Image. If Save To File Path is set, the downloaded file is also written there.

Blocking version: [Download Image From URL](#)

C++: `UDownloadImageFromURLAsyncAction::DownloadImageFromURLAsync`

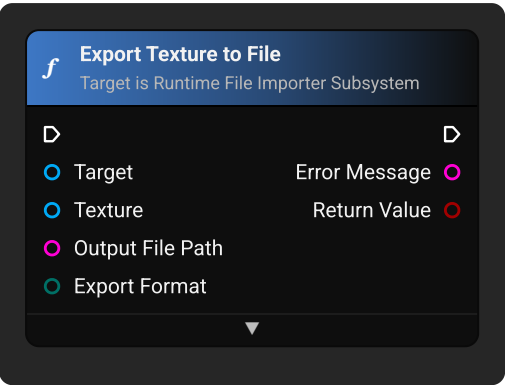
INPUTS

Pin	Type	Description
URL	String	The http(s) address to download. Google Docs, Sheets, Slides and Drive share links are converted automatically.
Image Format ADVANCED	ERuntime Image Format	Format of the image. Auto Detect reads it from the contents. Default: Auto Detect
Save to File Path ADVANCED	String	Optional. Also writes the downloaded file there, unmodified. Default: (empty)
Timeout Seconds ADVANCED	Float	Gives up (and fails) if the download takes longer than this. Default: 60.0

OUTPUTS

Pin	Type	Description
On Progress	Exec	Fires as the operation advances (Percentage from 0 to 100). Other outputs are not set yet.
On Success	Exec	Fires when the operation has finished successfully.
On Failure	Exec	Fires if the operation failed or was cancelled; Error Message says why.
Image	Runtime Image Object Reference	The Runtime Image.
Percentage	Float	Progress from 0 to 100.
Bytes Received	Integer	Bytes downloaded so far.
Total Bytes	Integer	Size of the download in bytes, or 0 while the server hasn't said.
Error Message	String	Empty on success; otherwise why it failed.
Async Action	Download Image from URLAsync Action Object Reference	This node's operation. Call Cancel on it to stop the operation (no more events fire).

Export Texture to File FUNCTION



Writes any texture to an image file: Texture 2D (imported or an asset), render target, video texture (its current frame), cube (as a panorama). Textures whose pixels are only on the GPU (cooked assets, render targets) are read back from it, which needs a renderer.

Target: Runtime File Importer Subsystem
Async version: [Export Texture To File \(Async\)](#)
C++: URuntimeFileImporterSubsystem::ExportTextureToFile

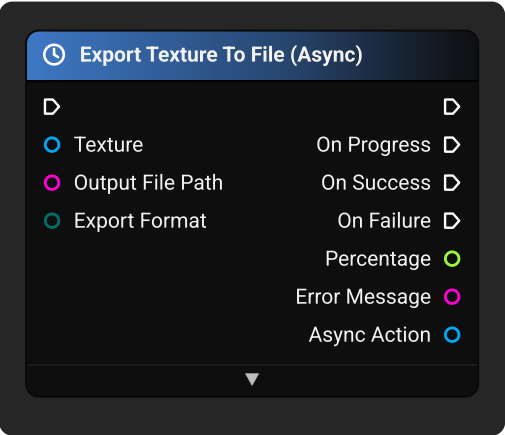
INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Texture	Texture Object Reference	Any texture: Texture 2D, render target, cube, media or video texture.
Output File Path	String	Full path of the file to write. Missing folders are created; an existing file is replaced.
Export Format	ERuntime Image Export Format	Format to write.
Options	ADVANCED OPTIONAL Runtime Image Export Options	How textures and images are written to image files.

OUTPUTS

Pin	Type	Description
Error Message	String	Empty on success; otherwise why it failed.
Return Value	Boolean	True on success. False if it failed (Error Message says why).

Export Texture To File (Async) ASYNC



Writes any texture to an image file without blocking the game thread (GPU-only pixels are read back without waiting).

Blocking version: [Export Texture to File](#)
C++: UExportTextureAsyncAction::ExportTextureToFileAsync

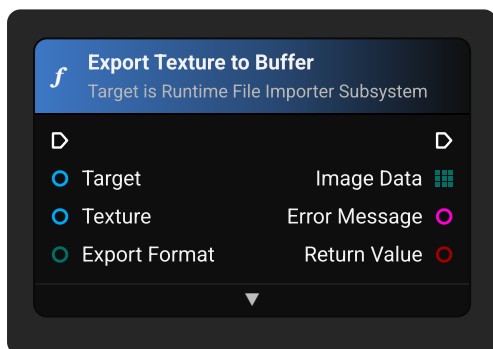
INPUTS

Pin	Type	Description
Texture	Texture Object Reference	Any texture: Texture 2D, render target, cube, media or video texture.

Pin	Type	Description
Output File Path	String	Full path of the file to write. Missing folders are created; an existing file is replaced.
Export Format	ERuntime Image Export Format	Format to write.
Options ADVANCED OPTIONAL	Runtime Image Export Options	How textures and images are written to image files.
OUTPUTS		
Pin	Type	Description
On Progress	Exec	Fires as the operation advances (Percentage from 0 to 100). Other outputs are not set yet.
On Success	Exec	Fires when the operation has finished successfully.
On Failure	Exec	Fires if the operation failed or was cancelled; Error Message says why.
Percentage	Float	Progress from 0 to 100.
Error Message	String	Empty on success; otherwise why it failed.
Async Action	Export Texture Async Action Object Reference	This node's operation. Call Cancel on it to stop the operation (no more events fire).

Export Texture to Buffer

FUNCTION



Encodes any texture into image file bytes (e.g. to send over the network or save in a save game).

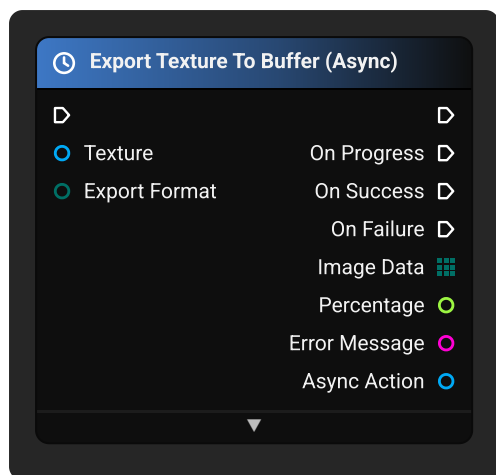
Target: Runtime File Importer Subsystem

Async version: [Export Texture To Buffer \(Async\)](#)

C++: `URuntimeFileImporterSubsystem::ExportTextureToBuffer`

INPUTS		
Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Texture	Texture Object Reference	Any texture: Texture 2D, render target, cube, media or video texture.
Export Format	ERuntime Image Export Format	Format to write.
Options ADVANCED OPTIONAL	Runtime Image Export Options	How textures and images are written to image files.
OUTPUTS		
Pin	Type	Description
Image Data	Array of Byte	Bytes of an image file.
Error Message	String	Empty on success; otherwise why it failed.
Return Value	Boolean	True on success. False if it failed (Error Message says why).

Export Texture To Buffer (Async) ASYNC



Encodes any texture into image file bytes without blocking the game thread.

Blocking version: [Export Texture to Buffer](#)

C++: `UEncodeTextureAsyncAction::ExportTextureToBufferAsync`

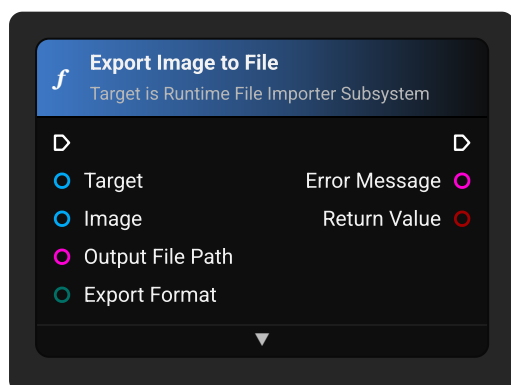
INPUTS

Pin	Type	Description
Texture	Texture Object Reference	Any texture: Texture 2D, render target, cube, media or video texture.
Export Format	ERuntime Image Export Format	Format to write.
Options	ADVANCED OPTIONAL Runtime Image Export Options	How textures and images are written to image files.

OUTPUTS

Pin	Type	Description
On Progress	Exec	Fires as the operation advances (Percentage from 0 to 100). Other outputs are not set yet.
On Success	Exec	Fires when the operation has finished successfully.
On Failure	Exec	Fires if the operation failed or was cancelled; Error Message says why.
Image Data	Array of Byte	Bytes of an image file.
Percentage	Float	Progress from 0 to 100.
Error Message	String	Empty on success; otherwise why it failed.
Async Action	Encode Texture Async Action Object Reference	This node's operation. Call Cancel on it to stop the operation (no more events fire).

Export Image to File FUNCTION



Writes a Runtime Image to an image file.

Target: Runtime File Importer Subsystem

Async version: [Export Image To File \(Async\)](#)

C++: `URuntimeFileImporterSubsystem::ExportImageToFile`

INPUTS

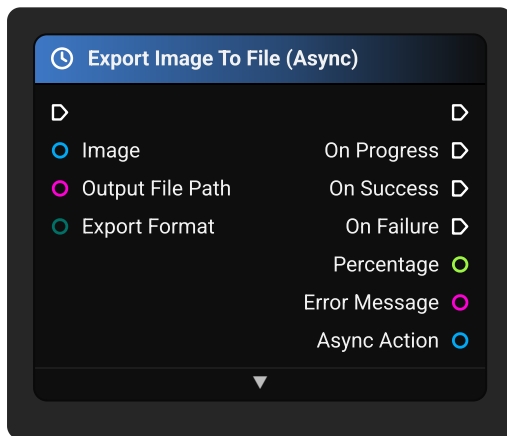
Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Image	Runtime Image Object Reference	The Runtime Image.
Output File Path	String	Full path of the file to write. Missing folders are created; an existing file is replaced.
Export Format	ERuntime Image Export Format	Format to write.
Options	ADVANCED OPTIONAL Runtime Image Export Options	How textures and images are written to image files.

OUTPUTS

Pin	Type	Description
Error Message	String	Empty on success; otherwise why it failed.
Return Value	Boolean	True on success. False if it failed (Error Message says why).

Export Image To File (Async)

ASYNC



Writes a Runtime Image to an image file without blocking the game thread.

Blocking version: [Export Image to File](#)

C++: `UExportTextureAsyncAction::ExportImageToFileAsync`

INPUTS

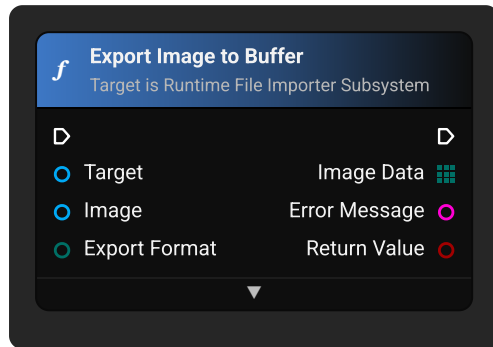
Pin	Type	Description
Image	Runtime Image Object Reference	The Runtime Image.
Output File Path	String	Full path of the file to write. Missing folders are created; an existing file is replaced.
Export Format	ERuntime Image Export Format	Format to write.
Options	ADVANCED OPTIONAL Runtime Image Export Options	How textures and images are written to image files.

OUTPUTS

Pin	Type	Description
On Progress	Exec	Fires as the operation advances (Percentage from 0 to 100). Other outputs are not set yet.
On Success	Exec	Fires when the operation has finished successfully.
On Failure	Exec	Fires if the operation failed or was cancelled; Error Message says why.
Percentage	Float	Progress from 0 to 100.

Pin	Type	Description
Error Message	String	Empty on success; otherwise why it failed.
Async Action	Export Texture Async Action Object Reference	This node's operation. Call Cancel on it to stop the operation (no more events fire).

Export Image to Buffer FUNCTION



Encodes a Runtime Image into image file bytes.

Target: Runtime File Importer Subsystem

Async version: [Export Image To Buffer \(Async\)](#)

C++: `URuntimeFileImporterSubsystem::ExportImageToBuffer`

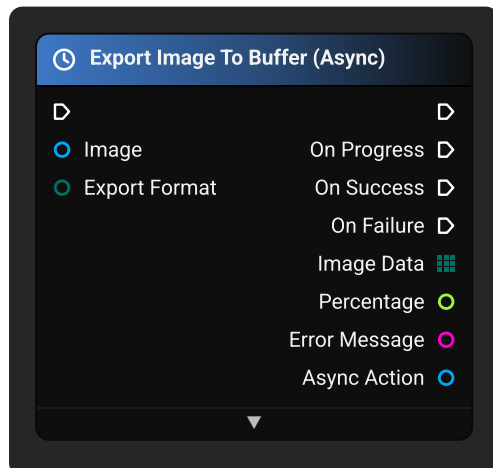
INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Image	Runtime Image Object Reference	The Runtime Image.
Export Format	ERuntime Image Export Format	Format to write.
Options	Runtime Image Export Options	How textures and images are written to image files.

OUTPUTS

Pin	Type	Description
Image Data	Array of Byte	Bytes of an image file.
Error Message	String	Empty on success; otherwise why it failed.
Return Value	Boolean	True on success. False if it failed (Error Message says why).

Export Image To Buffer (Async) ASYNC



Encodes a Runtime Image into image file bytes without blocking the game thread.

Blocking version: [Export Image to Buffer](#)

C++: `UEncodeTextureAsyncAction::ExportImageToBufferAsync`

INPUTS

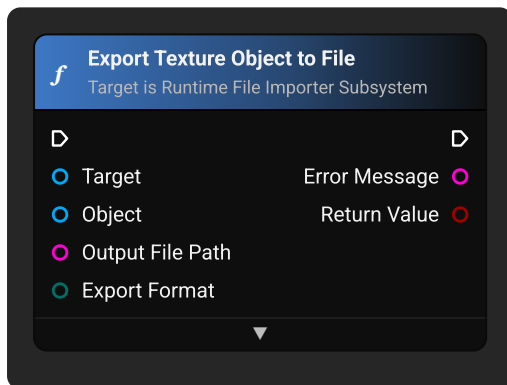
Pin	Type	Description
Image	Runtime Image Object Reference	The Runtime Image.
Export Format	ERuntime Image Export Format	Format to write.
Options	Runtime Image Export Options	How textures and images are written to image files.

OUTPUTS

Pin	Type	Description
On Progress	Exec	Fires as the operation advances (Percentage from 0 to 100). Other outputs are not set yet.
On Success	Exec	Fires when the operation has finished successfully.
On Failure	Exec	Fires if the operation failed or was cancelled; Error Message says why.
Image Data	Array of Byte	Bytes of an image file.
Percentage	Float	Progress from 0 to 100.
Error Message	String	Empty on success; otherwise why it failed.
Async Action	Encode Texture Async Action Object Reference	This node's operation. Call Cancel on it to stop the operation (no more events fire).

Export Texture Object to File

FUNCTION



Writes any texture object (texture, Runtime Image, or a Runtime Video's current frame; detected automatically) to an image file.

Target: Runtime File Importer Subsystem

Async version: [Export Texture Object To File \(Async\)](#)

C++: `URuntimeFileImporterSubsystem::ExportTextureObjectToFile`

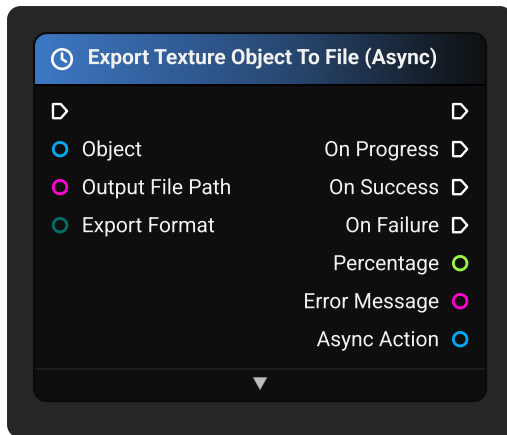
INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Object	Object Object Reference	Any supported object (detected automatically).
Output File Path	String	Full path of the file to write. Missing folders are created; an existing file is replaced.
Export Format	ERuntime Image Export Format	Format to write.
Options	Runtime Image Export Options	How textures and images are written to image files.

OUTPUTS

Pin	Type	Description
Error Message	String	Empty on success; otherwise why it failed.
Return Value	Boolean	True on success. False if it failed (Error Message says why).

Export Texture Object To File (Async) ASYNC



Writes any texture object (texture, Runtime Image, Runtime Video frame) to an image file without blocking the game thread.

Blocking version: [Export Texture Object to File](#)

C++: `UExportTextureAsyncAction::ExportTextureObjectToFileAsync`

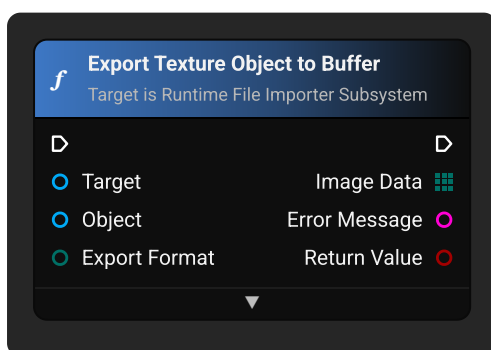
INPUTS

Pin	Type	Description
Object	Object Object Reference	Any supported object (detected automatically).
Output File Path	String	Full path of the file to write. Missing folders are created; an existing file is replaced.
Export Format	ERuntime Image Export Format	Format to write.
Options	ADVANCED OPTIONAL Runtime Image Export Options	How textures and images are written to image files.

OUTPUTS

Pin	Type	Description
On Progress	Exec	Fires as the operation advances (Percentage from 0 to 100). Other outputs are not set yet.
On Success	Exec	Fires when the operation has finished successfully.
On Failure	Exec	Fires if the operation failed or was cancelled; Error Message says why.
Percentage	Float	Progress from 0 to 100.
Error Message	String	Empty on success; otherwise why it failed.
Async Action	Export Texture Async Action Object Reference	This node's operation. Call Cancel on it to stop the operation (no more events fire).

Export Texture Object to Buffer FUNCTION



Encodes any texture object into image file bytes.

Target: Runtime File Importer Subsystem

Async version: [Export Texture Object To Buffer \(Async\)](#)

C++: `URuntimeFileImporterSubsystem::ExportTextureObjectToBuffer`

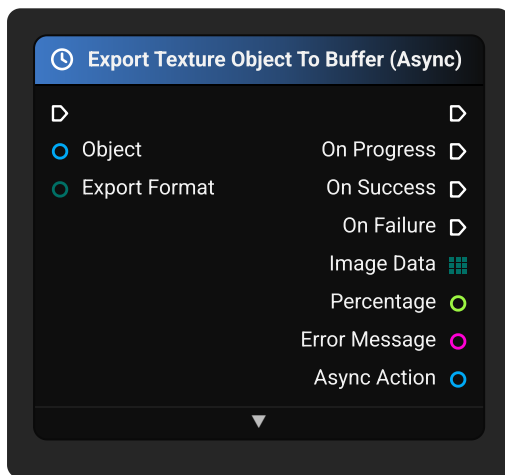
INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Object	Object Object Reference	Any supported object (detected automatically).
Export Format	ERuntime Image Export Format	Format to write.
Options	ADVANCED OPTIONAL Runtime Image Export Options	How textures and images are written to image files.

OUTPUTS

Pin	Type	Description
Image Data	Array of Byte	Bytes of an image file.
Error Message	String	Empty on success; otherwise why it failed.
Return Value	Boolean	True on success. False if it failed (Error Message says why).

Export Texture Object To Buffer (Async) ASYNC



Encodes any texture object into image file bytes without blocking the game thread.

Blocking version: [Export Texture Object to Buffer](#)

C++: `UEncodeTextureAsyncAction::ExportTextureObjectToBufferAsync`

INPUTS

Pin	Type	Description
Object	Object Object Reference	Any supported object (detected automatically).
Export Format	ERuntime Image Export Format	Format to write.
Options	ADVANCED OPTIONAL Runtime Image Export Options	How textures and images are written to image files.

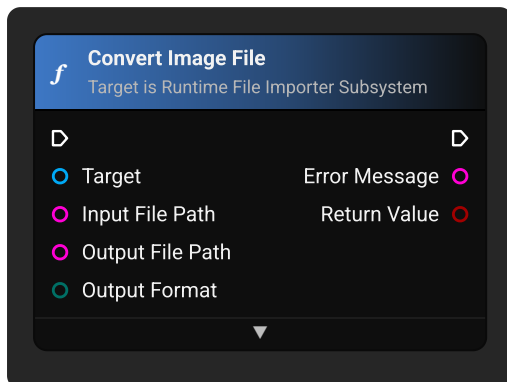
OUTPUTS

Pin	Type	Description
On Progress	Exec	Fires as the operation advances (Percentage from 0 to 100). Other outputs are not set yet.
On Success	Exec	Fires when the operation has finished successfully.
On Failure	Exec	Fires if the operation failed or was cancelled; Error Message says why.
Image Data	Array of Byte	Bytes of an image file.
Percentage	Float	Progress from 0 to 100.
Error Message	String	Empty on success; otherwise why it failed.

Pin	Type	Description
Async Action	 Encode Texture Async Action Object Reference	This node's operation. Call Cancel on it to stop the operation (no more events fire).

Convert Image File

FUNCTION



Converts an image file to another format (e.g. PNG to JPEG, EXR to HDR), scaling it down to Max Size (0: keep its size).

Target: Runtime File Importer Subsystem

Async version: [Convert Image File \(Async\)](#)

C++: `URuntimeFileImporterSubsystem::ConvertImageFile`

INPUTS

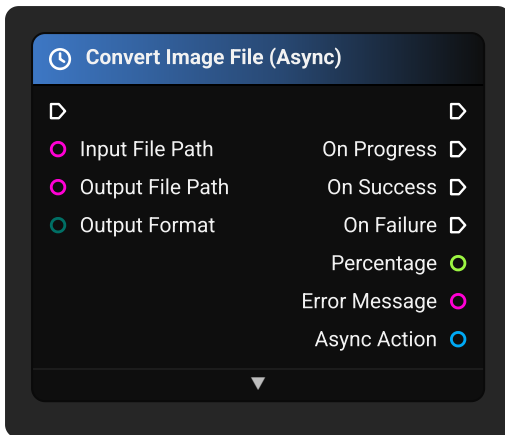
Pin	Type	Description
Target	 Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Input File Path	 String	Full path of the file to convert.
Output File Path	 String	Full path of the file to write. Missing folders are created; an existing file is replaced.
Output Format	 ERuntime Image Export Format	Format to write.
Input Format ADVANCED	 ERuntime Image Format	Format of the input. Auto Detect reads it from the contents.
Options ADVANCED OPTIONAL	 Runtime Image Export Options	How textures and images are written to image files.
Max Size ADVANCED	 Integer	Largest width or height of the result; bigger images are scaled down. 0: no limit. Default: 0

OUTPUTS

Pin	Type	Description
Error Message	 String	Empty on success; otherwise why it failed.
Return Value	 Boolean	True on success. False if it failed (Error Message says why).

Convert Image File (Async)

ASYNC



Converts an image file to another format without blocking the game thread.

Blocking version: [Convert Image File](#)

C++: `UExportTextureAsyncAction::ConvertImageFileAsync`

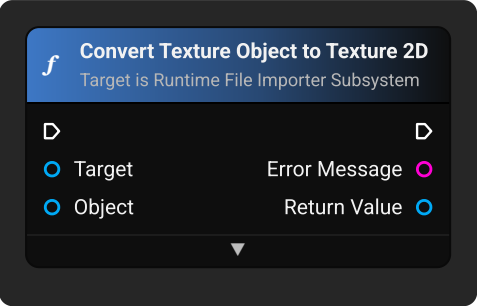
INPUTS

Pin	Type	Description
Input File Path	String	Full path of the file to convert.
Output File Path	String	Full path of the file to write. Missing folders are created; an existing file is replaced.
Output Format	ERuntime Image Export Format	Format to write.
Input Format ADVANCED	ERuntime Image Format	Format of the input. Auto Detect reads it from the contents.
Options ADVANCED OPTIONAL	Runtime Image Export Options	How textures and images are written to image files.
Max Size ADVANCED	Integer	Largest width or height of the result; bigger images are scaled down. 0: no limit. Default: 0

OUTPUTS

Pin	Type	Description
On Progress	Exec	Fires as the operation advances (Percentage from 0 to 100). Other outputs are not set yet.
On Success	Exec	Fires when the operation has finished successfully.
On Failure	Exec	Fires if the operation failed or was cancelled; Error Message says why.
Percentage	Float	Progress from 0 to 100.
Error Message	String	Empty on success; otherwise why it failed.
Async Action	Export Texture Async Action Object Reference	This node's operation. Call Cancel on it to stop the operation (no more events fire).

Convert Texture Object to Texture 2D FUNCTION

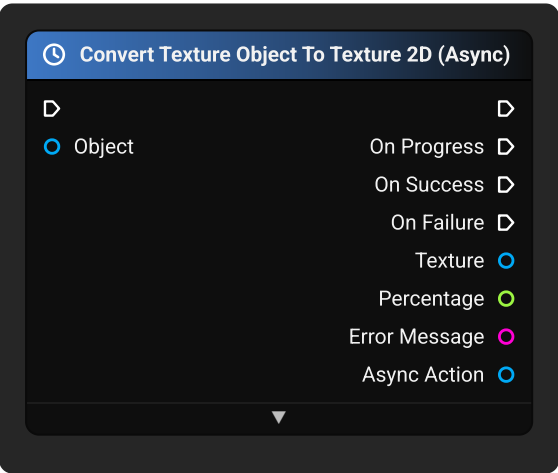


Makes a Texture 2D from any texture object (texture of any kind, Runtime Image, Runtime Video frame; cubes become a panorama), e.g. to keep a copy of a render target or a video frame, or to change its size or pixel format.

Target: Runtime File Importer Subsystem
Async version: [Convert Texture Object To Texture 2D \(Async\)](#)
C++: URuntimeFileImporterSubsystem::ConvertTextureObjectToTexture2D

INPUTS		
Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Object	Object Object Reference	Any supported object (detected automatically).
Options	ADVANCED OPTIONAL Runtime Texture Convert Options	How Convert Texture ...
OUTPUTS		
Pin	Type	Description
Error Message	String	Empty on success; otherwise why it failed.
Return Value	Texture 2D Object Reference	The result, or None/empty if it failed (Error Message says why).

Convert Texture Object To Texture 2D (Async) ASYNC



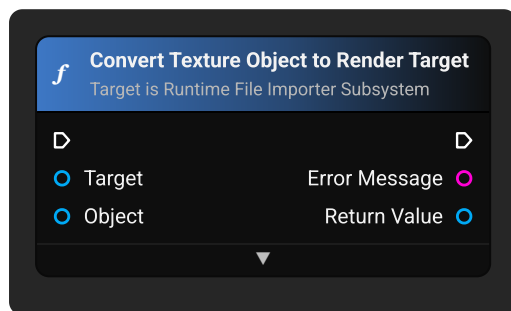
Makes a Texture 2D from any texture object (cubes become a panorama) without blocking the game thread.

Blocking version: [Convert Texture Object to Texture 2D](#)
C++:
UConvertToTexture2DAsyncAction::ConvertTextureObjectToTexture2D Async

INPUTS		
Pin	Type	Description
Object	Object Object Reference	Any supported object (detected automatically).
Options	ADVANCED OPTIONAL Runtime Texture Convert Options	How Convert Texture ...
OUTPUTS		
Pin	Type	Description
On Progress	Exec	Fires as the operation advances (Percentage from 0 to 100). Other outputs are not set yet.

Pin	Type	Description
On Success	 Exec	Fires when the operation has finished successfully.
On Failure	 Exec	Fires if the operation failed or was cancelled; Error Message says why.
Texture	 Texture 2D Object Reference	The Texture 2D.
Percentage	 Float	Progress from 0 to 100.
Error Message	 String	Empty on success; otherwise why it failed.
Async Action	 Convert to Texture 2D Async Action Object Reference	This node's operation. Call Cancel on it to stop the operation (no more events fire).

Convert Texture Object to Render Target FUNCTION



Makes a render target holding a texture object's pixels, e.g. to draw on it with the engine's Draw ... To Render Target nodes.

Target: Runtime File Importer Subsystem

Async version: [Convert Texture Object To Render Target \(Async\)](#)

C++:

```
URuntimeFileImporterSubsystem::ConvertTextureObjectToRenderTarget
```

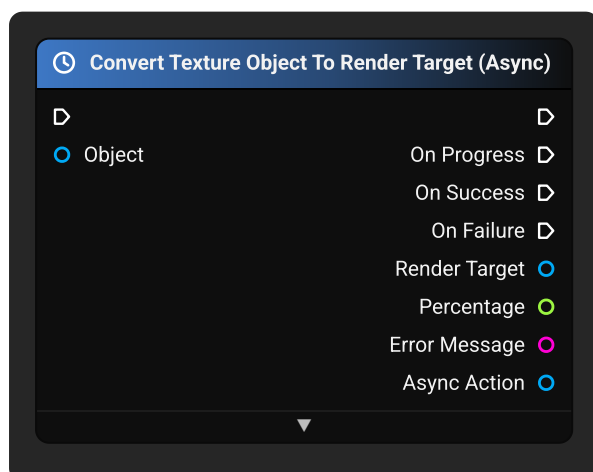
INPUTS

Pin	Type	Description
Target	 Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Object	 Object Object Reference	Any supported object (detected automatically).
Options	ADVANCED OPTIONAL  Runtime Texture Convert Options	How Convert Texture ...

OUTPUTS

Pin	Type	Description
Error Message	 String	Empty on success; otherwise why it failed.
Return Value	 Texture Render Target 2D Object Reference	The result, or None/empty if it failed (Error Message says why).

Convert Texture Object To Render Target (Async) ASYNC



Makes a render target holding a texture object's pixels without blocking the game thread.

Blocking version: [Convert Texture Object to Render Target](#)

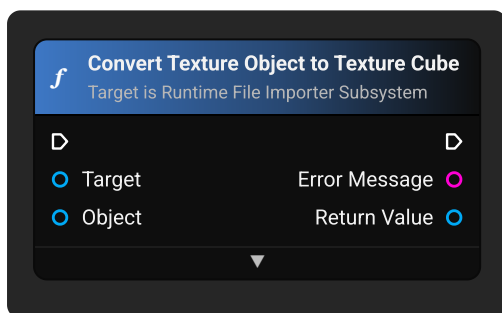
C++:

```
UConvertToRenderTargetAsyncAction::ConvertTextureObjectToRenderTargetAsync
```

INPUTS		
Pin	Type	Description
Object	 Object Object Reference	Any supported object (detected automatically).
Options	ADVANCED OPTIONAL  Runtime Texture Convert Options	How Convert Texture ...
OUTPUTS		
Pin	Type	Description
On Progress	 Exec	Fires as the operation advances (Percentage from 0 to 100). Other outputs are not set yet.
On Success	 Exec	Fires when the operation has finished successfully.
On Failure	 Exec	Fires if the operation failed or was cancelled; Error Message says why.
Render Target	 Texture Render Target 2D Object Reference	The Texture Render Target 2D made.
Percentage	 Float	Progress from 0 to 100.
Error Message	 String	Empty on success; otherwise why it failed.
Async Action	 Convert to Render Target Async Action Object Reference	This node's operation. Call Cancel on it to stop the operation (no more events fire).

Convert Texture Object to Texture Cube

FUNCTION



Makes a Texture Cube from a panorama (equirectangular, 2:1) texture object, e.g. an HDRI for a sky light or sky sphere.

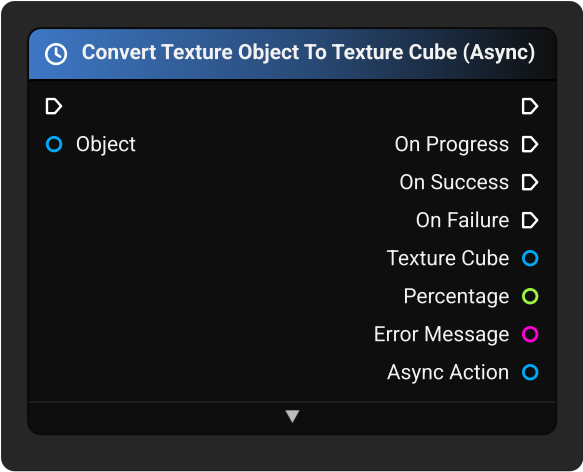
Target: Runtime File Importer Subsystem

Async version: [Convert Texture Object To Texture Cube \(Async\)](#)

C++: `URuntimeFileImporterSubsystem::ConvertTextureObjectToTextureCube`

INPUTS		
Pin	Type	Description
Target	 Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Object	 Object Object Reference	Any supported object (detected automatically).
Options	ADVANCED OPTIONAL  Runtime Texture Convert Options	How Convert Texture ...
OUTPUTS		
Pin	Type	Description
Error Message	 String	Empty on success; otherwise why it failed.
Return Value	 Texture Cube Object Reference	The result, or None/empty if it failed (Error Message says why).

Convert Texture Object To Texture Cube (Async) ASYNC



Makes a Texture Cube from a panorama texture object (e.g. an HDRI) without blocking the game thread.

Blocking version: [Convert Texture Object to Texture Cube](#)

C++:

```
UConvertToTextureCubeAsyncAction::ConvertTextureObjectToTextureCubeAsync
```

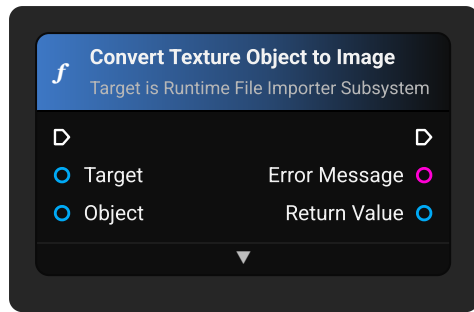
INPUTS

Pin	Type	Description
Object	Object Object Reference	Any supported object (detected automatically).
Options ADVANCED OPTIONAL	Runtime Texture Convert Options	How Convert Texture ...

OUTPUTS

Pin	Type	Description
On Progress	Exec	Fires as the operation advances (Percentage from 0 to 100). Other outputs are not set yet.
On Success	Exec	Fires when the operation has finished successfully.
On Failure	Exec	Fires if the operation failed or was cancelled; Error Message says why.
Texture Cube	Texture Cube Object Reference	The Texture Cube made.
Percentage	Float	Progress from 0 to 100.
Error Message	String	Empty on success; otherwise why it failed.
Async Action	Convert to Texture Cube Async Action Object Reference	This node's operation. Call Cancel on it to stop the operation (no more events fire).

Convert Texture Object to Image FUNCTION



Makes a Runtime Image (editable pixels) of any texture object.

Target: Runtime File Importer Subsystem

Async version: [Convert Texture Object To Image \(Async\)](#)

C++: `URuntimeFileImporterSubsystem::ConvertTextureObjectToImage`

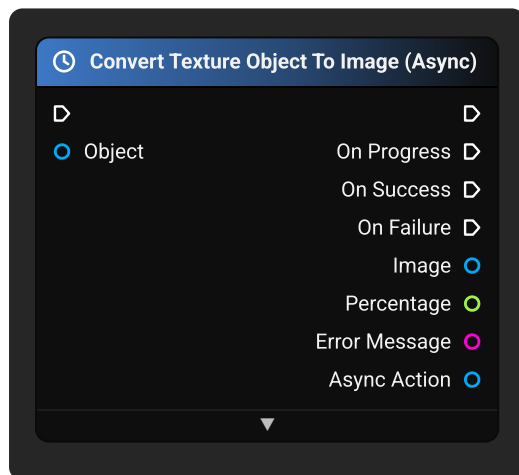
INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Object	Object Object Reference	Any supported object (detected automatically).
Options	ADVANCED OPTIONAL Runtime Texture Convert Options	How Convert Texture ...

OUTPUTS

Pin	Type	Description
Error Message	String	Empty on success; otherwise why it failed.
Return Value	Runtime Image Object Reference	The result, or None/empty if it failed (Error Message says why).

Convert Texture Object To Image (Async) ASYNC



Makes a Runtime Image (editable pixels) of any texture object without blocking the game thread.

Blocking version: [Convert Texture Object to Image](#)

C++: `UConvertToImageAsyncAction::ConvertTextureObjectToImageAsync`

INPUTS

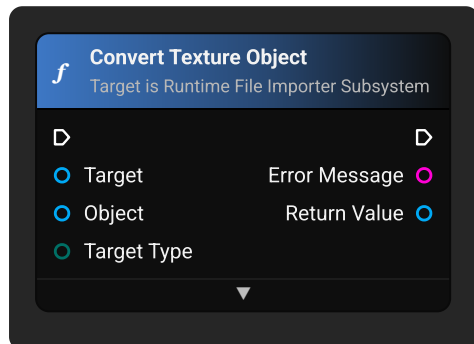
Pin	Type	Description
Object	Object Object Reference	Any supported object (detected automatically).
Options	ADVANCED OPTIONAL Runtime Texture Convert Options	How Convert Texture ...

OUTPUTS

Pin	Type	Description
On Progress	Exec	Fires as the operation advances (Percentage from 0 to 100). Other outputs are not set yet.
On Success	Exec	Fires when the operation has finished successfully.

Pin	Type	Description
On Failure	 Exec	Fires if the operation failed or was cancelled; Error Message says why.
Image	 Runtime Image Object Reference	The Runtime Image.
Percentage	 Float	Progress from 0 to 100.
Error Message	 String	Empty on success; otherwise why it failed.
Async Action	 Convert to Image Async Action Object Reference	This node's operation. Call Cancel on it to stop the operation (no more events fire).

Convert Texture Object FUNCTION



Converts any texture object (Texture 2D, cube, render target, video texture, Texture 2D Dynamic, Runtime Image, Runtime Video; detected automatically, see Get Texture Object Type) into Target Type. Cast the result to that type.

Target: Runtime File Importer Subsystem

Async version: [Convert Texture Object \(Async\)](#)

C++: `URuntimeFileImporterSubsystem::ConvertTextureObject`

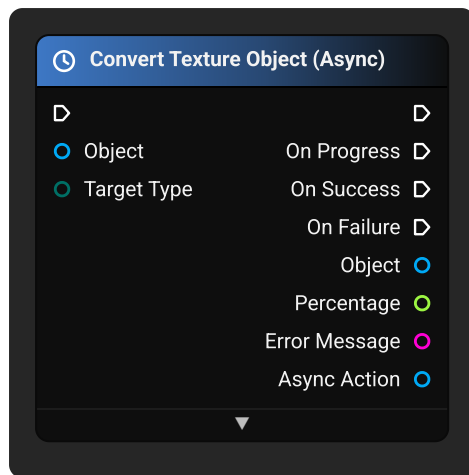
INPUTS

Pin	Type	Description
Target	 Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Object	 Object Object Reference	Any supported object (detected automatically).
Target Type	 ERuntime Texture Convert Target	What to convert into.
Options	ADVANCED OPTIONAL  Runtime Texture Convert Options	How Convert Texture ...

OUTPUTS

Pin	Type	Description
Error Message	 String	Empty on success; otherwise why it failed.
Return Value	 Object Object Reference	The result, or None/empty if it failed (Error Message says why).

Convert Texture Object (Async) ASYNC



Converts any texture object into Target Type without blocking the game thread. Cast the result to that type.

Blocking version: [Convert Texture Object](#)

C++: `UConvertTextureObjectAsyncAction::ConvertTextureObjectAsync`

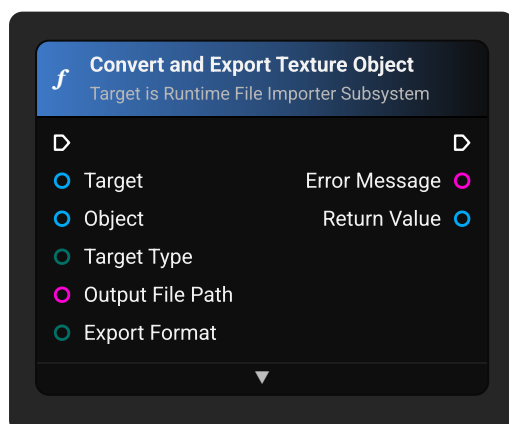
INPUTS

Pin	Type	Description
Object	Object Object Reference	Any supported object (detected automatically).
Target Type	ERuntime Texture Convert Target	What to convert into.
Options ADVANCED OPTIONAL	Runtime Texture Convert Options	How Convert Texture ...

OUTPUTS

Pin	Type	Description
On Progress	Exec	Fires as the operation advances (Percentage from 0 to 100). Other outputs are not set yet.
On Success	Exec	Fires when the operation has finished successfully.
On Failure	Exec	Fires if the operation failed or was cancelled; Error Message says why.
Object	Object Object Reference	Any supported object (detected automatically).
Percentage	Float	Progress from 0 to 100.
Error Message	String	Empty on success; otherwise why it failed.
Async Action	Convert Texture Object Async Action Object Reference	This node's operation. Call Cancel on it to stop the operation (no more events fire).

Convert and Export Texture Object FUNCTION



Converts any texture object into Target Type, then writes the result to an image file (cubes as a panorama). Returns the converted object.

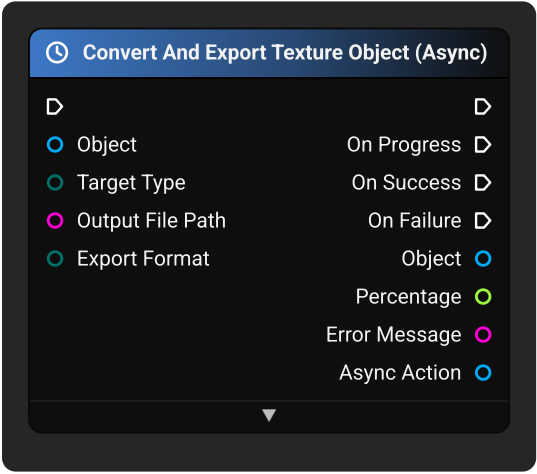
Target: Runtime File Importer Subsystem

Async version: [Convert And Export Texture Object \(Async\)](#)

C++: `URuntimeFileImporterSubsystem::ConvertAndExportTextureObject`

INPUTS		
Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Object	Object Object Reference	Any supported object (detected automatically).
Target Type	ERuntime Texture Convert Target	What to convert into.
Output File Path	String	Full path of the file to write. Missing folders are created; an existing file is replaced.
Export Format	ERuntime Image Export Format	Format to write.
Options	ADVANCED OPTIONAL Runtime Texture Convert Options	How Convert Texture ...
Export Options	ADVANCED OPTIONAL Runtime Image Export Options	How textures and images are written to image files.
OUTPUTS		
Pin	Type	Description
Error Message	String	Empty on success; otherwise why it failed.
Return Value	Object Object Reference	The converted object.

Convert And Export Texture Object (Async) ASYNC



Converts any texture object into Target Type, then writes it to an image file. Outputs the converted object too.

Blocking version: [Convert and Export Texture Object](#)

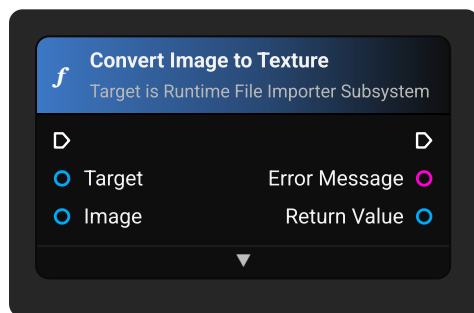
C++:
 UConvertTextureObjectAsyncAction::ConvertAndExportTextureObjectAs
 ync

INPUTS		
Pin	Type	Description
Object	Object Object Reference	Any supported object (detected automatically).
Target Type	ERuntime Texture Convert Target	What to convert into.
Output File Path	String	Full path of the file to write. Missing folders are created; an existing file is replaced.
Export Format	ERuntime Image Export Format	Format to write.
Options	ADVANCED OPTIONAL Runtime Texture Convert Options	How Convert Texture ...

Pin	Type	Description
Export Options	ADVANCED OPTIONAL Runtime Image Export Options	How textures and images are written to image files.
OUTPUTS		
Pin	Type	Description
On Progress	Exec	Fires as the operation advances (Percentage from 0 to 100). Other outputs are not set yet.
On Success	Exec	Fires when the operation has finished successfully.
On Failure	Exec	Fires if the operation failed or was cancelled; Error Message says why.
Object	Object Object Reference	Any supported object (detected automatically).
Percentage	Float	Progress from 0 to 100.
Error Message	String	Empty on success; otherwise why it failed.
Async Action	Convert Texture Object Async Action Object Reference	This node's operation. Call Cancel on it to stop the operation (no more events fire).

Convert Image to Texture

FUNCTION



Makes a texture of a Runtime Image, with the import options' settings (filter, mipmaps, ...; sRGB comes from the image).

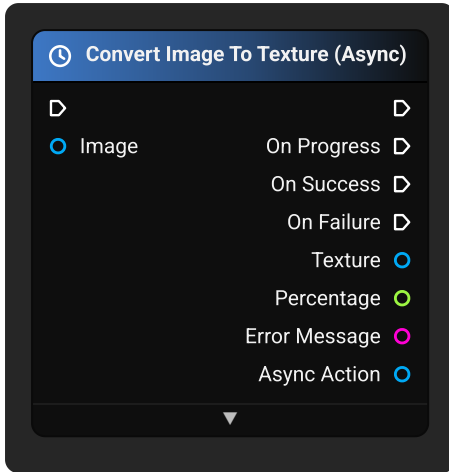
Target: Runtime File Importer Subsystem

Async version: [Convert Image To Texture \(Async\)](#)

C++: `URuntimeFileImporterSubsystem::ConvertImageToTexture`

INPUTS		
Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Image	Runtime Image Object Reference	The Runtime Image.
Options	ADVANCED OPTIONAL Runtime Texture Import Options	How imported (and converted) images become textures.
OUTPUTS		
Pin	Type	Description
Error Message	String	Empty on success; otherwise why it failed.
Return Value	Texture 2D Object Reference	The result, or None/empty if it failed (Error Message says why).

Convert Image To Texture (Async) ASYNC



Makes a texture of a Runtime Image (with the import options' settings) without blocking the game thread.

Blocking version: [Convert Image to Texture](#)

C++: `UIImageToTextureAsyncAction::ConvertImageToTextureAsync`

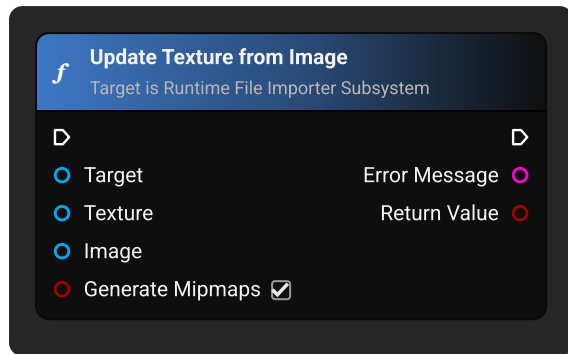
INPUTS

Pin	Type	Description
Image	Runtime Image Object Reference	The Runtime Image.
Options ADVANCED OPTIONAL	Runtime Texture Import Options	How imported (and converted) images become textures.

OUTPUTS

Pin	Type	Description
On Progress	Exec	Fires as the operation advances (Percentage from 0 to 100). Other outputs are not set yet.
On Success	Exec	Fires when the operation has finished successfully.
On Failure	Exec	Fires if the operation failed or was cancelled; Error Message says why.
Texture	Texture Object Reference	Any texture: Texture 2D, render target, cube, media or video texture.
Percentage	Float	Progress from 0 to 100.
Error Message	String	Empty on success; otherwise why it failed.
Async Action	Image to Texture Async Action Object Reference	This node's operation. Call Cancel on it to stop the operation (no more events fire).

Update Texture from Image FUNCTION



Puts a Runtime Image's pixels into an existing Texture 2D or render target (resizing it if needed): everything using the texture (materials, widgets) shows them right away. Texture 2Ds become uncompressed.

Target: Runtime File Importer Subsystem

Async version: [Update Texture From Image \(Async\)](#)

C++: `URuntimeFileImporterSubsystem::UpdateTextureFromImage`

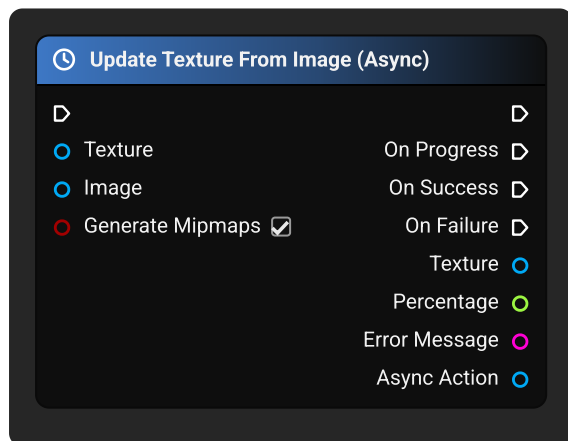
INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Texture	Texture Object Reference	Any texture: Texture 2D, render target, cube, media or video texture.
Image	Runtime Image Object Reference	The Runtime Image.
Generate Mipmaps	Boolean	Make mipmaps (smaller versions used at a distance). Default: <code>true</code>

OUTPUTS

Pin	Type	Description
Error Message	String	Empty on success; otherwise why it failed.
Return Value	Boolean	True on success. False if it failed (Error Message says why).

Update Texture From Image (Async) ASYNC



Puts a Runtime Image's pixels into an existing Texture 2D or render target without blocking the game thread.

Blocking version: [Update Texture from Image](#)

C++: `UIImageToTextureAsyncAction::UpdateTextureFromImageAsync`

INPUTS

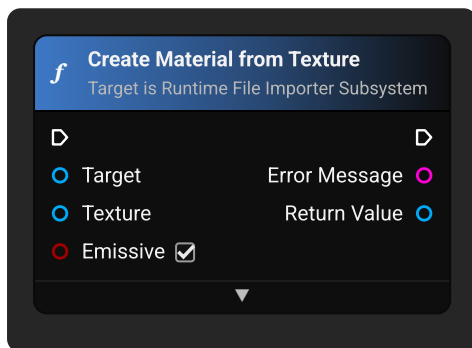
Pin	Type	Description
Texture	Texture Object Reference	Any texture: Texture 2D, render target, cube, media or video texture.
Image	Runtime Image Object Reference	The Runtime Image.
Generate Mipmaps	Boolean	Make mipmaps (smaller versions used at a distance). Default: <code>true</code>

OUTPUTS

Pin	Type	Description
On Progress	 Exec	Fires as the operation advances (Percentage from 0 to 100). Other outputs are not set yet.
On Success	 Exec	Fires when the operation has finished successfully.
On Failure	 Exec	Fires if the operation failed or was cancelled; Error Message says why.
Texture	 Texture Object Reference	Any texture: Texture 2D, render target, cube, media or video texture.
Percentage	 Float	Progress from 0 to 100.
Error Message	 String	Empty on success; otherwise why it failed.
Async Action	 Image to Texture Async Action Object Reference	This node's operation. Call Cancel on it to stop the operation (no more events fire).

Create Material from Texture

FUNCTION



A material showing a texture (any 2D texture, render target or video texture) on meshes: lit, or Emissive for a screen that glows (unlit look, e.g. videos). Translucent uses the texture's alpha.

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::CreateMaterialFromTexture`

INPUTS

Pin	Type	Description
Target	 Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Texture	 Texture Object Reference	Any texture: Texture 2D, render target, cube, media or video texture.
Emissive	 Boolean	Make the texture glow (unlit) instead of being lit. Default: <code>true</code>
Translucent	 Boolean	Use the texture's alpha for transparency. Default: <code>false</code>
Two Sided	 Boolean	Draw both sides of each triangle. Default: <code>false</code>

OUTPUTS

Pin	Type	Description
Error Message	 String	Empty on success; otherwise why it failed.
Return Value	 Material Instance Dynamic Object Reference	The result, or None/empty if it failed (Error Message says why).

Get Texture Details

FUNCTION

f

Get Texture Details

Target is Runtime File Importer Subsystem

Target

Object

Details

Error Message

Return Value

Everything about a texture object (any texture, Runtime Image or Runtime Video): size, pixel format, mips, sRGB, compression settings, filter, address modes, texture group, streaming, memory, render target and video settings. Change them with the Editing nodes.

Target: Runtime File Importer Subsystem
C++: `URuntimeFileImporterSubsystem::GetTextureDetails`

INPUTS		
Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Object	Object Object Reference	Any supported object (detected automatically).
OUTPUTS		
Pin	Type	Description
Details	Runtime Texture Details	Everything about a texture (any kind: 2D, cube, render target, video texture, ...).
Error Message	String	Empty on success; otherwise why it failed.
Return Value	Boolean	True on success. False if it failed (Error Message says why).

Get Texture Object Details as String

FUNCTION

f

Get Texture Object Details as String

Target is Runtime File Importer Subsystem

Target

Object

Return Value

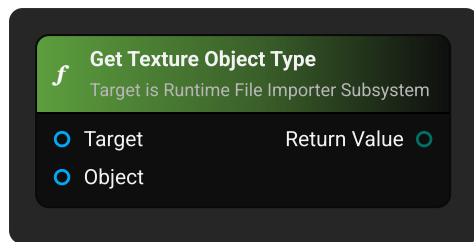
Readable, multi-line details of a texture object, e.g. to print.

Target: Runtime File Importer Subsystem
C++: `URuntimeFileImporterSubsystem::GetTextureObjectDetailsAsString`

INPUTS		
Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Object	Object Object Reference	Any supported object (detected automatically).
OUTPUTS		
Pin	Type	Description
Return Value	String	Readable, multi-line details of a texture object, e.g.

Get Texture Object Type

PURE



What kind of texture object something is (Texture 2D, cube, render target, video texture, Runtime Image, Runtime Video), or None.

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::GetTextureObjectType`

INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Object	Object Object Reference	Any supported object (detected automatically).

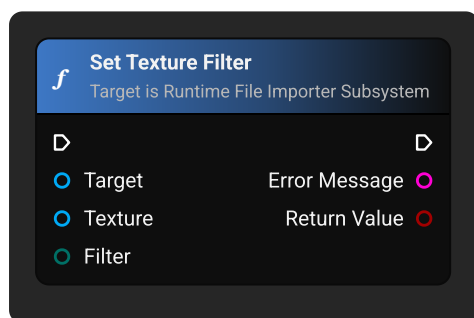
OUTPUTS

Pin	Type	Description
Return Value	ERuntime Texture Object Type	What kind of texture object something is (Texture 2D, cube, render target, video texture, Runtime Image, Runtime Video), or None.

Editing 8 nodes

Set Texture Filter

FUNCTION



Texture filtering (Nearest for pixel art, Bilinear, Trilinear, Default: the texture group's), applied right away. Render targets are cleared.

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::SetTextureFilter`

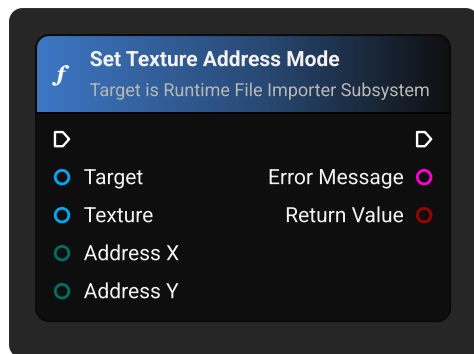
INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Texture	Texture Object Reference	Any texture: Texture 2D, render target, cube, media or video texture.
Filter	Texture Filter	How the texture is sampled.

OUTPUTS

Pin	Type	Description
Error Message	String	Empty on success; otherwise why it failed.
Return Value	Boolean	True on success. False if it failed (Error Message says why).

Set Texture Address Mode FUNCTION



What happens outside the 0-1 UV range (Wrap, Clamp, Mirror), applied right away. Render targets are cleared.

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::SetTextureAddressMode`

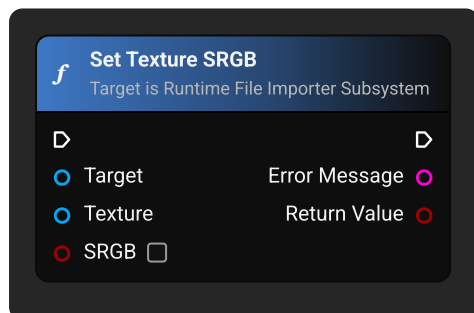
INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Texture	Texture Object Reference	Any texture: Texture 2D, render target, cube, media or video texture.
Address X	Texture Address	What happens past the texture's left and right edges.
Address Y	Texture Address	What happens past the texture's top and bottom edges.

OUTPUTS

Pin	Type	Description
Error Message	String	Empty on success; otherwise why it failed.
Return Value	Boolean	True on success. False if it failed (Error Message says why).

Set Texture SRGB FUNCTION



Whether the texture's colors are sRGB (on for colors, off for data like normal maps and masks), applied right away.

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::SetTextureSRGB`

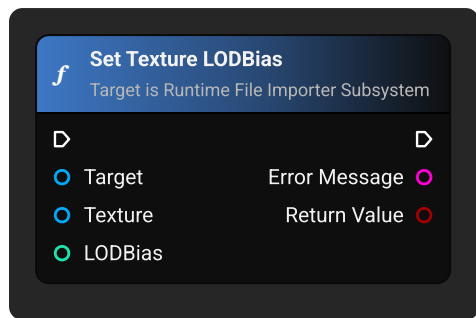
INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Texture	Texture Object Reference	Any texture: Texture 2D, render target, cube, media or video texture.
SRGB	Boolean	Colors are sRGB (most images). Off for data such as normal maps or masks.

OUTPUTS

Pin	Type	Description
Error Message	String	Empty on success; otherwise why it failed.
Return Value	Boolean	True on success. False if it failed (Error Message says why).

Set Texture LODBias FUNCTION



Skips the most detailed mips (1: half resolution, 2: quarter, ...), e.g. to save memory.

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::SetTextureLODBias`

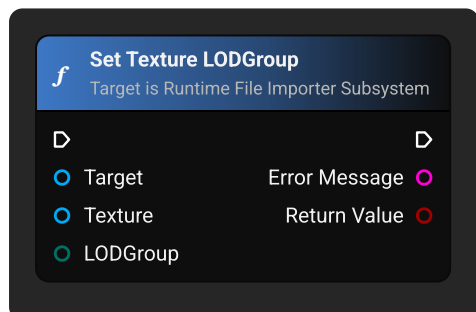
INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Texture	Texture Object Reference	Any texture: Texture 2D, render target, cube, media or video texture.
LODBias	Integer	How many of the most detailed mips to skip.

OUTPUTS

Pin	Type	Description
Error Message	String	Empty on success; otherwise why it failed.
Return Value	Boolean	True on success. False if it failed (Error Message says why).

Set Texture LODGroup FUNCTION



Texture group (World, UI, Character, ...): its filtering and the device profile's LOD bias apply.

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::SetTextureLODGroup`

INPUTS

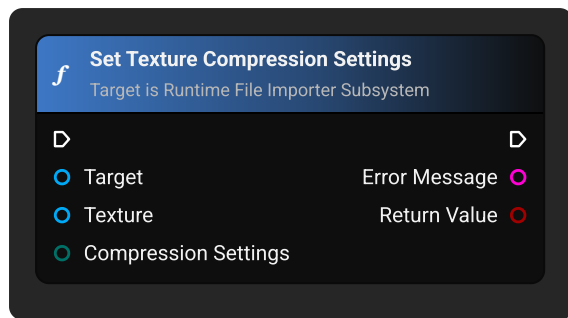
Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Texture	Texture Object Reference	Any texture: Texture 2D, render target, cube, media or video texture.
LODGroup	Texture Group	Texture group (sets mip and filter settings).

OUTPUTS

Pin	Type	Description
Error Message	String	Empty on success; otherwise why it failed.
Return Value	Boolean	True on success. False if it failed (Error Message says why).

Set Texture Compression Settings

FUNCTION



How materials treat the texture (e.g. Normalmap to sample it as Normal). Runtime textures keep their pixel format.

Target: Runtime File Importer Subsystem

C++:

```
URuntimeFileImporterSubsystem::SetTextureCompressionSettings
```

INPUTS

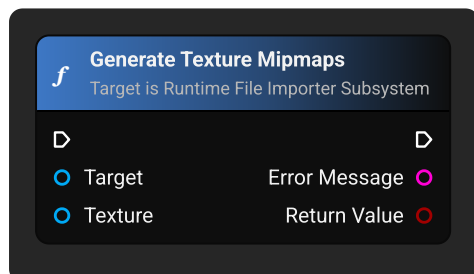
Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Texture	Texture Object Reference	Any texture: Texture 2D, render target, cube, media or video texture.
Compression Settings	Texture Compression Settings	How the texture is stored on the GPU.

OUTPUTS

Pin	Type	Description
Error Message	String	Empty on success; otherwise why it failed.
Return Value	Boolean	True on success. False if it failed (Error Message says why).

Generate Texture Mipmaps

FUNCTION



(Re)builds the mipmaps of a Texture 2D whose pixels are in memory (e.g. imported without mipmaps), so it looks smooth at a distance.

Target: Runtime File Importer Subsystem

Async version: [Generate Texture Mipmaps \(Async\)](#)

```
C++: URuntimeFileImporterSubsystem::GenerateTextureMipmaps
```

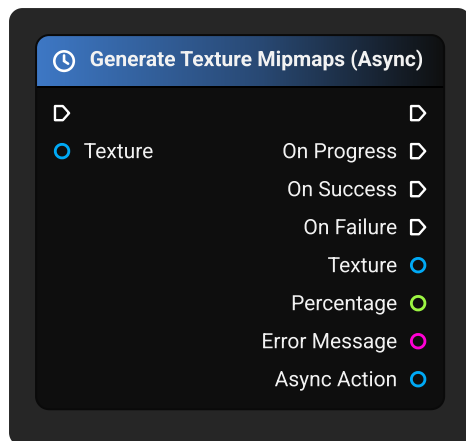
INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Texture	Texture 2D Object Reference	The Texture 2D.

OUTPUTS

Pin	Type	Description
Error Message	String	Empty on success; otherwise why it failed.
Return Value	Boolean	True on success. False if it failed (Error Message says why).

Generate Texture Mipmaps (Async) ASYNC



(Re)builds the mipmaps of a Texture 2D whose pixels are in memory, without blocking the game thread.

Blocking version: [Generate Texture Mipmaps](#)

C++: `UImageToTextureAsyncAction::GenerateTextureMipmapsAsync`

INPUTS

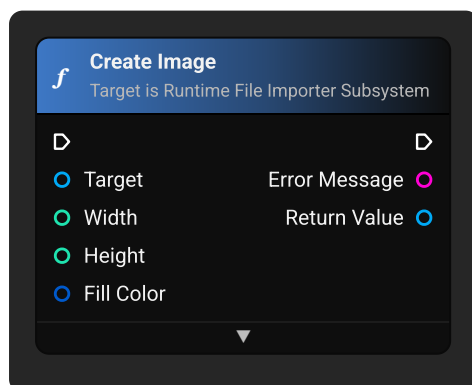
Pin	Type	Description
Texture	Texture 2D Object Reference	The Texture 2D.

OUTPUTS

Pin	Type	Description
On Progress	Exec	Fires as the operation advances (Percentage from 0 to 100). Other outputs are not set yet.
On Success	Exec	Fires when the operation has finished successfully.
On Failure	Exec	Fires if the operation failed or was cancelled; Error Message says why.
Texture	Texture Object Reference	Any texture: Texture 2D, render target, cube, media or video texture.
Percentage	Float	Progress from 0 to 100.
Error Message	String	Empty on success; otherwise why it failed.
Async Action	Image to Texture Async Action Object Reference	This node's operation. Call Cancel on it to stop the operation (no more events fire).

Image 31 nodes

Create Image FUNCTION



A new Runtime Image of Width x Height pixels filled with Fill Color (8-bit sRGB, or 32-bit float with HDR).

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::CreateImage`

INPUTS

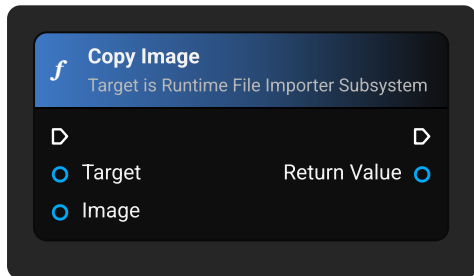
Pin	Type	Description
Target	 Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Width	 Integer	Width in pixels.
Height	 Integer	Height in pixels.
Fill Color	 Linear Color	Color of every pixel of the new image.
HDR ADVANCED	 Boolean	Store 32-bit float colors (values above 1), e.g. for EXR/HDR. Default: <code>false</code>

OUTPUTS

Pin	Type	Description
Error Message	 String	Empty on success; otherwise why it failed.
Return Value	 Runtime Image Object Reference	The result, or None/empty if it failed (Error Message says why).

Copy Image

FUNCTION



A copy of a Runtime Image (edit one, keep the other).

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::CopyImage`

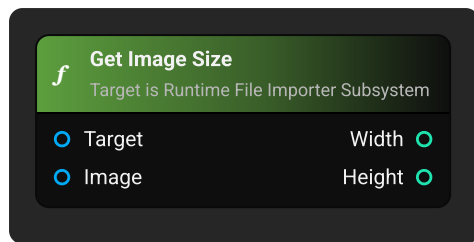
INPUTS

Pin	Type	Description
Target	 Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Image	 Runtime Image Object Reference	The Runtime Image.

OUTPUTS

Pin	Type	Description
Return Value	 Runtime Image Object Reference	A copy of a Runtime Image (edit one, keep the other).

Get Image Size PURE



Size of a Runtime Image in pixels. Same as its Width and Height variables.

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::GetImageSize`

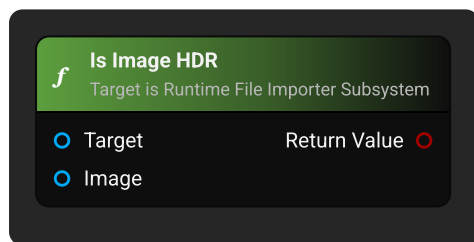
INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Image	Runtime Image Object Reference	The Runtime Image.

OUTPUTS

Pin	Type	Description
Width	Integer	Width in pixels.
Height	Integer	Height in pixels.

Is Image HDR PURE



True if a Runtime Image has float (HDR) pixels. Same as its Is HDR variable.

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::IsImageHDR`

INPUTS

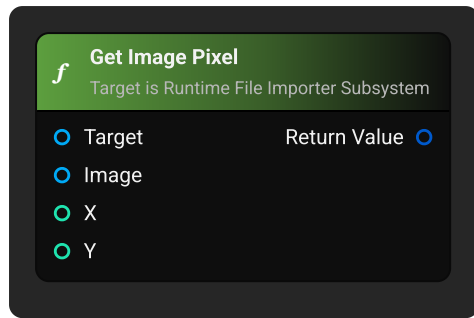
Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Image	Runtime Image Object Reference	The Runtime Image.

OUTPUTS

Pin	Type	Description
Return Value	Boolean	True if a Runtime Image has float (HDR) pixels.

Get Image Pixel

PURE



The color of a pixel (linear, as materials see it). (0, 0) is the top-left corner.

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::GetImagePixel`

INPUTS

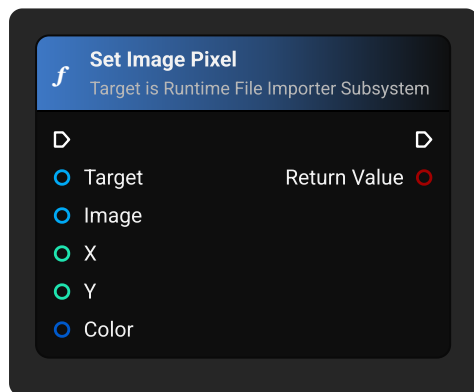
Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Image	Runtime Image Object Reference	The Runtime Image.
X	Integer	Column, from the left (0-based).
Y	Integer	Row, from the top (0-based).

OUTPUTS

Pin	Type	Description
Return Value	Linear Color	The color of a pixel (linear, as materials see it).

Set Image Pixel

FUNCTION



Sets the color of a pixel. Returns false if it's outside the image.

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::SetImagePixel`

INPUTS

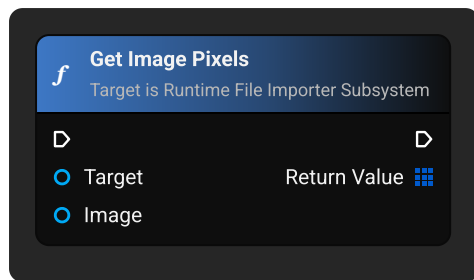
Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Image	Runtime Image Object Reference	The Runtime Image.
X	Integer	Column, from the left (0-based).
Y	Integer	Row, from the top (0-based).
Color	Linear Color	The color.

OUTPUTS

Pin	Type	Description
Return Value	Boolean	False if it's outside the image.

Get Image Pixels

FUNCTION



Every pixel's color, row by row from the top-left (index = Y * Width + X).

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::GetImagePixels`

INPUTS

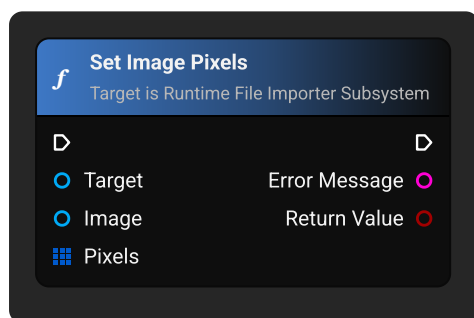
Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Image	Runtime Image Object Reference	The Runtime Image.

OUTPUTS

Pin	Type	Description
Return Value	Array of Linear Color	Every pixel's color, row by row from the top-left (index = Y * Width + X).

Set Image Pixels

FUNCTION



Replaces every pixel (Pixels must have Width * Height colors, row by row from the top-left).

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::SetImagePixels`

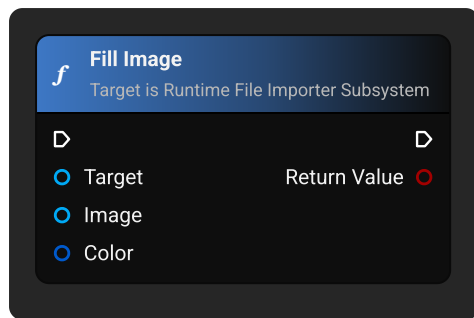
INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Image	Runtime Image Object Reference	The Runtime Image.
Pixels	Array of Linear Color	Every pixel, row by row from the top-left (Width x Height colors).

OUTPUTS

Pin	Type	Description
Error Message	String	Empty on success; otherwise why it failed.
Return Value	Boolean	True on success. False if it failed (Error Message says why).

Fill Image FUNCTION



Fills the whole image with a color.

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::FillImage`

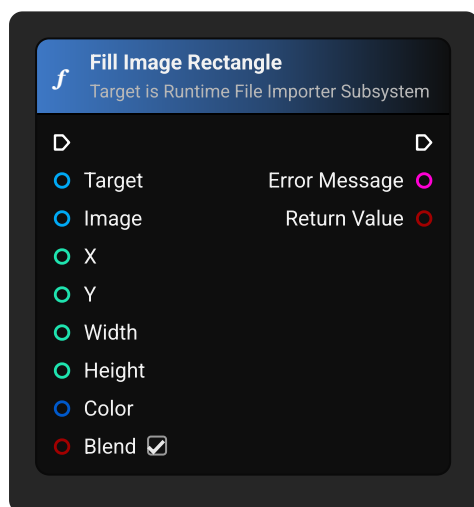
INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Image	Runtime Image Object Reference	The Runtime Image.
Color	Linear Color	The color.

OUTPUTS

Pin	Type	Description
Return Value	Boolean	Fills the whole image with a color.

Fill Image Rectangle FUNCTION



Fills a rectangle with a color (blended with what's under it by the color's alpha when Blend is on).

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::FillImageRectangle`

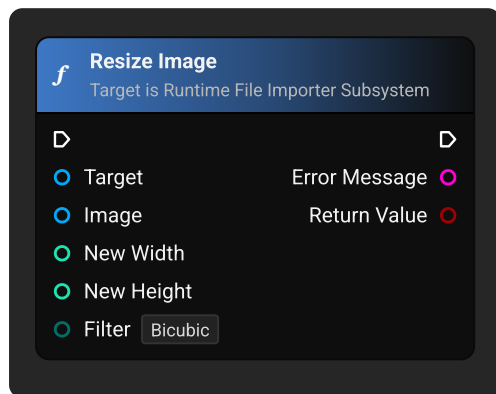
INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Image	Runtime Image Object Reference	The Runtime Image.
X	Integer	Column, from the left (0-based).
Y	Integer	Row, from the top (0-based).
Width	Integer	Width in pixels.
Height	Integer	Height in pixels.
Color	Linear Color	The color.

Pin	Type	Description
Blend	 Boolean	Blend the color over the pixels using its alpha (off: replace them). Default: <code>true</code>
OUTPUTS		
Pin	Type	Description
Error Message	 String	Empty on success; otherwise why it failed.
Return Value	 Boolean	True on success. False if it failed (Error Message says why).

Resize Image

FUNCTION



Scales an image to a new size. Use the Async version for large images.

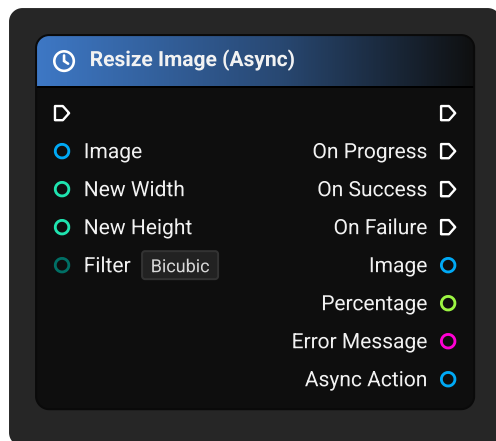
Target: Runtime File Importer Subsystem

Async version: [Resize Image \(Async\)](#)

C++: `URuntimeFileImporterSubsystem::ResizeImage`

INPUTS		
Pin	Type	Description
Target	 Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Image	 Runtime Image Object Reference	The Runtime Image.
New Width	 Integer	New width in pixels.
New Height	 Integer	New height in pixels.
Filter	 ERuntime Image Resize Filter	How pixels are interpolated. Default: <code>Bicubic</code>
OUTPUTS		
Pin	Type	Description
Error Message	 String	Empty on success; otherwise why it failed.
Return Value	 Boolean	True on success. False if it failed (Error Message says why).

Resize Image (Async) ASYNC



Scales an image without blocking the game thread. Don't change the image until it's done. Outputs the same image.

Blocking version: [Resize Image](#)

C++: `UEditImageAsyncAction::ResizeImageAsync`

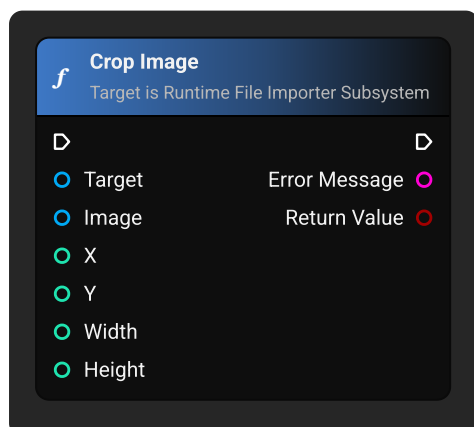
INPUTS

Pin	Type	Description
Image	Runtime Image Object Reference	The Runtime Image.
New Width	Integer	New width in pixels.
New Height	Integer	New height in pixels.
Filter	ERuntime Image Resize Filter	How pixels are interpolated. Default: Bicubic

OUTPUTS

Pin	Type	Description
On Progress	Exec	Fires as the operation advances (Percentage from 0 to 100). Other outputs are not set yet.
On Success	Exec	Fires when the operation has finished successfully.
On Failure	Exec	Fires if the operation failed or was cancelled; Error Message says why.
Image	Runtime Image Object Reference	The Runtime Image.
Percentage	Float	Progress from 0 to 100.
Error Message	String	Empty on success; otherwise why it failed.
Async Action	Edit Image Async Action Object Reference	This node's operation. Call Cancel on it to stop the operation (no more events fire).

Crop Image FUNCTION



Keeps only a rectangle of the image.

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::CropImage`

INPUTS

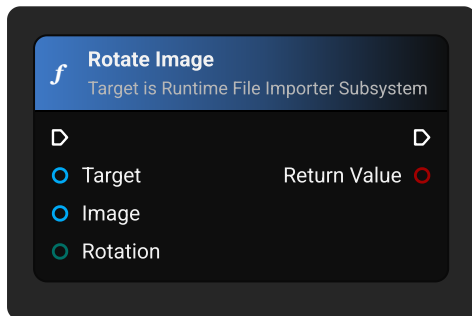
Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Image	Runtime Image Object Reference	The Runtime Image.
X	Integer	Column, from the left (0-based).
Y	Integer	Row, from the top (0-based).
Width	Integer	Width in pixels.
Height	Integer	Height in pixels.

OUTPUTS

Pin	Type	Description
Error Message	String	Empty on success; otherwise why it failed.
Return Value	Boolean	True on success. False if it failed (Error Message says why).

Rotate Image

FUNCTION



Rotates an image clockwise by 90, 180 or 270 degrees.

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::RotateImage`

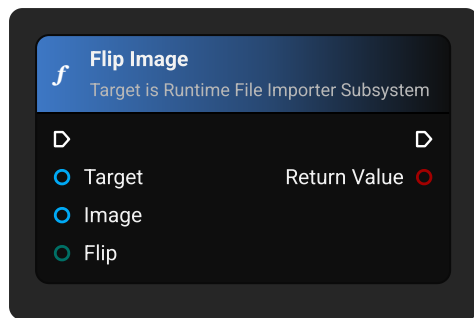
INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Image	Runtime Image Object Reference	The Runtime Image.
Rotation	ERuntime Image Rotation	Clockwise rotations.

OUTPUTS

Pin	Type	Description
Return Value	Boolean	Rotates an image clockwise by 90, 180 or 270 degrees.

Flip Image FUNCTION



Mirrors an image horizontally or vertically.

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::FlipImage`

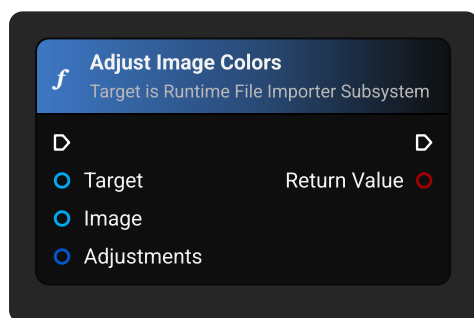
INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Image	Runtime Image Object Reference	The Runtime Image.
Flip	ERuntime Image Flip	Which way to flip.

OUTPUTS

Pin	Type	Description
Return Value	Boolean	Mirrors an image horizontally or vertically.

Adjust Image Colors FUNCTION



Brightness, contrast, saturation, hue, gamma, exposure and tint. Use the Async version for large images.

Target: Runtime File Importer Subsystem

Async version: [Adjust Image Colors \(Async\)](#)

C++: `URuntimeFileImporterSubsystem::AdjustImageColors`

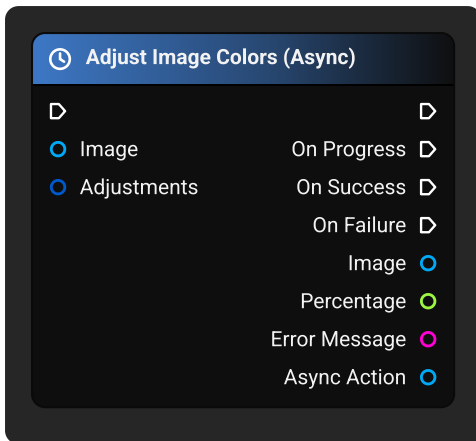
INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Image	Runtime Image Object Reference	The Runtime Image.
Adjustments OPTIONAL	Runtime Image Adjustments	Brightness, contrast, saturation, hue, gamma, exposure and tint. The defaults change nothing.

OUTPUTS

Pin	Type	Description
Return Value	Boolean	Brightness, contrast, saturation, hue, gamma, exposure and tint.

Adjust Image Colors (Async) ASYNC



Adjusts an image's colors without blocking the game thread. Outputs the same image.

Blocking version: [Adjust Image Colors](#)

C++: `UEditImageAsyncAction::AdjustImageColorsAsync`

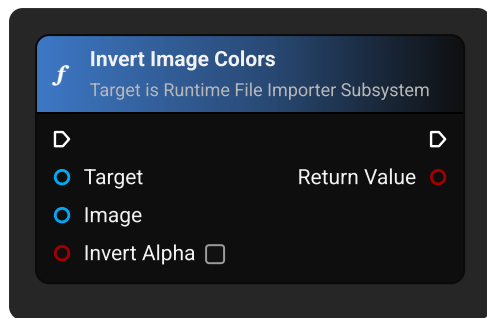
INPUTS

Pin	Type	Description
Image	Runtime Image Object Reference	The Runtime Image.
Adjustments OPTIONAL	Runtime Image Adjustments	Brightness, contrast, saturation, hue, gamma, exposure and tint. The defaults change nothing.

OUTPUTS

Pin	Type	Description
On Progress	Exec	Fires as the operation advances (Percentage from 0 to 100). Other outputs are not set yet.
On Success	Exec	Fires when the operation has finished successfully.
On Failure	Exec	Fires if the operation failed or was cancelled; Error Message says why.
Image	Runtime Image Object Reference	The Runtime Image.
Percentage	Float	Progress from 0 to 100.
Error Message	String	Empty on success; otherwise why it failed.
Async Action	Edit Image Async Action Object Reference	This node's operation. Call Cancel on it to stop the operation (no more events fire).

Invert Image Colors FUNCTION



Negative of the image (alpha too, if Invert Alpha).

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::InvertImageColors`

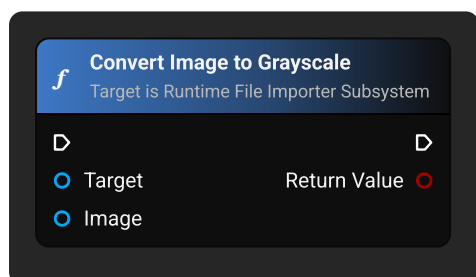
INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Image	Runtime Image Object Reference	The Runtime Image.
Invert Alpha	Boolean	Invert the alpha channel too. Default: <code>false</code>

OUTPUTS

Pin	Type	Description
Return Value	Boolean	Negative of the image (alpha too, if Invert Alpha).

Convert Image to Grayscale FUNCTION



Removes the colors, keeping the brightness.

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::ConvertImageToGrayscale`

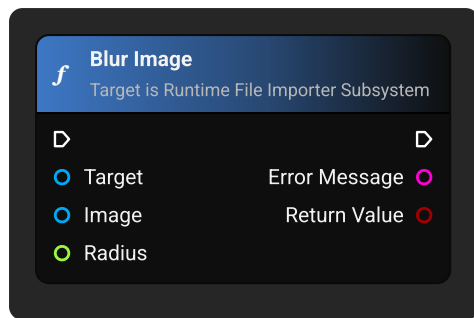
INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Image	Runtime Image Object Reference	The Runtime Image.

OUTPUTS

Pin	Type	Description
Return Value	Boolean	Removes the colors, keeping the brightness.

Blur Image FUNCTION



Gaussian blur, Radius in pixels. Use the Async version for large images.

Target: Runtime File Importer Subsystem

Async version: [Blur Image \(Async\)](#)

C++: `URuntimeFileImporterSubsystem::BlurImage`

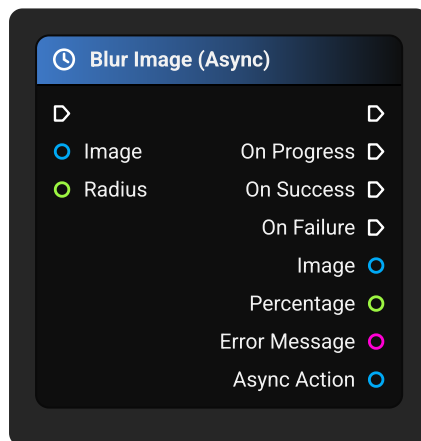
INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Image	Runtime Image Object Reference	The Runtime Image.
Radius	Float	Radius of the shape.

OUTPUTS

Pin	Type	Description
Error Message	String	Empty on success; otherwise why it failed.
Return Value	Boolean	True on success. False if it failed (Error Message says why).

Blur Image (Async) ASYNC



Blurs an image without blocking the game thread. Outputs the same image.

Blocking version: [Blur Image](#)

C++: `UEditImageAsyncAction::BlurImageAsync`

INPUTS

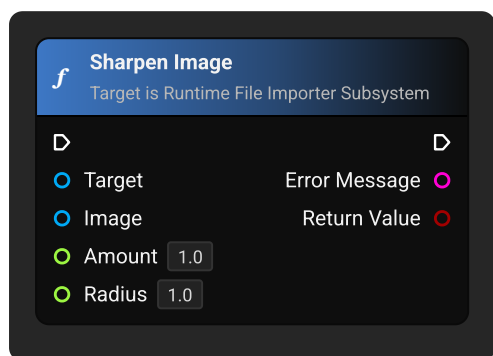
Pin	Type	Description
Image	Runtime Image Object Reference	The Runtime Image.
Radius	Float	Radius of the shape.

OUTPUTS

Pin	Type	Description
On Progress	Exec	Fires as the operation advances (Percentage from 0 to 100). Other outputs are not set yet.
On Success	Exec	Fires when the operation has finished successfully.
On Failure	Exec	Fires if the operation failed or was cancelled; Error Message says why.

Pin	Type	Description
Image	Runtime Image Object Reference	The Runtime Image.
Percentage	Float	Progress from 0 to 100.
Error Message	String	Empty on success; otherwise why it failed.
Async Action	Edit Image Async Action Object Reference	This node's operation. Call Cancel on it to stop the operation (no more events fire).

Sharpen Image FUNCTION



Sharpens details (unsharp mask: Amount 0-3, Radius in pixels). Use the Async version for large images.

Target: Runtime File Importer Subsystem

Async version: [Sharpen Image \(Async\)](#)

C++: `URuntimeFileImporterSubsystem::SharpenImage`

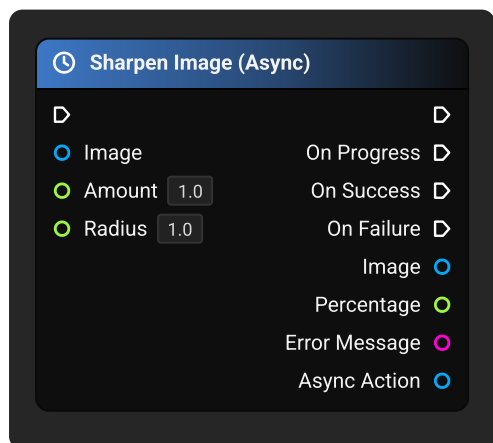
INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Image	Runtime Image Object Reference	The Runtime Image.
Amount	Float	Strength of the effect. Default: 1.0
Radius	Float	Radius of the effect in pixels. Default: 1.0

OUTPUTS

Pin	Type	Description
Error Message	String	Empty on success; otherwise why it failed.
Return Value	Boolean	True on success. False if it failed (Error Message says why).

Sharpen Image (Async) ASYNC



Sharpens an image without blocking the game thread. Outputs the same image.

Blocking version: [Sharpen Image](#)

C++: `UEditImageAsyncAction::SharpenImageAsync`

INPUTS

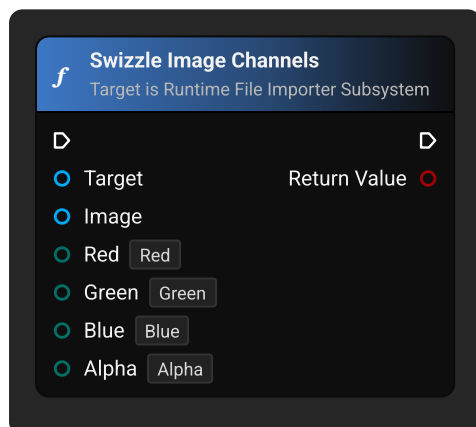
Pin	Type	Description
Image	Runtime Image Object Reference	The Runtime Image.
Amount	Float	Strength of the effect. Default: 1.0
Radius	Float	Radius of the effect in pixels. Default: 1.0

OUTPUTS

Pin	Type	Description
On Progress	Exec	Fires as the operation advances (Percentage from 0 to 100). Other outputs are not set yet.
On Success	Exec	Fires when the operation has finished successfully.
On Failure	Exec	Fires if the operation failed or was cancelled; Error Message says why.
Image	Runtime Image Object Reference	The Runtime Image.
Percentage	Float	Progress from 0 to 100.
Error Message	String	Empty on success; otherwise why it failed.
Async Action	Edit Image Async Action Object Reference	This node's operation. Call Cancel on it to stop the operation (no more events fire).

Swizzle Image Channels

FUNCTION



Rearranges the channels: each output channel takes another channel (or 0 or 1), e.g. Alpha = Red to make a mask. Values are copied as stored.

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::SwizzleImageChannels`

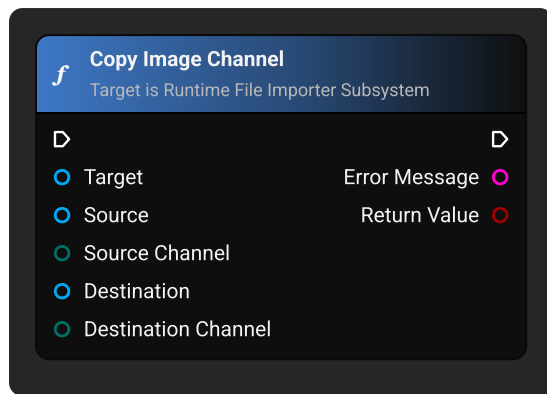
INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Image	Runtime Image Object Reference	The Runtime Image.
Red	ERuntime Image Channel Source	Source of the new red channel. Default: Red
Green	ERuntime Image Channel Source	Source of the new green channel. Default: Green
Blue	ERuntime Image Channel Source	Source of the new blue channel. Default: Blue
Alpha	ERuntime Image Channel Source	Source of the new alpha channel. Default: Alpha

OUTPUTS

Pin	Type	Description
Return Value	Boolean	Rearranges the channels: each output channel takes another channel (or 0 or 1), e.g.

Copy Image Channel FUNCTION



Copies a channel of one image into a channel of another of the same size, e.g. to pack roughness, metallic and occlusion into one texture.

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::CopyImageChannel`

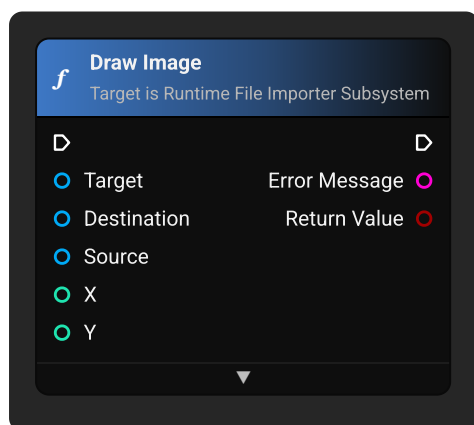
INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Source	Runtime Image Object Reference	The image the channel is read from.
Source Channel	ERuntime Image Channel	Channel read from Source.
Destination	Runtime Image Object Reference	The image the channel is written to (changed in place, same size as Source).
Destination Channel	ERuntime Image Channel	Channel written in Destination.

OUTPUTS

Pin	Type	Description
Error Message	String	Empty on success; otherwise why it failed.
Return Value	Boolean	True on success. False if it failed (Error Message says why).

Draw Image FUNCTION



Draws an image onto another with its top-left corner at (X, Y), e.g. a logo or watermark.

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::DrawImage`

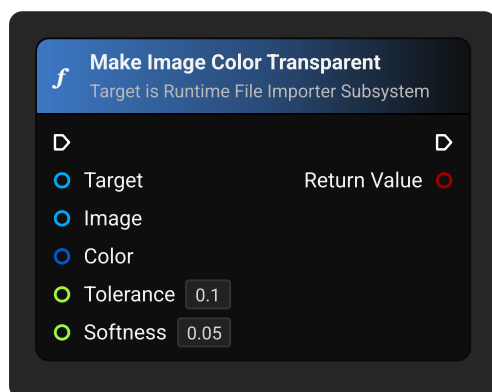
INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Destination	Runtime Image Object Reference	The image that is drawn on (changed in place).
Source	Runtime Image Object Reference	The image drawn onto Destination.

Pin	Type	Description
X	 Integer	Column, from the left (0-based).
Y	 Integer	Row, from the top (0-based).
Blend Mode	 ERuntime Image Blend Mode	How the colors are combined. Default: Alpha Blend
Opacity	 Float	Opacity of the drawn image (0 to 1). Default: 1.0
OUTPUTS		
Pin	Type	Description
Error Message	 String	Empty on success; otherwise why it failed.
Return Value	 Boolean	True on success. False if it failed (Error Message says why).

Make Image Color Transparent

FUNCTION



Makes pixels close to a color transparent (chroma key / green screen): Tolerance 0-1, fading over Softness.

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::MakeImageColorTransparent`

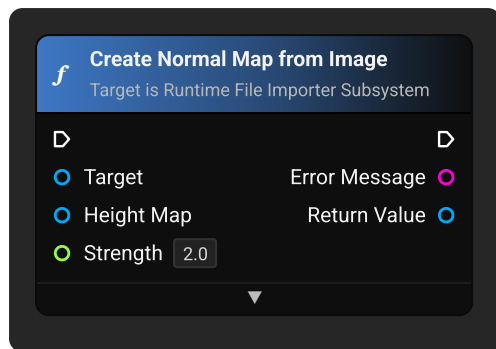
INPUTS

Pin	Type	Description
Target	 Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Image	 Runtime Image Object Reference	The Runtime Image.
Color	 Linear Color	The color.
Tolerance	 Float	How different a color may be and still count as Color (0: exactly; 1: any). Default: 0.1
Softness	 Float	Width of the fade between transparent and opaque, as a color difference. Default: 0.05

OUTPUTS

Pin	Type	Description
Return Value	 Boolean	Makes pixels close to a color transparent (chroma key / green screen): Tolerance 0-1, fading over Softness.

Create Normal Map from Image FUNCTION



A normal map (for materials' Normal input) from an image used as a height map. Use the Async version for large images.

Target: Runtime File Importer Subsystem

Async version: [Create Normal Map From Image \(Async\)](#)

C++: `URuntimeFileImporterSubsystem::CreateNormalMapFromImage`

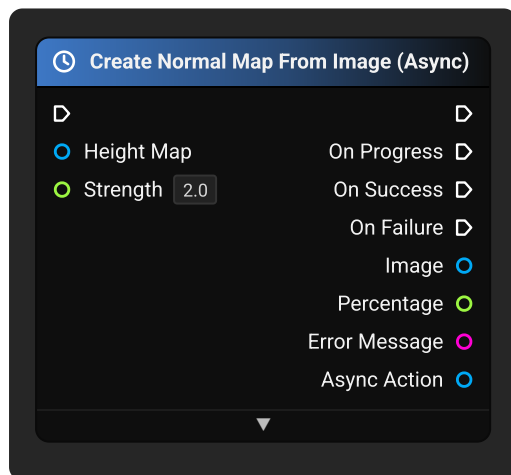
INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Height Map	Runtime Image Object Reference	Grayscale image: brighter is higher.
Strength	Float	How pronounced the bumps are. Default: 2.0
Direct X ADVANCED	Boolean	DirectX normal map (green down, as Unreal expects). Off: OpenGL (green up). Default: true

OUTPUTS

Pin	Type	Description
Error Message	String	Empty on success; otherwise why it failed.
Return Value	Runtime Image Object Reference	The result, or None/empty if it failed (Error Message says why).

Create Normal Map From Image (Async) ASYNC



Makes a normal map from a height map image without blocking the game thread. Outputs the new image.

Blocking version: [Create Normal Map from Image](#)

C++: `UEditImageAsyncAction::CreateNormalMapFromImageAsync`

INPUTS

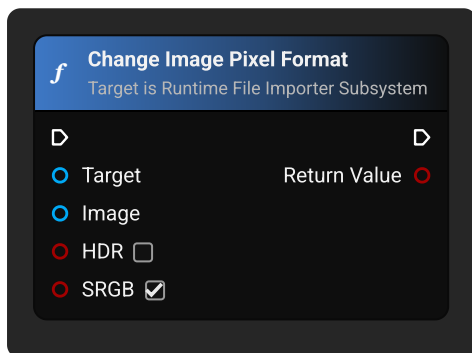
Pin	Type	Description
Height Map	Runtime Image Object Reference	Grayscale image: brighter is higher.
Strength	Float	How pronounced the bumps are. Default: 2.0
Direct X ADVANCED	Boolean	DirectX normal map (green down, as Unreal expects). Off: OpenGL (green up). Default: true

OUTPUTS

Pin	Type	Description
On Progress	 Exec	Fires as the operation advances (Percentage from 0 to 100). Other outputs are not set yet.
On Success	 Exec	Fires when the operation has finished successfully.
On Failure	 Exec	Fires if the operation failed or was cancelled; Error Message says why.
Image	 Runtime Image Object Reference	The Runtime Image.
Percentage	 Float	Progress from 0 to 100.
Error Message	 String	Empty on success; otherwise why it failed.
Async Action	 Edit Image Async Action Object Reference	This node's operation. Call Cancel on it to stop the operation (no more events fire).

Change Image Pixel Format

FUNCTION



Changes how pixels are stored: 8-bit or float (HDR), sRGB colors or linear data. Colors are kept.

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::ChangeImagePixelFormat`

INPUTS

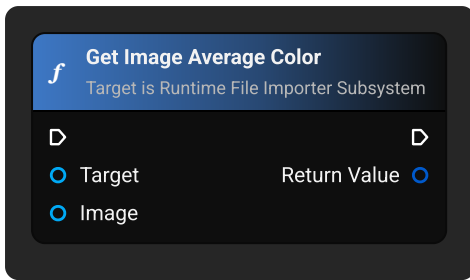
Pin	Type	Description
Target	 Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Image	 Runtime Image Object Reference	The Runtime Image.
HDR	 Boolean	Store 32-bit float colors (values above 1), e.g. for EXR/HDR.
SRGB	 Boolean	Colors are sRGB (most images). Off for data such as normal maps or masks. Default: <code>true</code>

OUTPUTS

Pin	Type	Description
Return Value	 Boolean	Changes how pixels are stored: 8-bit or float (HDR), sRGB colors or linear data.

Get Image Average Color

FUNCTION



The average color of an image.

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::GetImageAverageColor`

INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Image	Runtime Image Object Reference	The Runtime Image.

OUTPUTS

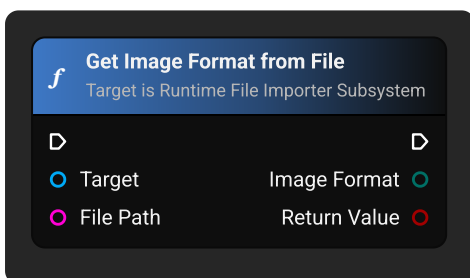
Pin	Type	Description
Return Value	Linear Color	The average color of an image.

Utilities

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Get Image Format from File

FUNCTION



Detects a file's image format from its contents (falling back to its extension). Returns false if it isn't recognized.

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::GetImageFormatFromFile`

INPUTS

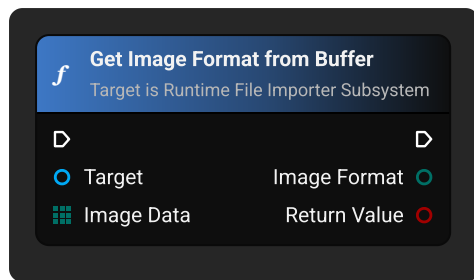
Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
File Path	String	Full path of the file on disk.

OUTPUTS

Pin	Type	Description
Image Format	ERuntime Image Format	Format of the image. Auto Detect reads it from the contents.
Return Value	Boolean	False if it isn't recognized.

Get Image Format from Buffer

FUNCTION



Detects the image format of bytes in memory. Returns false if it isn't recognized.

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::GetImageFormatFromBuffer`

INPUTS

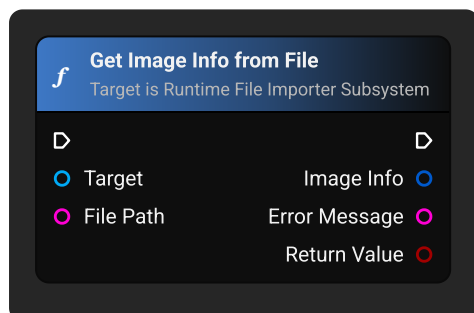
Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Image Data	Array of Byte	Bytes of an image file.

OUTPUTS

Pin	Type	Description
Image Format	ERuntime Image Format	Format of the image. Auto Detect reads it from the contents.
Return Value	Boolean	False if it isn't recognized.

Get Image Info from File

FUNCTION



Reads an image file's format, size, channels, bit depth and frames without importing it.

Target: Runtime File Importer Subsystem

Async version: [Get Image Info From File \(Async\)](#)

C++: `URuntimeFileImporterSubsystem::GetImageInfoFromFile`

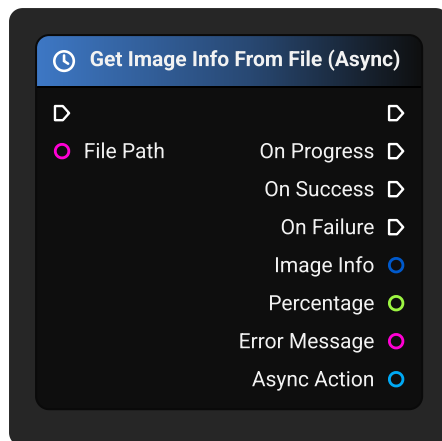
INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
File Path	String	Full path of the file on disk.

OUTPUTS

Pin	Type	Description
Image Info	Runtime Image Info	Size, format, bit depth and alpha of the image.
Error Message	String	Empty on success; otherwise why it failed.
Return Value	Boolean	True on success. False if it failed (Error Message says why).

Get Image Info From File (Async) ASYNC



Reads an image file's format, size, channels, bit depth and frames without importing it or blocking the game thread.

Blocking version: [Get Image Info from File](#)

C++: `UGetImageInfoAsyncAction::GetImageInfoFromFileAsync`

INPUTS

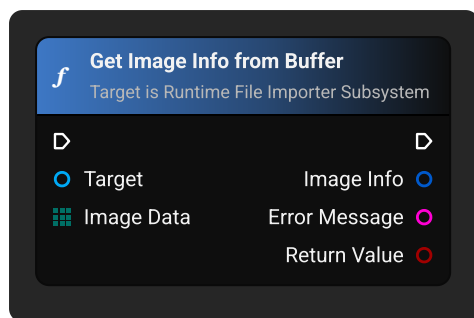
Pin	Type	Description
File Path	 String	Full path of the file on disk.

OUTPUTS

Pin	Type	Description
On Progress	 Exec	Fires as the operation advances (Percentage from 0 to 100). Other outputs are not set yet.
On Success	 Exec	Fires when the operation has finished successfully.
On Failure	 Exec	Fires if the operation failed or was cancelled; Error Message says why.
Image Info	 Runtime Image Info	Size, format, bit depth and alpha of the image.
Percentage	 Float	Progress from 0 to 100.
Error Message	 String	Empty on success; otherwise why it failed.
Async Action	 Get Image Info Async Action Object Reference	This node's operation. Call Cancel on it to stop the operation (no more events fire).

Get Image Info from Buffer

FUNCTION



Reads the info of an image in memory without importing it.

Target: Runtime File Importer Subsystem

Async version: [Get Image Info From Buffer \(Async\)](#)

C++: `URuntimeFileImporterSubsystem::GetImageInfoFromBuffer`

INPUTS

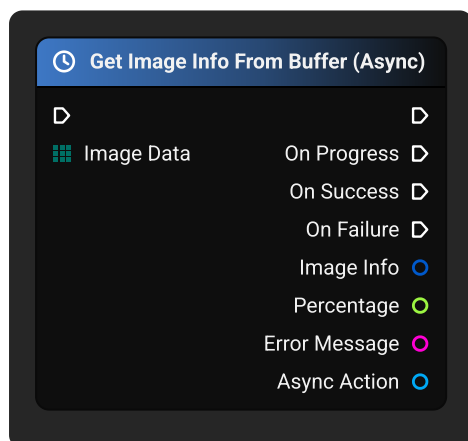
Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Image Data	Array of Byte	Bytes of an image file.

OUTPUTS

Pin	Type	Description
Image Info	Runtime Image Info	Size, format, bit depth and alpha of the image.
Error Message	String	Empty on success; otherwise why it failed.
Return Value	Boolean	True on success. False if it failed (Error Message says why).

Get Image Info From Buffer (Async)

ASYNC



Reads the info of an image in memory without blocking the game thread.

Blocking version: [Get Image Info from Buffer](#)

C++: `UGetImageInfoAsyncAction::GetImageInfoFromBufferAsync`

INPUTS

Pin	Type	Description
Image Data	Array of Byte	Bytes of an image file.

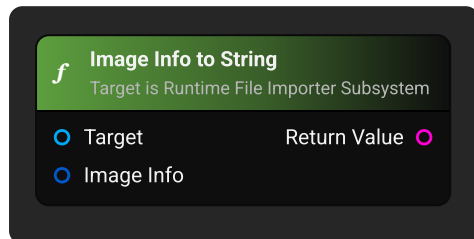
OUTPUTS

Pin	Type	Description
On Progress	Exec	Fires as the operation advances (Percentage from 0 to 100). Other outputs are not set yet.
On Success	Exec	Fires when the operation has finished successfully.
On Failure	Exec	Fires if the operation failed or was cancelled; Error Message says why.
Image Info	Runtime Image Info	Size, format, bit depth and alpha of the image.

Pin	Type	Description
Percentage	 Float	Progress from 0 to 100.
Error Message	 String	Empty on success; otherwise why it failed.
Async Action	 Get Image Info Async Action Object Reference	This node's operation. Call Cancel on it to stop the operation (no more events fire).

Image Info to String

PURE



A readable, multi-line description of image info.

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::ImageInfoToString`

INPUTS

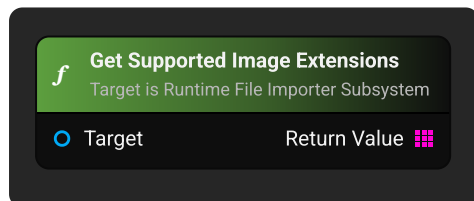
Pin	Type	Description
Target	 Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Image Info	 Runtime Image Info	Size, format, bit depth and alpha of the image.

OUTPUTS

Pin	Type	Description
Return Value	 String	A readable, multi-line description of image info.

Get Supported Image Extensions

PURE



File extensions of images that can be imported on this platform (without dots), e.g. for a file picker filter.

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::GetSupportedImageExtensions`

INPUTS

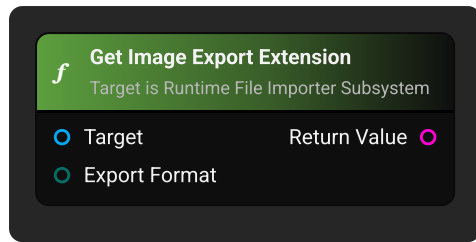
Pin	Type	Description
Target	 Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.

OUTPUTS

Pin	Type	Description
Return Value	 Array of String	File extensions of images that can be imported on this platform (without dots), e.g.

Get Image Export Extension

PURE



File extension written for an image export format, without the dot (e.g. "png").

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::GetImageExportExtension`

INPUTS

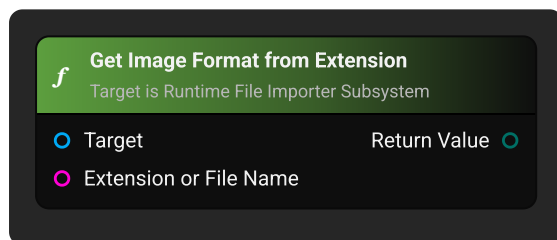
Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Export Format	ERuntime Image Export Format	Format to write.

OUTPUTS

Pin	Type	Description
Return Value	String	File extension written for an image export format, without the dot (e.g.

Get Image Format from Extension

PURE



Import format of a file extension or file name (Auto Detect if unknown).

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::GetImageFormatFromExtension`

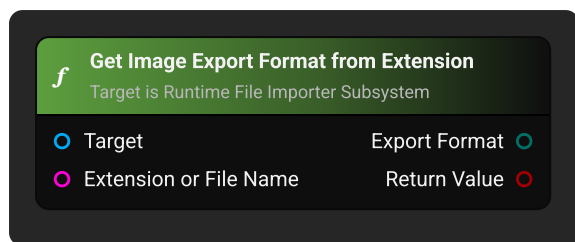
INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Extension or File Name	String	An extension (with or without the dot) or a file name / path / URL ending in one.

OUTPUTS

Pin	Type	Description
Return Value	ERuntime Image Format	Import format of a file extension or file name (Auto Detect if unknown).

Get Image Export Format from Extension PURE



Export format matching a file extension or file name. Returns false if there's none.

Target: Runtime File Importer Subsystem

C++:

`URuntimeFileImporterSubsystem::GetImageExportFormatFromExtension`

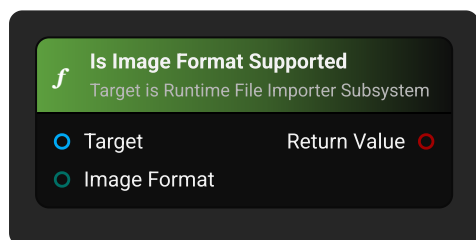
INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Extension or File Name	String	An extension (with or without the dot) or a file name / path / URL ending in one.

OUTPUTS

Pin	Type	Description
Export Format	ERuntime Image Export Format	Format to write.
Return Value	Boolean	False if there's none.

Is Image Format Supported PURE



False for formats this platform can't read (EXR and TIFF on mobile, ICNS outside Mac).

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::IsImageFormatSupported`

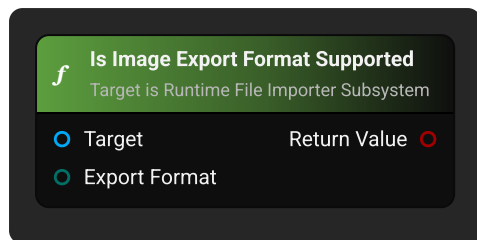
INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Image Format	ERuntime Image Format	Format of the image. Auto Detect reads it from the contents.

OUTPUTS

Pin	Type	Description
Return Value	Boolean	False for formats this platform can't read (EXR and TIFF on mobile, ICNS outside Mac).

Is Image Export Format Supported PURE



False for formats this platform can't write (EXR on mobile).

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::IsImageExportFormatSupported`

INPUTS

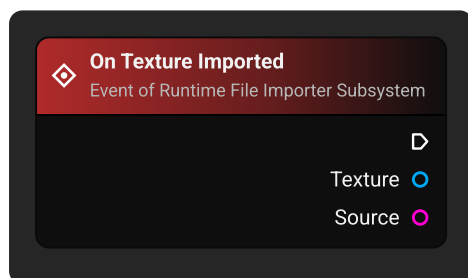
Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Export Format	ERuntime Image Export Format	Format to write.

OUTPUTS

Pin	Type	Description
Return Value	Boolean	False for formats this platform can't write (EXR on mobile).

Events 4 nodes

On Texture Imported EVENT



Any texture or image import succeeded. Texture is a Texture 2D or a Runtime Image; Source the file path or URL (empty for buffers).

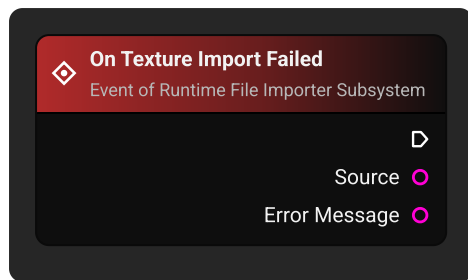
Bind it with "Bind Event to On Texture Imported" (or "Assign On Texture Imported") on the Runtime File Importer Subsystem.

C++: `URuntimeFileImporterSubsystem::OnTextureImported`

OUTPUTS

Pin	Type	Description
Texture	Object Object Reference	Any texture: Texture 2D, render target, cube, media or video texture.
Source	String	File path or URL the import came from (empty for buffers and strings).

On Texture Import Failed EVENT



Any texture or image import failed.

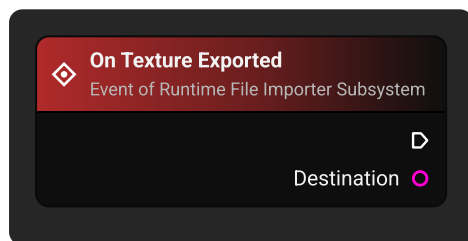
Bind it with "Bind Event to On Texture Import Failed" (or "Assign On Texture Import Failed") on the Runtime File Importer Subsystem.

```
C++: URuntimeFileImporterSubsystem::OnTextureImportFailed
```

OUTPUTS

Pin	Type	Description
Source	String	File path or URL the import came from (empty for buffers and strings).
Error Message	String	Empty on success; otherwise why it failed.

On Texture Exported EVENT



Any texture or image export (or image file conversion) succeeded. Destination is the output file path (empty for buffers).

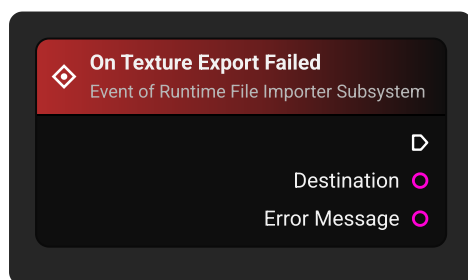
Bind it with "Bind Event to On Texture Exported" (or "Assign On Texture Exported") on the Runtime File Importer Subsystem.

```
C++: URuntimeFileImporterSubsystem::OnTextureExported
```

OUTPUTS

Pin	Type	Description
Destination	String	Output file path (empty when writing to a buffer).

On Texture Export Failed EVENT



Any texture or image export failed.

Bind it with "Bind Event to On Texture Export Failed" (or "Assign On Texture Export Failed") on the Runtime File Importer Subsystem.

```
C++: URuntimeFileImporterSubsystem::OnTextureExportFailed
```

OUTPUTS

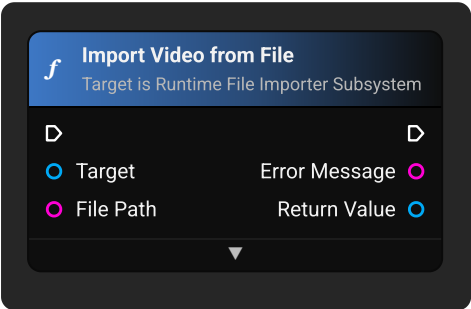
Pin	Type	Description
Destination	String	Output file path (empty when writing to a buffer).
Error Message	String	Empty on success; otherwise why it failed.

Video

Play videos from files, URLs or memory into textures, control playback, read video info, extract frames and thumbnails, convert video files, and record the screen or a render target to a video file.

Import 8 nodes

Import Video from File FUNCTION



Opens a video file and plays it into a texture (Runtime Video: show its Texture variable in a material or widget, or use Create Material From Texture). Returns right away; the video is ready when its On Video Opened event fires (the Async version waits for it). GIF and Motion JPEG AVI play on every platform with the plugin's own player, which also plays everything Windows decodes on Windows (MP4, MOV, WMV, MKV, ...); other files play with the platform's player.

Target: Runtime File Importer Subsystem
 Async version: [Import Video From File \(Async\)](#)
 C++: `URuntimeFileImporterSubsystem::ImportVideoFromFile`

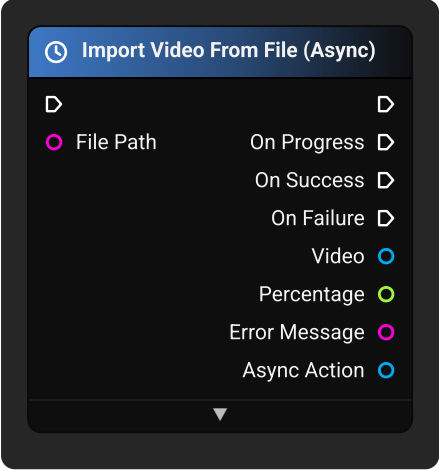
INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
File Path	String	Full path of the file on disk.
Options	Runtime Video Open Options	How videos are opened.

OUTPUTS

Pin	Type	Description
Error Message	String	Empty on success; otherwise why it failed.
Return Value	Runtime Video Object Reference	Right away; the video is ready when its On Video Opened event fires (the Async version waits for it).

Import Video From File (Async) ASYNC



Opens a video file and waits until it's ready to play (see Import Video From File).

Blocking version: [Import Video from File](#)

C++: `UImportVideoAsyncAction::ImportVideoFromFileAsync`

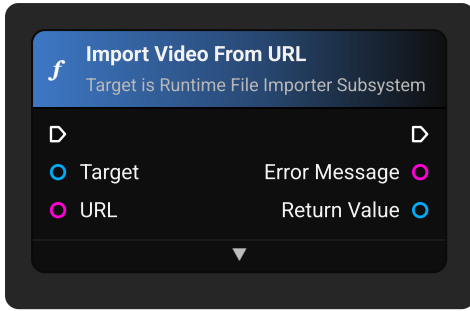
INPUTS

Pin	Type	Description
File Path	String	Full path of the file on disk.
Options ADVANCED OPTIONAL	Runtime Video Open Options	How videos are opened.

OUTPUTS

Pin	Type	Description
On Progress	Exec	Fires as the operation advances (Percentage from 0 to 100). Other outputs are not set yet.
On Success	Exec	The video is open: size, length and tracks are known (it plays now if Play On Open).
On Failure	Exec	Fires if the operation failed or was cancelled; Error Message says why.
Video	Runtime Video Object Reference	The Runtime Video.
Percentage	Float	Progress from 0 to 100.
Error Message	String	Empty on success; otherwise why it failed.
Async Action	Import Video Async Action Object Reference	This node's operation. Call Cancel on it to stop the operation (no more events fire).

Import Video From URL FUNCTION



Streams a video from a URL (http(s), rtsp, ... as the platform's player supports) without downloading it first. Returns right away, like Import Video From File.

Target: Runtime File Importer Subsystem

Async version: [Import Video From URL \(Async\)](#)

C++: `URuntimeFileImporterSubsystem::ImportVideoFromURL`

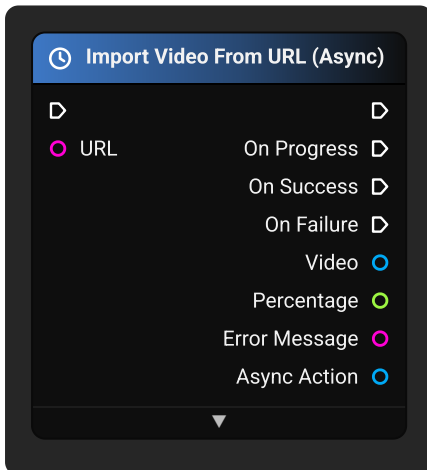
INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
URL	String	The http(s) address to download. Google Docs, Sheets, Slides and Drive share links are converted automatically.
Options ADVANCED OPTIONAL	Runtime Video Open Options	How videos are opened.

OUTPUTS

Pin	Type	Description
Error Message	String	Empty on success; otherwise why it failed.
Return Value	Runtime Video Object Reference	Right away, like Import Video From File.

Import Video From URL (Async) ASYNC



Streams a video from a URL and waits until it's ready to play.

Blocking version: [Import Video From URL](#)

C++: `UImportVideoAsyncAction::ImportVideoFromURLAsync`

INPUTS

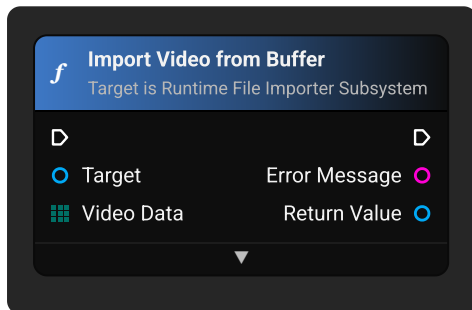
Pin	Type	Description
URL	String	The http(s) address to download. Google Docs, Sheets, Slides and Drive share links are converted automatically.
Options ADVANCED OPTIONAL	Runtime Video Open Options	How videos are opened.

OUTPUTS

Pin	Type	Description
On Progress	 Exec	Fires as the operation advances (Percentage from 0 to 100). Other outputs are not set yet.
On Success	 Exec	The video is open: size, length and tracks are known (it plays now if Play On Open).
On Failure	 Exec	Fires if the operation failed or was cancelled; Error Message says why.
Video	 Runtime Video Object Reference	The Runtime Video.
Percentage	 Float	Progress from 0 to 100.
Error Message	 String	Empty on success; otherwise why it failed.
Async Action	 Import Video Async Action Object Reference	This node's operation. Call Cancel on it to stop the operation (no more events fire).

Import Video from Buffer

FUNCTION



Plays a video from bytes in memory (written to a temporary file, as players read files). Returns right away, like Import Video From File.

Target: Runtime File Importer Subsystem

Async version: [Import Video From Buffer \(Async\)](#)

C++: `URuntimeFileImporterSubsystem::ImportVideoFromBuffer`

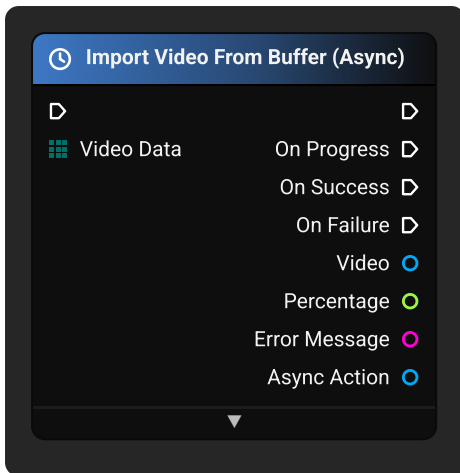
INPUTS

Pin	Type	Description
Target	 Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Video Data	 Array of Byte	Bytes of a video file.
Options	ADVANCED OPTIONAL  Runtime Video Open Options	How videos are opened.

OUTPUTS

Pin	Type	Description
Error Message	 String	Empty on success; otherwise why it failed.
Return Value	 Runtime Video Object Reference	Right away, like Import Video From File.

Import Video From Buffer (Async) ASYNC



Plays a video in memory (written to a temporary file without blocking) and waits until it's ready.

Blocking version: [Import Video from Buffer](#)

C++: `UImportVideoAsyncAction::ImportVideoFromBufferAsync`

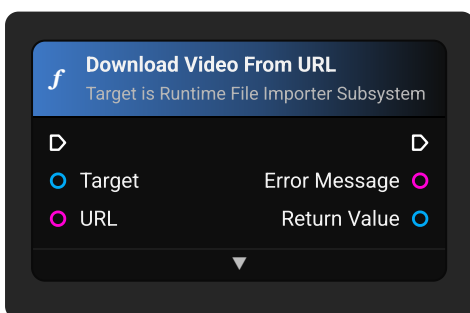
INPUTS

Pin	Type	Description
Video Data	Array of Byte	Bytes of a video file.
Options ADVANCED OPTIONAL	Runtime Video Open Options	How videos are opened.

OUTPUTS

Pin	Type	Description
On Progress	Exec	Fires as the operation advances (Percentage from 0 to 100). Other outputs are not set yet.
On Success	Exec	The video is open: size, length and tracks are known (it plays now if Play On Open).
On Failure	Exec	Fires if the operation failed or was cancelled; Error Message says why.
Video	Runtime Video Object Reference	The Runtime Video.
Percentage	Float	Progress from 0 to 100.
Error Message	String	Empty on success; otherwise why it failed.
Async Action	Import Video Async Action Object Reference	This node's operation. Call Cancel on it to stop the operation (no more events fire).

Download Video From URL FUNCTION



Downloads a whole video file over HTTP(S), then plays it (Import Video From URL streams instead). FREEZES the game until the download finishes; prefer the Async version. Save To File Path keeps the file (else a temporary file is used).

Target: Runtime File Importer Subsystem

Async version: [Download Video From URL \(Async\)](#)

C++: `URuntimeFileImporterSubsystem::DownloadVideoFromURL`

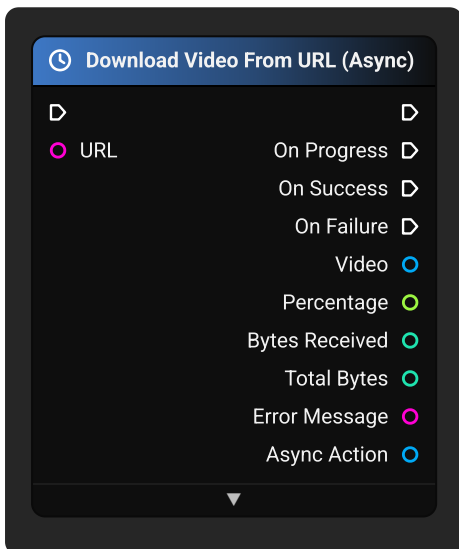
INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.

Pin	Type	Description
URL	 String	The http(s) address to download. Google Docs, Sheets, Slides and Drive share links are converted automatically.
Options	ADVANCED OPTIONAL  Runtime Video Open Options	How videos are opened.
Save to File Path	ADVANCED  String	Optional. Also writes the downloaded file there, unmodified. Default: (empty)
Timeout Seconds	ADVANCED  Float	Gives up (and fails) if the download takes longer than this. Default: 120.0
OUTPUTS		
Pin	Type	Description
Error Message	 String	Empty on success; otherwise why it failed.
Return Value	 Runtime Video Object Reference	The result, or None/empty if it failed (Error Message says why).

Download Video From URL (Async)

ASYNC



Downloads a whole video file over HTTP(S), then opens it and waits until it's ready. Save To File Path keeps the file.

Blocking version: [Download Video From URL](#)

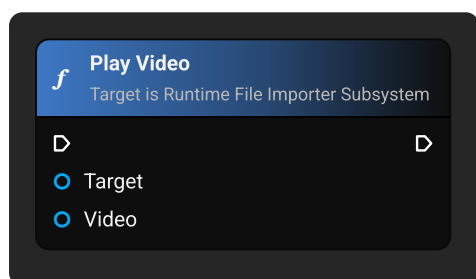
C++: UDownloadVideoFromURLAsyncAction::DownloadVideoFromURLAsync

INPUTS		
Pin	Type	Description
URL	 String	The http(s) address to download. Google Docs, Sheets, Slides and Drive share links are converted automatically.
Options	ADVANCED OPTIONAL  Runtime Video Open Options	How videos are opened.
Save to File Path	ADVANCED  String	Optional. Also writes the downloaded file there, unmodified. Default: (empty)
Timeout Seconds	ADVANCED  Float	Gives up (and fails) if the download takes longer than this. Default: 300.0
OUTPUTS		
Pin	Type	Description
On Progress	 Exec	Fires repeatedly while downloading. Percentage (0-100) and Total Bytes are 0 if the server doesn't report the file size.
On Success	 Exec	Fires when the operation has finished successfully.
On Failure	 Exec	Fires if the operation failed or was cancelled; Error Message says why.

Pin	Type	Description
Video	 Runtime Video Object Reference	The Runtime Video.
Percentage	 Float	Progress from 0 to 100.
Bytes Received	 Integer	Bytes downloaded so far.
Total Bytes	 Integer	Size of the download in bytes, or 0 while the server hasn't said.
Error Message	 String	Empty on success; otherwise why it failed.
Async Action	 Download Video from URL Async Action Object Reference	This node's operation. Call Cancel on it to stop the operation (no more events fire).

Playback 25 nodes

Play Video FUNCTION



Plays (or resumes) a video; a finished video starts over.

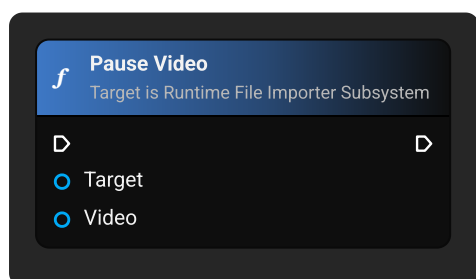
Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::PlayVideo`

INPUTS

Pin	Type	Description
Target	 Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Video	 Runtime Video Object Reference	The Runtime Video.

Pause Video FUNCTION



Pauses a video. Same as setting its Paused variable.

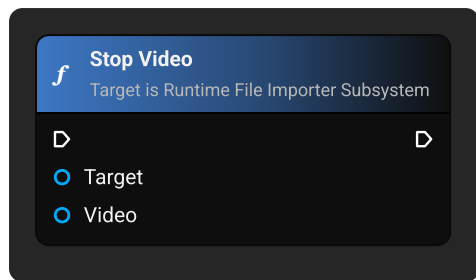
Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::PauseVideo`

INPUTS

Pin	Type	Description
Target	 Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Video	 Runtime Video Object Reference	The Runtime Video.

Stop Video FUNCTION



Pauses a video and goes back to the start.

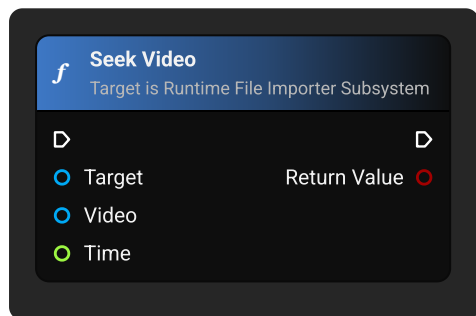
Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::StopVideo`

INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Video	Runtime Video Object Reference	The Runtime Video.

Seek Video FUNCTION



Jumps to a time in seconds. Same as setting its Time variable. Returns false if the video can't seek.

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::SeekVideo`

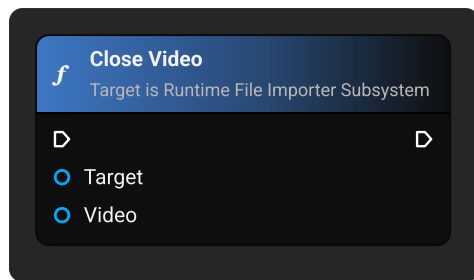
INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Video	Runtime Video Object Reference	The Runtime Video.
Time	Float	Time in seconds.

OUTPUTS

Pin	Type	Description
Return Value	Boolean	False if the video can't seek.

Close Video FUNCTION



Stops a video and releases its player, decoder and sound.

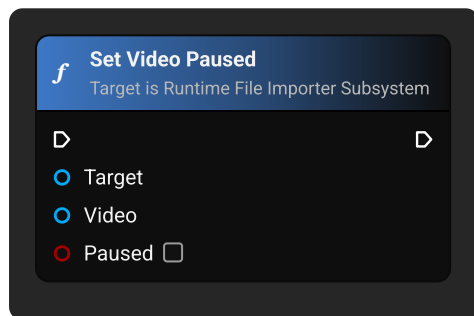
Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::CloseVideo`

INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Video	Runtime Video Object Reference	The Runtime Video.

Set Video Paused FUNCTION



Pauses or resumes a video. Same as setting its Paused variable.

Target: Runtime File Importer Subsystem

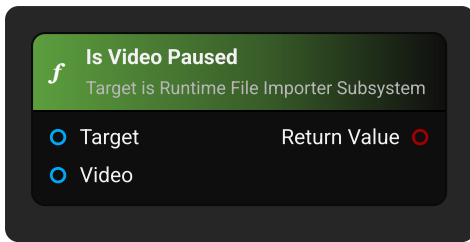
C++: `URuntimeFileImporterSubsystem::SetVideoPaused`

INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Video	Runtime Video Object Reference	The Runtime Video.
Paused	Boolean	True to pause, false to resume.

Is Video Paused

PURE



Whether a video is paused. Same as its Paused variable.

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::IsVideoPaused`

INPUTS

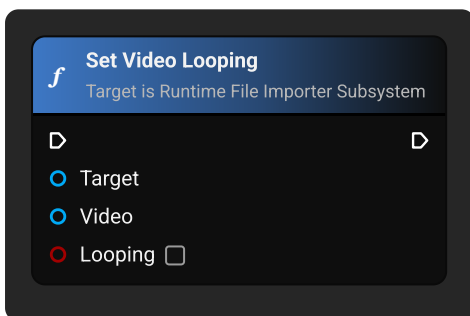
Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Video	Runtime Video Object Reference	The Runtime Video.

OUTPUTS

Pin	Type	Description
Return Value	Boolean	Whether a video is paused.

Set Video Looping

FUNCTION



Sets whether a video starts over at its end. Same as setting its Looping variable.

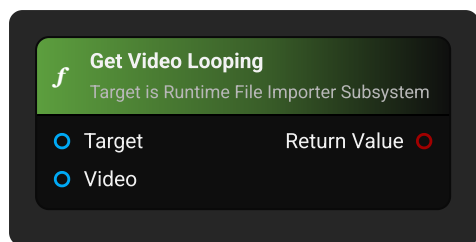
Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::SetVideoLooping`

INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Video	Runtime Video Object Reference	The Runtime Video.
Looping	Boolean	Start again from the beginning when the end is reached.

Get Video Looping PURE



Whether a video loops. Same as its Looping variable.

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::GetVideoLooping`

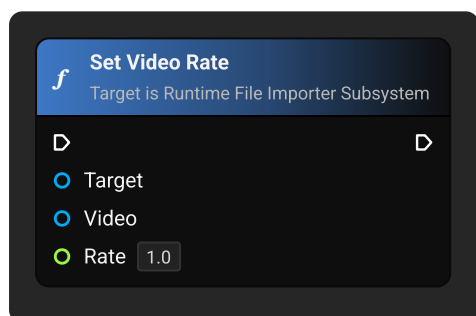
INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Video	Runtime Video Object Reference	The Runtime Video.

OUTPUTS

Pin	Type	Description
Return Value	Boolean	Whether a video loops.

Set Video Rate FUNCTION



Sets a video's playback speed (1: normal). Same as setting its Rate variable.

Target: Runtime File Importer Subsystem

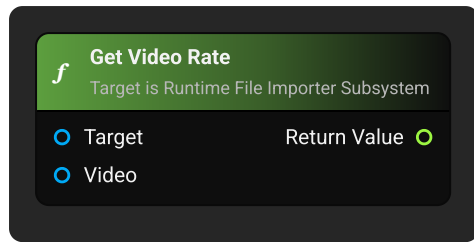
C++: `URuntimeFileImporterSubsystem::SetVideoRate`

INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Video	Runtime Video Object Reference	The Runtime Video.
Rate	Float	Playback speed (1: normal, 2: double, negative: backwards if supported). Default: 1.0

Get Video Rate

PURE



A video's playback speed. Same as its Rate variable.

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::GetVideoRate`

INPUTS

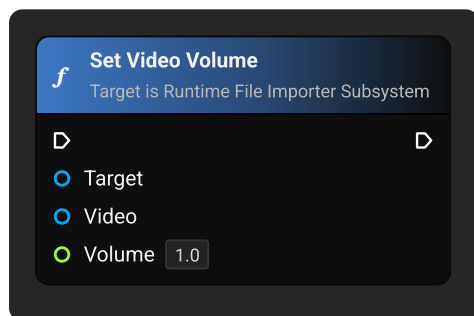
Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Video	Runtime Video Object Reference	The Runtime Video.

OUTPUTS

Pin	Type	Description
Return Value	Float	A video's playback speed.

Set Video Volume

FUNCTION



Sets a video's sound volume multiplier. Same as setting its Volume variable.

Target: Runtime File Importer Subsystem

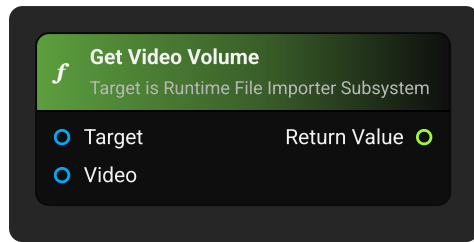
C++: `URuntimeFileImporterSubsystem::SetVideoVolume`

INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Video	Runtime Video Object Reference	The Runtime Video.
Volume	Float	Volume multiplier (1: unchanged, 0: silent). Default: 1.0

Get Video Volume

PURE



A video's sound volume multiplier. Same as its Volume variable.

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::GetVideoVolume`

INPUTS

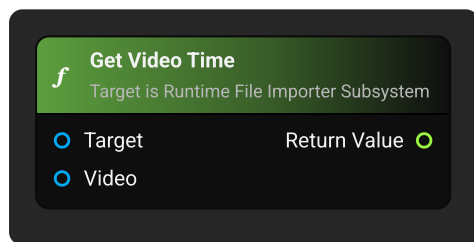
Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Video	Runtime Video Object Reference	The Runtime Video.

OUTPUTS

Pin	Type	Description
Return Value	Float	A video's sound volume multiplier.

Get Video Time

PURE



A video's playback position in seconds. Same as its Time variable.

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::GetVideoTime`

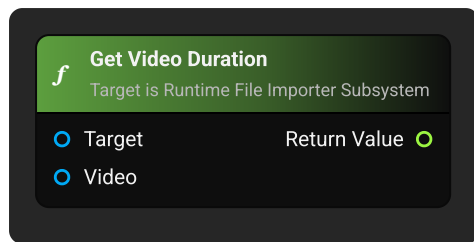
INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Video	Runtime Video Object Reference	The Runtime Video.

OUTPUTS

Pin	Type	Description
Return Value	Float	A video's playback position in seconds.

Get Video Duration PURE



A video's length in seconds. Same as its Duration variable.

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::GetVideoDuration`

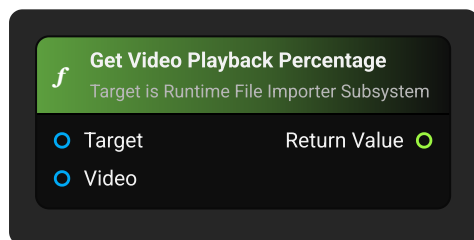
INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Video	Runtime Video Object Reference	The Runtime Video.

OUTPUTS

Pin	Type	Description
Return Value	Float	A video's length in seconds.

Get Video Playback Percentage PURE



A video's progress from 0 to 100. Same as its Playback Percentage variable.

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::GetVideoPlaybackPercentage`

INPUTS

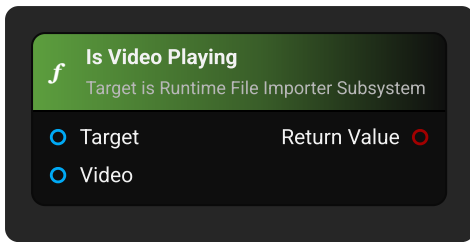
Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Video	Runtime Video Object Reference	The Runtime Video.

OUTPUTS

Pin	Type	Description
Return Value	Float	A video's progress from 0 to 100.

Is Video Playing

PURE



True while a video plays. Same as its Is Playing variable.

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::IsVideoPlaying`

INPUTS

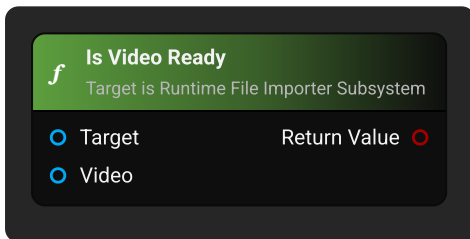
Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Video	Runtime Video Object Reference	The Runtime Video.

OUTPUTS

Pin	Type	Description
Return Value	Boolean	True while a video plays.

Is Video Ready

PURE



True once a video is open. Same as its Is Ready variable.

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::IsVideoReady`

INPUTS

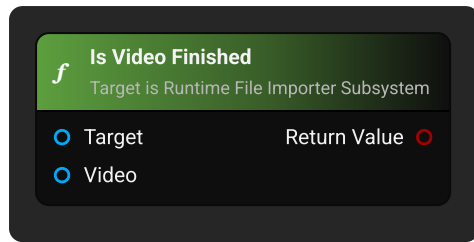
Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Video	Runtime Video Object Reference	The Runtime Video.

OUTPUTS

Pin	Type	Description
Return Value	Boolean	True once a video is open.

Is Video Finished

PURE



True once a non-looping video has played to the end. Same as its Is Finished variable.

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::IsVideoFinished`

INPUTS

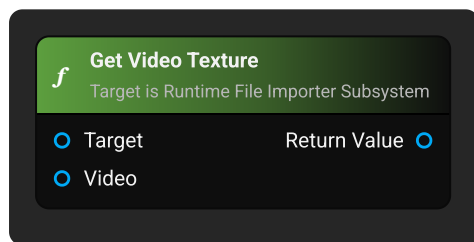
Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Video	Runtime Video Object Reference	The Runtime Video.

OUTPUTS

Pin	Type	Description
Return Value	Boolean	True once a non-looping video has played to the end.

Get Video Texture

PURE



The texture a video plays into. Same as its Texture variable.

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::GetVideoTexture`

INPUTS

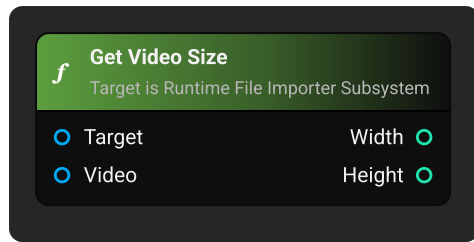
Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Video	Runtime Video Object Reference	The Runtime Video.

OUTPUTS

Pin	Type	Description
Return Value	Texture Object Reference	The texture a video plays into.

Get Video Size

PURE



A video's size in pixels. Same as its Width and Height variables.

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::GetVideoSize`

INPUTS

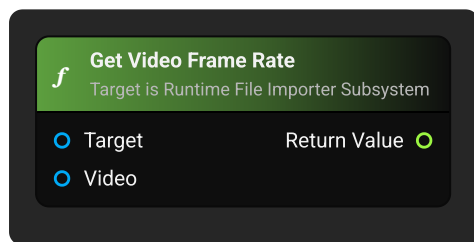
Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Video	Runtime Video Object Reference	The Runtime Video.

OUTPUTS

Pin	Type	Description
Width	Integer	Width in pixels.
Height	Integer	Height in pixels.

Get Video Frame Rate

PURE



A video's frames per second. Same as its Frame Rate variable.

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::GetVideoFrameRate`

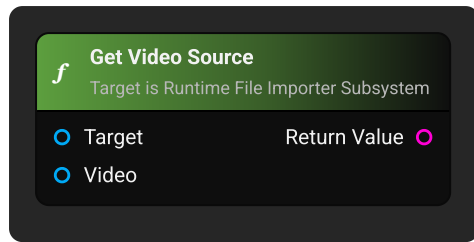
INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Video	Runtime Video Object Reference	The Runtime Video.

OUTPUTS

Pin	Type	Description
Return Value	Float	A video's frames per second.

Get Video Source PURE



File path or URL a video plays from. Same as its Source variable.

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::GetVideoSource`

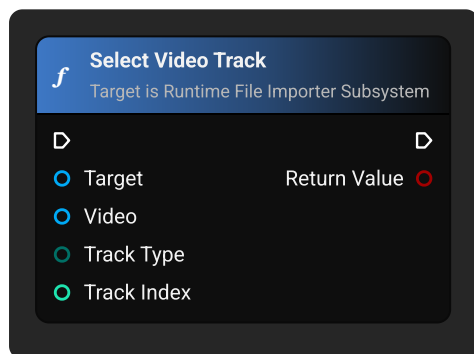
INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Video	Runtime Video Object Reference	The Runtime Video.

OUTPUTS

Pin	Type	Description
Return Value	String	File path or URL a video plays from.

Select Video Track FUNCTION



Selects a video's audio, video or caption track (see Get Video Details' track lists; -1: none).

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::SelectVideoTrack`

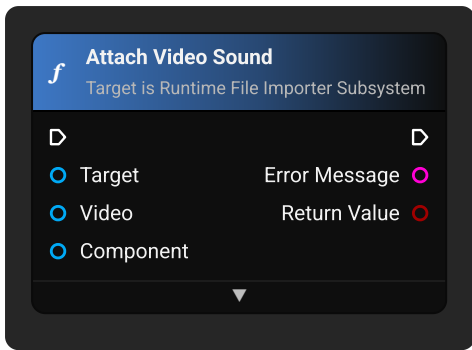
INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Video	Runtime Video Object Reference	The Runtime Video.
Track Type	ERuntime Video Track Type	Video or audio (or caption) tracks.
Track Index	Integer	Index of the track to use (-1: none).

OUTPUTS

Pin	Type	Description
Return Value	Boolean	Selects a video's audio, video or caption track (see Get Video Details' track lists; -1: none).

Attach Video Sound FUNCTION



Makes a video's sound come from a component in the level (3D, with Attenuation when set); no component makes it 2D again.

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::AttachVideoSound`

INPUTS

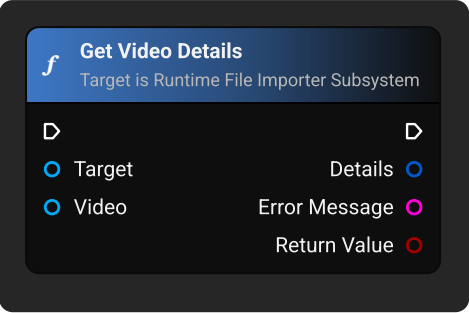
Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Video	Runtime Video Object Reference	The Runtime Video.
Component	Scene Component Object Reference	The component the sound follows (spatial audio). Empty: the sound plays in 2D again.
Attenuation ADVANCED	Sound Attenuation Object Reference	How the sound fades with distance. Empty: the engine default. Default: <code>None</code>

OUTPUTS

Pin	Type	Description
Error Message	String	Empty on success; otherwise why it failed.
Return Value	Boolean	True on success. False if it failed (Error Message says why).

Get Video Details

FUNCTION



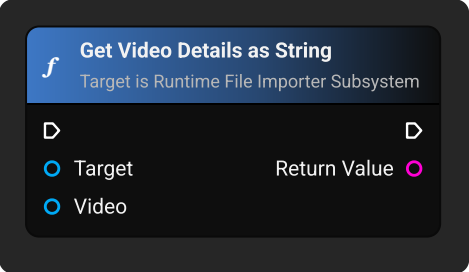
Everything about a playing video: player, state, size, frame rate, sound format, and its video, audio and caption tracks.

Target: Runtime File Importer Subsystem
C++: `URuntimeFileImporterSubsystem::GetVideoDetails`

INPUTS		
Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Video	Runtime Video Object Reference	The Runtime Video.
OUTPUTS		
Pin	Type	Description
Details	Runtime Video Details	Everything about a Runtime Video as it plays.
Error Message	String	Empty on success; otherwise why it failed.
Return Value	Boolean	True on success. False if it failed (Error Message says why).

Get Video Details as String

FUNCTION

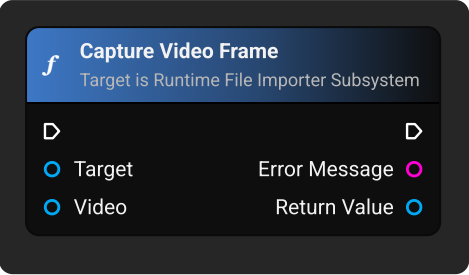


Readable, multi-line details of a playing video, e.g. to print.

Target: Runtime File Importer Subsystem
C++: `URuntimeFileImporterSubsystem::GetVideoDetailsAsString`

INPUTS		
Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Video	Runtime Video Object Reference	The Runtime Video.
OUTPUTS		
Pin	Type	Description
Return Value	String	Readable, multi-line details of a playing video, e.g.

Capture Video Frame FUNCTION



A Texture 2D copy of the frame a video shows now (e.g. a snapshot).

Target: Runtime File Importer Subsystem

Async version: [Capture Video Frame \(Async\)](#)

C++: `URuntimeFileImporterSubsystem::CaptureVideoFrame`

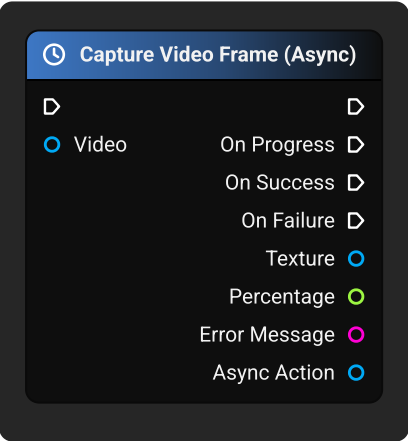
INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Video	Runtime Video Object Reference	The Runtime Video.

OUTPUTS

Pin	Type	Description
Error Message	String	Empty on success; otherwise why it failed.
Return Value	Texture 2D Object Reference	The result, or None/empty if it failed (Error Message says why).

Capture Video Frame (Async) ASYNC



A Texture 2D copy of the frame a video shows now, without blocking the game thread.

Blocking version: [Capture Video Frame](#)

C++: `UConvertToTexture2DAsyncAction::CaptureVideoFrameAsync`

INPUTS

Pin	Type	Description
Video	Runtime Video Object Reference	The Runtime Video.

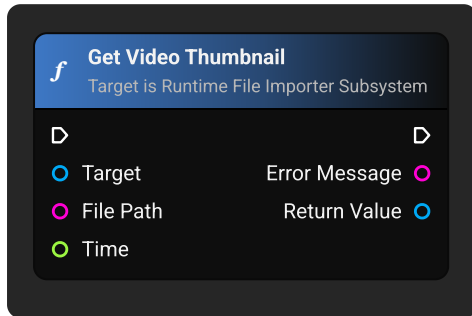
OUTPUTS

Pin	Type	Description
On Progress	Exec	Fires as the operation advances (Percentage from 0 to 100). Other outputs are not set yet.
On Success	Exec	Fires when the operation has finished successfully.
On Failure	Exec	Fires if the operation failed or was cancelled; Error Message says why.
Texture	Texture 2D Object Reference	The Texture 2D.

Pin	Type	Description
Percentage	 Float	Progress from 0 to 100.
Error Message	 String	Empty on success; otherwise why it failed.
Async Action	 Convert to Texture 2D Async Action Object Reference	This node's operation. Call Cancel on it to stop the operation (no more events fire).

Get Video Thumbnail

FUNCTION



The frame of a video file at Time (seconds) as a texture, e.g. for thumbnails in a video browser. GIF and Motion JPEG AVI on every platform; other formats on Windows.

Target: Runtime File Importer Subsystem

Async version: [Get Video Thumbnail \(Async\)](#)

C++: `URuntimeFileImporterSubsystem::GetVideoThumbnail`

INPUTS

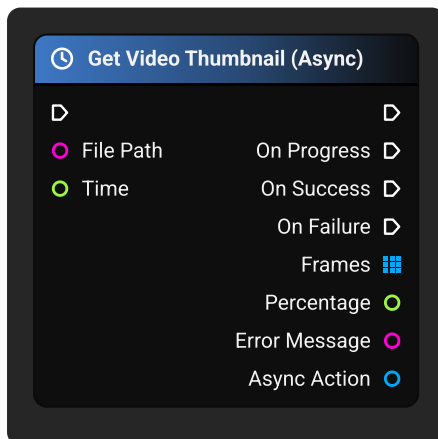
Pin	Type	Description
Target	 Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
File Path	 String	Full path of the file on disk.
Time	 Float	Time in seconds.

OUTPUTS

Pin	Type	Description
Error Message	 String	Empty on success; otherwise why it failed.
Return Value	 Texture 2D Object Reference	The result, or None/empty if it failed (Error Message says why).

Get Video Thumbnail (Async)

ASYNC



The frame of a video file at Time as a texture (the array's only element), without blocking the game thread.

Blocking version: [Get Video Thumbnail](#)

C++: `UExtractVideoFramesAsyncAction::GetVideoThumbnailAsync`

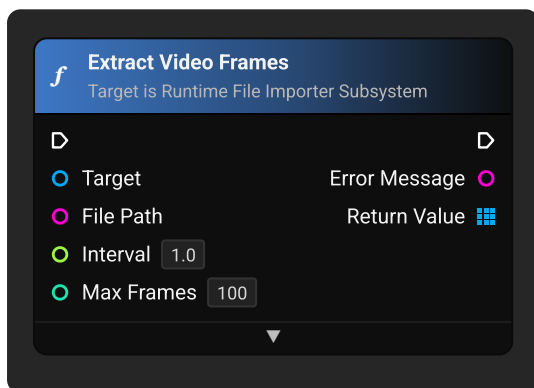
INPUTS

Pin	Type	Description
File Path	 String	Full path of the file on disk.
Time	 Float	Time in seconds.

OUTPUTS

Pin	Type	Description
On Progress	 Exec	Fires as the operation advances (Percentage from 0 to 100). Other outputs are not set yet.
On Success	 Exec	Fires when the operation has finished successfully.
On Failure	 Exec	Fires if the operation failed or was cancelled; Error Message says why.
Frames	 Array of Texture 2D Object Reference	The frames, in order (textures, render targets or Runtime Images, all the same size).
Percentage	 Float	Progress from 0 to 100.
Error Message	 String	Empty on success; otherwise why it failed.
Async Action	 Extract Video Frames Async Action Object Reference	This node's operation. Call Cancel on it to stop the operation (no more events fire).

Extract Video Frames FUNCTION



Frames of a video file as textures: every frame between Start Time and End Time (0: the end), or one every Interval seconds, at most Max Frames. Same platform support as Get Video Thumbnail. Uses a lot of memory for many frames.

Target: Runtime File Importer Subsystem

Async version: [Extract Video Frames \(Async\)](#)

C++: `URuntimeFileImporterSubsystem::ExtractVideoFrames`

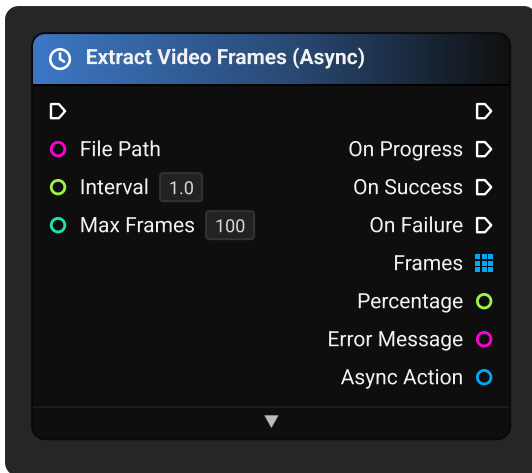
INPUTS

Pin	Type	Description
Target	 Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
File Path	 String	Full path of the file on disk.
Interval	 Float	Seconds between frames. Default: 1.0
Max Frames	 Integer	Most frames to extract (0: no limit). Default: 100
Start Time ADVANCED	 Float	Start of the range, in seconds. Default: 0.0
End Time ADVANCED	 Float	End of the range, in seconds. 0: the end. Default: 0.0

OUTPUTS

Pin	Type	Description
Error Message	 String	Empty on success; otherwise why it failed.
Return Value	 Array of Texture 2D Object Reference	The result, or None/empty if it failed (Error Message says why).

Extract Video Frames (Async) ASYNC



Frames of a video file as textures (one every Interval seconds, at most Max Frames) without blocking the game thread.

Blocking version: [Extract Video Frames](#)

C++: `UExtractVideoFramesAsyncAction::ExtractVideoFramesAsync`

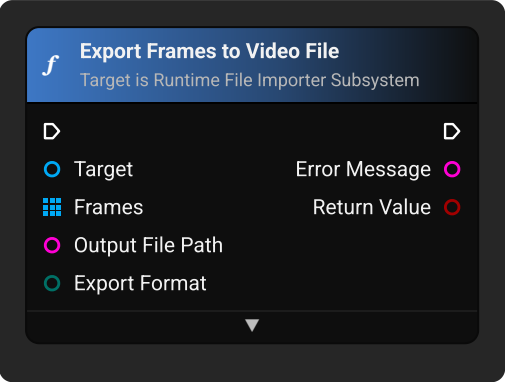
INPUTS

Pin	Type	Description
File Path	String	Full path of the file on disk.
Interval	Float	Seconds between frames. Default: 1.0
Max Frames	Integer	Most frames to extract (0: no limit). Default: 100
Start Time ADVANCED	Float	Start of the range, in seconds. Default: 0.0
End Time ADVANCED	Float	End of the range, in seconds. 0: the end. Default: 0.0

OUTPUTS

Pin	Type	Description
On Progress	Exec	Fires as the operation advances (Percentage from 0 to 100). Other outputs are not set yet.
On Success	Exec	Fires when the operation has finished successfully.
On Failure	Exec	Fires if the operation failed or was cancelled; Error Message says why.
Frames	Array of Texture 2D Object Reference	The frames, in order (textures, render targets or Runtime Images, all the same size).
Percentage	Float	Progress from 0 to 100.
Error Message	String	Empty on success; otherwise why it failed.
Async Action	Extract Video Frames Async Action Object Reference	This node's operation. Call Cancel on it to stop the operation (no more events fire).

Export Frames to Video File FUNCTION



Writes frames (textures, render targets, Runtime Images, video frames; e.g. screenshots or a drawing's steps) as a video: MP4 (Windows), AVI, animated GIF or numbered images, at Options' frame rate, with Options' Audio.

Target: Runtime File Importer Subsystem
Async version: [Export Frames To Video File \(Async\)](#)
C++: `URuntimeFileImporterSubsystem::ExportFramesToVideoFile`

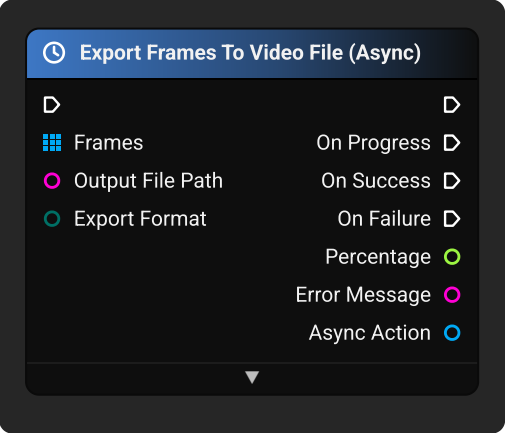
INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Frames	Array of Object Object Reference	The frames, in order (textures, render targets or Runtime Images, all the same size).
Output File Path	String	Full path of the file to write. Missing folders are created; an existing file is replaced.
Export Format	ERuntime Video Export Format	Format to write.
Options ADVANCED OPTIONAL	Runtime Video Export Options	How videos are written.

OUTPUTS

Pin	Type	Description
Error Message	String	Empty on success; otherwise why it failed.
Return Value	Boolean	True on success. False if it failed (Error Message says why).

Export Frames To Video File (Async) ASYNC



Writes frames (textures, render targets, Runtime Images, video frames) as a video without blocking the game thread.

Blocking version: [Export Frames to Video File](#)
C++: `UExportVideoAsyncAction::ExportFramesToVideoFileAsync`

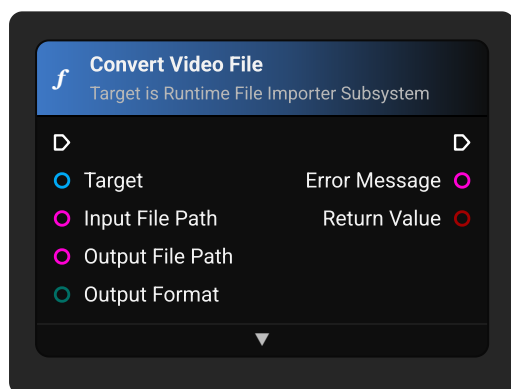
INPUTS

Pin	Type	Description
Frames	Array of Object Object Reference	The frames, in order (textures, render targets or Runtime Images, all the same size).

Pin	Type	Description
Output File Path	 String	Full path of the file to write. Missing folders are created; an existing file is replaced.
Export Format	 ERuntime Video Export Format	Format to write.
Options ADVANCED OPTIONAL	 Runtime Video Export Options	How videos are written.
OUTPUTS		
Pin	Type	Description
On Progress	 Exec	Fires as the operation advances (Percentage from 0 to 100). Other outputs are not set yet.
On Success	 Exec	Fires when the operation has finished successfully.
On Failure	 Exec	Fires if the operation failed or was cancelled; Error Message says why.
Percentage	 Float	Progress from 0 to 100.
Error Message	 String	Empty on success; otherwise why it failed.
Async Action	 Export Video Async Action Object Reference	This node's operation. Call Cancel on it to stop the operation (no more events fire).

Convert Video File

FUNCTION



Converts a video file to another format (e.g. MP4 to GIF, GIF to MP4, a part of a video, frames to images), sound included. GIF and Motion JPEG AVI sources on every platform; other formats on Windows.

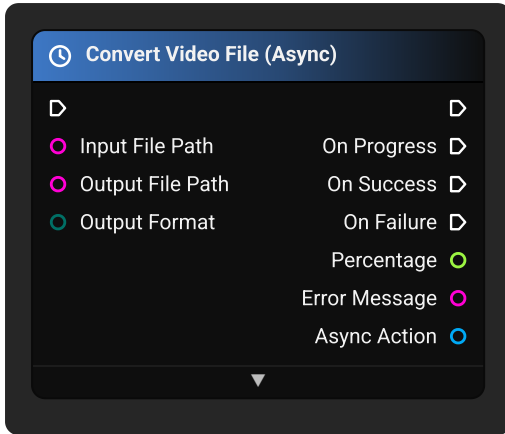
Target: Runtime File Importer Subsystem

Async version: [Convert Video File \(Async\)](#)

C++: `URuntimeFileImporterSubsystem::ConvertVideoFile`

INPUTS		
Pin	Type	Description
Target	 Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Input File Path	 String	Full path of the file to convert.
Output File Path	 String	Full path of the file to write. Missing folders are created; an existing file is replaced.
Output Format	 ERuntime Video Export Format	Format to write.
Options ADVANCED OPTIONAL	 Runtime Video Export Options	How videos are written.
OUTPUTS		
Pin	Type	Description
Error Message	 String	Empty on success; otherwise why it failed.
Return Value	 Boolean	True on success. False if it failed (Error Message says why).

Convert Video File (Async) ASYNC



Converts a video file to another format without blocking the game thread.

Blocking version: [Convert Video File](#)

C++: `UExportVideoAsyncAction::ConvertVideoFileAsync`

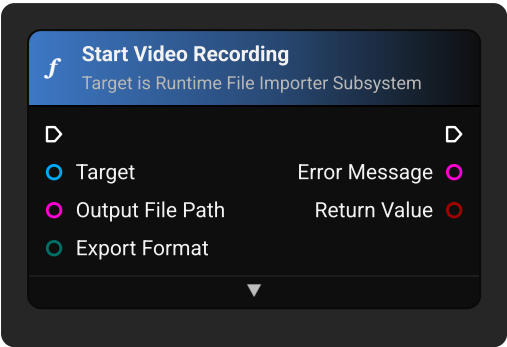
INPUTS

Pin	Type	Description
Input File Path	String	Full path of the file to convert.
Output File Path	String	Full path of the file to write. Missing folders are created; an existing file is replaced.
Output Format	ERuntime Video Export Format	Format to write.
Options ADVANCED OPTIONAL	Runtime Video Export Options	How videos are written.

OUTPUTS

Pin	Type	Description
On Progress	Exec	Fires as the operation advances (Percentage from 0 to 100). Other outputs are not set yet.
On Success	Exec	Fires when the operation has finished successfully.
On Failure	Exec	Fires if the operation failed or was cancelled; Error Message says why.
Percentage	Float	Progress from 0 to 100.
Error Message	String	Empty on success; otherwise why it failed.
Async Action	Export Video Async Action Object Reference	This node's operation. Call Cancel on it to stop the operation (no more events fire).

Start Video Recording FUNCTION



Starts recording the game's screen (Source empty) or a texture / render target / video (Source) into a video file, with the game's sound (MP4, AVI) if Record Audio. One recording runs at a time; Stop Video Recording completes the file.

Target: Runtime File Importer Subsystem
C++: URuntimeFileImporterSubsystem::StartVideoRecording

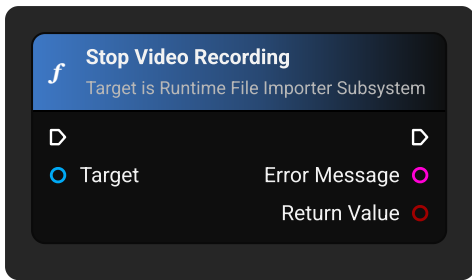
INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Output File Path	String	Full path of the file to write. Missing folders are created; an existing file is replaced.
Export Format	ERuntime Video Export Format	Format to write.
Source ADVANCED	Object Object Reference	What to record: a Texture Render Target 2D or any texture. Empty records the game viewport (the screen).
Options ADVANCED OPTIONAL	Runtime Video Export Options	How videos are written.
Record Audio ADVANCED	Boolean	Also record the game's sound. Default: <code>true</code>

OUTPUTS

Pin	Type	Description
Error Message	String	Empty on success; otherwise why it failed.
Return Value	Boolean	True on success. False if it failed (Error Message says why).

Stop Video Recording FUNCTION



Stops recording and completes the file, waiting for the frames still being encoded (the Async version doesn't wait).

Target: Runtime File Importer Subsystem

Async version: [Stop Video Recording \(Async\)](#)

C++: `URuntimeFileImporterSubsystem::StopVideoRecording`

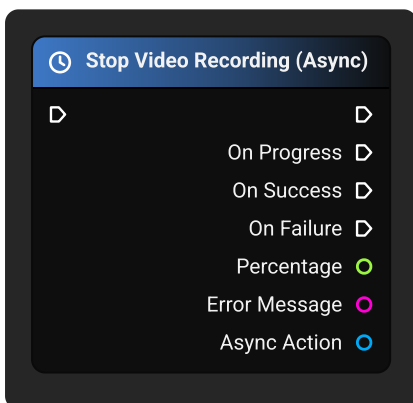
INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.

OUTPUTS

Pin	Type	Description
Error Message	String	Empty on success; otherwise why it failed.
Return Value	Boolean	True on success. False if it failed (Error Message says why).

Stop Video Recording (Async) ASYNC



Stops recording and fires On Success once the file is complete, without blocking the game thread.

Blocking version: [Stop Video Recording](#)

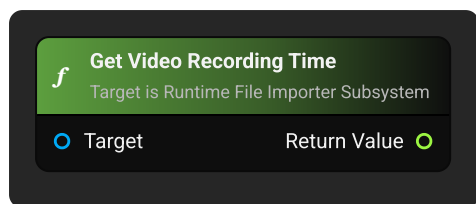
C++: `UExportVideoAsyncAction::StopVideoRecordingAsync`

OUTPUTS

Pin	Type	Description
On Progress	Exec	Fires as the operation advances (Percentage from 0 to 100). Other outputs are not set yet.
On Success	Exec	Fires when the operation has finished successfully.
On Failure	Exec	Fires if the operation failed or was cancelled; Error Message says why.
Percentage	Float	Progress from 0 to 100.
Error Message	String	Empty on success; otherwise why it failed.
Async Action	Export Video Async Action Object Reference	This node's operation. Call Cancel on it to stop the operation (no more events fire).

Get Video Recording Time

PURE



Seconds recorded so far by the current (or last) video recording.

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::GetVideoRecordingTime`

INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.

OUTPUTS

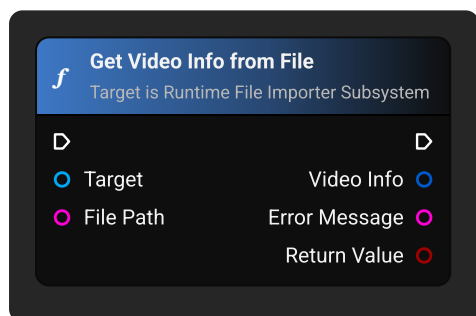
Pin	Type	Description
Return Value	Float	Seconds recorded so far by the current (or last) video recording.

Utilities

8 nodes

Get Video Info from File

FUNCTION



Reads a video file's format, length, size, frame rate, codecs and sound without playing it. Containers are read on every platform; Windows fills in what they don't say.

Target: Runtime File Importer Subsystem

Async version: [Get Video Info From File \(Async\)](#)

C++: `URuntimeFileImporterSubsystem::GetVideoInfoFromFile`

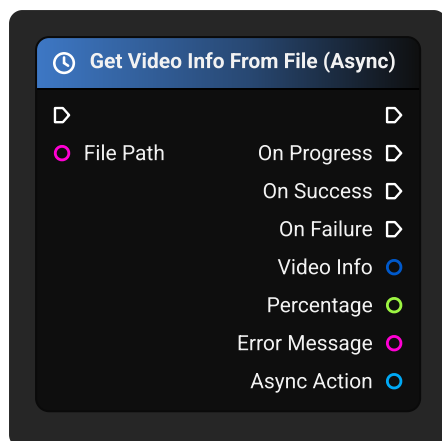
INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
File Path	String	Full path of the file on disk.

OUTPUTS

Pin	Type	Description
Video Info	Runtime Video Info	Duration, size, frame rate, codecs and tracks of the video.
Error Message	String	Empty on success; otherwise why it failed.
Return Value	Boolean	True on success. False if it failed (Error Message says why).

Get Video Info From File (Async) ASYNC



Reads a video file's format, length, size, frame rate, codecs and sound without blocking the game thread.

Blocking version: [Get Video Info from File](#)

C++: `UGetVideoInfoAsyncAction::GetVideoInfoFromFileAsync`

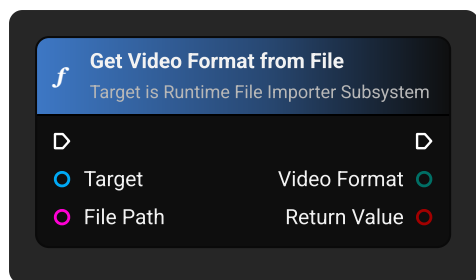
INPUTS

Pin	Type	Description
File Path	String	Full path of the file on disk.

OUTPUTS

Pin	Type	Description
On Progress	Exec	Fires as the operation advances (Percentage from 0 to 100). Other outputs are not set yet.
On Success	Exec	Fires when the operation has finished successfully.
On Failure	Exec	Fires if the operation failed or was cancelled; Error Message says why.
Video Info	Runtime Video Info	Duration, size, frame rate, codecs and tracks of the video.
Percentage	Float	Progress from 0 to 100.
Error Message	String	Empty on success; otherwise why it failed.
Async Action	Get Video Info Async Action Object Reference	This node's operation. Call Cancel on it to stop the operation (no more events fire).

Get Video Format from File FUNCTION



Detects a file's video format from its contents (falling back to its extension). Returns false if it isn't recognized.

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::GetVideoFormatFromFile`

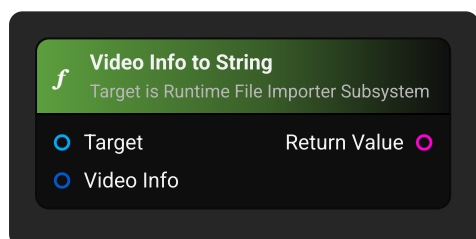
INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
File Path	String	Full path of the file on disk.

OUTPUTS

Pin	Type	Description
Video Format	ERuntime Video Format	Format of the video.
Return Value	Boolean	False if it isn't recognized.

Video Info to String PURE



A readable, multi-line description of video info.

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::VideoInfoToString`

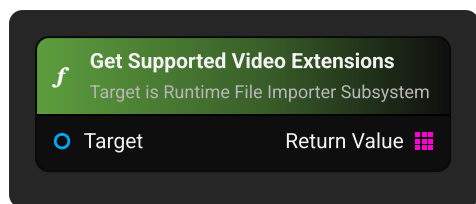
INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Video Info	Runtime Video Info	Duration, size, frame rate, codecs and tracks of the video.

OUTPUTS

Pin	Type	Description
Return Value	String	A readable, multi-line description of video info.

Get Supported Video Extensions PURE



File extensions of videos the platform's player usually plays (without dots), e.g. for a file picker filter.

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::GetSupportedVideoExtensions`

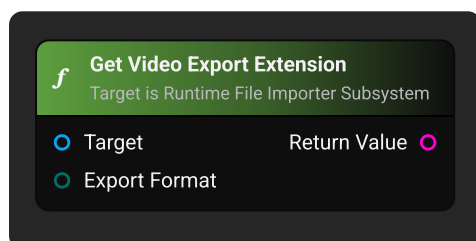
INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.

OUTPUTS

Pin	Type	Description
Return Value	Array of String	File extensions of videos the platform's player usually plays (without dots), e.g.

Get Video Export Extension PURE



File extension written for a video export format, without the dot (e.g. "mp4").

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::GetVideoExportExtension`

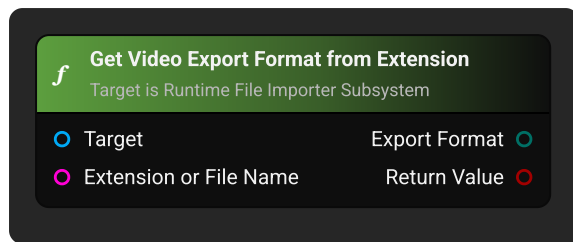
INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Export Format	ERuntime Video Export Format	Format to write.

OUTPUTS

Pin	Type	Description
Return Value	String	File extension written for a video export format, without the dot (e.g.

Get Video Export Format from Extension PURE



Export format matching a file extension or file name. Returns false if there's none.

Target: Runtime File Importer Subsystem

C++:

```
URuntimeFileImporterSubsystem::GetVideoExportFormatFromExtension
```

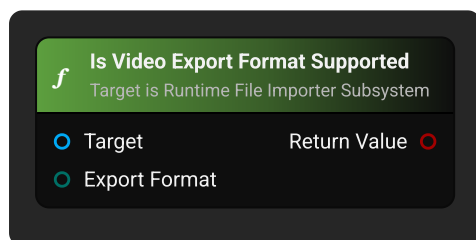
INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Extension or File Name	String	An extension (with or without the dot) or a file name / path / URL ending in one.

OUTPUTS

Pin	Type	Description
Export Format	ERuntime Video Export Format	Format to write.
Return Value	Boolean	False if there's none.

Is Video Export Format Supported PURE



False for formats this platform can't write (MP4 outside Windows).

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::IsVideoExportFormatSupported`

INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Export Format	ERuntime Video Export Format	Format to write.

OUTPUTS

Pin	Type	Description
Return Value	Boolean	False for formats this platform can't write (MP4 outside Windows).

On Video Imported

EVENT

On Video Imported

Event of Runtime File Importer Subsystem

Video

Source

Any video opened (from a file, URL or buffer).

Bind it with "Bind Event to On Video Imported" (or "Assign On Video Imported") on the Runtime File Importer Subsystem.

```
C++: URuntimeFileImporterSubsystem::OnVideoImported
```

OUTPUTS		
Pin	Type	Description
Video	Runtime Video Object Reference	The Runtime Video.
Source	String	File path or URL the import came from (empty for buffers and strings).

On Video Import Failed

EVENT

On Video Import Failed

Event of Runtime File Importer Subsystem

Source

Error Message

Any video failed to open.

Bind it with "Bind Event to On Video Import Failed" (or "Assign On Video Import Failed") on the Runtime File Importer Subsystem.

```
C++: URuntimeFileImporterSubsystem::OnVideoImportFailed
```

OUTPUTS		
Pin	Type	Description
Source	String	File path or URL the import came from (empty for buffers and strings).
Error Message	String	Empty on success; otherwise why it failed.

On Video Exported

EVENT

On Video Exported

Event of Runtime File Importer Subsystem

Destination

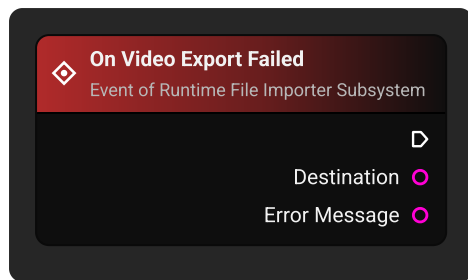
Any video export, conversion or recording was written.

Bind it with "Bind Event to On Video Exported" (or "Assign On Video Exported") on the Runtime File Importer Subsystem.

```
C++: URuntimeFileImporterSubsystem::OnVideoExported
```

OUTPUTS		
Pin	Type	Description
Destination	String	Output file path (empty when writing to a buffer).

On Video Export Failed EVENT



Any video export, conversion or recording failed.

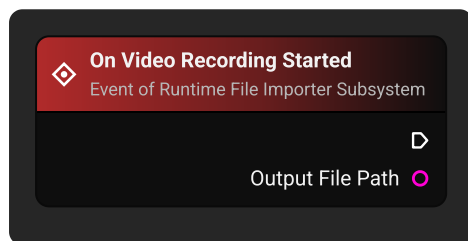
Bind it with "Bind Event to On Video Export Failed" (or "Assign On Video Export Failed") on the Runtime File Importer Subsystem.

```
C++: URuntimeFileImporterSubsystem::OnVideoExportFailed
```

OUTPUTS

Pin	Type	Description
Destination	String	Output file path (empty when writing to a buffer).
Error Message	String	Empty on success; otherwise why it failed.

On Video Recording Started EVENT



A video recording started (it's written when stopped: On Video Exported / Failed).

Bind it with "Bind Event to On Video Recording Started" (or "Assign On Video Recording Started") on the Runtime File Importer Subsystem.

```
C++: URuntimeFileImporterSubsystem::OnVideoRecordingStarted
```

OUTPUTS

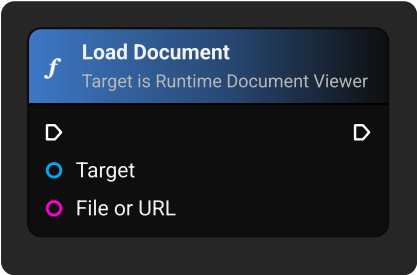
Pin	Type	Description
Output File Path	String	Full path of the file to write. Missing folders are created; an existing file is replaced.

Text

Read and write text in any encoding; structured data (JSON, XML, CSV, YAML, INI, spreadsheets) as Runtime Data, Data Tables and structs; String Tables; subtitles; and formatted documents (Word, PDF, OpenDocument, HTML, Markdown, Google Docs links) shown by the Runtime Document Viewer widget.

Document 31 nodes

Load Document FUNCTION



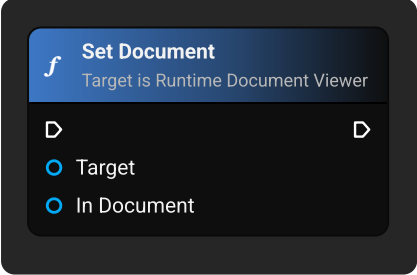
Loads and shows a document from a file path or URL (Google Docs / Sheets / Slides / Drive and Dropbox links work), in the background.

Target: Runtime Document Viewer
C++: `URuntimeDocumentViewer::LoadDocument`

INPUTS

Pin	Type	Description
Target	Runtime Document Viewer Object Reference	The Runtime Document Viewer to use.
File or URL	String	A file path, an http(s) URL or a Google Docs / Sheets / Slides / Drive link.

Set Document FUNCTION



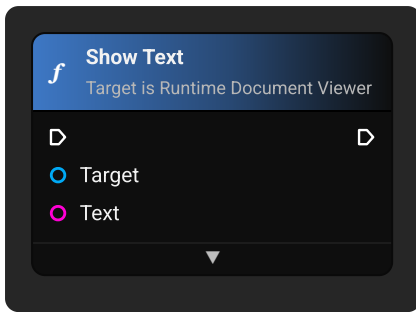
Shows a document made with the Document nodes (none: clears the viewer).

Target: Runtime Document Viewer
C++: `URuntimeDocumentViewer::SetDocument`

INPUTS

Pin	Type	Description
Target	Runtime Document Viewer Object Reference	The Runtime Document Viewer to use.
In Document	Runtime Document Object Reference	The document to show.

Show Text FUNCTION



Shows text, e.g. Markdown or HTML written in the game (Format Auto Detect: from the text).

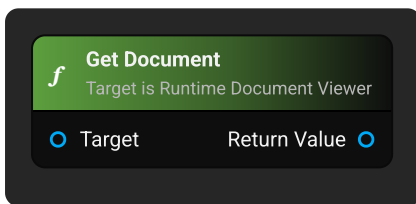
Target: Runtime Document Viewer

C++: `URuntimeDocumentViewer::ShowText`

INPUTS

Pin	Type	Description
Target	Runtime Document Viewer Object Reference	The Runtime Document Viewer to use.
Text	String	The text.
Format ADVANCED	ERuntime Text Format	Format of the text (markup is converted). Auto Detect reads it from the contents. Default: Auto Detect

Get Document PURE



The document shown (none while loading or after a failure).

Target: Runtime Document Viewer

C++: `URuntimeDocumentViewer::GetDocument`

INPUTS

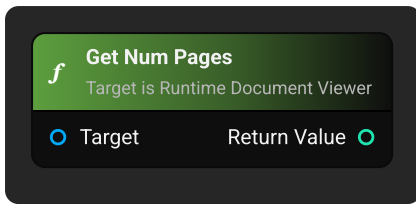
Pin	Type	Description
Target	Runtime Document Viewer Object Reference	The Runtime Document Viewer to use.

OUTPUTS

Pin	Type	Description
Return Value	Runtime Document Object Reference	The document shown (none while loading or after a failure).

Get Num Pages

PURE



Number of pages of the shown document (0 if none is shown).

Target: Runtime Document Viewer

C++: `URuntimeDocumentViewer::GetNumPages`

INPUTS

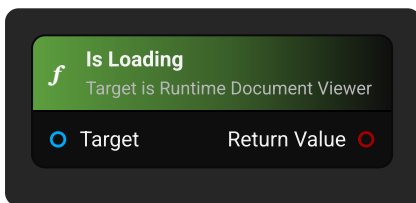
Pin	Type	Description
Target	Runtime Document Viewer Object Reference	The Runtime Document Viewer to use.

OUTPUTS

Pin	Type	Description
Return Value	Integer	Number of pages of the shown document (0 if none is shown).

Is Loading

PURE



True while Load Document (or Source) is loading.

Target: Runtime Document Viewer

C++: `URuntimeDocumentViewer::IsLoading`

INPUTS

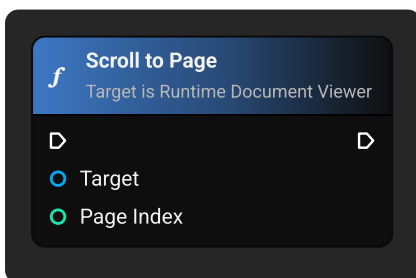
Pin	Type	Description
Target	Runtime Document Viewer Object Reference	The Runtime Document Viewer to use.

OUTPUTS

Pin	Type	Description
Return Value	Boolean	True while Load Document (or Source) is loading.

Scroll to Page

FUNCTION



Scrolls so a page (0 is the first) is at the top.

Target: Runtime Document Viewer

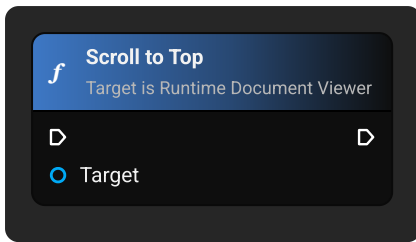
C++: `URuntimeDocumentViewer::ScrollToPage`

INPUTS

Pin	Type	Description
Target	Runtime Document Viewer Object Reference	The Runtime Document Viewer to use.
Page Index	Integer	The page to scroll to (0-based).

Scroll to Top

FUNCTION



Scrolls to the start of the document.

Target: Runtime Document Viewer

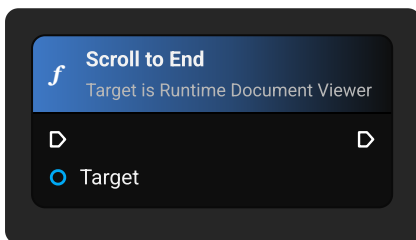
C++: `URuntimeDocumentViewer::ScrollToTop`

INPUTS

Pin	Type	Description
Target	Runtime Document Viewer Object Reference	The Runtime Document Viewer to use.

Scroll to End

FUNCTION



Scrolls to the end of the document.

Target: Runtime Document Viewer

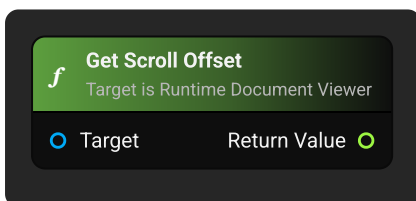
C++: `URuntimeDocumentViewer::ScrollToEnd`

INPUTS

Pin	Type	Description
Target	Runtime Document Viewer Object Reference	The Runtime Document Viewer to use.

Get Scroll Offset

PURE



Scroll position in Slate units from the top.

Target: Runtime Document Viewer

C++: `URuntimeDocumentViewer::GetScrollOffset`

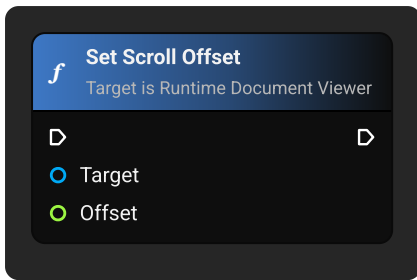
INPUTS

Pin	Type	Description
Target	Runtime Document Viewer Object Reference	The Runtime Document Viewer to use.

OUTPUTS

Pin	Type	Description
Return Value	Float	Scroll position in Slate units from the top.

Set Scroll Offset FUNCTION



Scrolls to Offset (Slate units from the top), e.g. to restore a position read with Get Scroll Offset.

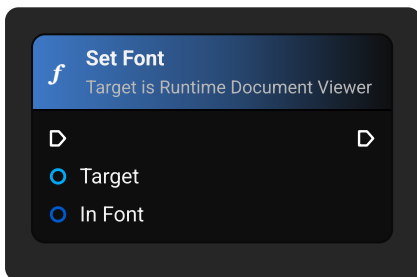
Target: Runtime Document Viewer

C++: `URuntimeDocumentViewer::SetScrollOffset`

INPUTS

Pin	Type	Description
Target	Runtime Document Viewer Object Reference	The Runtime Document Viewer to use.
Offset	Float	Distance from the top of the document, in Slate units.

Set Font FUNCTION



Changes the font (empty: Roboto 16) and redraws the document.

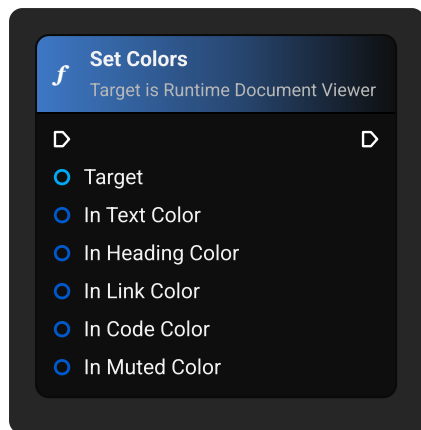
Target: Runtime Document Viewer

C++: `URuntimeDocumentViewer::SetFont`

INPUTS

Pin	Type	Description
Target	Runtime Document Viewer Object Reference	The Runtime Document Viewer to use.
In Font	Slate Font Info	Font of the body text (headings are scaled from it).

Set Colors FUNCTION



Changes the colors and redraws the document.

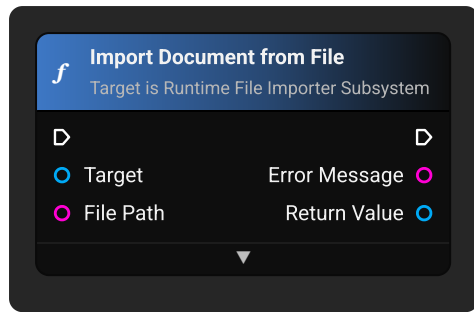
Target: Runtime Document Viewer

C++: `URuntimeDocumentViewer::SetColors`

INPUTS

Pin	Type	Description
Target	 Runtime Document Viewer Object Reference	The Runtime Document Viewer to use.
In Text Color	 Slate Color	Color of the body text.
In Heading Color	 Slate Color	Color of headings.
In Link Color	 Slate Color	Color of links.
In Code Color	 Slate Color	Color of code.
In Muted Color	 Slate Color	Color of secondary text (quotes, captions, page numbers).

Import Document from File FUNCTION



Reads a document with its formatting (headings, bold, italic, lists, tables, links) and pages, to show it in a widget (the Runtime Document Viewer, or a Rich Text Block with its Rich Text): Word (DOCX), OpenDocument (ODT), PDF, HTML, RTF, Markdown, spreadsheets (as tables), subtitles and any text file. Blocks until finished; use the Async version for big files.

Target: Runtime File Importer Subsystem

Async version: [Import Document From File \(Async\)](#)

C++: `URuntimeFileImporterSubsystem::ImportDocumentFromFile`

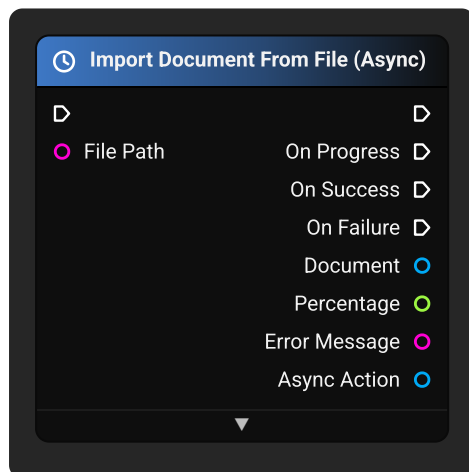
INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
File Path	String	Full path of the file on disk.
Format ADVANCED	ERuntime Text Format	Format of the text (markup is converted). Auto Detect reads it from the contents. Default: <code>Auto Detect</code>
Encoding ADVANCED	ERuntime Text Encoding	Character encoding. Auto Detect: from the byte order mark, else from the bytes (UTF-8 when writing). Default: <code>Auto Detect</code>

OUTPUTS

Pin	Type	Description
Error Message	String	Empty on success; otherwise why it failed.
Return Value	Runtime Document Object Reference	The result, or None/empty if it failed (Error Message says why).

Import Document From File (Async) ASYNC



Reads a document (Word, OpenDocument, PDF, HTML, RTF, Markdown, spreadsheet, text) with its formatting without blocking the game thread.

Blocking version: [Import Document from File](#)

C++: `UImportDocumentAsyncAction::ImportDocumentFromFileAsync`

INPUTS

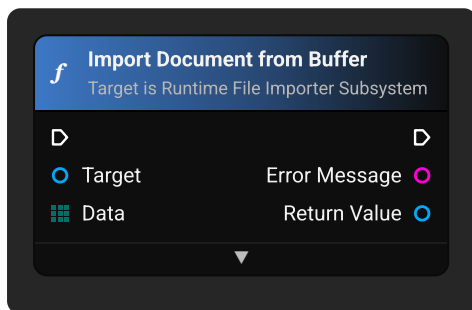
Pin	Type	Description
File Path	String	Full path of the file on disk.
Format ADVANCED	ERuntime Text Format	Format of the text (markup is converted). Auto Detect reads it from the contents. Default: <code>Auto Detect</code>
Encoding ADVANCED	ERuntime Text Encoding	Character encoding. Auto Detect: from the byte order mark, else from the bytes (UTF-8 when writing). Default: <code>Auto Detect</code>

OUTPUTS

Pin	Type	Description
On Progress	 Exec	Fires as the operation advances (Percentage from 0 to 100). Other outputs are not set yet.
On Success	 Exec	Fires when the operation has finished successfully.
On Failure	 Exec	Fires if the operation failed or was cancelled; Error Message says why.
Document	 Runtime Document Object Reference	The Runtime Document.
Percentage	 Float	Progress from 0 to 100.
Error Message	 String	Empty on success; otherwise why it failed.
Async Action	 Import Document Async Action Object Reference	This node's operation. Call Cancel on it to stop the operation (no more events fire).

Import Document from Buffer

FUNCTION



Reads a document from bytes in memory.

Target: Runtime File Importer Subsystem

Async version: [Import Document From Buffer \(Async\)](#)

C++: `URuntimeFileImporterSubsystem::ImportDocumentFromBuffer`

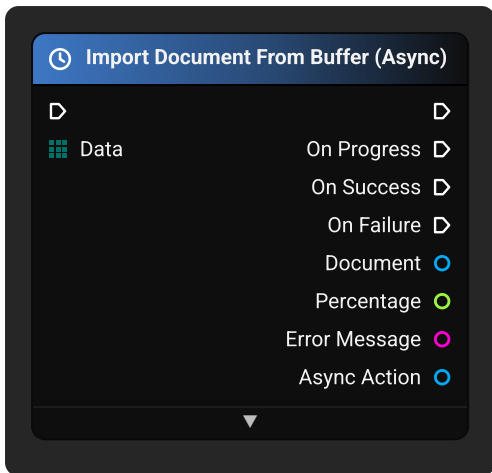
INPUTS

Pin	Type	Description
Target	 Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Data	 Array of Byte	Bytes of a data file.
Format ADVANCED	 ERuntime Text Format	Format of the text (markup is converted). Auto Detect reads it from the contents. Default: <code>Auto Detect</code>
Encoding ADVANCED	 ERuntime Text Encoding	Character encoding. Auto Detect: from the byte order mark, else from the bytes (UTF-8 when writing). Default: <code>Auto Detect</code>

OUTPUTS

Pin	Type	Description
Error Message	 String	Empty on success; otherwise why it failed.
Return Value	 Runtime Document Object Reference	The result, or None/empty if it failed (Error Message says why).

Import Document From Buffer (Async) ASYNC



Reads a document from bytes in memory without blocking the game thread.

Blocking version: [Import Document from Buffer](#)

C++: `UImportDocumentAsyncAction::ImportDocumentFromBufferAsync`

INPUTS

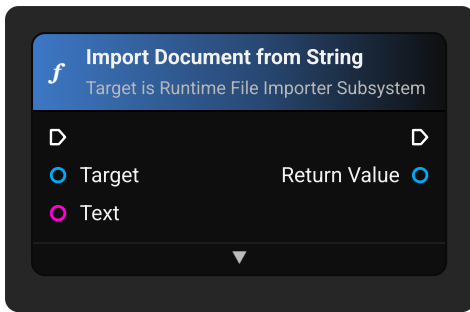
Pin	Type	Description
Data	Array of Byte	Bytes of a data file.
Format ADVANCED	ERuntime Text Format	Format of the text (markup is converted). Auto Detect reads it from the contents. Default: <code>Auto Detect</code>
Encoding ADVANCED	ERuntime Text Encoding	Character encoding. Auto Detect: from the byte order mark, else from the bytes (UTF-8 when writing). Default: <code>Auto Detect</code>

OUTPUTS

Pin	Type	Description
On Progress	Exec	Fires as the operation advances (Percentage from 0 to 100). Other outputs are not set yet.
On Success	Exec	Fires when the operation has finished successfully.
On Failure	Exec	Fires if the operation failed or was cancelled; Error Message says why.
Document	Runtime Document Object Reference	The Runtime Document.
Percentage	Float	Progress from 0 to 100.
Error Message	String	Empty on success; otherwise why it failed.
Async Action	Import Document Async Action Object Reference	This node's operation. Call Cancel on it to stop the operation (no more events fire).

Import Document from String

FUNCTION



Makes a document of text, e.g. Markdown or HTML written in the game (Format Auto Detect: from the text; plain text stays as it is).

Target: Runtime File Importer Subsystem

Async version: [Import Document From String \(Async\)](#)

C++: `URuntimeFileImporterSubsystem::ImportDocumentFromString`

INPUTS

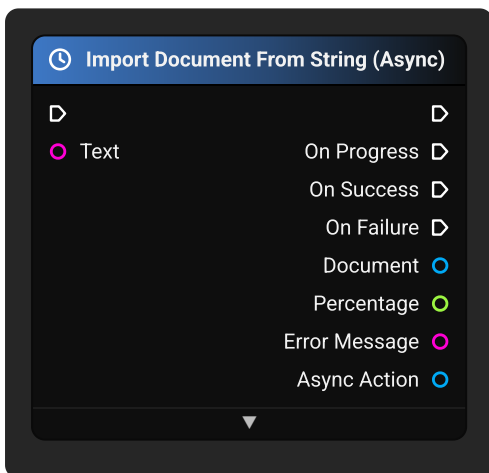
Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Text	String	The text.
Format ADVANCED	ERuntime Text Format	Format of the text (markup is converted). Auto Detect reads it from the contents. Default: <code>Auto Detect</code>

OUTPUTS

Pin	Type	Description
Return Value	Runtime Document Object Reference	Makes a document of text, e.g.

Import Document From String (Async)

ASYNC



Makes a document of text (e.g. long Markdown or HTML) without blocking the game thread.

Blocking version: [Import Document from String](#)

C++: `UImportDocumentAsyncAction::ImportDocumentFromStringAsync`

INPUTS

Pin	Type	Description
Text	String	The text.
Format ADVANCED	ERuntime Text Format	Format of the text (markup is converted). Auto Detect reads it from the contents. Default: <code>Auto Detect</code>

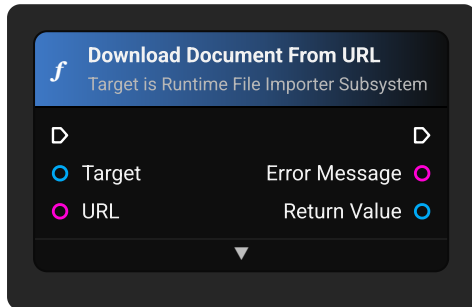
OUTPUTS

Pin	Type	Description
On Progress	Exec	Fires as the operation advances (Percentage from 0 to 100). Other outputs are not set yet.
On Success	Exec	Fires when the operation has finished successfully.

Pin	Type	Description
On Failure	 Exec	Fires if the operation failed or was cancelled; Error Message says why.
Document	 Runtime Document Object Reference	The Runtime Document.
Percentage	 Float	Progress from 0 to 100.
Error Message	 String	Empty on success; otherwise why it failed.
Async Action	 Import Document Async Action Object Reference	This node's operation. Call Cancel on it to stop the operation (no more events fire).

Download Document From URL

FUNCTION



Downloads a document and reads it. Share links work directly: Google Docs (read as Word), Sheets (as Excel), Slides (as PDF) and Drive files, and Dropbox; they must be shared with "Anyone with the link". FREEZES the game until the download finishes; prefer the Async version. Save To File Path keeps the downloaded file.

Target: Runtime File Importer Subsystem

Async version: [Download Document From URL \(Async\)](#)

C++: `URuntimeFileImporterSubsystem::DownloadDocumentFromURL`

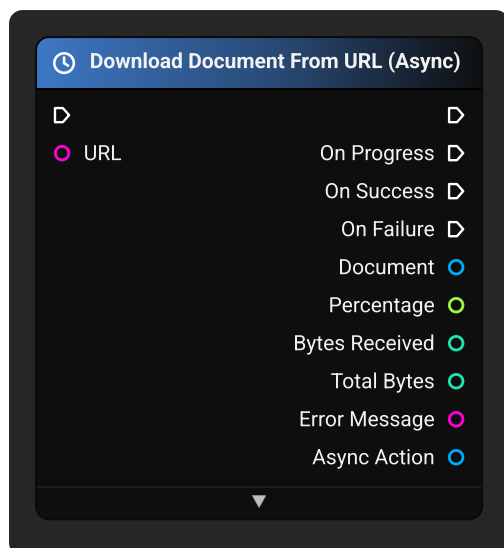
INPUTS

Pin	Type	Description
Target	 Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
URL	 String	The http(s) address to download. Google Docs, Sheets, Slides and Drive share links are converted automatically.
Save to File Path ADVANCED	 String	Optional. Also writes the downloaded file there, unmodified. Default: (empty)
Timeout Seconds ADVANCED	 Float	Gives up (and fails) if the download takes longer than this. Default: 60.0

OUTPUTS

Pin	Type	Description
Error Message	 String	Empty on success; otherwise why it failed.
Return Value	 Runtime Document Object Reference	The result, or None/empty if it failed (Error Message says why).

Download Document From URL (Async) ASYNC



Downloads a document and reads it with its formatting. Google Docs / Sheets / Slides / Drive and Dropbox share links work directly (shared with "Anyone with the link"). If Save To File Path is set, the downloaded file is also written there.

Blocking version: [Download Document From URL](#)

C++:

UDownloadDocumentFromURLAsyncAction::DownloadDocumentFromURLAsync

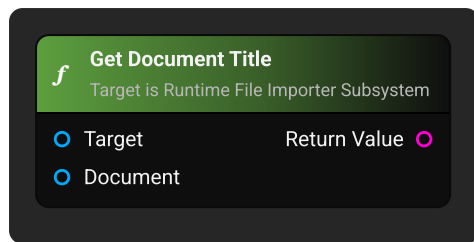
INPUTS

Pin	Type	Description
URL	String	The http(s) address to download. Google Docs, Sheets, Slides and Drive share links are converted automatically.
Save to File Path ADVANCED	String	Optional. Also writes the downloaded file there, unmodified. Default: (empty)
Timeout Seconds ADVANCED	Float	Gives up (and fails) if the download takes longer than this. Default: 60.0

OUTPUTS

Pin	Type	Description
On Progress	Exec	Fires repeatedly while downloading. Percentage (0-100) and Total Bytes are 0 if the server doesn't report the file size.
On Success	Exec	Fires when the operation has finished successfully.
On Failure	Exec	Fires if the operation failed or was cancelled; Error Message says why.
Document	Runtime Document Object Reference	The Runtime Document.
Percentage	Float	Progress from 0 to 100.
Bytes Received	Integer	Bytes downloaded so far.
Total Bytes	Integer	Size of the download in bytes, or 0 while the server hasn't said.
Error Message	String	Empty on success; otherwise why it failed.
Async Action	Download Document from URLAsync Action Object Reference	This node's operation. Call Cancel on it to stop the operation (no more events fire).

Get Document Title PURE



A document's title (its Title property, else its first heading). Same as its Title variable.

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::GetDocumentTitle`

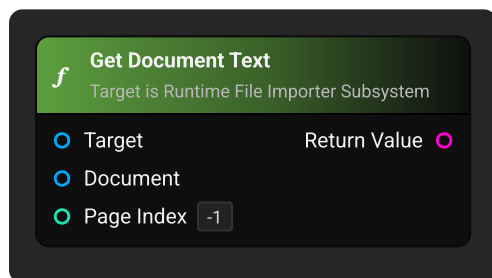
INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Document	Runtime Document Object Reference	The Runtime Document.

OUTPUTS

Pin	Type	Description
Return Value	String	A document's title (its Title property, else its first heading).

Get Document Text PURE



A document's plain text: one page (0 is the first), or all of them (-1). Same as its Text variable for all pages.

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::GetDocumentText`

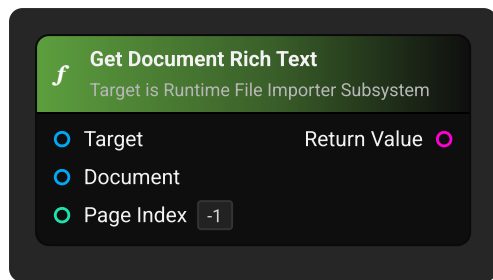
INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Document	Runtime Document Object Reference	The Runtime Document.
Page Index	Integer	Page to read (0-based). -1: the whole document. Default: -1

OUTPUTS

Pin	Type	Description
Return Value	String	A document's plain text: one page (0 is the first), or all of them (-1).

Get Document Rich Text PURE



A document's rich text markup for a Rich Text Block: one page, or all of them (-1). Same as its Rich Text variable for all pages.

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::GetDocumentRichText`

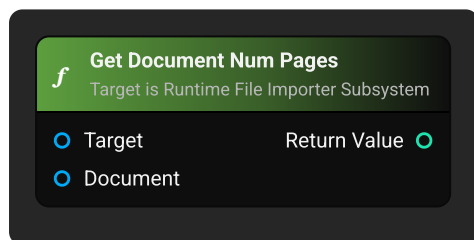
INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Document	Runtime Document Object Reference	The Runtime Document.
Page Index	Integer	Page to read (0-based). -1: the whole document. Default: -1

OUTPUTS

Pin	Type	Description
Return Value	String	A document's rich text markup for a Rich Text Block: one page, or all of them (-1).

Get Document Num Pages PURE



A document's pages (PDF pages; 1 for other documents). Same as its Num Pages variable.

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::GetDocumentNumPages`

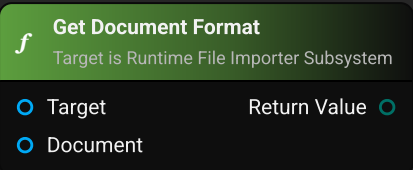
INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Document	Runtime Document Object Reference	The Runtime Document.

OUTPUTS

Pin	Type	Description
Return Value	Integer	A document's pages (PDF pages; 1 for other documents).

PURE

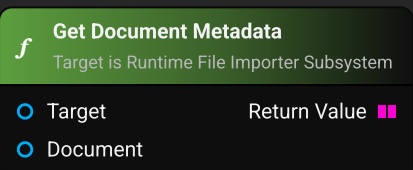


```
C++: URuntimeFileImporterSubsystem::GetDocumentFormat
```

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Document	Runtime Document Object Reference	The Runtime Document.

Pin	Type	Description
Return Value	ERuntimeTextFormat	The format a document was read from.

PURE



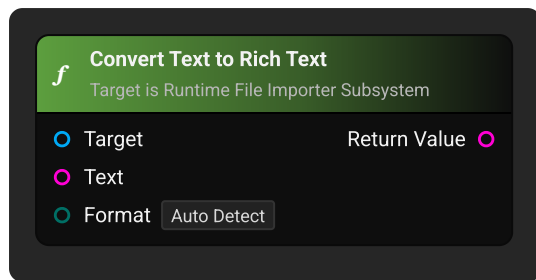
C++: URuntimeFileImporterSubsystem::GetDocumentMetadata

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Document	Runtime Document Object Reference	The Runtime Document.

Pin	Type	Description
Return Value	Map of String to String	A document's properties (Title, Author, Subject, Keywords, Created, Modified, ...).

Convert Text to Rich Text

PURE



Rich text markup of text in a format (Markdown, HTML, RTF, ...; Auto Detect: from the text), e.g. to show Markdown in a Rich Text Block.

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::ConvertTextToRichText`

INPUTS

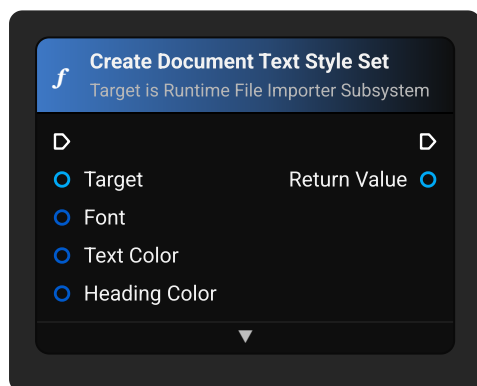
Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Text	String	The text.
Format	ERuntime Text Format	Format of the text (markup is converted). Auto Detect reads it from the contents. Default: Auto Detect

OUTPUTS

Pin	Type	Description
Return Value	String	Rich text markup of text in a format (Markdown, HTML, RTF, ...; Auto Detect: from the text), e.g.

Create Document Text Style Set

FUNCTION



A Text Style Set for a Rich Text Block showing document rich text: rows Default, h1-h4, b, i, bi, code, quote and small, from Font (empty: Roboto 16) and the colors. Assign it with the Rich Text Block's Set Text Style Set.

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::CreateDocumentTextStyleSet`

INPUTS

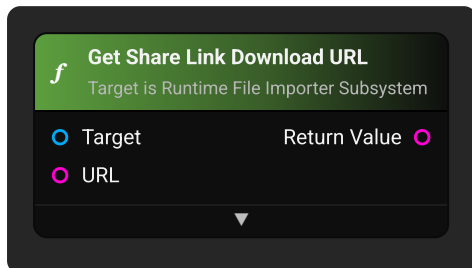
Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Font	Slate Font Info	Font family, typeface and size of the body text.
Text Color	Slate Color	Color of the body text.
Heading Color	Slate Color	Color of headings.
Code Color	ADVANCED Slate Color	Color of code.
Muted Color	ADVANCED Slate Color	Color of secondary text (quotes, captions).

OUTPUTS

Pin	Type	Description
Return Value	 Data Table Object Reference	A Text Style Set for a Rich Text Block showing document rich text: rows Default, h1-h4, b, i, bi, code, quote and small, from Font (empty: Roboto 16) and the colors.

Get Share Link Download URL

PURE



The direct download link of a share link (Google Docs / Sheets / Slides / Drive, Dropbox), as the Download nodes use it. With For Data, a Sheets link to one sheet (gid) downloads that sheet as CSV. Other links are returned unchanged.

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::GetShareLinkDownloadURL`

INPUTS

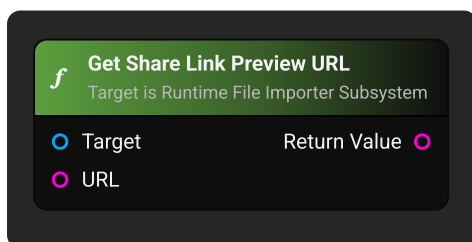
Pin	Type	Description
Target	 Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
URL	 String	The http(s) address to download. Google Docs, Sheets, Slides and Drive share links are converted automatically.
For Data	 Boolean	For Google Sheets: export as CSV (for Import Data) instead of a document format. Default: <code>false</code>

OUTPUTS

Pin	Type	Description
Return Value	 String	The direct download link of a share link (Google Docs / Sheets / Slides / Drive, Dropbox), as the Download nodes use it.

Get Share Link Preview URL

PURE



Google's embeddable preview page of a Google Docs / Sheets / Slides / Drive link (empty for other links), e.g. for the engine's Web Browser widget.

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::GetShareLinkPreviewURL`

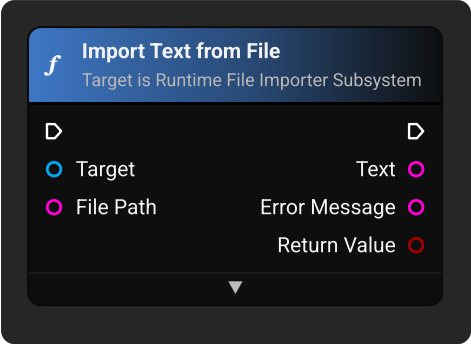
INPUTS

Pin	Type	Description
Target	 Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
URL	 String	The http(s) address to download. Google Docs, Sheets, Slides and Drive share links are converted automatically.

OUTPUTS

Pin	Type	Description
Return Value	 String	Google's embeddable preview page of a Google Docs / Sheets / Slides / Drive link (empty for other links), e.g.

Import Text from File FUNCTION



Reads a text file as text, in any encoding (detected: UTF-8, UTF-16, UTF-32, Windows-1252). Documents come as their text: Word (DOCX), OpenDocument (ODT), spreadsheets (XLSX, ODS: tab-separated rows), and with Options' Convert To Plain Text, HTML, RTF and Markdown. Blocks until finished; use the Async version for large files.

Target: Runtime File Importer Subsystem
Async version: [Import Text From File \(Async\)](#)
C++: URuntimeFileImporterSubsystem::ImportTextFromFile

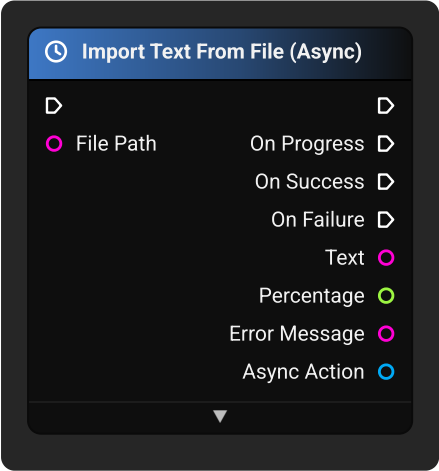
INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
File Path	String	Full path of the file on disk.
Options	ADVANCED OPTIONAL Runtime Text Import Options	How text files are read.

OUTPUTS

Pin	Type	Description
Text	String	The text.
Error Message	String	Empty on success; otherwise why it failed.
Return Value	Boolean	True on success. False if it failed (Error Message says why).

Import Text From File (Async) ASYNC



Reads a text file (any encoding; documents and spreadsheets as their text) without blocking the game thread.

Blocking version: [Import Text from File](#)
C++: UImportTextAsyncAction::ImportTextFromFileAsync

INPUTS

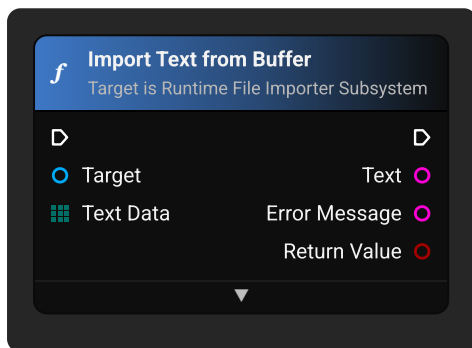
Pin	Type	Description
File Path	String	Full path of the file on disk.
Options	ADVANCED OPTIONAL Runtime Text Import Options	How text files are read.

OUTPUTS

Pin	Type	Description
On Progress	 Exec	Fires as the operation advances (Percentage from 0 to 100). Other outputs are not set yet.
On Success	 Exec	Fires when the operation has finished successfully.
On Failure	 Exec	Fires if the operation failed or was cancelled; Error Message says why.
Text	 String	The text.
Percentage	 Float	Progress from 0 to 100.
Error Message	 String	Empty on success; otherwise why it failed.
Async Action	 Import Text Async Action Object Reference	This node's operation. Call Cancel on it to stop the operation (no more events fire).

Import Text from Buffer

FUNCTION



Reads text from bytes in memory (e.g. received over the network), like Import Text From File.

Target: Runtime File Importer Subsystem

Async version: [Import Text From Buffer \(Async\)](#)

C++: `URuntimeFileImporterSubsystem::ImportTextFromBuffer`

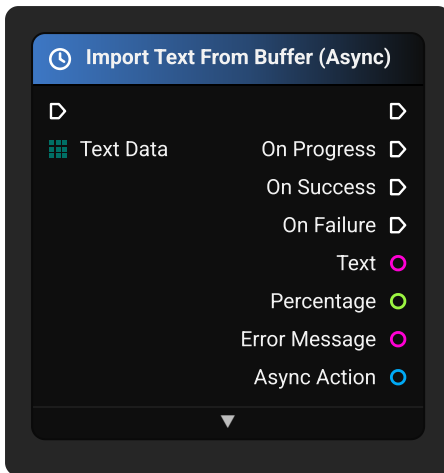
INPUTS

Pin	Type	Description
Target	 Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Text Data	 Array of Byte	Bytes of a text file.
Options	ADVANCED OPTIONAL  Runtime Text Import Options	How text files are read.

OUTPUTS

Pin	Type	Description
Text	 String	The text.
Error Message	 String	Empty on success; otherwise why it failed.
Return Value	 Boolean	True on success. False if it failed (Error Message says why).

Import Text From Buffer (Async) ASYNC



Reads text from bytes in memory without blocking the game thread.

Blocking version: [Import Text from Buffer](#)

C++: `UImportTextAsyncAction::ImportTextFromBufferAsync`

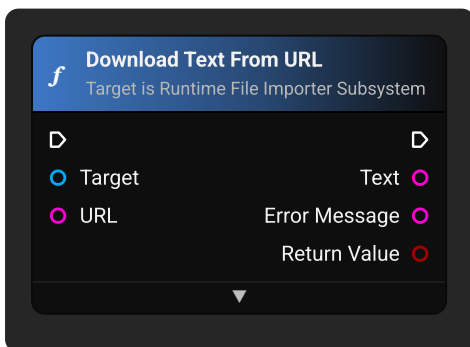
INPUTS

Pin	Type	Description
Text Data	Array of Byte	Bytes of a text file.
Options ADVANCED OPTIONAL	Runtime Text Import Options	How text files are read.

OUTPUTS

Pin	Type	Description
On Progress	Exec	Fires as the operation advances (Percentage from 0 to 100). Other outputs are not set yet.
On Success	Exec	Fires when the operation has finished successfully.
On Failure	Exec	Fires if the operation failed or was cancelled; Error Message says why.
Text	String	The text.
Percentage	Float	Progress from 0 to 100.
Error Message	String	Empty on success; otherwise why it failed.
Async Action	Import Text Async Action Object Reference	This node's operation. Call Cancel on it to stop the operation (no more events fire).

Download Text From URL FUNCTION



Downloads a text file over HTTP(S) and reads it. Google Docs / Sheets / Slides / Drive and Dropbox share links work directly (shared with "Anyone with the link"). This FREEZES the game until the download finishes or times out; prefer the Async version. If Save To File Path is set, the downloaded file is also written there, unmodified.

Target: Runtime File Importer Subsystem

Async version: [Download Text From URL \(Async\)](#)

C++: `URuntimeFileImporterSubsystem::DownloadTextFromURL`

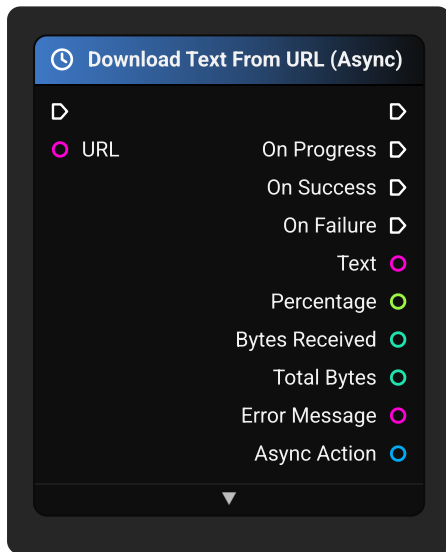
INPUTS

Pin	Type	Description
Target	 Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
URL	 String	The http(s) address to download. Google Docs, Sheets, Slides and Drive share links are converted automatically.
Options ADVANCED OPTIONAL	 Runtime Text Import Options	How text files are read.
Save to File Path ADVANCED	 String	Optional. Also writes the downloaded file there, unmodified. Default: (empty)
Timeout Seconds ADVANCED	 Float	Gives up (and fails) if the download takes longer than this. Default: 30.0

OUTPUTS

Pin	Type	Description
Text	 String	The text.
Error Message	 String	Empty on success; otherwise why it failed.
Return Value	 Boolean	True on success. False if it failed (Error Message says why).

Download Text From URL (Async) ASYNC



Downloads a text file over HTTP(S) and reads it. If Save To File Path is set, the downloaded file is also written there.

Blocking version: [Download Text From URL](#)

C++: UDownloadTextFromURLAsyncAction::DownloadTextFromURLAsync

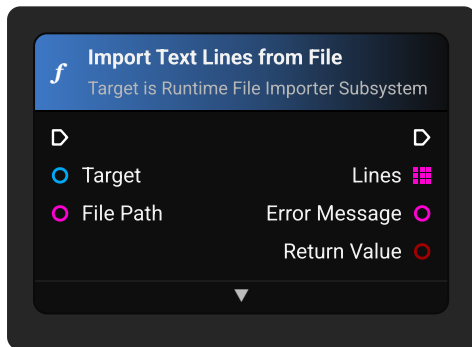
INPUTS

Pin	Type	Description
URL	 String	The http(s) address to download. Google Docs, Sheets, Slides and Drive share links are converted automatically.
Options ADVANCED OPTIONAL	 Runtime Text Import Options	How text files are read.
Save to File Path ADVANCED	 String	Optional. Also writes the downloaded file there, unmodified. Default: (empty)
Timeout Seconds ADVANCED	 Float	Gives up (and fails) if the download takes longer than this. Default: 60.0

OUTPUTS

Pin	Type	Description
On Progress	 Exec	Fires repeatedly while downloading. Percentage (0-100) and Total Bytes are 0 if the server doesn't report the file size.
On Success	 Exec	Fires when the operation has finished successfully.
On Failure	 Exec	Fires if the operation failed or was cancelled; Error Message says why.
Text	 String	The text.
Percentage	 Float	Progress from 0 to 100.
Bytes Received	 Integer	Bytes downloaded so far.
Total Bytes	 Integer	Size of the download in bytes, or 0 while the server hasn't said.
Error Message	 String	Empty on success; otherwise why it failed.
Async Action	 Download Text from URL Async Action Object Reference	This node's operation. Call Cancel on it to stop the operation (no more events fire).

Import Text Lines from File FUNCTION



Reads a text file as lines (any line break style; a final line break doesn't add an empty line).

Target: Runtime File Importer Subsystem

Async version: [Import Text Lines From File \(Async\)](#)

C++: `URuntimeFileImporterSubsystem::ImportTextLinesFromFile`

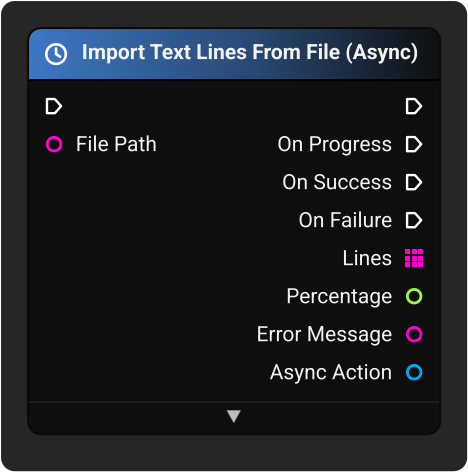
INPUTS

Pin	Type	Description
Target	 Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
File Path	 String	Full path of the file on disk.
Options ADVANCED OPTIONAL	 Runtime Text Import Options	How text files are read.
Remove Empty Lines ADVANCED	 Boolean	Leave out empty lines. Default: <code>false</code>

OUTPUTS

Pin	Type	Description
Lines	 Array of String	The lines of text.
Error Message	 String	Empty on success; otherwise why it failed.
Return Value	 Boolean	True on success. False if it failed (Error Message says why).

Import Text Lines From File (Async) ASYNC



Reads a text file as lines without blocking the game thread.

Blocking version: [Import Text Lines from File](#)

C++: `UImportTextLinesAsyncAction::ImportTextLinesFromFileAsync`

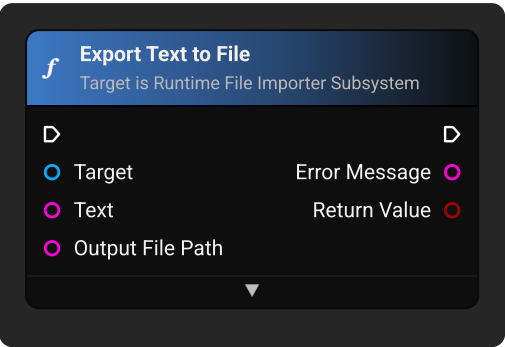
INPUTS

Pin	Type	Description
File Path	String	Full path of the file on disk.
Options ADVANCED OPTIONAL	Runtime Text Import Options	How text files are read.
Remove Empty Lines ADVANCED	Boolean	Leave out empty lines. Default: <code>false</code>

OUTPUTS

Pin	Type	Description
On Progress	Exec	Fires as the operation advances (Percentage from 0 to 100). Other outputs are not set yet.
On Success	Exec	Fires when the operation has finished successfully.
On Failure	Exec	Fires if the operation failed or was cancelled; Error Message says why.
Lines	Array of String	The lines of text.
Percentage	Float	Progress from 0 to 100.
Error Message	String	Empty on success; otherwise why it failed.
Async Action	Import Text Lines Async Action Object Reference	This node's operation. Call Cancel on it to stop the operation (no more events fire).

Export Text to File FUNCTION



Writes text to a file in Options' encoding (UTF-8 by default), or adds it to the file's end with Options' Append.

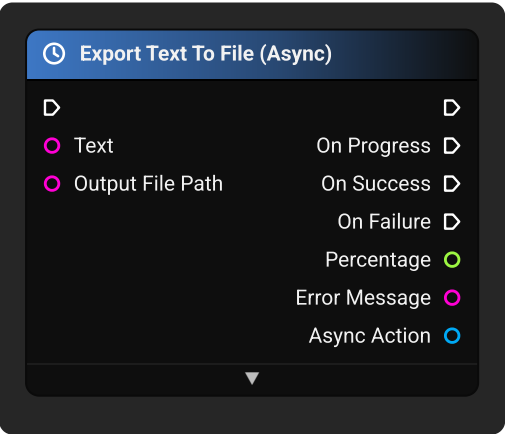
Target: Runtime File Importer Subsystem

Async version: [Export Text To File \(Async\)](#)

C++: `URuntimeFileImporterSubsystem::ExportTextToFile`

INPUTS		
Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Text	String	The text.
Output File Path	String	Full path of the file to write. Missing folders are created; an existing file is replaced.
Options	ADVANCED OPTIONAL Runtime Text Export Options	How text is written.
OUTPUTS		
Pin	Type	Description
Error Message	String	Empty on success; otherwise why it failed.
Return Value	Boolean	True on success. False if it failed (Error Message says why).

Export Text To File (Async) ASYNC



Writes text to a file (or adds it to its end) without blocking the game thread.

Blocking version: [Export Text to File](#)

C++: `UExportTextAsyncAction::ExportTextToFileAsync`

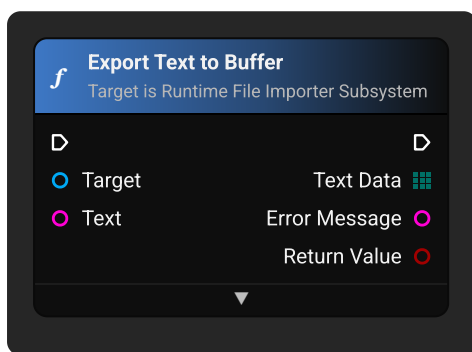
INPUTS		
Pin	Type	Description
Text	String	The text.
Output File Path	String	Full path of the file to write. Missing folders are created; an existing file is replaced.
Options	ADVANCED OPTIONAL Runtime Text Export Options	How text is written.

OUTPUTS

Pin	Type	Description
On Progress	 Exec	Fires as the operation advances (Percentage from 0 to 100). Other outputs are not set yet.
On Success	 Exec	Fires when the operation has finished successfully.
On Failure	 Exec	Fires if the operation failed or was cancelled; Error Message says why.
Percentage	 Float	Progress from 0 to 100.
Error Message	 String	Empty on success; otherwise why it failed.
Async Action	 Export Text Async Action Object Reference	This node's operation. Call Cancel on it to stop the operation (no more events fire).

Export Text to Buffer

FUNCTION



Encodes text into bytes (e.g. to send over the network).

Target: Runtime File Importer Subsystem

Async version: [Export Text To Buffer \(Async\)](#)

C++: `URuntimeFileImporterSubsystem::ExportTextToBuffer`

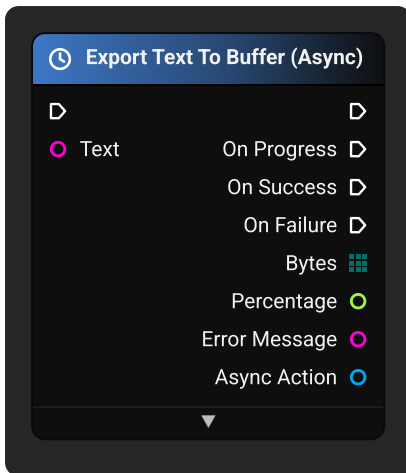
INPUTS

Pin	Type	Description
Target	 Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Text	 String	The text.
Options	 Runtime Text Export Options	How text is written.

OUTPUTS

Pin	Type	Description
Text Data	 Array of Byte	Bytes of a text file.
Error Message	 String	Empty on success; otherwise why it failed.
Return Value	 Boolean	True on success. False if it failed (Error Message says why).

Export Text To Buffer (Async) ASYNC



Encodes text into bytes without blocking the game thread.

Blocking version: [Export Text to Buffer](#)

C++: `UEncodeTextAsyncAction::ExportTextToBufferAsync`

INPUTS

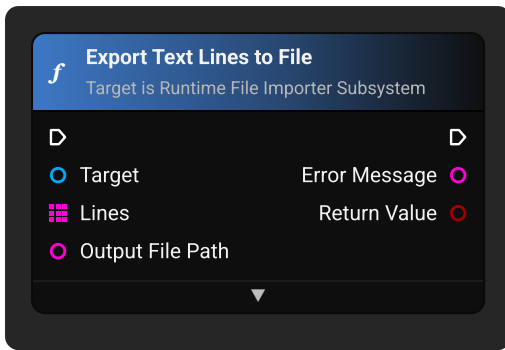
Pin	Type	Description
Text	String	The text.
Options ADVANCED OPTIONAL	Runtime Text Export Options	How text is written.

OUTPUTS

Pin	Type	Description
On Progress	Exec	Fires as the operation advances (Percentage from 0 to 100). Other outputs are not set yet.
On Success	Exec	Fires when the operation has finished successfully.
On Failure	Exec	Fires if the operation failed or was cancelled; Error Message says why.
Bytes	Array of Byte	The bytes.
Percentage	Float	Progress from 0 to 100.
Error Message	String	Empty on success; otherwise why it failed.
Async Action	Encode Text Async Action Object Reference	This node's operation. Call Cancel on it to stop the operation (no more events fire).

Export Text Lines to File

FUNCTION



Writes lines to a file, one per line (or adds them to its end with Options' Append).

Target: Runtime File Importer Subsystem

Async version: [Export Text Lines To File \(Async\)](#)

C++: `URuntimeFileImporterSubsystem::ExportTextLinesToFile`

INPUTS

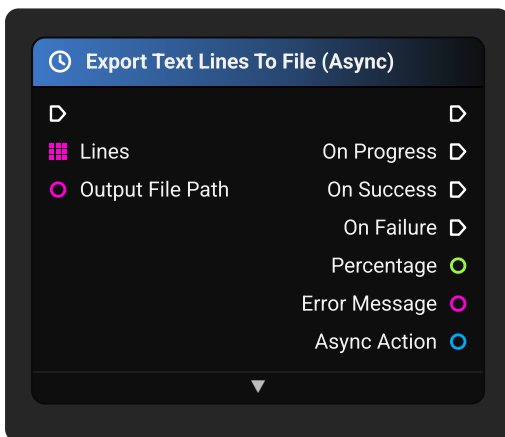
Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Lines	Array of String	The lines of text.
Output File Path	String	Full path of the file to write. Missing folders are created; an existing file is replaced.
Options	Runtime Text Export Options	How text is written.

OUTPUTS

Pin	Type	Description
Error Message	String	Empty on success; otherwise why it failed.
Return Value	Boolean	True on success. False if it failed (Error Message says why).

Export Text Lines To File (Async)

ASYNC



Writes lines to a file without blocking the game thread.

Blocking version: [Export Text Lines to File](#)

C++: `UExportTextAsyncAction::ExportTextLinesToFileAsync`

INPUTS

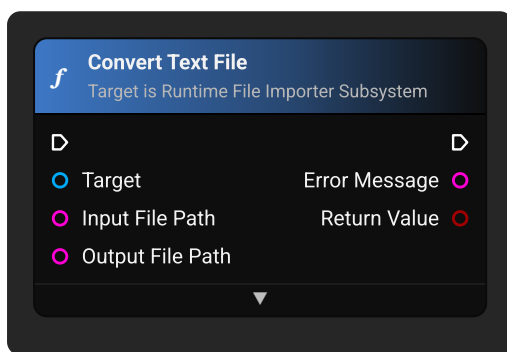
Pin	Type	Description
Lines	Array of String	The lines of text.
Output File Path	String	Full path of the file to write. Missing folders are created; an existing file is replaced.
Options	Runtime Text Export Options	How text is written.

OUTPUTS

Pin	Type	Description
On Progress	 Exec	Fires as the operation advances (Percentage from 0 to 100). Other outputs are not set yet.
On Success	 Exec	Fires when the operation has finished successfully.
On Failure	 Exec	Fires if the operation failed or was cancelled; Error Message says why.
Percentage	 Float	Progress from 0 to 100.
Error Message	 String	Empty on success; otherwise why it failed.
Async Action	 Export Text Async Action Object Reference	This node's operation. Call Cancel on it to stop the operation (no more events fire).

Convert Text File

FUNCTION



Rewrites a text file in another encoding or line break style; documents (HTML, RTF, DOCX, ODT, spreadsheets) become plain text.

Target: Runtime File Importer Subsystem

Async version: [Convert Text File \(Async\)](#)

C++: `URuntimeFileImporterSubsystem::ConvertTextFile`

INPUTS

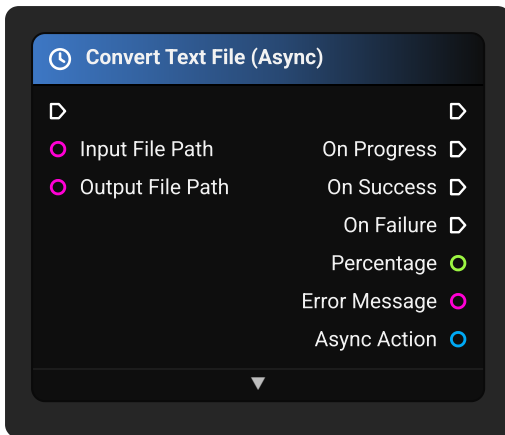
Pin	Type	Description
Target	 Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Input File Path	 String	Full path of the file to convert.
Output File Path	 String	Full path of the file to write. Missing folders are created; an existing file is replaced.
Import Options	ADVANCED OPTIONAL  Runtime Text Import Options	How text files are read.
Export Options	ADVANCED OPTIONAL  Runtime Text Export Options	How text is written.

OUTPUTS

Pin	Type	Description
Error Message	 String	Empty on success; otherwise why it failed.
Return Value	 Boolean	True on success. False if it failed (Error Message says why).

Convert Text File (Async)

ASYNC



Rewrites a text file in another encoding or line break style (documents as plain text) without blocking the game thread.

Blocking version: [Convert Text File](#)

C++: `UExportTextAsyncAction::ConvertTextFileAsync`

INPUTS

Pin	Type	Description
Input File Path	String	Full path of the file to convert.
Output File Path	String	Full path of the file to write. Missing folders are created; an existing file is replaced.
Import Options	ADVANCED OPTIONAL Runtime Text Import Options	How text files are read.
Export Options	ADVANCED OPTIONAL Runtime Text Export Options	How text is written.

OUTPUTS

Pin	Type	Description
On Progress	Exec	Fires as the operation advances (Percentage from 0 to 100). Other outputs are not set yet.
On Success	Exec	Fires when the operation has finished successfully.
On Failure	Exec	Fires if the operation failed or was cancelled; Error Message says why.
Percentage	Float	Progress from 0 to 100.
Error Message	String	Empty on success; otherwise why it failed.
Async Action	Export Text Async Action Object Reference	This node's operation. Call Cancel on it to stop the operation (no more events fire).

Split Text Into Lines

PURE

f

Split Text Into Lines

Target is Runtime File Importer Subsystem

Target

Return Value

Text

Splits text into lines at any line break (LF, CRLF, CR).

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::SplitTextIntoLines`

INPUTS		
Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Text	String	The text.
Remove Empty Lines	Boolean	Leave out empty lines. Default: <code>false</code>
Trim Lines	Boolean	Remove spaces at the start and end of each line. Default: <code>false</code>
OUTPUTS		
Pin	Type	Description
Return Value	Array of String	Splits text into lines at any line break (LF, CRLF, CR).

Convert Line Endings

PURE

f

Convert Line Endings

Target is Runtime File Importer Subsystem

Target

Return Value

Text

Line Ending

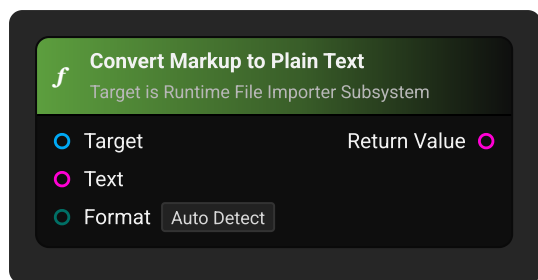
Converts every line break to one style (e.g. CRLF for Windows programs).

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::ConvertLineEndings`

INPUTS		
Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Text	String	The text.
Line Ending	ERuntime Line Ending	The line breaks to use.
OUTPUTS		
Pin	Type	Description
Return Value	String	Converts every line break to one style (e.g.

Convert Markup to Plain Text PURE



Plain text of HTML, RTF or Markdown (Format Auto Detect: detected), e.g. to show web or rich text in a text widget.

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::ConvertMarkupToPlainText`

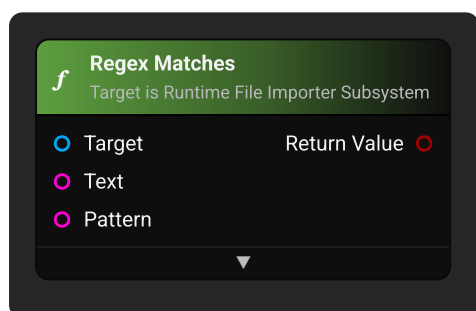
INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Text	String	The text.
Format	<code>ERuntimeTextFormat</code>	Format of the text (markup is converted). Auto Detect reads it from the contents. Default: <code>Auto Detect</code>

OUTPUTS

Pin	Type	Description
Return Value	String	Plain text of HTML, RTF or Markdown (Format Auto Detect: detected), e.g.

Regex Matches PURE



True if the regular expression (ICU syntax, e.g. `"^[A-Z]{3}-[0-9]{4}$"`) matches somewhere in the text (use `^` and `$` to match all of it).

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::RegexMatches`

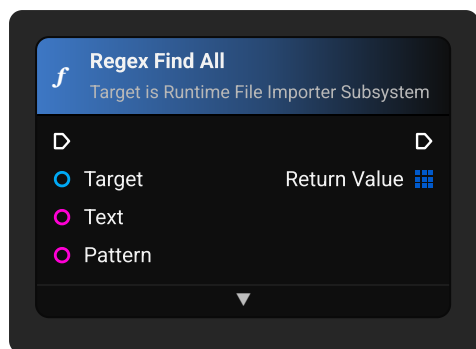
INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Text	String	The text.
Pattern	String	Regular expression (ICU syntax).
Ignore Case	Boolean	Match upper and lower case alike. Default: <code>false</code>

OUTPUTS

Pin	Type	Description
Return Value	Boolean	True if the regular expression (ICU syntax, e.g.

Regex Find All FUNCTION



Every match of a regular expression, with its position and capture groups (Max Matches 0: all).

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::RegexFindAll`

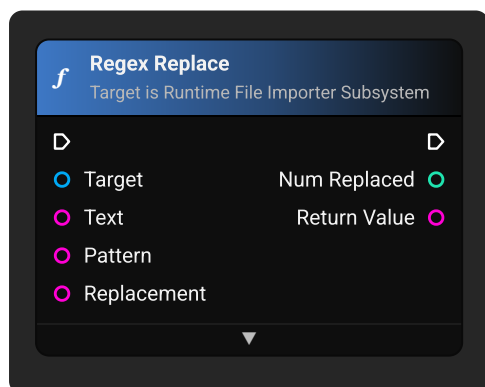
INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Text	String	The text.
Pattern	String	Regular expression (ICU syntax).
Ignore Case ADVANCED	Boolean	Match upper and lower case alike. Default: <code>false</code>
Max Matches ADVANCED	Integer	Most matches to return (0: all). Default: <code>0</code>

OUTPUTS

Pin	Type	Description
Return Value	Array of Runtime Regex Match	Every match of a regular expression, with its position and capture groups (Max Matches 0: all).

Regex Replace FUNCTION



Replaces every match of a regular expression; Replacement can use the capture groups (\$1, \$2, ...; \$0: the whole match; \$\$: a dollar sign).

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::RegexReplace`

INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Text	String	The text.
Pattern	String	Regular expression (ICU syntax).
Replacement	String	Text that replaces each match. \$1, \$2, ... insert the pattern's groups.
Ignore Case ADVANCED	Boolean	Match upper and lower case alike. Default: <code>false</code>

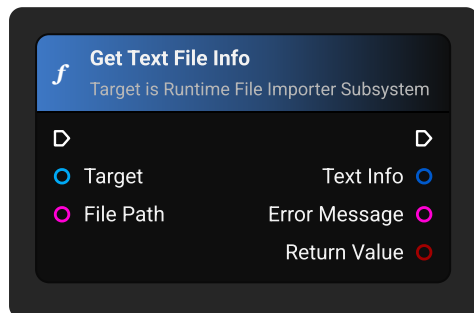
OUTPUTS

Pin	Type	Description
Num Replaced	Integer	How many matches were replaced.
Return Value	String	Replaces every match of a regular expression; Replacement can use the capture groups (\$1, \$2, ...; \$0: the whole match; \$\$: a dollar sign).

Utilities 15 nodes

Get Text File Info

FUNCTION



Reads a text file's format, encoding, size, character / word / line counts, line breaks and document properties without importing it.

Target: Runtime File Importer Subsystem

Async version: [Get Text File Info \(Async\)](#)

C++: `URuntimeFileImporterSubsystem::GetTextFileInfo`

INPUTS

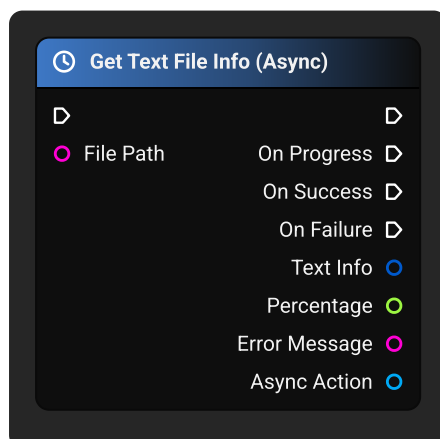
Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
File Path	String	Full path of the file on disk.

OUTPUTS

Pin	Type	Description
Text Info	Runtime Text Info	Format, encoding, line endings and counts of the text.
Error Message	String	Empty on success; otherwise why it failed.
Return Value	Boolean	True on success. False if it failed (Error Message says why).

Get Text File Info (Async)

ASYNC



Reads a text file's format, encoding, counts, line breaks and document properties without blocking the game thread.

Blocking version: [Get Text File Info](#)

C++: `UGetTextInfoAsyncAction::GetTextFileInfoAsync`

INPUTS

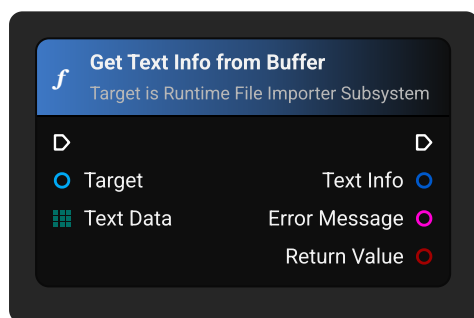
Pin	Type	Description
File Path	 String	Full path of the file on disk.

OUTPUTS

Pin	Type	Description
On Progress	 Exec	Fires as the operation advances (Percentage from 0 to 100). Other outputs are not set yet.
On Success	 Exec	Fires when the operation has finished successfully.
On Failure	 Exec	Fires if the operation failed or was cancelled; Error Message says why.
Text Info	 Runtime Text Info	Format, encoding, line endings and counts of the text.
Percentage	 Float	Progress from 0 to 100.
Error Message	 String	Empty on success; otherwise why it failed.
Async Action	 Get Text Info Async Action Object Reference	This node's operation. Call Cancel on it to stop the operation (no more events fire).

Get Text Info from Buffer

FUNCTION



Info of text file bytes in memory.

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::GetTextInfoFromBuffer`

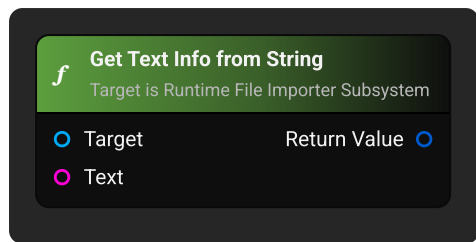
INPUTS

Pin	Type	Description
Target	 Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Text Data	 Array of Byte	Bytes of a text file.

OUTPUTS

Pin	Type	Description
Text Info	 Runtime Text Info	Format, encoding, line endings and counts of the text.
Error Message	 String	Empty on success; otherwise why it failed.
Return Value	 Boolean	True on success. False if it failed (Error Message says why).

Get Text Info from String PURE



Character, word and line counts, line breaks and format of a string.

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::GetTextInfoFromString`

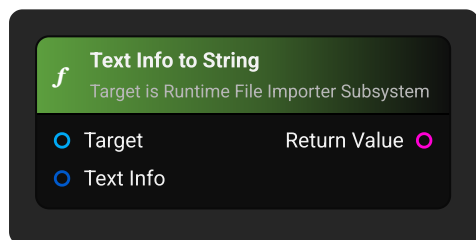
INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Text	String	The text.

OUTPUTS

Pin	Type	Description
Return Value	Runtime Text Info	Character, word and line counts, line breaks and format of a string.

Text Info to String PURE



A readable, multi-line description of text info.

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::TextInfoToString`

INPUTS

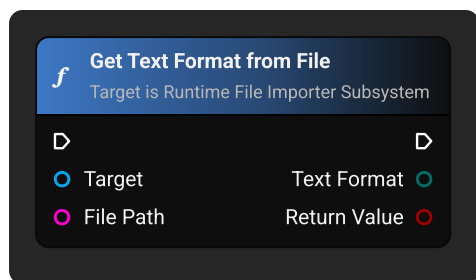
Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Text Info	Runtime Text Info	Format, encoding, line endings and counts of the text.

OUTPUTS

Pin	Type	Description
Return Value	String	A readable, multi-line description of text info.

Get Text Format from File

FUNCTION



Detects a file's text format from its contents (falling back to its extension). Returns false if the file can't be read.

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::GetTextFormatFromFile`

INPUTS

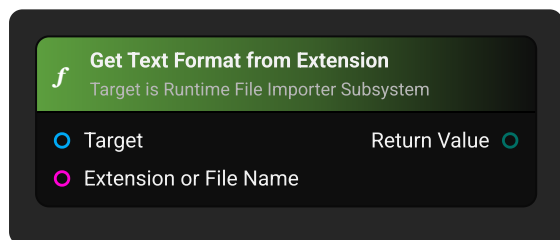
Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
File Path	String	Full path of the file on disk.

OUTPUTS

Pin	Type	Description
Text Format	ERuntime Text Format	Format of the text.
Return Value	Boolean	False if the file can't be read.

Get Text Format from Extension

PURE



Text format of a file extension or file name (Auto Detect if unknown).

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::GetTextFormatFromExtension`

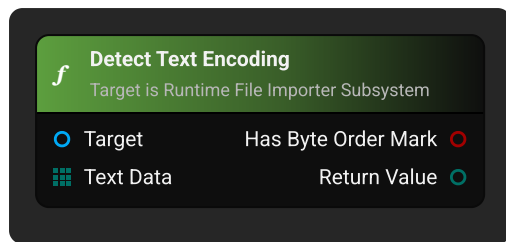
INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Extension or File Name	String	An extension (with or without the dot) or a file name / path / URL ending in one.

OUTPUTS

Pin	Type	Description
Return Value	ERuntime Text Format	Text format of a file extension or file name (Auto Detect if unknown).

Detect Text Encoding PURE



Encoding of text bytes (from the byte order mark, else from the bytes).

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::DetectTextEncoding`

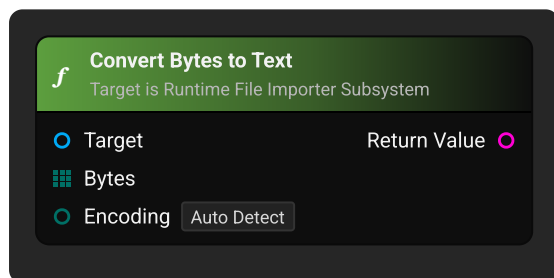
INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Text Data	Array of Byte	Bytes of a text file.

OUTPUTS

Pin	Type	Description
Has Byte Order Mark	Boolean	True if the bytes start with a byte order mark.
Return Value	ERuntime Text Encoding	Encoding of text bytes (from the byte order mark, else from the bytes).

Convert Bytes to Text PURE



Decodes bytes into text (a byte order mark is skipped).

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::ConvertBytesToText`

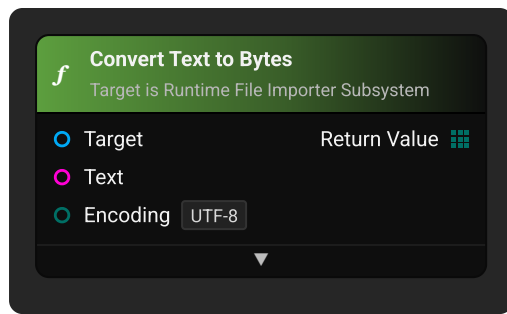
INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Bytes	Array of Byte	The bytes.
Encoding	ERuntime Text Encoding	Character encoding. Auto Detect: from the byte order mark, else from the bytes (UTF-8 when writing). Default: Auto Detect

OUTPUTS

Pin	Type	Description
Return Value	String	Decodes bytes into text (a byte order mark is skipped).

Convert Text to Bytes PURE



Encodes text into bytes.

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::ConvertTextToBytes`

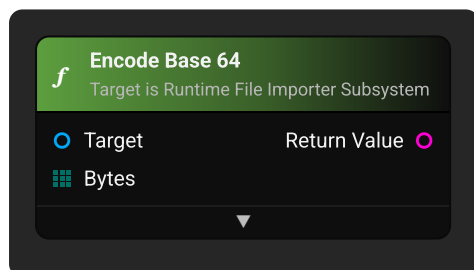
INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Text	String	The text.
Encoding	ERuntime Text Encoding	Character encoding. Auto Detect: from the byte order mark, else from the bytes (UTF-8 when writing). Default: <code>UTF-8</code>
Byte Order Mark ADVANCED	Boolean	Start with a byte order mark (BOM). Default: <code>false</code>

OUTPUTS

Pin	Type	Description
Return Value	Array of Byte	Encodes text into bytes.

Encode Base 64 PURE



Bytes as Base64 text (e.g. to put a file in JSON); URL Safe uses - and _ without padding.

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::EncodeBase64`

INPUTS

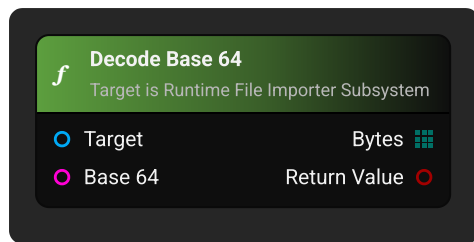
Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Bytes	Array of Byte	The bytes.
Url Safe ADVANCED	Boolean	Use the URL-safe alphabet (- and _ instead of + and /) without padding. Default: <code>false</code>

OUTPUTS

Pin	Type	Description
Return Value	String	Bytes as Base64 text (e.g.

Decode Base 64

PURE



Bytes of Base64 text (standard or URL-safe, padding optional, a "data:...;base64," prefix allowed). Returns false if it isn't Base64.

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::DecodeBase64`

INPUTS

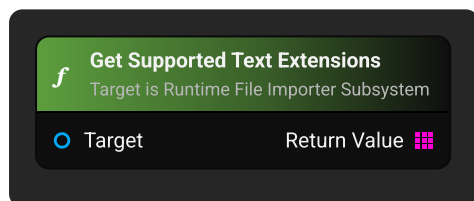
Pin	Type	Description
Target	 Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Base 64	 String	Base64 text.

OUTPUTS

Pin	Type	Description
Bytes	 Array of Byte	The bytes.
Return Value	 Boolean	False if it isn't Base64.

Get Supported Text Extensions

PURE



File extensions the Text nodes read (without dots), e.g. for a file picker filter.

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::GetSupportedTextExtensions`

INPUTS

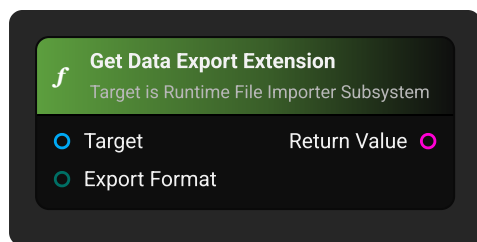
Pin	Type	Description
Target	 Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.

OUTPUTS

Pin	Type	Description
Return Value	 Array of String	File extensions the Text nodes read (without dots), e.g.

Get Data Export Extension

PURE



File extension written for a data export format, without the dot (e.g. "json").

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::GetDataExportExtension`

INPUTS

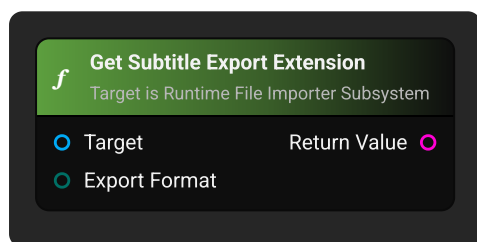
Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Export Format	ERuntime Data Export Format	Format to write.

OUTPUTS

Pin	Type	Description
Return Value	String	File extension written for a data export format, without the dot (e.g.

Get Subtitle Export Extension

PURE



File extension written for a subtitle export format, without the dot (e.g. "srt").

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::GetSubtitleExportExtension`

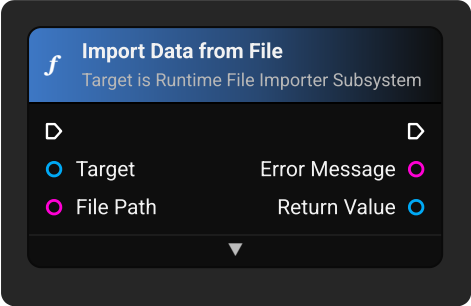
INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Export Format	ERuntime Subtitle Export Format	Format to write.

OUTPUTS

Pin	Type	Description
Return Value	String	File extension written for a subtitle export format, without the dot (e.g.

Import Data from File FUNCTION

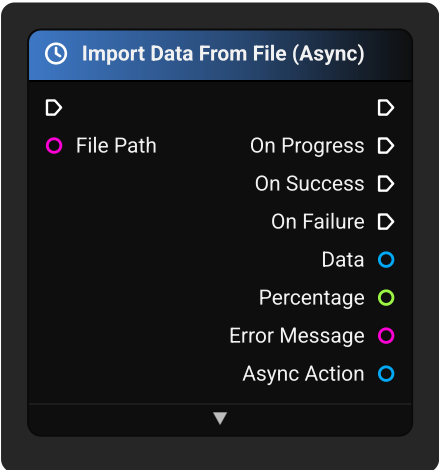


Reads a data file (JSON, XML, CSV, TSV, INI, Excel XLSX, OpenDocument ODS) as Runtime Data. CSV, TSV and spreadsheets become an array of rows (objects keyed by the header row, see Options); XML elements objects; INI sections objects.

Target: Runtime File Importer Subsystem
Async version: [Import Data From File \(Async\)](#)
C++: URuntimeFileImporterSubsystem::ImportDataFromFile

INPUTS		
Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
File Path	String	Full path of the file on disk.
Format ADVANCED	ERuntime Data Format	Format of the data. Auto Detect reads it from the contents.
Options ADVANCED OPTIONAL	Runtime Data Import Options	How structured data files are read into Runtime Data.
OUTPUTS		
Pin	Type	Description
Error Message	String	Empty on success; otherwise why it failed.
Return Value	Runtime Data Object Reference	The result, or None/empty if it failed (Error Message says why).

Import Data From File (Async) ASYNC



Reads a data file (JSON, XML, CSV, TSV, INI, XLSX, ODS) as Runtime Data without blocking the game thread.

Blocking version: [Import Data from File](#)
C++: UImportDataAsyncAction::ImportDataFromFileAsync

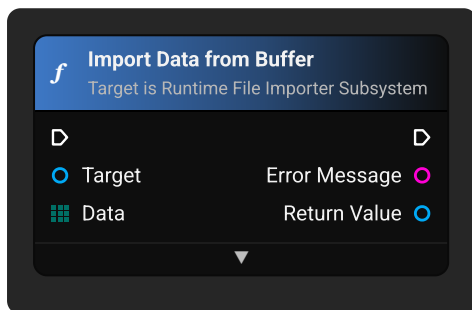
INPUTS		
Pin	Type	Description
File Path	String	Full path of the file on disk.
Format ADVANCED	ERuntime Data Format	Format of the data. Auto Detect reads it from the contents.
Options ADVANCED OPTIONAL	Runtime Data Import Options	How structured data files are read into Runtime Data.

OUTPUTS

Pin	Type	Description
On Progress	 Exec	Fires as the operation advances (Percentage from 0 to 100). Other outputs are not set yet.
On Success	 Exec	Fires when the operation has finished successfully.
On Failure	 Exec	Fires if the operation failed or was cancelled; Error Message says why.
Data	 Runtime Data Object Reference	The Runtime Data object.
Percentage	 Float	Progress from 0 to 100.
Error Message	 String	Empty on success; otherwise why it failed.
Async Action	 Import Data Async Action Object Reference	This node's operation. Call Cancel on it to stop the operation (no more events fire).

Import Data from Buffer

FUNCTION



Reads data file bytes in memory as Runtime Data.

Target: Runtime File Importer Subsystem

Async version: [Import Data From Buffer \(Async\)](#)

C++: `URuntimeFileImporterSubsystem::ImportDataFromBuffer`

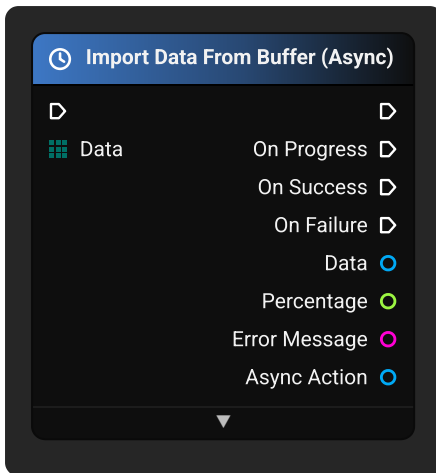
INPUTS

Pin	Type	Description
Target	 Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Data	 Array of Byte	Bytes of a data file.
Format ADVANCED	 ERuntime Data Format	Format of the data. Auto Detect reads it from the contents.
Options ADVANCED OPTIONAL	 Runtime Data Import Options	How structured data files are read into Runtime Data.

OUTPUTS

Pin	Type	Description
Error Message	 String	Empty on success; otherwise why it failed.
Return Value	 Runtime Data Object Reference	The result, or None/empty if it failed (Error Message says why).

Import Data From Buffer (Async) ASYNC



Reads data file bytes in memory as Runtime Data without blocking the game thread.

Blocking version: [Import Data from Buffer](#)

C++: `UImportDataAsyncAction::ImportDataFromBufferAsync`

INPUTS

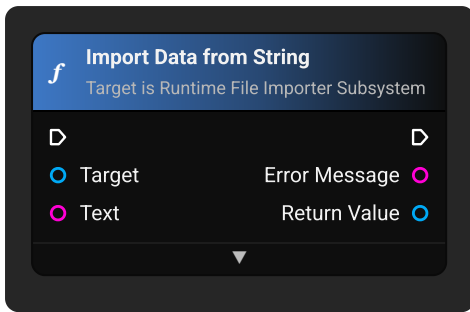
Pin	Type	Description
Data	Array of Byte	Bytes of a data file.
Format ADVANCED	ERuntime Data Format	Format of the data. Auto Detect reads it from the contents.
Options ADVANCED OPTIONAL	Runtime Data Import Options	How structured data files are read into Runtime Data.

OUTPUTS

Pin	Type	Description
On Progress	Exec	Fires as the operation advances (Percentage from 0 to 100). Other outputs are not set yet.
On Success	Exec	Fires when the operation has finished successfully.
On Failure	Exec	Fires if the operation failed or was cancelled; Error Message says why.
Data	Runtime Data Object Reference	The Runtime Data object.
Percentage	Float	Progress from 0 to 100.
Error Message	String	Empty on success; otherwise why it failed.
Async Action	Import Data Async Action Object Reference	This node's operation. Call Cancel on it to stop the operation (no more events fire).

Import Data from String

FUNCTION



Parses data text (e.g. a JSON string from a web API) as Runtime Data.

Target: Runtime File Importer Subsystem

Async version: [Import Data From String \(Async\)](#)

C++: `URuntimeFileImporterSubsystem::ImportDataFromString`

INPUTS

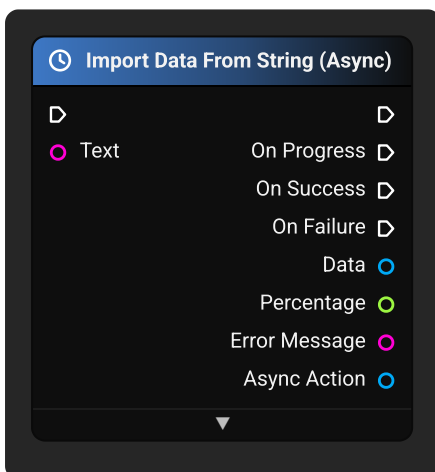
Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Text	String	The text.
Format ADVANCED	ERuntime Data Format	Format of the data. Auto Detect reads it from the contents.
Options ADVANCED OPTIONAL	Runtime Data Import Options	How structured data files are read into Runtime Data.

OUTPUTS

Pin	Type	Description
Error Message	String	Empty on success; otherwise why it failed.
Return Value	Runtime Data Object Reference	The result, or None/empty if it failed (Error Message says why).

Import Data From String (Async)

ASYNC



Parses data text (e.g. a large JSON string) as Runtime Data without blocking the game thread.

Blocking version: [Import Data from String](#)

C++: `UImportDataAsyncAction::ImportDataFromStringAsync`

INPUTS

Pin	Type	Description
Text	String	The text.
Format ADVANCED	ERuntime Data Format	Format of the data. Auto Detect reads it from the contents.
Options ADVANCED OPTIONAL	Runtime Data Import Options	How structured data files are read into Runtime Data.

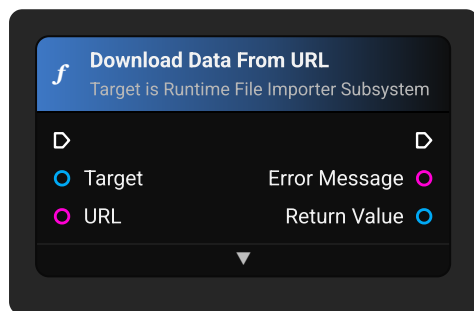
OUTPUTS

Pin	Type	Description
On Progress	Exec	Fires as the operation advances (Percentage from 0 to 100). Other outputs are not set yet.

Pin	Type	Description
On Success	 Exec	Fires when the operation has finished successfully.
On Failure	 Exec	Fires if the operation failed or was cancelled; Error Message says why.
Data	 Runtime Data Object Reference	The Runtime Data object.
Percentage	 Float	Progress from 0 to 100.
Error Message	 String	Empty on success; otherwise why it failed.
Async Action	 Import Data Async Action Object Reference	This node's operation. Call Cancel on it to stop the operation (no more events fire).

Download Data From URL

FUNCTION



Downloads a data file (or a web API's response) and reads it. Google Sheets links work directly (all sheets as Excel, or one sheet as CSV when the link points to it), as do Drive and Dropbox links. FREEZES the game until done; prefer the Async version.

Target: Runtime File Importer Subsystem

Async version: [Download Data From URL \(Async\)](#)

C++: `URuntimeFileImporterSubsystem::DownloadDataFromURL`

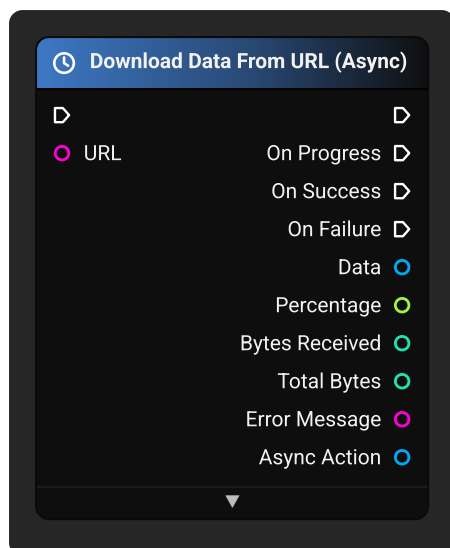
INPUTS

Pin	Type	Description
Target	 Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
URL	 String	The http(s) address to download. Google Docs, Sheets, Slides and Drive share links are converted automatically.
Format ADVANCED	 ERuntime Data Format	Format of the data. Auto Detect reads it from the contents.
Options ADVANCED OPTIONAL	 Runtime Data Import Options	How structured data files are read into Runtime Data.
Save to File Path ADVANCED	 String	Optional. Also writes the downloaded file there, unmodified. Default: (empty)
Timeout Seconds ADVANCED	 Float	Gives up (and fails) if the download takes longer than this. Default: 30.0

OUTPUTS

Pin	Type	Description
Error Message	 String	Empty on success; otherwise why it failed.
Return Value	 Runtime Data Object Reference	The result, or None/empty if it failed (Error Message says why).

Download Data From URL (Async) ASYNC



Downloads a data file (or a web API's response) and reads it as Runtime Data. If Save To File Path is set, the downloaded file is also written there.

Blocking version: [Download Data From URL](#)

C++: `UDownloadDataFromURLAsyncAction::DownloadDataFromURLAsync`

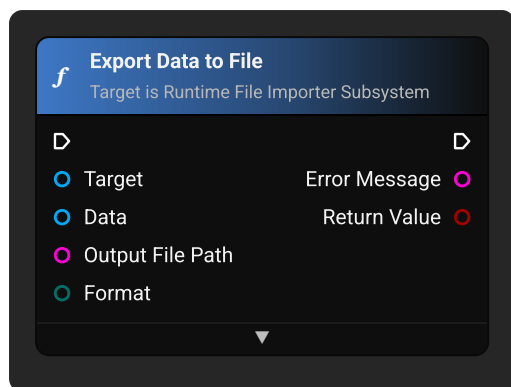
INPUTS

Pin	Type	Description
URL	String	The http(s) address to download. Google Docs, Sheets, Slides and Drive share links are converted automatically.
Format ADVANCED	ERuntime Data Format	Format of the data. Auto Detect reads it from the contents.
Options ADVANCED OPTIONAL	Runtime Data Import Options	How structured data files are read into Runtime Data.
Save to File Path ADVANCED	String	Optional. Also writes the downloaded file there, unmodified. Default: (empty)
Timeout Seconds ADVANCED	Float	Gives up (and fails) if the download takes longer than this. Default: 60.0

OUTPUTS

Pin	Type	Description
On Progress	Exec	Fires repeatedly while downloading. Percentage (0-100) and Total Bytes are 0 if the server doesn't report the file size.
On Success	Exec	Fires when the operation has finished successfully.
On Failure	Exec	Fires if the operation failed or was cancelled; Error Message says why.
Data	Runtime Data Object Reference	The Runtime Data object.
Percentage	Float	Progress from 0 to 100.
Bytes Received	Integer	Bytes downloaded so far.
Total Bytes	Integer	Size of the download in bytes, or 0 while the server hasn't said.
Error Message	String	Empty on success; otherwise why it failed.
Async Action	Download Data from URLAsync Action Object Reference	This node's operation. Call Cancel on it to stop the operation (no more events fire).

Export Data to File FUNCTION



Writes Runtime Data to a file. CSV and TSV: arrays of objects become rows with a column per key; arrays of arrays rows; objects of objects rows with a Name column. INI: object fields that are objects become sections. XML: "@name" fields become attributes, "#text" text, arrays repeated elements.

Target: Runtime File Importer Subsystem

Async version: [Export Data To File \(Async\)](#)

C++: `URuntimeFileImporterSubsystem::ExportDataToFile`

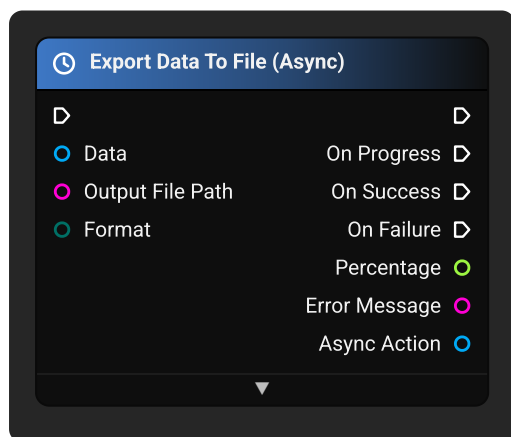
INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Data	Runtime Data Object Reference	The Runtime Data object.
Output File Path	String	Full path of the file to write. Missing folders are created; an existing file is replaced.
Format	ERuntime Data Export Format	Format to write.
Options ADVANCED OPTIONAL	Runtime Data Export Options	How Runtime Data is written.

OUTPUTS

Pin	Type	Description
Error Message	String	Empty on success; otherwise why it failed.
Return Value	Boolean	True on success. False if it failed (Error Message says why).

Export Data To File (Async) ASYNC



Writes Runtime Data to a file without blocking the game thread (the data is copied when the node starts).

Blocking version: [Export Data to File](#)

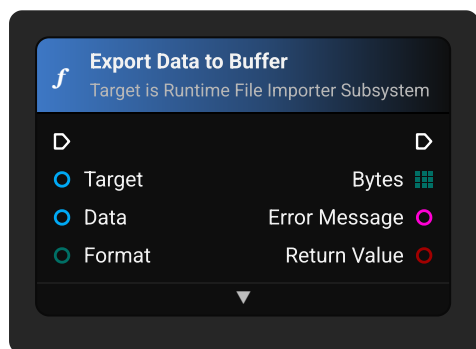
C++: `UExportTextAsyncAction::ExportDataToFileAsync`

INPUTS

Pin	Type	Description
Data	Runtime Data Object Reference	The Runtime Data object.
Output File Path	String	Full path of the file to write. Missing folders are created; an existing file is replaced.

Pin	Type	Description
Format	ERuntime Data Export Format	Format to write.
Options	ADVANCED OPTIONAL Runtime Data Export Options	How Runtime Data is written.
OUTPUTS		
Pin	Type	Description
On Progress	Exec	Fires as the operation advances (Percentage from 0 to 100). Other outputs are not set yet.
On Success	Exec	Fires when the operation has finished successfully.
On Failure	Exec	Fires if the operation failed or was cancelled; Error Message says why.
Percentage	Float	Progress from 0 to 100.
Error Message	String	Empty on success; otherwise why it failed.
Async Action	Export Text Async Action Object Reference	This node's operation. Call Cancel on it to stop the operation (no more events fire).

Export Data to Buffer FUNCTION



Encodes Runtime Data into file bytes.

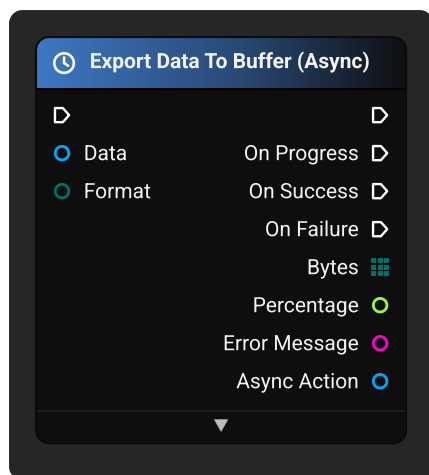
Target: Runtime File Importer Subsystem

Async version: [Export Data To Buffer \(Async\)](#)

C++: `URuntimeFileImporterSubsystem::ExportDataToBuffer`

INPUTS		
Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Data	Runtime Data Object Reference	The Runtime Data object.
Format	ERuntime Data Export Format	Format to write.
Options	ADVANCED OPTIONAL Runtime Data Export Options	How Runtime Data is written.
OUTPUTS		
Pin	Type	Description
Bytes	Array of Byte	The bytes.
Error Message	String	Empty on success; otherwise why it failed.
Return Value	Boolean	True on success. False if it failed (Error Message says why).

Export Data To Buffer (Async) ASYNC



Encodes Runtime Data into file bytes without blocking the game thread (the data is copied when the node starts).

Blocking version: [Export Data to Buffer](#)

C++: `UEncodeTextAsyncAction::ExportDataToBufferAsync`

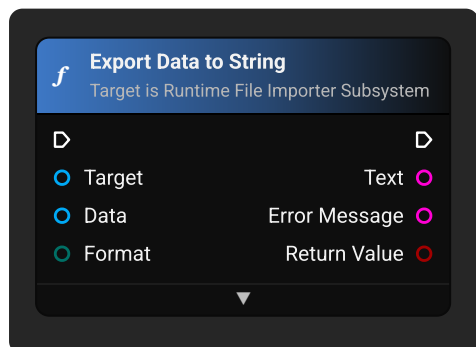
INPUTS

Pin	Type	Description
Data	Runtime Data Object Reference	The Runtime Data object.
Format	ERuntime Data Export Format	Format to write.
Options ADVANCED OPTIONAL	Runtime Data Export Options	How Runtime Data is written.

OUTPUTS

Pin	Type	Description
On Progress	Exec	Fires as the operation advances (Percentage from 0 to 100). Other outputs are not set yet.
On Success	Exec	Fires when the operation has finished successfully.
On Failure	Exec	Fires if the operation failed or was cancelled; Error Message says why.
Bytes	Array of Byte	The bytes.
Percentage	Float	Progress from 0 to 100.
Error Message	String	Empty on success; otherwise why it failed.
Async Action	Encode Text Async Action Object Reference	This node's operation. Call Cancel on it to stop the operation (no more events fire).

Export Data to String FUNCTION



Writes Runtime Data as text (e.g. a JSON string to send to a web API).

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::ExportDataToString`

INPUTS

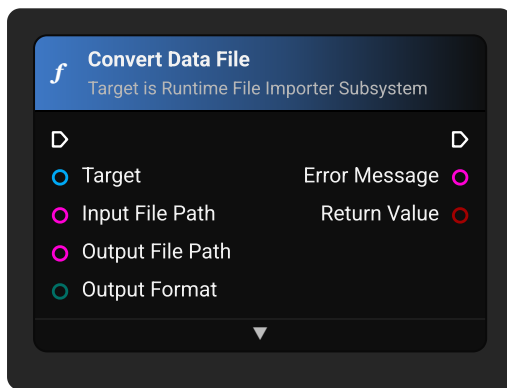
Pin	Type	Description
Target	 Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Data	 Runtime Data Object Reference	The Runtime Data object.
Format	 ERuntime Data Export Format	Format to write.
Options	 Runtime Data Export Options	How Runtime Data is written.

OUTPUTS

Pin	Type	Description
Text	 String	The text.
Error Message	 String	Empty on success; otherwise why it failed.
Return Value	 Boolean	True on success. False if it failed (Error Message says why).

Convert Data File

FUNCTION



Converts a data file to another format (e.g. Excel or CSV to JSON, XML to JSON, JSON to CSV).

Target: Runtime File Importer Subsystem

Async version: [Convert Data File \(Async\)](#)

C++: `URuntimeFileImporterSubsystem::ConvertDataFile`

INPUTS

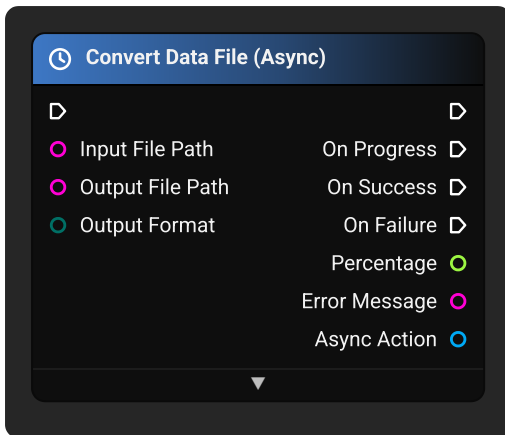
Pin	Type	Description
Target	 Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Input File Path	 String	Full path of the file to convert.
Output File Path	 String	Full path of the file to write. Missing folders are created; an existing file is replaced.
Output Format	 ERuntime Data Export Format	Format to write.
Input Format	 ERuntime Data Format	Format of the input. Auto Detect reads it from the contents.
Import Options	 Runtime Data Import Options	How structured data files are read into Runtime Data.
Export Options	 Runtime Data Export Options	How Runtime Data is written.

OUTPUTS

Pin	Type	Description
Error Message	 String	Empty on success; otherwise why it failed.
Return Value	 Boolean	True on success. False if it failed (Error Message says why).

Convert Data File (Async)

ASYNC



Converts a data file to another format without blocking the game thread.

Blocking version: [Convert Data File](#)

C++: `UExportTextAsyncAction::ConvertDataFileAsync`

INPUTS

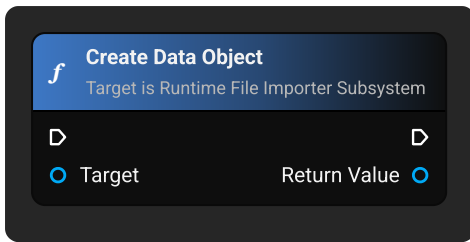
Pin	Type	Description
Input File Path	String	Full path of the file to convert.
Output File Path	String	Full path of the file to write. Missing folders are created; an existing file is replaced.
Output Format	ERuntime Data Export Format	Format to write.
Input Format ADVANCED	ERuntime Data Format	Format of the input. Auto Detect reads it from the contents.
Import Options ADVANCED OPTIONAL	Runtime Data Import Options	How structured data files are read into Runtime Data.
Export Options ADVANCED OPTIONAL	Runtime Data Export Options	How Runtime Data is written.

OUTPUTS

Pin	Type	Description
On Progress	Exec	Fires as the operation advances (Percentage from 0 to 100). Other outputs are not set yet.
On Success	Exec	Fires when the operation has finished successfully.
On Failure	Exec	Fires if the operation failed or was cancelled; Error Message says why.
Percentage	Float	Progress from 0 to 100.
Error Message	String	Empty on success; otherwise why it failed.
Async Action	Export Text Async Action Object Reference	This node's operation. Call Cancel on it to stop the operation (no more events fire).

Create Data Object

FUNCTION



An empty object, to fill with Set Data ... nodes.

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::CreateDataObject`

INPUTS

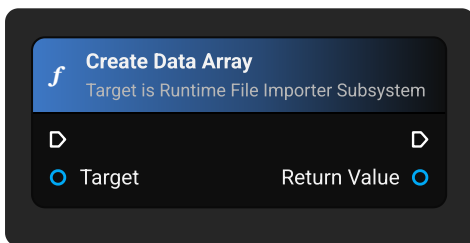
Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.

OUTPUTS

Pin	Type	Description
Return Value	Runtime Data Object Reference	An empty object, to fill with Set Data ...

Create Data Array

FUNCTION



An empty array, to fill with Add Data Item or Set Data ... at "[0]", "[1]", ...

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::CreateDataArray`

INPUTS

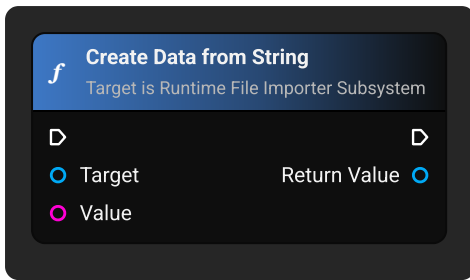
Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.

OUTPUTS

Pin	Type	Description
Return Value	Runtime Data Object Reference	An empty array, to fill with Add Data Item or Set Data ...

Create Data from String

FUNCTION



Makes a data object holding a string (e.g. to Set Data Value at a path).

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::CreateDataFromString`

INPUTS

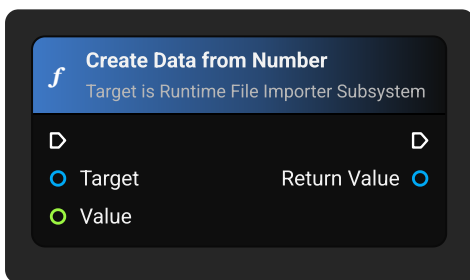
Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Value	String	The value.

OUTPUTS

Pin	Type	Description
Return Value	Runtime Data Object Reference	Makes a data object holding a string (e.g.

Create Data from Number

FUNCTION



Makes a data object holding a number.

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::CreateDataFromNumber`

INPUTS

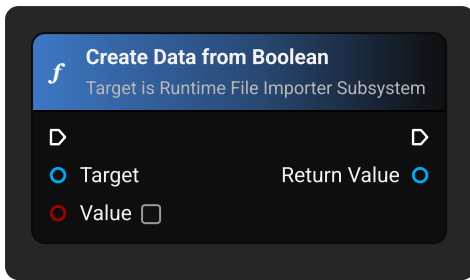
Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Value	Float	The number.

OUTPUTS

Pin	Type	Description
Return Value	Runtime Data Object Reference	Makes a data object holding a number.

Create Data from Boolean

FUNCTION



Makes a data object holding a boolean.

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::CreateDataFromBoolean`

INPUTS

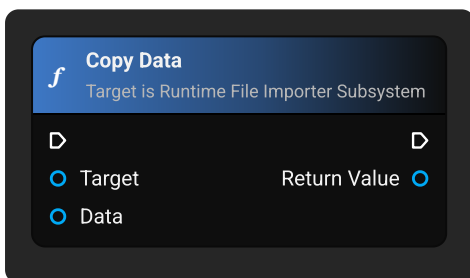
Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Value	Boolean	The boolean value.

OUTPUTS

Pin	Type	Description
Return Value	Runtime Data Object Reference	Makes a data object holding a boolean.

Copy Data

FUNCTION



An independent copy of Runtime Data (changing one doesn't change the other).

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::CopyData`

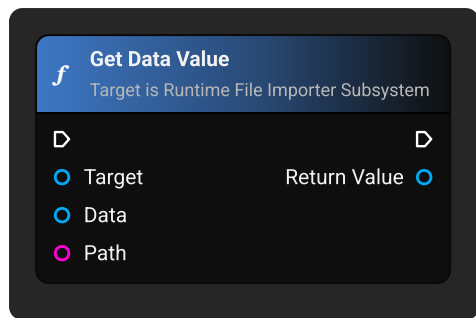
INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Data	Runtime Data Object Reference	The Runtime Data object.

OUTPUTS

Pin	Type	Description
Return Value	Runtime Data Object Reference	An independent copy of Runtime Data (changing one doesn't change the other).

Get Data Value FUNCTION



The value at a path, as Runtime Data sharing it (changes through either show in both); none if there's nothing there.

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::GetDataValue`

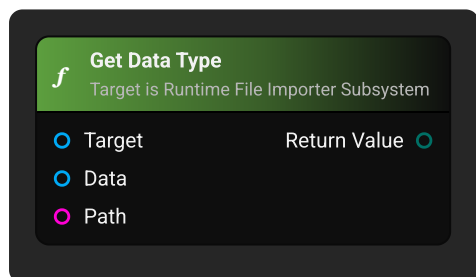
INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Data	Runtime Data Object Reference	The Runtime Data object.
Path	String	Path of a value inside the data, e.g. "player.inventory[2].name" (empty: the root).

OUTPUTS

Pin	Type	Description
Return Value	Runtime Data Object Reference	The value at a path, as Runtime Data sharing it (changes through either show in both); none if there's nothing there.

Get Data Type PURE



Kind of the value at a path (None if there's nothing there). Same as the Runtime Data's Type variable for an empty path.

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::GetDataType`

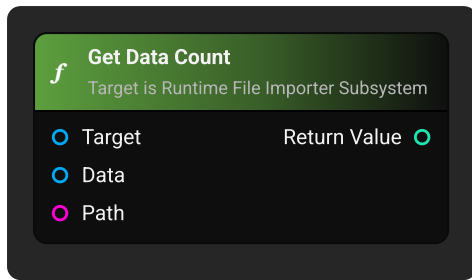
INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Data	Runtime Data Object Reference	The Runtime Data object.
Path	String	Path of a value inside the data, e.g. "player.inventory[2].name" (empty: the root).

OUTPUTS

Pin	Type	Description
Return Value	ERuntime Data Type	Kind of the value at a path (None if there's nothing there).

Get Data Count PURE



Items of the array or fields of the object at a path (0 for other values). Same as the Runtime Data's Count variable for an empty path.

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::GetDataCount`

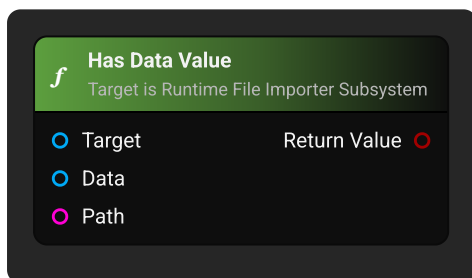
INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Data	Runtime Data Object Reference	The Runtime Data object.
Path	String	Path of a value inside the data, e.g. "player.inventory[2].name" (empty: the root).

OUTPUTS

Pin	Type	Description
Return Value	Integer	Items of the array or fields of the object at a path (0 for other values).

Has Data Value PURE



True if there's a value (even null) at a path.

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::HasDataValue`

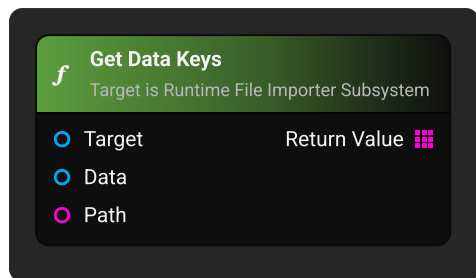
INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Data	Runtime Data Object Reference	The Runtime Data object.
Path	String	Path of a value inside the data, e.g. "player.inventory[2].name" (empty: the root).

OUTPUTS

Pin	Type	Description
Return Value	Boolean	True if there's a value (even null) at a path.

Get Data Keys PURE



Field names of the object at a path, in order.

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::GetDataKeys`

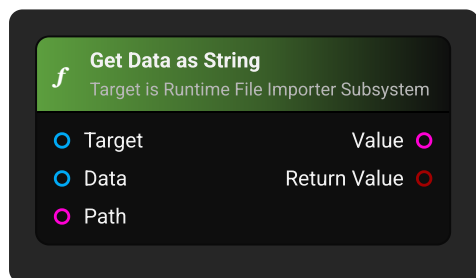
INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Data	Runtime Data Object Reference	The Runtime Data object.
Path	String	Path of a value inside the data, e.g. "player.inventory[2].name" (empty: the root).

OUTPUTS

Pin	Type	Description
Return Value	Array of String	Field names of the object at a path, in order.

Get Data as String PURE



The value at a path as text (numbers and booleans converted, arrays and objects as JSON). False if there's nothing there.

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::GetDataAsString`

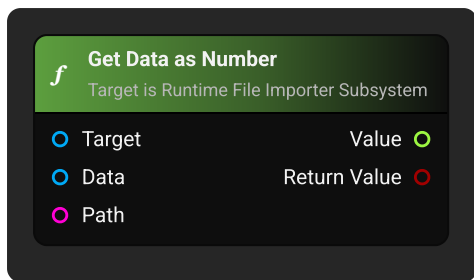
INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Data	Runtime Data Object Reference	The Runtime Data object.
Path	String	Path of a value inside the data, e.g. "player.inventory[2].name" (empty: the root).

OUTPUTS

Pin	Type	Description
Value	String	The value.
Return Value	Boolean	The value at a path as text (numbers and booleans converted, arrays and objects as JSON).

Get Data as Number PURE



The value at a path as a number (strings that are numbers and booleans converted). False if it isn't one.

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::GetDataAsNumber`

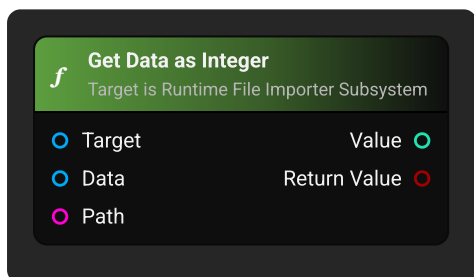
INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Data	Runtime Data Object Reference	The Runtime Data object.
Path	String	Path of a value inside the data, e.g. "player.inventory[2].name" (empty: the root).

OUTPUTS

Pin	Type	Description
Value	Float	The number.
Return Value	Boolean	The value at a path as a number (strings that are numbers and booleans converted).

Get Data as Integer PURE



The value at a path as an integer (rounded). False if it isn't a number.

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::GetDataAsInteger`

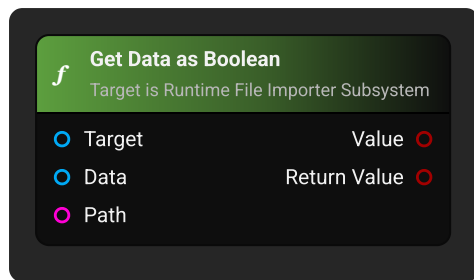
INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Data	Runtime Data Object Reference	The Runtime Data object.
Path	String	Path of a value inside the data, e.g. "player.inventory[2].name" (empty: the root).

OUTPUTS

Pin	Type	Description
Value	Integer	The integer.
Return Value	Boolean	The value at a path as an integer (rounded).

Get Data as Boolean PURE



The value at a path as a boolean (numbers, and strings like true/false/yes/no/1/0 converted). False if it isn't one.

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::GetDataAsBoolean`

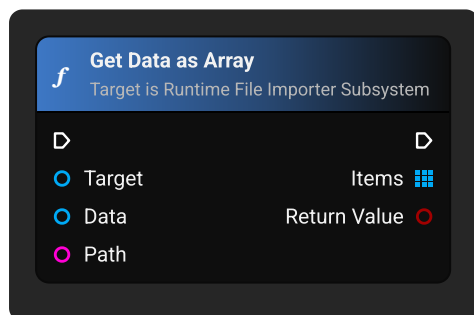
INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Data	Runtime Data Object Reference	The Runtime Data object.
Path	String	Path of a value inside the data, e.g. "player.inventory[2].name" (empty: the root).

OUTPUTS

Pin	Type	Description
Value	Boolean	The boolean value.
Return Value	Boolean	The value at a path as a boolean (numbers, and strings like true/false/yes/no/1/0 converted).

Get Data as Array FUNCTION



The items of the array at a path, each as Runtime Data sharing it (an object's field values for an object).

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::GetDataAsArray`

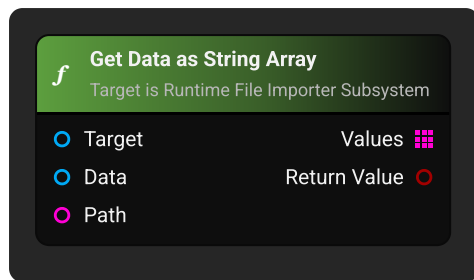
INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Data	Runtime Data Object Reference	The Runtime Data object.
Path	String	Path of a value inside the data, e.g. "player.inventory[2].name" (empty: the root).

OUTPUTS

Pin	Type	Description
Items	Array of Runtime Data Object Reference	The items of the array (or the values of the object) at Path, as data objects.
Return Value	Boolean	The items of the array at a path, each as Runtime Data sharing it (an object's field values for an object).

Get Data as String Array PURE



The items of the array at a path as text.

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::GetDataAsStringArray`

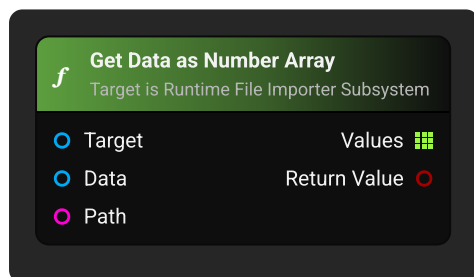
INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Data	Runtime Data Object Reference	The Runtime Data object.
Path	String	Path of a value inside the data, e.g. "player.inventory[2].name" (empty: the root).

OUTPUTS

Pin	Type	Description
Values	Array of String	The strings.
Return Value	Boolean	The items of the array at a path as text.

Get Data as Number Array PURE



The items of the array at a path as numbers (False if one isn't a number; it's 0 then).

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::GetDataAsNumberArray`

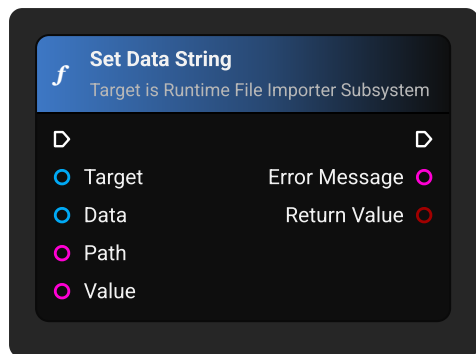
INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Data	Runtime Data Object Reference	The Runtime Data object.
Path	String	Path of a value inside the data, e.g. "player.inventory[2].name" (empty: the root).

OUTPUTS

Pin	Type	Description
Values	Array of Float	The numbers.
Return Value	Boolean	The items of the array at a path as numbers (False if one isn't a number; it's 0 then).

Set Data String FUNCTION



Stores a string at a path, creating the objects and arrays on the way (an index equal to an array's length adds an item).

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::SetDataString`

INPUTS

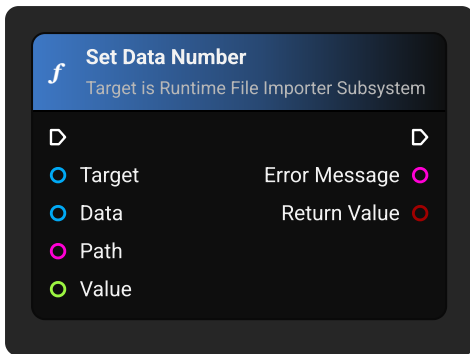
Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Data	Runtime Data Object Reference	The Runtime Data object.
Path	String	Path of a value inside the data, e.g. "player.inventory[2].name" (empty: the root).
Value	String	The value.

OUTPUTS

Pin	Type	Description
Error Message	String	Empty on success; otherwise why it failed.
Return Value	Boolean	True on success. False if it failed (Error Message says why).

Set Data Number

FUNCTION



Stores a number at a path (see Set Data String).

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::SetDataNumber`

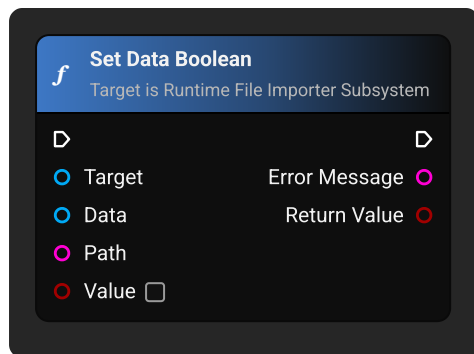
INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Data	Runtime Data Object Reference	The Runtime Data object.
Path	String	Path of a value inside the data, e.g. "player.inventory[2].name" (empty: the root).
Value	Float	The number.

OUTPUTS

Pin	Type	Description
Error Message	String	Empty on success; otherwise why it failed.
Return Value	Boolean	True on success. False if it failed (Error Message says why).

Set Data Boolean FUNCTION



Stores a boolean at a path (see Set Data String).

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::SetDataBoolean`

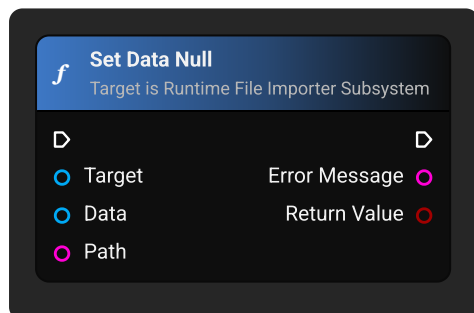
INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Data	Runtime Data Object Reference	The Runtime Data object.
Path	String	Path of a value inside the data, e.g. "player.inventory[2].name" (empty: the root).
Value	Boolean	The boolean value.

OUTPUTS

Pin	Type	Description
Error Message	String	Empty on success; otherwise why it failed.
Return Value	Boolean	True on success. False if it failed (Error Message says why).

Set Data Null FUNCTION



Stores null at a path (see Set Data String).

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::SetDataNull`

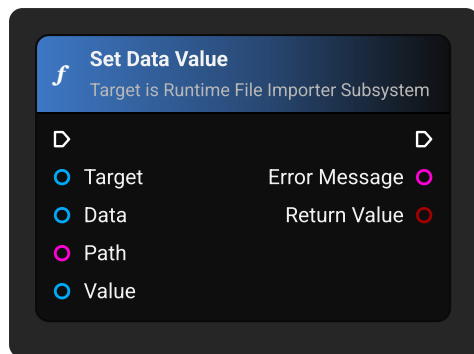
INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Data	Runtime Data Object Reference	The Runtime Data object.
Path	String	Path of a value inside the data, e.g. "player.inventory[2].name" (empty: the root).

OUTPUTS

Pin	Type	Description
Error Message	String	Empty on success; otherwise why it failed.
Return Value	Boolean	True on success. False if it failed (Error Message says why).

Set Data Value FUNCTION



Stores a copy of other Runtime Data (any value, object or array) at a path (see Set Data String).

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::SetDataValue`

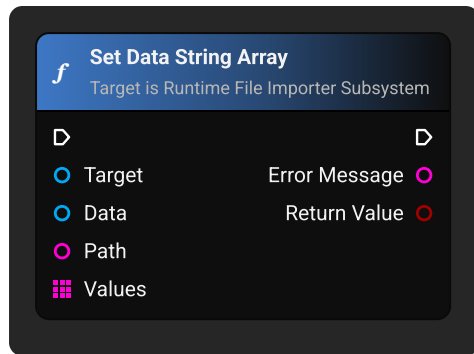
INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Data	Runtime Data Object Reference	The Runtime Data object.
Path	String	Path of a value inside the data, e.g. "player.inventory[2].name" (empty: the root).
Value	Runtime Data Object Reference	The data to put at Path.

OUTPUTS

Pin	Type	Description
Error Message	String	Empty on success; otherwise why it failed.
Return Value	Boolean	True on success. False if it failed (Error Message says why).

Set Data String Array FUNCTION



Stores an array of strings at a path (see Set Data String).

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::SetDataStringArray`

INPUTS

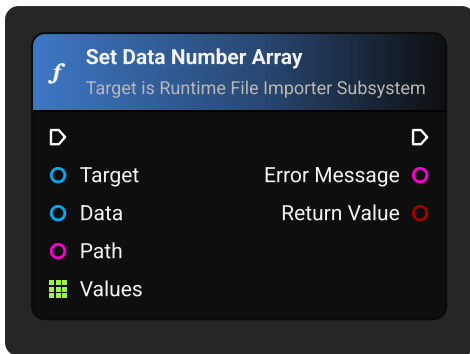
Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Data	Runtime Data Object Reference	The Runtime Data object.
Path	String	Path of a value inside the data, e.g. "player.inventory[2].name" (empty: the root).
Values	Array of String	The strings.

OUTPUTS

Pin	Type	Description
Error Message	String	Empty on success; otherwise why it failed.
Return Value	Boolean	True on success. False if it failed (Error Message says why).

Set Data Number Array

FUNCTION



Stores an array of numbers at a path (see Set Data String).

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::SetDataNumberArray`

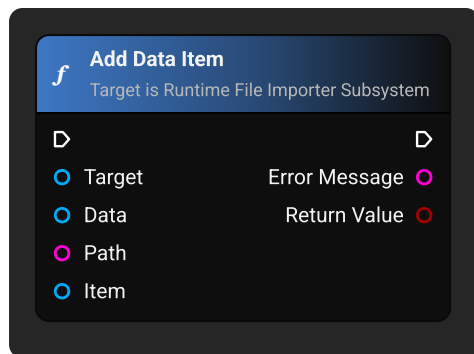
INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Data	Runtime Data Object Reference	The Runtime Data object.
Path	String	Path of a value inside the data, e.g. "player.inventory[2].name" (empty: the root).
Values	Array of Float	The numbers.

OUTPUTS

Pin	Type	Description
Error Message	String	Empty on success; otherwise why it failed.
Return Value	Boolean	True on success. False if it failed (Error Message says why).

Add Data Item FUNCTION



Adds a copy of Item to the end of the array at a path (made if there's nothing there).

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::AddDataItem`

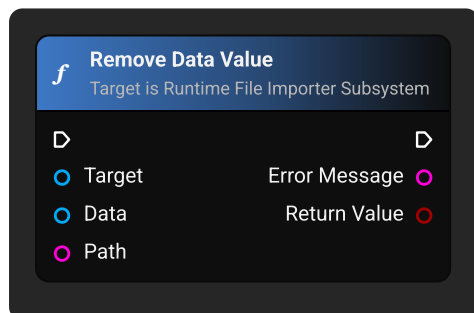
INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Data	Runtime Data Object Reference	The Runtime Data object.
Path	String	Path of a value inside the data, e.g. "player.inventory[2].name" (empty: the root).
Item	Runtime Data Object Reference	The data to add at the end of the array.

OUTPUTS

Pin	Type	Description
Error Message	String	Empty on success; otherwise why it failed.
Return Value	Boolean	True on success. False if it failed (Error Message says why).

Remove Data Value FUNCTION



Removes the field or array item at a path.

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::RemoveDataValue`

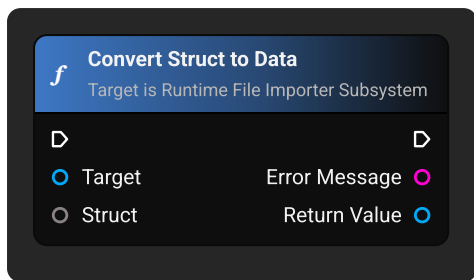
INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Data	Runtime Data Object Reference	The Runtime Data object.
Path	String	Path of a value inside the data, e.g. "player.inventory[2].name" (empty: the root).

OUTPUTS

Pin	Type	Description
Error Message	String	Empty on success; otherwise why it failed.
Return Value	Boolean	True on success. False if it failed (Error Message says why).

Convert Struct to Data FUNCTION



Makes Runtime Data of any struct (Blueprint or C++): an object with a field per member (named as in Blueprints), nested structs as objects, arrays and sets as arrays, maps as objects, enums as names, object references as paths. Export it as JSON to save the struct (Export Data To File / String).

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::ConvertStructToData`

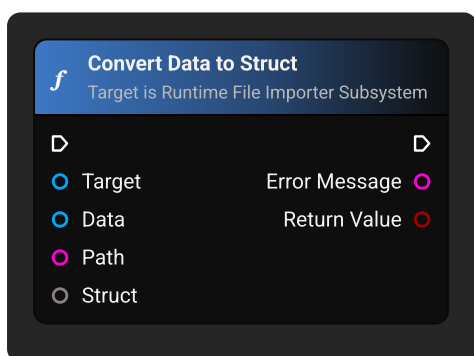
INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Struct	Any Struct	Any struct variable (Blueprint or C++). Its members become the fields of the data object.

OUTPUTS

Pin	Type	Description
Error Message	String	Empty on success; otherwise why it failed.
Return Value	Runtime Data Object Reference	The result, or None/empty if it failed (Error Message says why).

Convert Data to Struct FUNCTION



Fills a struct variable (Blueprint or C++) from the data at a path: fields are matched by member name (ignoring case), converting where it makes sense (numbers from strings, vectors from [x, y, z] or "(X=1,Y=2,Z=3)", enums from names, objects from paths). Members missing from the data keep their value. Returns false, listing the problems in Error Message, if some values couldn't be read (the others are still set).

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::ConvertDataToStruct`

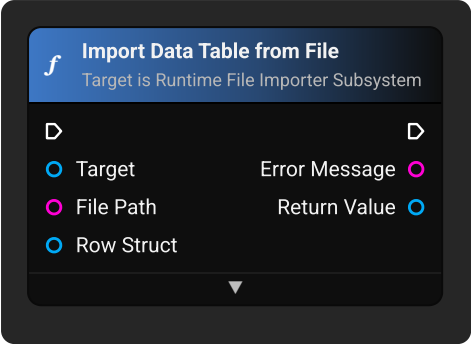
INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Data	Runtime Data Object Reference	The Runtime Data object.
Path	String	Path of the object inside the data to read from (empty: the root).
Struct	Any Struct	The struct variable to fill. Members are matched by name (ignoring case); members missing from the data keep their value.

OUTPUTS

Pin	Type	Description
Error Message	String	Empty on success; otherwise why it failed.
Return Value	Boolean	False, listing the problems in Error Message, if some values couldn't be read (the others are still set).

Import Data Table from File FUNCTION



Reads a data file (CSV or JSON as the editor exports them, TSV, XML, INI, Excel, OpenDocument) into a new Data Table of Row Struct, e.g. for moddable game data. Each row's name comes from Row Name Field (empty: a Name, RowName, ---, Key or ID column, else the first one); the other columns fill the members of the same name. Values that can't be read are logged as warnings and keep their default.

Target: Runtime File Importer Subsystem
Async version: [Import Data Table From File \(Async\)](#)
C++: URuntimeFileImporterSubsystem::ImportDataTableFromFile

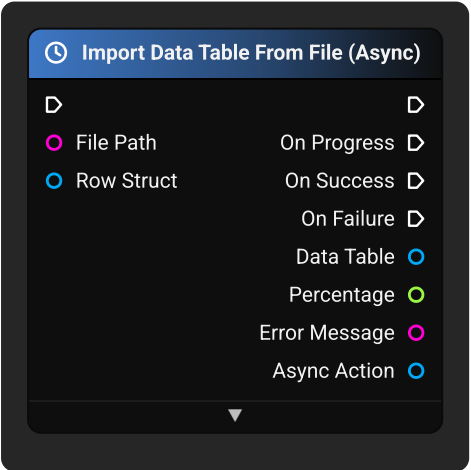
INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
File Path	String	Full path of the file on disk.
Row Struct	Script Struct Object Reference	Struct of the rows (the Data Table's row type).
Format ADVANCED	ERuntime Data Format	Format of the data. Auto Detect reads it from the contents.
Options ADVANCED OPTIONAL	Runtime Data Import Options	How structured data files are read into Runtime Data.
Row Name Field ADVANCED	String	Field (column) holding each row's name. Empty: "Name", "RowName" or "ID" if present, else the first field. Default: (empty)

OUTPUTS

Pin	Type	Description
Error Message	String	Empty on success; otherwise why it failed.
Return Value	Data Table Object Reference	The result, or None/empty if it failed (Error Message says why).

Import Data Table From File (Async) ASYNC



Reads a data file in the background, then fills a new Data Table of Row Struct with it (see Import Data Table From File).

Blocking version: [Import Data Table from File](#)
C++: UImportDataTableAsyncAction::ImportDataTableFromFileAsync

INPUTS

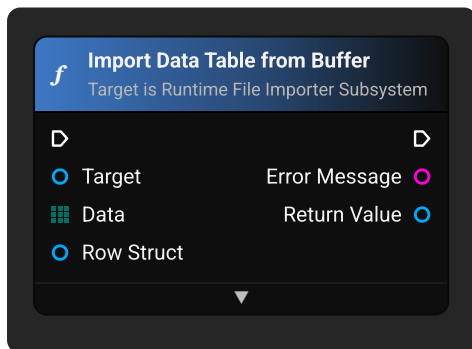
Pin	Type	Description
File Path	 String	Full path of the file on disk.
Row Struct	 Script Struct Object Reference	Struct of the rows (the Data Table's row type).
Format ADVANCED	 ERuntime Data Format	Format of the data. Auto Detect reads it from the contents.
Options ADVANCED OPTIONAL	 Runtime Data Import Options	How structured data files are read into Runtime Data.
Row Name Field ADVANCED	 String	Field (column) holding each row's name. Empty: "Name", "RowName" or "ID" if present, else the first field. Default: (empty)

OUTPUTS

Pin	Type	Description
On Progress	 Exec	Fires as the operation advances (Percentage from 0 to 100). Other outputs are not set yet.
On Success	 Exec	Fires when the operation has finished successfully.
On Failure	 Exec	Fires if the operation failed or was cancelled; Error Message says why.
Data Table	 Data Table Object Reference	The Data Table.
Percentage	 Float	Progress from 0 to 100.
Error Message	 String	Empty on success; otherwise why it failed.
Async Action	 Import Data Table Async Action Object Reference	This node's operation. Call Cancel on it to stop the operation (no more events fire).

Import Data Table from Buffer

FUNCTION



Reads data file bytes in memory into a new Data Table of Row Struct.

Target: Runtime File Importer Subsystem

Async version: [Import Data Table From Buffer \(Async\)](#)

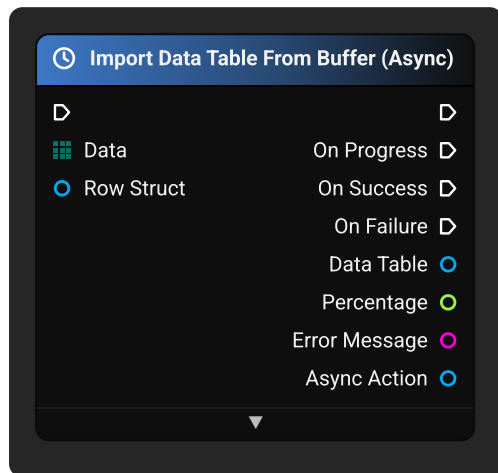
C++: `URuntimeFileImporterSubsystem::ImportDataTableFromBuffer`

INPUTS

Pin	Type	Description
Target	 Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Data	 Array of Byte	Bytes of a data file.
Row Struct	 Script Struct Object Reference	Struct of the rows (the Data Table's row type).
Format ADVANCED	 ERuntime Data Format	Format of the data. Auto Detect reads it from the contents.
Options ADVANCED OPTIONAL	 Runtime Data Import Options	How structured data files are read into Runtime Data.

Pin	Type	Description
Row Name Field ADVANCED	 String	Field (column) holding each row's name. Empty: "Name", "RowName" or "ID" if present, else the first field. Default: (empty)
OUTPUTS		
Pin	Type	Description
Error Message	 String	Empty on success; otherwise why it failed.
Return Value	 Data Table Object Reference	The result, or None/empty if it failed (Error Message says why).

Import Data Table From Buffer (Async) ASYNC



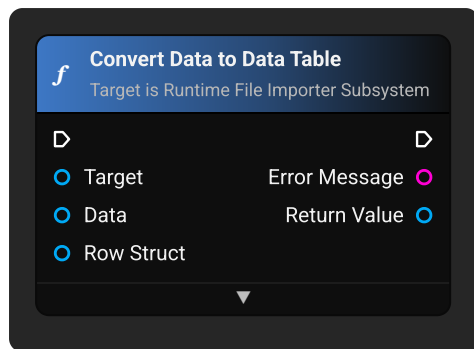
Reads data file bytes in the background, then fills a new Data Table of Row Struct with them.

Blocking version: [Import Data Table from Buffer](#)

C++: `UImportDataTableAsyncAction::ImportDataTableFromBufferAsync`

INPUTS		
Pin	Type	Description
Data	 Array of Byte	Bytes of a data file.
Row Struct	 Script Struct Object Reference	Struct of the rows (the Data Table's row type).
Format ADVANCED	 ERuntime Data Format	Format of the data. Auto Detect reads it from the contents.
Options ADVANCED OPTIONAL	 Runtime Data Import Options	How structured data files are read into Runtime Data.
Row Name Field ADVANCED	 String	Field (column) holding each row's name. Empty: "Name", "RowName" or "ID" if present, else the first field. Default: (empty)
OUTPUTS		
Pin	Type	Description
On Progress	 Exec	Fires as the operation advances (Percentage from 0 to 100). Other outputs are not set yet.
On Success	 Exec	Fires when the operation has finished successfully.
On Failure	 Exec	Fires if the operation failed or was cancelled; Error Message says why.
Data Table	 Data Table Object Reference	The Data Table.
Percentage	 Float	Progress from 0 to 100.
Error Message	 String	Empty on success; otherwise why it failed.
Async Action	 Import Data Table Async Action Object Reference	This node's operation. Call Cancel on it to stop the operation (no more events fire).

Convert Data to Data Table FUNCTION



A new Data Table of Row Struct from Runtime Data rows (an array of row objects, or an object of row objects keyed by row name).

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::ConvertDataToDataTable`

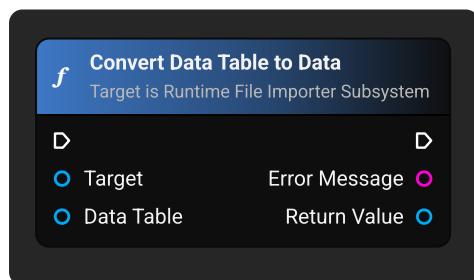
INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Data	Runtime Data Object Reference	The Runtime Data object.
Row Struct	Script Struct Object Reference	Struct of the rows (the Data Table's row type).
Row Name Field ADVANCED	String	Field (column) holding each row's name. Empty: "Name", "RowName" or "ID" if present, else the first field. Default: (empty)

OUTPUTS

Pin	Type	Description
Error Message	String	Empty on success; otherwise why it failed.
Return Value	Data Table Object Reference	The result, or None/empty if it failed (Error Message says why).

Convert Data Table to Data FUNCTION



A Data Table's rows as Runtime Data: an array of objects, each with the row's Name and its members.

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::ConvertDataTableToData`

INPUTS

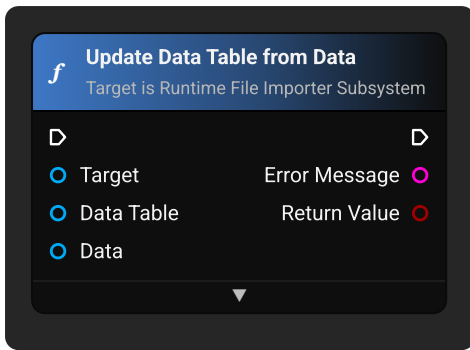
Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Data Table	Data Table Object Reference	The Data Table.

OUTPUTS

Pin	Type	Description
Error Message	String	Empty on success; otherwise why it failed.
Return Value	Runtime Data Object Reference	The result, or None/empty if it failed (Error Message says why).

Update Data Table from Data

FUNCTION



Changes an existing Data Table (an asset too, until the game ends) from Runtime Data rows: rows with the same name are updated (members missing from the data keep their value), others added. Remove Other Rows: the table holds only the data's rows.

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::UpdateDataTableFromData`

INPUTS

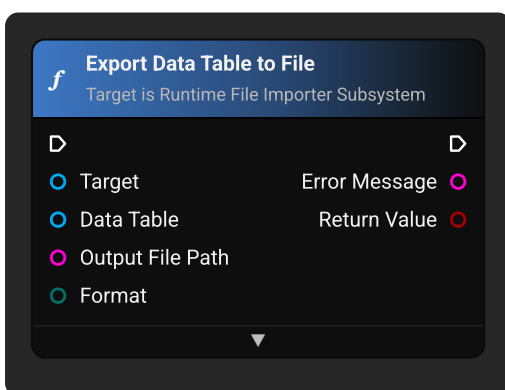
Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Data Table	Data Table Object Reference	The Data Table.
Data	Runtime Data Object Reference	The Runtime Data object.
Remove Other Rows	Boolean	Remove rows that are not in the data. Default: <code>false</code>
Row Name Field	String	Field (column) holding each row's name. Empty: "Name", "RowName" or "ID" if present, else the first field. Default: <code>(empty)</code>

OUTPUTS

Pin	Type	Description
Error Message	String	Empty on success; otherwise why it failed.
Return Value	Boolean	True on success. False if it failed (Error Message says why).

Export Data Table to File

FUNCTION



Writes a Data Table to a file: CSV and TSV in the layout the editor imports (row names under "---"), JSON as an array of rows, or XML / INI.

Target: Runtime File Importer Subsystem

Async version: [Export Data Table To File \(Async\)](#)

C++: `URuntimeFileImporterSubsystem::ExportDataTableToFile`

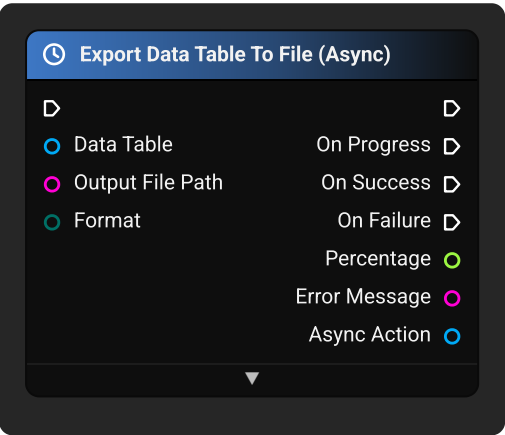
INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Data Table	Data Table Object Reference	The Data Table.

Pin	Type	Description
Output File Path	 String	Full path of the file to write. Missing folders are created; an existing file is replaced.
Format	 ERuntime Data Export Format	Format to write.
Options ADVANCED OPTIONAL	 Runtime Data Export Options	How Runtime Data is written.
OUTPUTS		
Pin	Type	Description
Error Message	 String	Empty on success; otherwise why it failed.
Return Value	 Boolean	True on success. False if it failed (Error Message says why).

Export Data Table To File (Async)

ASYNC



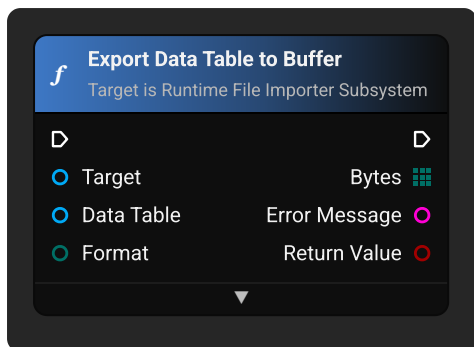
Writes a Data Table to a file (its rows are read when the node starts, written in the background).

Blocking version: [Export Data Table to File](#)

C++: `UExportTextAsyncAction::ExportDataTableToFileAsync`

INPUTS		
Pin	Type	Description
Data Table	 Data Table Object Reference	The Data Table.
Output File Path	 String	Full path of the file to write. Missing folders are created; an existing file is replaced.
Format	 ERuntime Data Export Format	Format to write.
Options ADVANCED OPTIONAL	 Runtime Data Export Options	How Runtime Data is written.
OUTPUTS		
Pin	Type	Description
On Progress	 Exec	Fires as the operation advances (Percentage from 0 to 100). Other outputs are not set yet.
On Success	 Exec	Fires when the operation has finished successfully.
On Failure	 Exec	Fires if the operation failed or was cancelled; Error Message says why.
Percentage	 Float	Progress from 0 to 100.
Error Message	 String	Empty on success; otherwise why it failed.
Async Action	 Export Text Async Action Object Reference	This node's operation. Call Cancel on it to stop the operation (no more events fire).

Export Data Table to Buffer FUNCTION



Encodes a Data Table into file bytes.

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::ExportDataTableToBuffer`

INPUTS

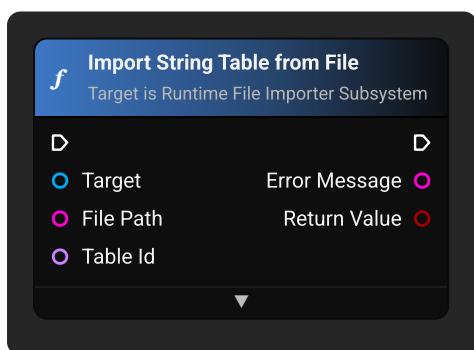
Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Data Table	Data Table Object Reference	The Data Table.
Format	ERuntime Data Export Format	Format to write.
Options	ADVANCED OPTIONAL Runtime Data Export Options	How Runtime Data is written.

OUTPUTS

Pin	Type	Description
Bytes	Array of Byte	The bytes.
Error Message	String	Empty on success; otherwise why it failed.
Return Value	Boolean	True on success. False if it failed (Error Message says why).

Localization 6 nodes

Import String Table from File FUNCTION



Registers a String Table from a file, e.g. translations or mod text: CSV / spreadsheet rows with Key and SourceString columns (and an optional Comment), or a JSON object of "key": "text". Use it with the engine's Text From String Table (Table Id, Key) or Make Literal Text. Importing again into the same Table Id updates it (Replace Entries: removes the keys the file doesn't have). Table Ids of String Table assets can't be used.

Target: Runtime File Importer Subsystem

Async version: [Import String Table From File \(Async\)](#)

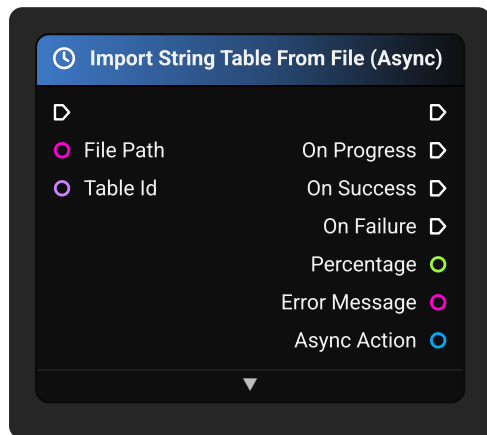
C++: `URuntimeFileImporterSubsystem::ImportStringTableFromFile`

INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
File Path	String	Full path of the file on disk.
Table Id	Name	ID of the String Table (used by Text values that refer to it).
Namespace	ADVANCED String	Namespace of the table's entries. Default: (empty)

Pin	Type	Description
Replace Entries ADVANCED	 Boolean	Replace the table's existing entries (otherwise they are added to). Default: <code>true</code>
OUTPUTS		
Pin	Type	Description
Error Message	 String	Empty on success; otherwise why it failed.
Return Value	 Boolean	True on success. False if it failed (Error Message says why).

Import String Table From File (Async) ASYNC



Reads a String Table file (CSV, spreadsheet or JSON) in the background, then registers it under Table Id.

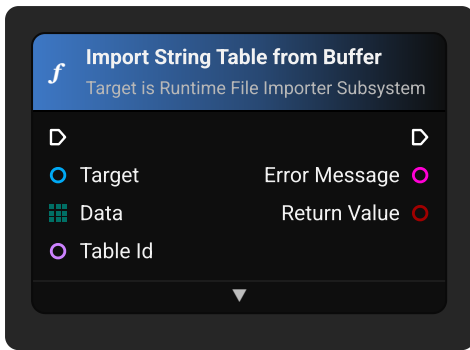
Blocking version: [Import String Table from File](#)

C++: `UImportStringTableAsyncAction::ImportStringTableFromFileAsync`

INPUTS		
Pin	Type	Description
File Path	 String	Full path of the file on disk.
Table Id	 Name	ID of the String Table (used by Text values that refer to it).
Namespace ADVANCED	 String	Namespace of the table's entries. Default: <code>(empty)</code>
Replace Entries ADVANCED	 Boolean	Replace the table's existing entries (otherwise they are added to). Default: <code>true</code>
OUTPUTS		
Pin	Type	Description
On Progress	 Exec	Fires as the operation advances (Percentage from 0 to 100). Other outputs are not set yet.
On Success	 Exec	Fires when the operation has finished successfully.
On Failure	 Exec	Fires if the operation failed or was cancelled; Error Message says why.
Percentage	 Float	Progress from 0 to 100.
Error Message	 String	Empty on success; otherwise why it failed.
Async Action	 Import String Table Async Action Object Reference	This node's operation. Call Cancel on it to stop the operation (no more events fire).

Import String Table from Buffer

FUNCTION



Registers a String Table from data file bytes in memory (see Import String Table From File).

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::ImportStringTableFromBuffer`

INPUTS

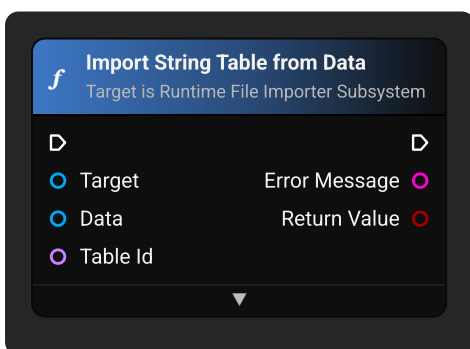
Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Data	Array of Byte	Bytes of a data file.
Table Id	Name	ID of the String Table (used by Text values that refer to it).
Format ADVANCED	ERuntime Data Format	Format of the data. Auto Detect reads it from the contents. Default: <code>Auto Detect</code>
Namespace ADVANCED	String	Namespace of the table's entries. Default: <code>(empty)</code>
Replace Entries ADVANCED	Boolean	Replace the table's existing entries (otherwise they are added to). Default: <code>true</code>

OUTPUTS

Pin	Type	Description
Error Message	String	Empty on success; otherwise why it failed.
Return Value	Boolean	True on success. False if it failed (Error Message says why).

Import String Table from Data

FUNCTION



Registers a String Table from Runtime Data (an object of key: text, or rows with Key and SourceString).

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::ImportStringTableFromData`

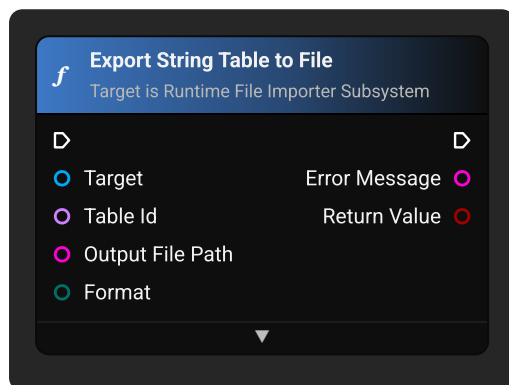
INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Data	Runtime Data Object Reference	The Runtime Data object.
Table Id	Name	ID of the String Table (used by Text values that refer to it).
Namespace ADVANCED	String	Namespace of the table's entries. Default: <code>(empty)</code>

Pin	Type	Description
Replace Entries	ADVANCED  Boolean	Replace the table's existing entries (otherwise they are added to). Default: <code>true</code>
OUTPUTS		
Pin	Type	Description
Error Message	 String	Empty on success; otherwise why it failed.
Return Value	 Boolean	True on success. False if it failed (Error Message says why).

Export String Table to File

FUNCTION



Writes a registered String Table (an asset or an imported one) to a file: rows of Key, SourceString and Comment (CSV: the layout the editor imports).

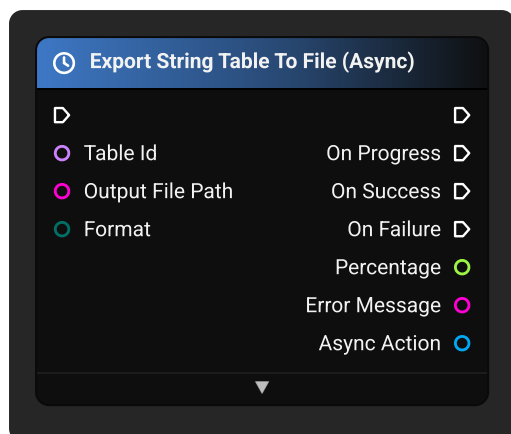
Target: Runtime File Importer Subsystem

Async version: [Export String Table To File \(Async\)](#)

C++: `URuntimeFileImporterSubsystem::ExportStringTableToFile`

INPUTS		
Pin	Type	Description
Target	 Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Table Id	 Name	ID of the String Table (used by Text values that refer to it).
Output File Path	 String	Full path of the file to write. Missing folders are created; an existing file is replaced.
Format	 ERuntime Data Export Format	Format to write.
Options	ADVANCED OPTIONAL  Runtime Data Export Options	How Runtime Data is written.
OUTPUTS		
Pin	Type	Description
Error Message	 String	Empty on success; otherwise why it failed.
Return Value	 Boolean	True on success. False if it failed (Error Message says why).

Export String Table To File (Async) ASYNC



Writes a registered String Table to a file without blocking the game thread.

Blocking version: [Export String Table to File](#)

C++: `UExportTextAsyncAction::ExportStringTableToFileAsync`

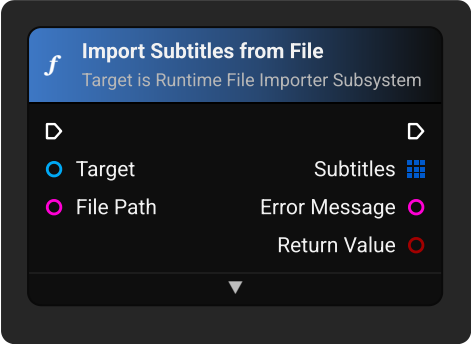
INPUTS

Pin	Type	Description
Table Id	Name	ID of the String Table (used by Text values that refer to it).
Output File Path	String	Full path of the file to write. Missing folders are created; an existing file is replaced.
Format	ERuntime Data Export Format	Format to write.
Options ADVANCED OPTIONAL	Runtime Data Export Options	How Runtime Data is written.

OUTPUTS

Pin	Type	Description
On Progress	Exec	Fires as the operation advances (Percentage from 0 to 100). Other outputs are not set yet.
On Success	Exec	Fires when the operation has finished successfully.
On Failure	Exec	Fires if the operation failed or was cancelled; Error Message says why.
Percentage	Float	Progress from 0 to 100.
Error Message	String	Empty on success; otherwise why it failed.
Async Action	Export Text Async Action Object Reference	This node's operation. Call Cancel on it to stop the operation (no more events fire).

Import Subtitles from File FUNCTION



Reads a subtitle file (SubRip SRT, WebVTT, SubStation Alpha ASS/SSA) as cues: start and end times and text (formatting tags removed).

Target: Runtime File Importer Subsystem

Async version: [Import Subtitles From File \(Async\)](#)

C++: `URuntimeFileImporterSubsystem::ImportSubtitlesFromFile`

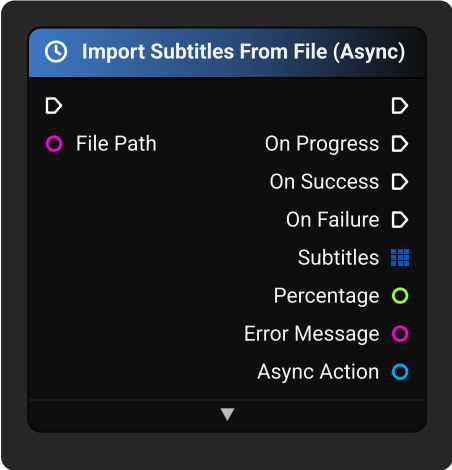
INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
File Path	String	Full path of the file on disk.
Format ADVANCED	ERuntime Subtitle Format	Format of the subtitles. Auto Detect reads it from the contents. Default: Auto Detect

OUTPUTS

Pin	Type	Description
Subtitles	Array of Runtime Subtitle Cue	The subtitle cues, in order.
Error Message	String	Empty on success; otherwise why it failed.
Return Value	Boolean	True on success. False if it failed (Error Message says why).

Import Subtitles From File (Async) ASYNC



Reads a subtitle file (SRT, WebVTT, ASS/SSA) without blocking the game thread.

Blocking version: [Import Subtitles from File](#)

C++: `UImportSubtitlesAsyncAction::ImportSubtitlesFromFileAsync`

INPUTS

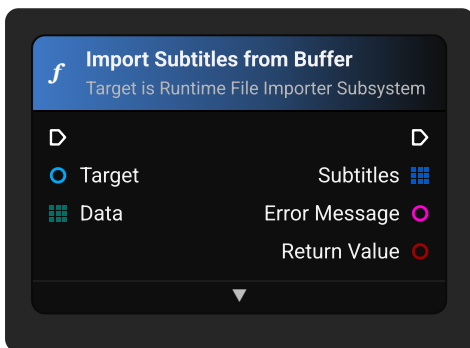
Pin	Type	Description
File Path	String	Full path of the file on disk.
Format ADVANCED	ERuntime Subtitle Format	Format of the subtitles. Auto Detect reads it from the contents. Default: Auto Detect

OUTPUTS

Pin	Type	Description
On Progress	 Exec	Fires as the operation advances (Percentage from 0 to 100). Other outputs are not set yet.
On Success	 Exec	Fires when the operation has finished successfully.
On Failure	 Exec	Fires if the operation failed or was cancelled; Error Message says why.
Subtitles	 Array of Runtime Subtitle Cue	The subtitle cues, in order.
Percentage	 Float	Progress from 0 to 100.
Error Message	 String	Empty on success; otherwise why it failed.
Async Action	 Import Subtitles Async Action Object Reference	This node's operation. Call Cancel on it to stop the operation (no more events fire).

Import Subtitles from Buffer

FUNCTION



Reads subtitles from file bytes in memory.

Target: Runtime File Importer Subsystem

Async version: [Import Subtitles From Buffer \(Async\)](#)

C++: `URuntimeFileImporterSubsystem::ImportSubtitlesFromBuffer`

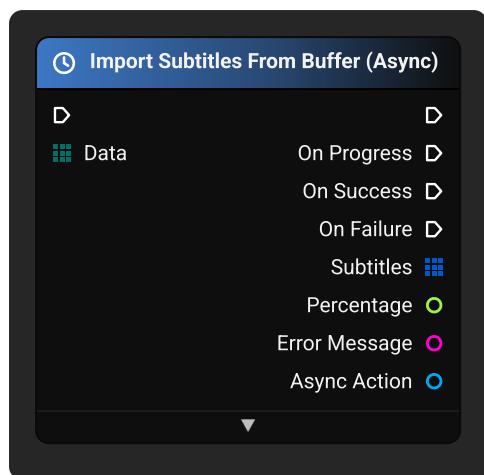
INPUTS

Pin	Type	Description
Target	 Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Data	 Array of Byte	Bytes of a data file.
Format	 ERuntime Subtitle Format	Format of the subtitles. Auto Detect reads it from the contents. Default: Auto Detect

OUTPUTS

Pin	Type	Description
Subtitles	 Array of Runtime Subtitle Cue	The subtitle cues, in order.
Error Message	 String	Empty on success; otherwise why it failed.
Return Value	 Boolean	True on success. False if it failed (Error Message says why).

Import Subtitles From Buffer (Async) ASYNC



Reads subtitles from file bytes in memory without blocking the game thread.

Blocking version: [Import Subtitles from Buffer](#)

C++: `UImportSubtitlesAsyncAction::ImportSubtitlesFromBufferAsync`

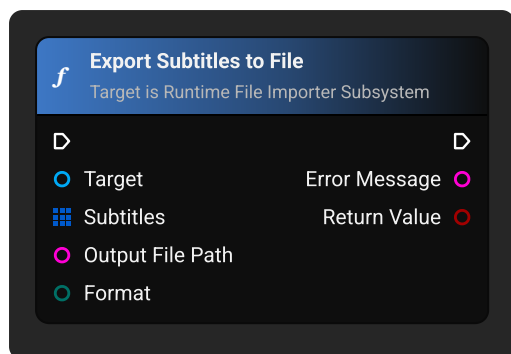
INPUTS

Pin	Type	Description
Data	Array of Byte	Bytes of a data file.
Format ADVANCED	ERuntime Subtitle Format	Format of the subtitles. Auto Detect reads it from the contents. Default: <code>Auto Detect</code>

OUTPUTS

Pin	Type	Description
On Progress	Exec	Fires as the operation advances (Percentage from 0 to 100). Other outputs are not set yet.
On Success	Exec	Fires when the operation has finished successfully.
On Failure	Exec	Fires if the operation failed or was cancelled; Error Message says why.
Subtitles	Array of Runtime Subtitle Cue	The subtitle cues, in order.
Percentage	Float	Progress from 0 to 100.
Error Message	String	Empty on success; otherwise why it failed.
Async Action	Import Subtitles Async Action Object Reference	This node's operation. Call Cancel on it to stop the operation (no more events fire).

Export Subtitles to File FUNCTION



Writes subtitles to a file (UTF-8).

Target: Runtime File Importer Subsystem

Async version: [Export Subtitles To File \(Async\)](#)

C++: `URuntimeFileImporterSubsystem::ExportSubtitlesToFile`

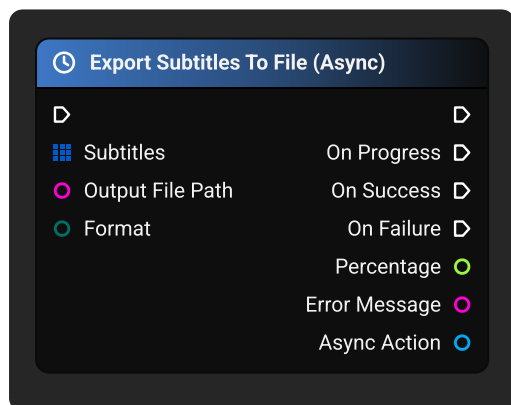
INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Subtitles	Array of Runtime Subtitle Cue	The subtitle cues, in order.
Output File Path	String	Full path of the file to write. Missing folders are created; an existing file is replaced.
Format	ERuntime Subtitle Export Format	Format of the subtitles. Auto Detect reads it from the contents.

OUTPUTS

Pin	Type	Description
Error Message	String	Empty on success; otherwise why it failed.
Return Value	Boolean	True on success. False if it failed (Error Message says why).

Export Subtitles To File (Async) ASYNC



Writes subtitles to a file without blocking the game thread.

Blocking version: [Export Subtitles to File](#)

C++: `UExportTextAsyncAction::ExportSubtitlesToFileAsync`

INPUTS

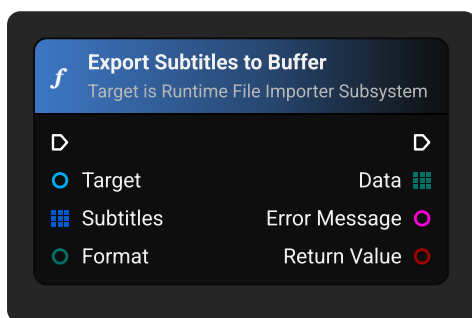
Pin	Type	Description
Subtitles	Array of Runtime Subtitle Cue	The subtitle cues, in order.
Output File Path	String	Full path of the file to write. Missing folders are created; an existing file is replaced.
Format	ERuntime Subtitle Export Format	Format of the subtitles. Auto Detect reads it from the contents.

OUTPUTS

Pin	Type	Description
On Progress	 Exec	Fires as the operation advances (Percentage from 0 to 100). Other outputs are not set yet.
On Success	 Exec	Fires when the operation has finished successfully.
On Failure	 Exec	Fires if the operation failed or was cancelled; Error Message says why.
Percentage	 Float	Progress from 0 to 100.
Error Message	 String	Empty on success; otherwise why it failed.
Async Action	 Export Text Async Action Object Reference	This node's operation. Call Cancel on it to stop the operation (no more events fire).

Export Subtitles to Buffer

FUNCTION



Encodes subtitles into file bytes (UTF-8).

Target: Runtime File Importer Subsystem

Async version: [Export Subtitles To Buffer \(Async\)](#)

C++: `URuntimeFileImporterSubsystem::ExportSubtitlesToBuffer`

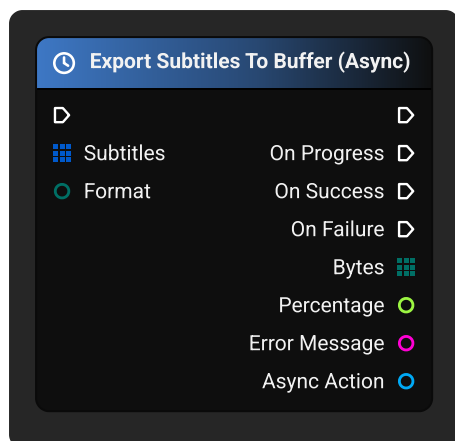
INPUTS

Pin	Type	Description
Target	 Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Subtitles	 Array of Runtime Subtitle Cue	The subtitle cues, in order.
Format	 ERuntime Subtitle Export Format	Format of the subtitles. Auto Detect reads it from the contents.

OUTPUTS

Pin	Type	Description
Data	 Array of Byte	Bytes of a data file.
Error Message	 String	Empty on success; otherwise why it failed.
Return Value	 Boolean	True on success. False if it failed (Error Message says why).

Export Subtitles To Buffer (Async) ASYNC



Encodes subtitles into file bytes without blocking the game thread.

Blocking version: [Export Subtitles to Buffer](#)

C++: `UEncodeTextAsyncAction::ExportSubtitlesToBufferAsync`

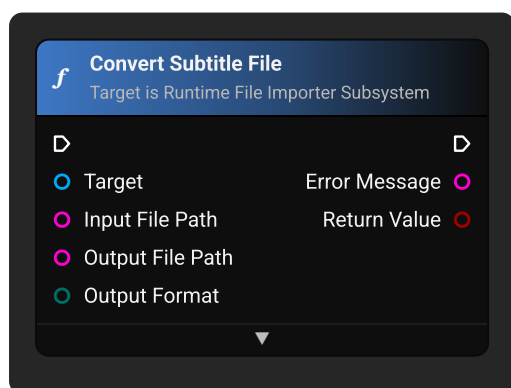
INPUTS

Pin	Type	Description
Subtitles	Array of Runtime Subtitle Cue	The subtitle cues, in order.
Format	ERuntime Subtitle Export Format	Format of the subtitles. Auto Detect reads it from the contents.

OUTPUTS

Pin	Type	Description
On Progress	Exec	Fires as the operation advances (Percentage from 0 to 100). Other outputs are not set yet.
On Success	Exec	Fires when the operation has finished successfully.
On Failure	Exec	Fires if the operation failed or was cancelled; Error Message says why.
Bytes	Array of Byte	The bytes.
Percentage	Float	Progress from 0 to 100.
Error Message	String	Empty on success; otherwise why it failed.
Async Action	Encode Text Async Action Object Reference	This node's operation. Call Cancel on it to stop the operation (no more events fire).

Convert Subtitle File FUNCTION



Converts a subtitle file to another format (e.g. SRT to WebVTT).

Target: Runtime File Importer Subsystem

Async version: [Convert Subtitle File \(Async\)](#)

C++: `URuntimeFileImporterSubsystem::ConvertSubtitleFile`

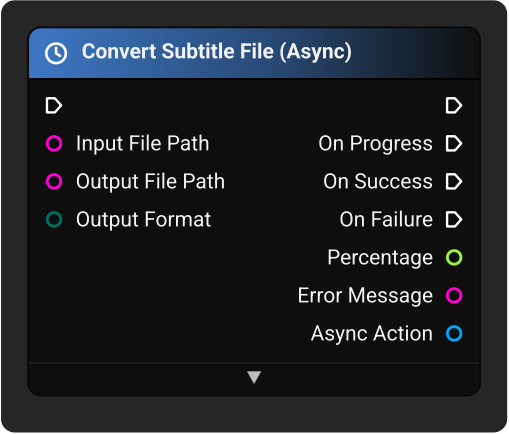
INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.

Pin	Type	Description
Input File Path	 String	Full path of the file to convert.
Output File Path	 String	Full path of the file to write. Missing folders are created; an existing file is replaced.
Output Format	 ERuntime Subtitle Export Format	Format to write.
Input Format ADVANCED	 ERuntime Subtitle Format	Format of the input. Auto Detect reads it from the contents. Default: Auto Detect
OUTPUTS		
Pin	Type	Description
Error Message	 String	Empty on success; otherwise why it failed.
Return Value	 Boolean	True on success. False if it failed (Error Message says why).

Convert Subtitle File (Async)

ASYNC



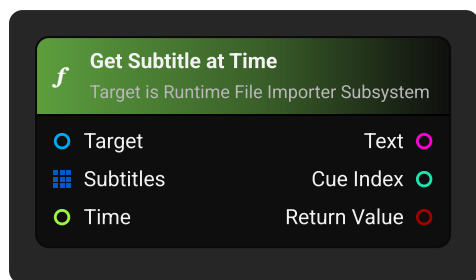
Converts a subtitle file to another format without blocking the game thread.

Blocking version: [Convert Subtitle File](#)

C++: `UExportTextAsyncAction::ConvertSubtitleFileAsync`

INPUTS		
Pin	Type	Description
Input File Path	 String	Full path of the file to convert.
Output File Path	 String	Full path of the file to write. Missing folders are created; an existing file is replaced.
Output Format	 ERuntime Subtitle Export Format	Format to write.
Input Format ADVANCED	 ERuntime Subtitle Format	Format of the input. Auto Detect reads it from the contents. Default: Auto Detect
OUTPUTS		
Pin	Type	Description
On Progress	 Exec	Fires as the operation advances (Percentage from 0 to 100). Other outputs are not set yet.
On Success	 Exec	Fires when the operation has finished successfully.
On Failure	 Exec	Fires if the operation failed or was cancelled; Error Message says why.
Percentage	 Float	Progress from 0 to 100.
Error Message	 String	Empty on success; otherwise why it failed.
Async Action	 Export Text Async Action Object Reference	This node's operation. Call Cancel on it to stop the operation (no more events fire).

Get Subtitle at Time PURE



The subtitle text shown at Time (seconds), e.g. a Runtime Video's Time each tick into a text widget; overlapping cues are on separate lines. Cue Index: the first one shown. Returns false (empty text) between subtitles.

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::GetSubtitleAtTime`

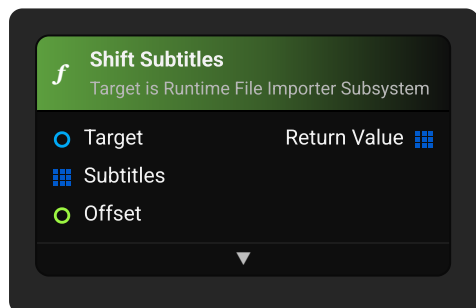
INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Subtitles	Array of Runtime Subtitle Cue	The subtitle cues, in order.
Time	Float	Time in seconds.

OUTPUTS

Pin	Type	Description
Text	String	The text.
Cue Index	Integer	Index of the cue shown at Time in Subtitles, or -1 if none is.
Return Value	Boolean	False (empty text) between subtitles.

Shift Subtitles PURE



Moves subtitles in time (Offset seconds, negative: earlier) and stretches them (Speed above 1: faster), e.g. to sync them to a video.

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::ShiftSubtitles`

INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Subtitles	Array of Runtime Subtitle Cue	The subtitle cues, in order.
Offset	Float	Seconds added to every start and end time (negative: earlier).
Speed ADVANCED	Float	Multiplies every time, e.g. 25/23.976 to fix subtitles timed for another frame rate. Default: 1.0

OUTPUTS

Pin	Type	Description
Return Value	Array of Runtime Subtitle Cue	Moves subtitles in time (Offset seconds, negative: earlier) and stretches them (Speed above 1: faster), e.g.

On Text Imported

EVENT

On Text Imported

Event of Runtime File Importer Subsystem

Object

Source

Any text, data, Data Table, String Table or subtitle import succeeded. Object is the Runtime Data or Data Table made (none for text, lines, string tables and subtitles); Source the file path or URL (empty for buffers and strings).

Bind it with "Bind Event to On Text Imported" (or "Assign On Text Imported") on the Runtime File Importer Subsystem.

```
C++: URuntimeFileImporterSubsystem::OnTextImported
```

OUTPUTS

Pin	Type	Description
Object	Object Object Reference	Any supported object (detected automatically).
Source	String	File path or URL the import came from (empty for buffers and strings).

On Text Import Failed

EVENT

On Text Import Failed

Event of Runtime File Importer Subsystem

Source

Error Message

Any text, data, Data Table, String Table or subtitle import failed.

Bind it with "Bind Event to On Text Import Failed" (or "Assign On Text Import Failed") on the Runtime File Importer Subsystem.

```
C++: URuntimeFileImporterSubsystem::OnTextImportFailed
```

OUTPUTS

Pin	Type	Description
Source	String	File path or URL the import came from (empty for buffers and strings).
Error Message	String	Empty on success; otherwise why it failed.

On Text Exported

EVENT

On Text Exported

Event of Runtime File Importer Subsystem

Destination

Any text, data, Data Table, String Table or subtitle export (or file conversion) succeeded. Destination is the output file path (empty for buffers).

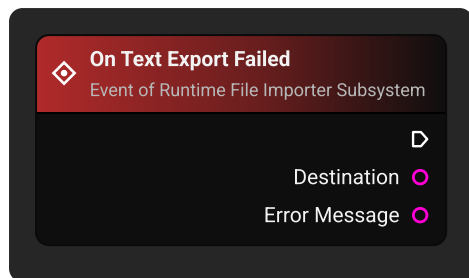
Bind it with "Bind Event to On Text Exported" (or "Assign On Text Exported") on the Runtime File Importer Subsystem.

```
C++: URuntimeFileImporterSubsystem::OnTextExported
```

OUTPUTS

Pin	Type	Description
Destination	String	Output file path (empty when writing to a buffer).

On Text Export Failed EVENT



Any text export or conversion failed.

Bind it with "Bind Event to On Text Export Failed" (or "Assign On Text Export Failed") on the Runtime File Importer Subsystem.

```
C++: URuntimeFileImporterSubsystem::OnTextExportFailed
```

OUTPUTS

Pin	Type	Description
Destination	 String	Output file path (empty when writing to a buffer).
Error Message	 String	Empty on success; otherwise why it failed.

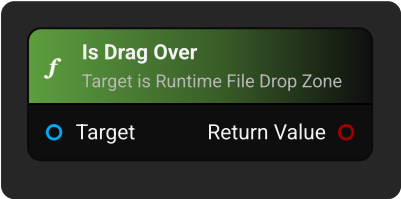
File

The system Open, Save and Folder dialogs, the Runtime File Drop Zone widget for files dragged onto the game window, loading, saving and downloading raw bytes, and file information that tells which nodes can import a file.

Drop Zone 2 nodes

Is Drag Over

PURE



True while accepted files are dragged over the area.

Target: Runtime File Drop Zone

C++: URuntimeFileDropZone::IsDragOver

INPUTS

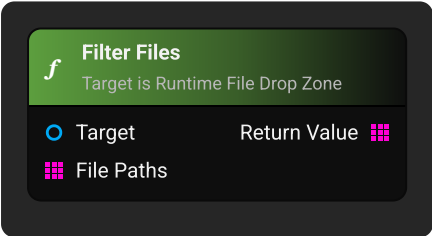
Pin	Type	Description
Target	Runtime File Drop Zone Object Reference	The Runtime File Drop Zone to use.

OUTPUTS

Pin	Type	Description
Return Value	Boolean	True while accepted files are dragged over the area.

Filter Files

PURE



The files of a list this area accepts (Allowed File Type, Extensions, Folders and Multiple Files applied).

Target: Runtime File Drop Zone

C++: URuntimeFileDropZone::FilterFiles

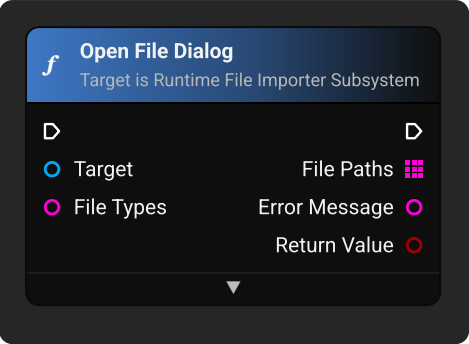
INPUTS

Pin	Type	Description
Target	Runtime File Drop Zone Object Reference	The Runtime File Drop Zone to use.
File Paths	Array of String	Full paths of the files.

OUTPUTS

Pin	Type	Description
Return Value	Array of String	The files of a list this area accepts (Allowed File Type, Extensions, Folders and Multiple Files applied).

Open File Dialog FUNCTION



Shows the system's Open dialog so the player can choose files to import (Windows). File Types limits what's shown: "Description|*.ext;*.ext2" pairs joined with | (Get File Dialog Filter makes them; empty: all files). The game pauses while the dialog is open; the Async version keeps it running. Returns false if the dialog was cancelled.

Target: Runtime File Importer Subsystem
Async version: [Open File Dialog \(Async\)](#)
C++: `URuntimeFileImporterSubsystem::OpenFileDialog`

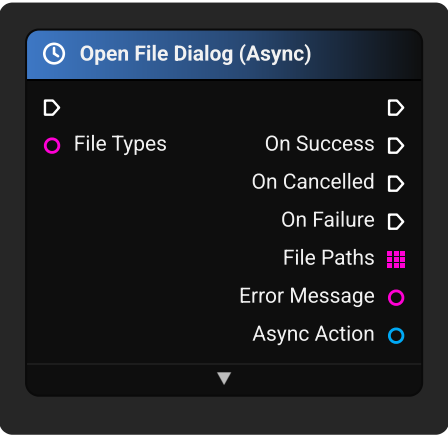
INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
File Types	String	Filter of the dialog, e.g. "Images *.png;*.jpg All Files *.*" (Make File Dialog Filter / Get File Dialog Filter build one). Empty: all files. Default: (empty)
Title ADVANCED	String	Title of the dialog window. Empty: the system default. Default: (empty)
Default Path ADVANCED	String	Folder the dialog opens in. Empty: the last one used. Default: (empty)
Allow Multiple ADVANCED	Boolean	Let the player choose more than one file. Default: false

OUTPUTS

Pin	Type	Description
File Paths	Array of String	Full paths of the files.
Error Message	String	Empty on success; otherwise why it failed.
Return Value	Boolean	False if the dialog was cancelled.

Open File Dialog (Async) ASYNC



Shows the system's Open dialog (Windows) while the game keeps running. File Types: see Open File Dialog.

Blocking version: [Open File Dialog](#)
C++: `UOpenFileDialogAsyncAction::OpenFileDialogAsync`

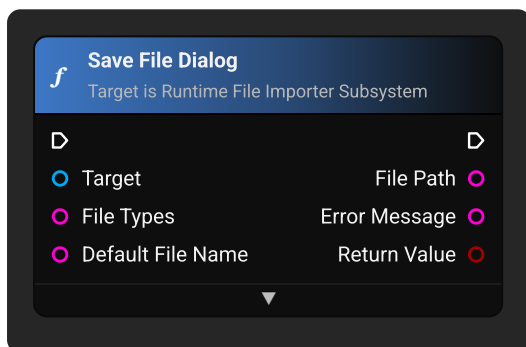
INPUTS

Pin	Type	Description
File Types	String	Filter of the dialog, e.g. "Images *.png;*.jpg All Files *.*" (Make File Dialog Filter / Get File Dialog Filter build one). Empty: all files. Default: (empty)

Pin	Type	Description
Title ADVANCED	String	Title of the dialog window. Empty: the system default. Default: (empty)
Default Path ADVANCED	String	Folder the dialog opens in. Empty: the last one used. Default: (empty)
Allow Multiple ADVANCED	Boolean	Let the player choose more than one file. Default: false
OUTPUTS		
Pin	Type	Description
On Success	Exec	Fires when the operation has finished successfully.
On Cancelled	Exec	The player closed the dialog without choosing.
On Failure	Exec	Fires if the operation failed or was cancelled; Error Message says why.
File Paths	Array of String	Full paths of the files.
Error Message	String	Empty on success; otherwise why it failed.
Async Action	Open File Dialog Async Action Object Reference	This node's operation. Call Cancel on it to stop the operation (no more events fire).

Save File Dialog

FUNCTION



Shows the system's Save dialog (Windows): the chosen type's extension is added if the name typed has none. Returns false if cancelled.

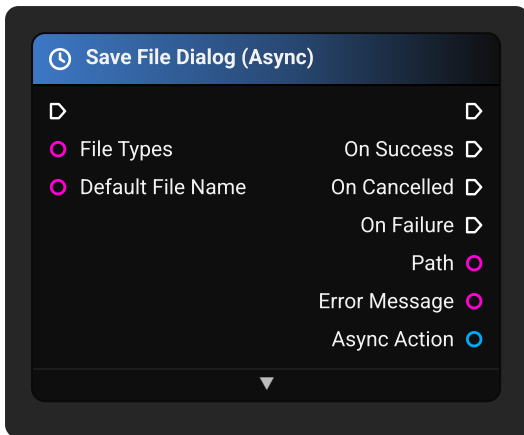
Target: Runtime File Importer Subsystem

Async version: [Save File Dialog \(Async\)](#)

C++: `URuntimeFileImporterSubsystem::SaveFileDialog`

INPUTS		
Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
File Types	String	Filter of the dialog, e.g. "Images .png;*.jpg All Files *.*" (Make File Dialog Filter / Get File Dialog Filter build one). Empty: all files. Default: (empty)
Default File Name	String	File name suggested in the dialog. Default: (empty)
Title ADVANCED	String	Title of the dialog window. Empty: the system default. Default: (empty)
Default Path ADVANCED	String	Folder the dialog opens in. Empty: the last one used. Default: (empty)
OUTPUTS		
Pin	Type	Description
File Path	String	Full path of the file on disk.
Error Message	String	Empty on success; otherwise why it failed.
Return Value	Boolean	False if cancelled.

Save File Dialog (Async) ASYNC



Shows the system's Save dialog (Windows) while the game keeps running.

Blocking version: [Save File Dialog](#)

C++: `UPathDialogAsyncAction::SaveFileDialogAsync`

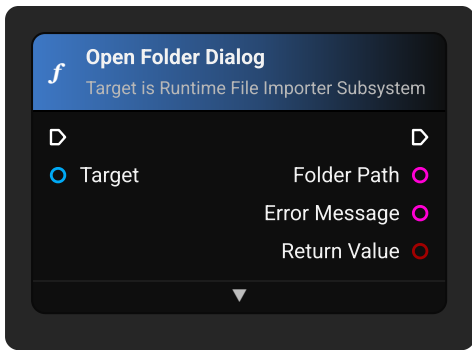
INPUTS

Pin	Type	Description
File Types	String	Filter of the dialog, e.g. "Images *.png;*.jpg All Files *.*" (Make File Dialog Filter / Get File Dialog Filter build one). Empty: all files. Default: (empty)
Default File Name	String	File name suggested in the dialog. Default: (empty)
Title ADVANCED	String	Title of the dialog window. Empty: the system default. Default: (empty)
Default Path ADVANCED	String	Folder the dialog opens in. Empty: the last one used. Default: (empty)

OUTPUTS

Pin	Type	Description
On Success	Exec	Fires when the operation has finished successfully.
On Cancelled	Exec	The player closed the dialog without choosing.
On Failure	Exec	Fires if the operation failed or was cancelled; Error Message says why.
Path	String	Path of a value inside the data, e.g. "player.inventory[2].name" (empty: the root).
Error Message	String	Empty on success; otherwise why it failed.
Async Action	Path Dialog Async Action Object Reference	This node's operation. Call Cancel on it to stop the operation (no more events fire).

Open Folder Dialog FUNCTION



Shows the system's folder picker (Windows), e.g. to choose where to extract an archive. Returns false if cancelled.

Target: Runtime File Importer Subsystem

Async version: [Open Folder Dialog \(Async\)](#)

C++: `URuntimeFileImporterSubsystem::OpenFolderDialog`

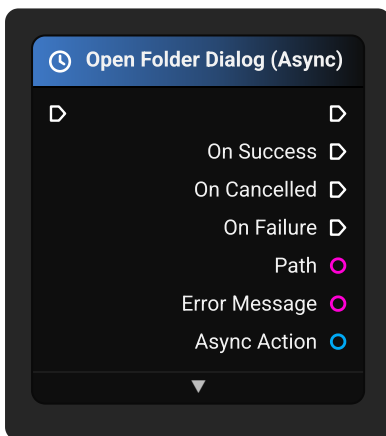
INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Title ADVANCED	String	Title of the dialog window. Empty: the system default. Default: (empty)
Default Path ADVANCED	String	Folder the dialog opens in. Empty: the last one used. Default: (empty)

OUTPUTS

Pin	Type	Description
Folder Path	String	The chosen folder.
Error Message	String	Empty on success; otherwise why it failed.
Return Value	Boolean	False if cancelled.

Open Folder Dialog (Async) ASYNC



Shows the system's folder picker (Windows) while the game keeps running.

Blocking version: [Open Folder Dialog](#)

C++: `UPathDialogAsyncAction::OpenFolderDialogAsync`

INPUTS

Pin	Type	Description
Title ADVANCED	String	Title of the dialog window. Empty: the system default. Default: (empty)
Default Path ADVANCED	String	Folder the dialog opens in. Empty: the last one used. Default: (empty)

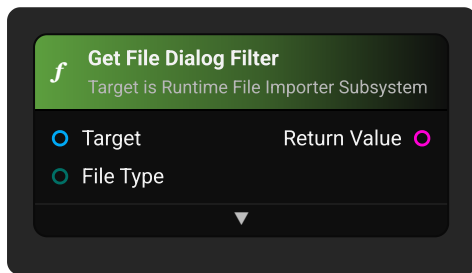
OUTPUTS

Pin	Type	Description
On Success	Exec	Fires when the operation has finished successfully.

Pin	Type	Description
On Cancelled	 Exec	The player closed the dialog without choosing.
On Failure	 Exec	Fires if the operation failed or was cancelled; Error Message says why.
Path	 String	Path of a value inside the data, e.g. "player.inventory[2].name" (empty: the root).
Error Message	 String	Empty on success; otherwise why it failed.
Async Action	 Path Dialog Async Action Object Reference	This node's operation. Call Cancel on it to stop the operation (no more events fire).

Get File Dialog Filter

PURE



File dialog filter of what a file type's nodes read, e.g. "Audio Files (*.mp3, *.wav, ...)|*.mp3;*.wav;...". Unknown: every supported file.

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::GetFileDialogFilter`

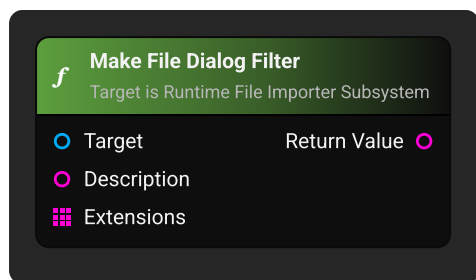
INPUTS

Pin	Type	Description
Target	 Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
File Type	 ERuntime File Type	The kind of file.
Add All Files ADVANCED	 Boolean	Also add an "All Files (*.*)" entry. Default: <code>true</code>

OUTPUTS

Pin	Type	Description
Return Value	 String	File dialog filter of what a file type's nodes read, e.g.

Make File Dialog Filter PURE



File dialog filter of extensions ("png", ".jpg" or "*.tga"), e.g. "Pictures (*.png, *.jpg)|*.png;*.jpg". Join several with |.

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::MakeFileDialogFilter`

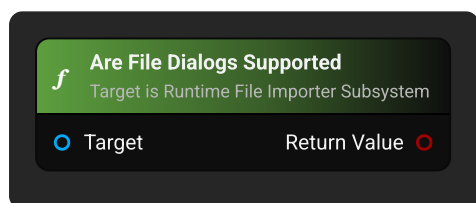
INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Description	String	Text shown in the dialog's file type list, e.g. "Images".
Extensions	Array of String	Extensions of the filter, with or without the dot (e.g. png, jpg).

OUTPUTS

Pin	Type	Description
Return Value	String	File dialog filter of extensions ("png", ".jpg" or "*.tga"), e.g.

Are File Dialogs Supported PURE



True where the file dialogs are available (Windows).

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::AreFileDialogsSupported`

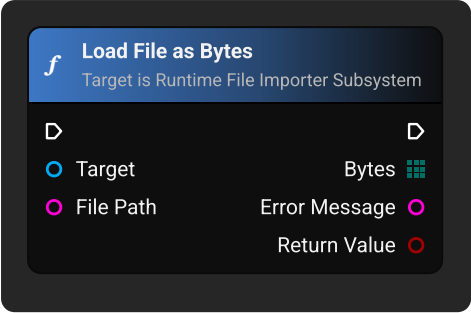
INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.

OUTPUTS

Pin	Type	Description
Return Value	Boolean	True where the file dialogs are available (Windows).

Load File as Bytes FUNCTION



Reads a whole file as bytes, e.g. for the Import ... From Buffer nodes or to send it over the network.

Target: Runtime File Importer Subsystem

Async version: [Load File As Bytes \(Async\)](#)

C++: URuntimeFileImporterSubsystem::LoadFileAsBytes

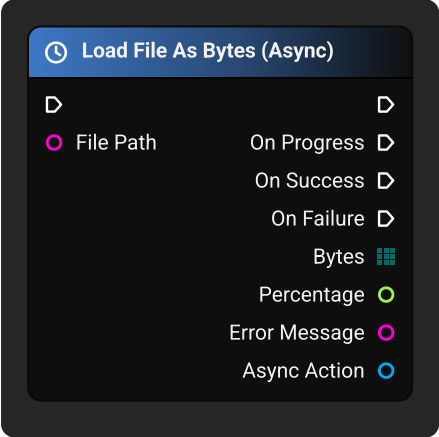
INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
File Path	String	Full path of the file on disk.

OUTPUTS

Pin	Type	Description
Bytes	Array of Byte	The bytes.
Error Message	String	Empty on success; otherwise why it failed.
Return Value	Boolean	True on success. False if it failed (Error Message says why).

Load File As Bytes (Async) ASYNC



Reads a whole file as bytes without blocking the game thread.

Blocking version: [Load File as Bytes](#)

C++: ULoadFileAsBytesAsyncAction::LoadFileAsBytesAsync

INPUTS

Pin	Type	Description
File Path	String	Full path of the file on disk.

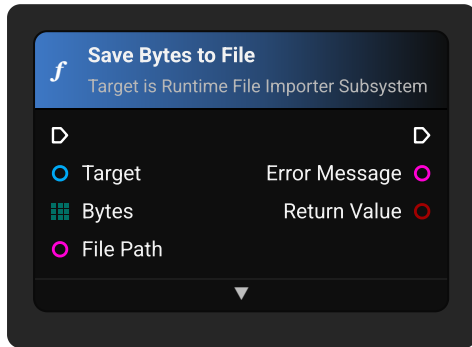
OUTPUTS

Pin	Type	Description
On Progress	Exec	Fires as the operation advances (Percentage from 0 to 100). Other outputs are not set yet.
On Success	Exec	Fires when the operation has finished successfully.
On Failure	Exec	Fires if the operation failed or was cancelled; Error Message says why.

Pin	Type	Description
Bytes	Array of Byte	The bytes.
Percentage	Float	Progress from 0 to 100.
Error Message	String	Empty on success; otherwise why it failed.
Async Action	Load File as Bytes Async Action Object Reference	This node's operation. Call Cancel on it to stop the operation (no more events fire).

Save Bytes to File

FUNCTION



Writes bytes to a file (making its folder), or adds them to its end with Append.

Target: Runtime File Importer Subsystem

Async version: [Save Bytes To File \(Async\)](#)

C++: `URuntimeFileImporterSubsystem::SaveBytesToFile`

INPUTS

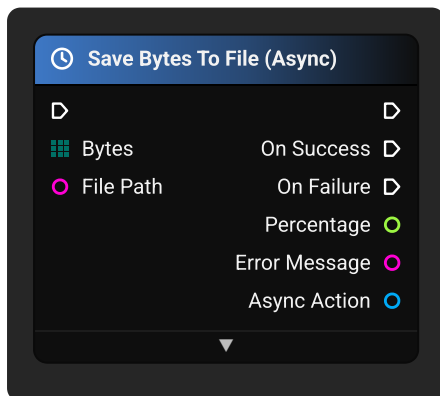
Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Bytes	Array of Byte	The bytes.
File Path	String	Full path of the file on disk.
Append ADVANCED	Boolean	Add to the end of an existing file instead of replacing it. Default: <code>false</code>

OUTPUTS

Pin	Type	Description
Error Message	String	Empty on success; otherwise why it failed.
Return Value	Boolean	True on success. False if it failed (Error Message says why).

Save Bytes To File (Async)

ASYNC



Writes bytes to a file (or adds them to its end with Append) without blocking the game thread.

Blocking version: [Save Bytes to File](#)

C++: `USaveBytesToFileAsyncAction::SaveBytesToFileAsync`

INPUTS

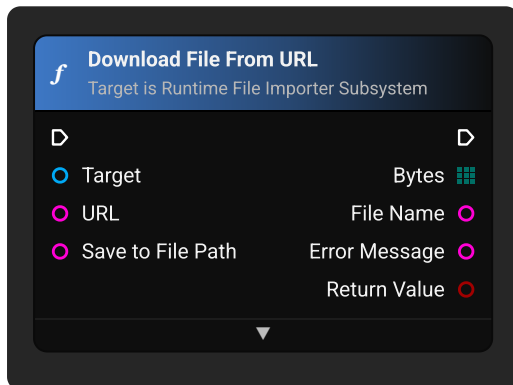
Pin	Type	Description
Bytes	 Array of Byte	The bytes.
File Path	 String	Full path of the file on disk.
Append ADVANCED	 Boolean	Add to the end of an existing file instead of replacing it. Default: <code>false</code>

OUTPUTS

Pin	Type	Description
On Success	 Exec	Fires when the operation has finished successfully.
On Failure	 Exec	Fires if the operation failed or was cancelled; Error Message says why.
Percentage	 Float	Progress from 0 to 100.
Error Message	 String	Empty on success; otherwise why it failed.
Async Action	 Save Bytes to File Async Action Object Reference	This node's operation. Call Cancel on it to stop the operation (no more events fire).

Download File From URL

FUNCTION



Downloads any file over HTTP(S) as bytes, saving it to Save To File Path if set (share links of Google Drive and Dropbox work). File Name is the name the server gave it. FREEZES the game until the download finishes; prefer the Async version.

Target: Runtime File Importer Subsystem

Async version: [Download File From URL \(Async\)](#)

C++: `URuntimeFileImporterSubsystem::DownloadFileFromURL`

INPUTS

Pin	Type	Description
Target	 Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
URL	 String	The http(s) address to download. Google Docs, Sheets, Slides and Drive share links are converted automatically.
Save to File Path	 String	Optional. Also writes the downloaded file there, unmodified. Default: <code>(empty)</code>
Timeout Seconds ADVANCED	 Float	Gives up (and fails) if the download takes longer than this. Default: <code>60.0</code>

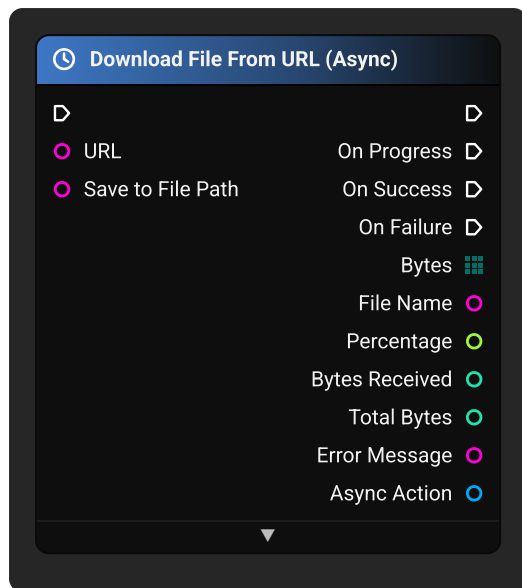
OUTPUTS

Pin	Type	Description
Bytes	 Array of Byte	The bytes.
File Name	 String	The file's name: from the server (Content-Disposition), else the URL, else made from the content type.
Error Message	 String	Empty on success; otherwise why it failed.

Pin	Type	Description
Return Value	 Boolean	True on success. False if it failed (Error Message says why).

Download File From URL (Async)

ASYNC



Downloads any file as bytes, saving it to Save To File Path if set. File Name is the name the server gave it.

Blocking version: [Download File From URL](#)

C++: `UDownloadFileAsyncAction::DownloadFileFromURLAsync`

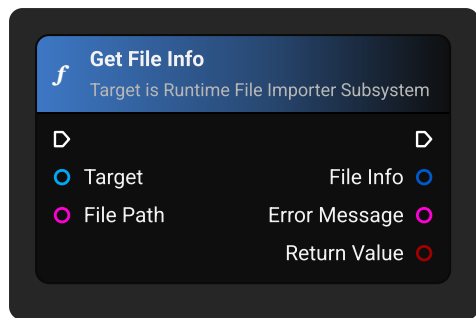
INPUTS

Pin	Type	Description
URL	 String	The http(s) address to download. Google Docs, Sheets, Slides and Drive share links are converted automatically.
Save to File Path	 String	Optional. Also writes the downloaded file there, unmodified. Default: (empty)
Timeout Seconds ADVANCED	 Float	Gives up (and fails) if the download takes longer than this. Default: 60.0

OUTPUTS

Pin	Type	Description
On Progress	 Exec	Fires repeatedly while downloading. Percentage (0-100) and Total Bytes are 0 if the server doesn't report the size.
On Success	 Exec	Fires when the operation has finished successfully.
On Failure	 Exec	Fires if the operation failed or was cancelled; Error Message says why.
Bytes	 Array of Byte	The bytes.
File Name	 String	The file name.
Percentage	 Float	Progress from 0 to 100.
Bytes Received	 Integer	Bytes downloaded so far.
Total Bytes	 Integer	Size of the download in bytes, or 0 while the server hasn't said.
Error Message	 String	Empty on success; otherwise why it failed.
Async Action	 Download File Async Action Object Reference	This node's operation. Call Cancel on it to stop the operation (no more events fire).

Get File Info FUNCTION



Size, dates and kind of a file or folder; File Type tells which nodes import it (detected from its contents).

Target: Runtime File Importer Subsystem

Async version: [Get File Info \(Async\)](#)

C++: `URuntimeFileImporterSubsystem::GetFileInfo`

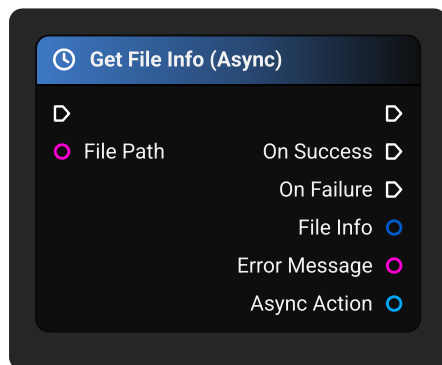
INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
File Path	String	Full path of the file on disk.

OUTPUTS

Pin	Type	Description
File Info	Runtime File Info	Size, dates and kind of the file, and which nodes import it.
Error Message	String	Empty on success; otherwise why it failed.
Return Value	Boolean	True on success. False if it failed (Error Message says why).

Get File Info (Async) ASYNC



Size, dates and kind of a file or folder, without blocking the game thread.

Blocking version: [Get File Info](#)

C++: `UGetFileInfoAsyncAction::GetFileInfoAsync`

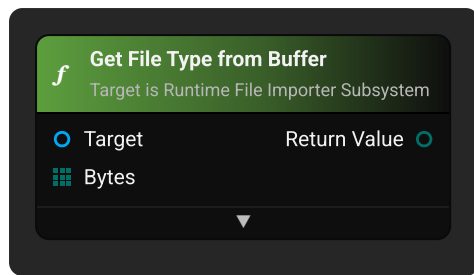
INPUTS

Pin	Type	Description
File Path	String	Full path of the file on disk.

OUTPUTS

Pin	Type	Description
On Success	Exec	Fires when the operation has finished successfully.
On Failure	Exec	Fires if the operation failed or was cancelled; Error Message says why.
File Info	Runtime File Info	Size, dates and kind of the file, and which nodes import it.
Error Message	String	Empty on success; otherwise why it failed.
Async Action	Get File Info Async Action Object Reference	This node's operation. Call Cancel on it to stop the operation (no more events fire).

Get File Type from Buffer PURE



Which nodes import bytes in memory, from their contents (File Name: an optional name or extension, used when unsure).

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::GetFileTypeFromBuffer`

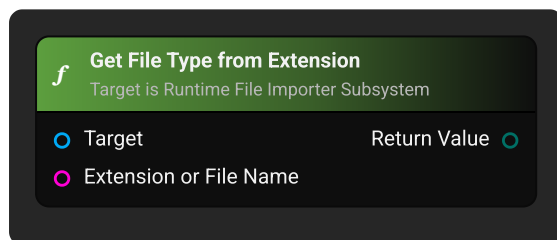
INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Bytes	Array of Byte	The bytes.
File Name ADVANCED	String	Optional file name (or URL) of the bytes. Its extension helps when the contents alone are ambiguous. Default: <code>(empty)</code>

OUTPUTS

Pin	Type	Description
Return Value	ERuntime File Type	Which nodes import bytes in memory, from their contents (File Name: an optional name or extension, used when unsure).

Get File Type from Extension PURE



Which nodes import a file of this extension or name (Unknown if none).

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::GetFileTypeFromExtension`

INPUTS

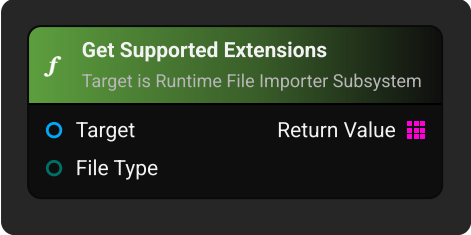
Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Extension or File Name	String	An extension (with or without the dot) or a file name / path / URL ending in one.

OUTPUTS

Pin	Type	Description
Return Value	ERuntime File Type	Which nodes import a file of this extension or name (Unknown if none).

Get Supported Extensions

PURE



File extensions (without dots) the nodes of a file type read on this platform.
Unknown: every supported extension.

Target: Runtime File Importer Subsystem
C++: URuntimeFileImporterSubsystem::GetSupportedExtensions

INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
File Type	ERuntime File Type	The kind of file.

OUTPUTS

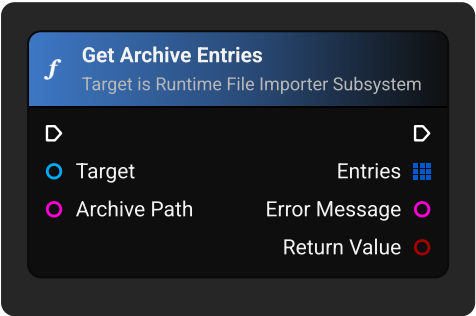
Pin	Type	Description
Return Value	Array of String	File extensions (without dots) the nodes of a file type read on this platform.

Archive

List, read and extract ZIP archives, create ZIP files from files and folders, and compress or decompress bytes with GZip, Zlib or Deflate.

General 19 nodes

Get Archive Entries FUNCTION



Lists the files and folders in a ZIP archive (names, sizes, dates) without extracting them.

Target: Runtime File Importer Subsystem
Async version: [Get Archive Entries \(Async\)](#)
C++: URuntimeFileImporterSubsystem::GetArchiveEntries

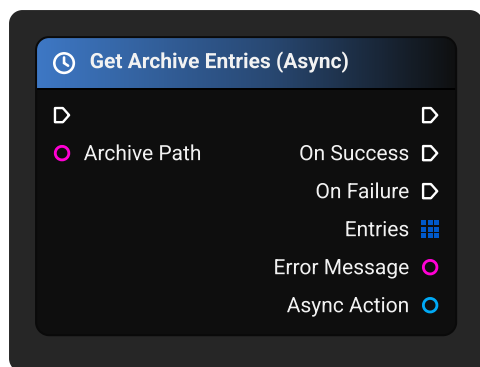
INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Archive Path	String	Full path of the ZIP file.

OUTPUTS

Pin	Type	Description
Entries	Array of Runtime Archive Entry	Every file and folder in the archive.
Error Message	String	Empty on success; otherwise why it failed.
Return Value	Boolean	True on success. False if it failed (Error Message says why).

Get Archive Entries (Async) ASYNC



Lists the files and folders in a ZIP archive without blocking the game thread.

Blocking version: [Get Archive Entries](#)

C++: `UGetArchiveEntriesAsyncAction::GetArchiveEntriesAsync`

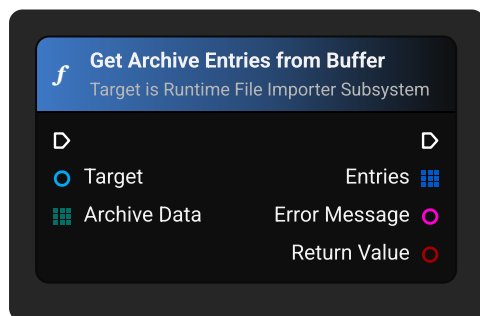
INPUTS

Pin	Type	Description
Archive Path	String	Full path of the ZIP file.

OUTPUTS

Pin	Type	Description
On Success	Exec	Fires when the operation has finished successfully.
On Failure	Exec	Fires if the operation failed or was cancelled; Error Message says why.
Entries	Array of Runtime Archive Entry	Every file and folder in the archive.
Error Message	String	Empty on success; otherwise why it failed.
Async Action	Get Archive Entries Async Action Object Reference	This node's operation. Call Cancel on it to stop the operation (no more events fire).

Get Archive Entries from Buffer FUNCTION



Lists the files and folders in a ZIP archive in memory.

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::GetArchiveEntriesFromBuffer`

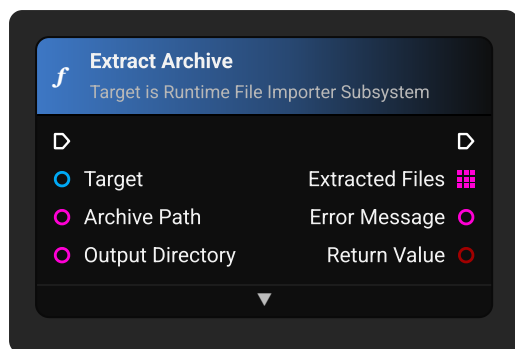
INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Archive Data	Array of Byte	Bytes of a ZIP archive.

OUTPUTS

Pin	Type	Description
Entries	Array of Runtime Archive Entry	Every file and folder in the archive.
Error Message	String	Empty on success; otherwise why it failed.
Return Value	Boolean	True on success. False if it failed (Error Message says why).

Extract Archive FUNCTION



Extracts a ZIP archive into a folder (made if missing), e.g. a downloaded mod or level. Overwrite: replace existing files (else they're kept). Entries whose path would leave the folder are skipped. Use the Async version for big archives.

Target: Runtime File Importer Subsystem

Async version: [Extract Archive \(Async\)](#)

C++: `URuntimeFileImporterSubsystem::ExtractArchive`

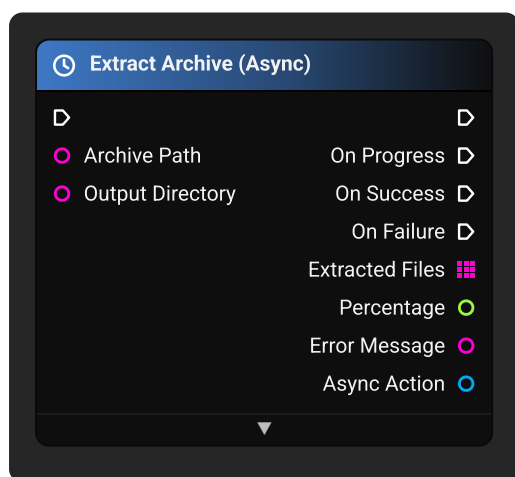
INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Archive Path	String	Full path of the ZIP file.
Output Directory	String	Folder the archive is extracted into (created if missing).
Overwrite ADVANCED	Boolean	Replace files that already exist (otherwise they are skipped). Default: <code>true</code>

OUTPUTS

Pin	Type	Description
Extracted Files	Array of String	Full paths of the files written.
Error Message	String	Empty on success; otherwise why it failed.
Return Value	Boolean	True on success. False if it failed (Error Message says why).

Extract Archive (Async) ASYNC



Extracts a ZIP archive into a folder without blocking the game thread (see Extract Archive).

Blocking version: [Extract Archive](#)

C++: `UExtractArchiveAsyncAction::ExtractArchiveAsync`

INPUTS

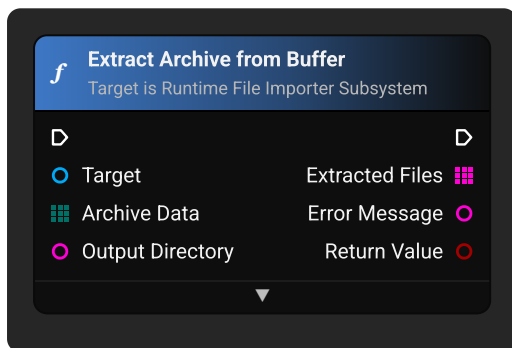
Pin	Type	Description
Archive Path	String	Full path of the ZIP file.
Output Directory	String	Folder the archive is extracted into (created if missing).
Overwrite ADVANCED	Boolean	Replace files that already exist (otherwise they are skipped). Default: <code>true</code>

OUTPUTS

Pin	Type	Description
On Progress	 Exec	Fires as the operation advances (Percentage from 0 to 100). Other outputs are not set yet.
On Success	 Exec	Fires when the operation has finished successfully.
On Failure	 Exec	Fires if the operation failed or was cancelled; Error Message says why.
Extracted Files	 Array of String	Full paths of the files written.
Percentage	 Float	Progress from 0 to 100.
Error Message	 String	Empty on success; otherwise why it failed.
Async Action	 Extract Archive Async Action Object Reference	This node's operation. Call Cancel on it to stop the operation (no more events fire).

Extract Archive from Buffer

FUNCTION



Extracts a ZIP archive in memory into a folder.

Target: Runtime File Importer Subsystem

Async version: [Extract Archive From Buffer \(Async\)](#)

C++: `URuntimeFileImporterSubsystem::ExtractArchiveFromBuffer`

INPUTS

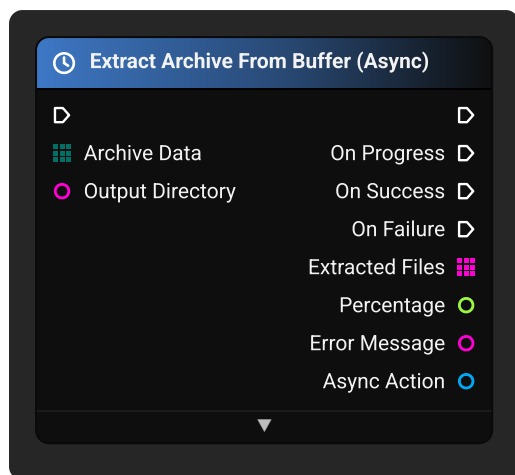
Pin	Type	Description
Target	 Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Archive Data	 Array of Byte	Bytes of a ZIP archive.
Output Directory	 String	Folder the archive is extracted into (created if missing).
Overwrite ADVANCED	 Boolean	Replace files that already exist (otherwise they are skipped). Default: <code>true</code>

OUTPUTS

Pin	Type	Description
Extracted Files	 Array of String	Full paths of the files written.
Error Message	 String	Empty on success; otherwise why it failed.
Return Value	 Boolean	True on success. False if it failed (Error Message says why).

Extract Archive From Buffer (Async)

ASYNC



Extracts a ZIP archive in memory into a folder without blocking the game thread.

Blocking version: [Extract Archive from Buffer](#)

C++: `UExtractArchiveAsyncAction::ExtractArchiveFromBufferAsync`

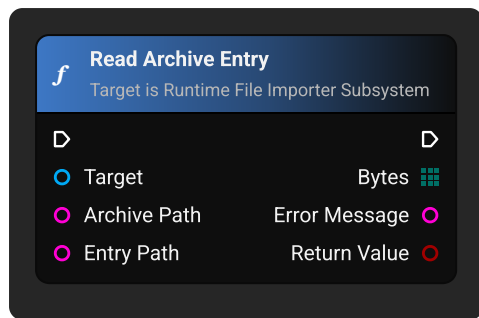
INPUTS

Pin	Type	Description
Archive Data	Array of Byte	Bytes of a ZIP archive.
Output Directory	String	Folder the archive is extracted into (created if missing).
Overwrite ADVANCED	Boolean	Replace files that already exist (otherwise they are skipped). Default: <code>true</code>

OUTPUTS

Pin	Type	Description
On Progress	Exec	Fires as the operation advances (Percentage from 0 to 100). Other outputs are not set yet.
On Success	Exec	Fires when the operation has finished successfully.
On Failure	Exec	Fires if the operation failed or was cancelled; Error Message says why.
Extracted Files	Array of String	Full paths of the files written.
Percentage	Float	Progress from 0 to 100.
Error Message	String	Empty on success; otherwise why it failed.
Async Action	Extract Archive Async Action Object Reference	This node's operation. Call Cancel on it to stop the operation (no more events fire).

Read Archive Entry FUNCTION



Extracts one file of a ZIP archive into memory (e.g. for an Import ... From Buffer node). Entry Path: as Get Archive Entries lists it.

Target: Runtime File Importer Subsystem

Async version: [Read Archive Entry \(Async\)](#)

C++: `URuntimeFileImporterSubsystem::ReadArchiveEntry`

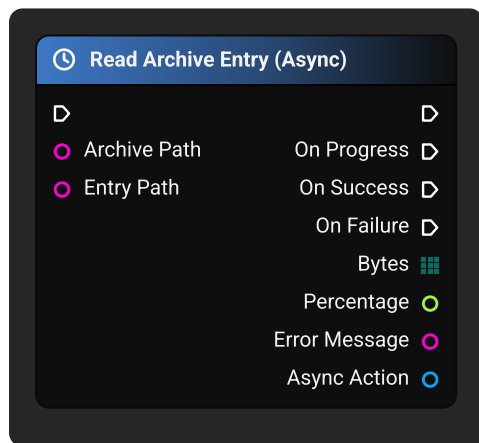
INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Archive Path	String	Full path of the ZIP file.
Entry Path	String	Path of the file inside the archive, e.g. "textures/wood.png" (forward slashes).

OUTPUTS

Pin	Type	Description
Bytes	Array of Byte	The bytes.
Error Message	String	Empty on success; otherwise why it failed.
Return Value	Boolean	True on success. False if it failed (Error Message says why).

Read Archive Entry (Async) ASYNC



Extracts one file of a ZIP archive into memory without blocking the game thread.

Blocking version: [Read Archive Entry](#)

C++: `UArchiveBytesAsyncAction::ReadArchiveEntryAsync`

INPUTS

Pin	Type	Description
Archive Path	String	Full path of the ZIP file.
Entry Path	String	Path of the file inside the archive, e.g. "textures/wood.png" (forward slashes).

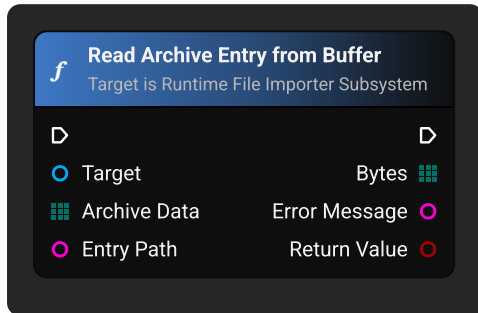
OUTPUTS

Pin	Type	Description
On Progress	Exec	Fires as the operation advances (Percentage from 0 to 100). Other outputs are not set yet.
On Success	Exec	Fires when the operation has finished successfully.

Pin	Type	Description
On Failure	Exec	Fires if the operation failed or was cancelled; Error Message says why.
Bytes	Array of Byte	The bytes.
Percentage	Float	Progress from 0 to 100.
Error Message	String	Empty on success; otherwise why it failed.
Async Action	Archive Bytes Async Action Object Reference	This node's operation. Call Cancel on it to stop the operation (no more events fire).

Read Archive Entry from Buffer

FUNCTION



Extracts one file of a ZIP archive in memory.

Target: Runtime File Importer Subsystem

Async version: [Read Archive Entry From Buffer \(Async\)](#)

C++: `URuntimeFileImporterSubsystem::ReadArchiveEntryFromBuffer`

INPUTS

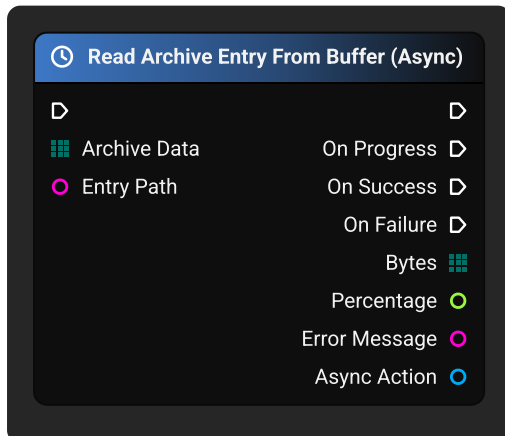
Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Archive Data	Array of Byte	Bytes of a ZIP archive.
Entry Path	String	Path of the file inside the archive, e.g. "textures/wood.png" (forward slashes).

OUTPUTS

Pin	Type	Description
Bytes	Array of Byte	The bytes.
Error Message	String	Empty on success; otherwise why it failed.
Return Value	Boolean	True on success. False if it failed (Error Message says why).

Read Archive Entry From Buffer (Async)

ASYNC



Extracts one file of a ZIP archive in memory without blocking the game thread.

Blocking version: [Read Archive Entry from Buffer](#)

C++: `UArchiveBytesAsyncAction::ReadArchiveEntryFromBufferAsync`

INPUTS

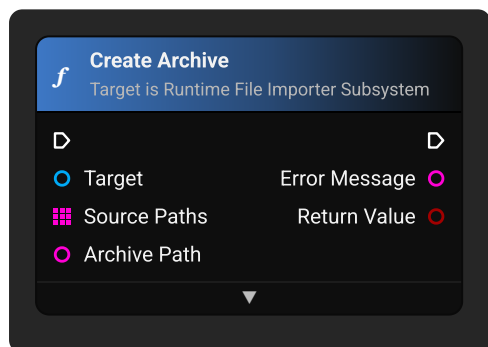
Pin	Type	Description
Archive Data	 Array of Byte	Bytes of a ZIP archive.
Entry Path	 String	Path of the file inside the archive, e.g. "textures/wood.png" (forward slashes).

OUTPUTS

Pin	Type	Description
On Progress	 Exec	Fires as the operation advances (Percentage from 0 to 100). Other outputs are not set yet.
On Success	 Exec	Fires when the operation has finished successfully.
On Failure	 Exec	Fires if the operation failed or was cancelled; Error Message says why.
Bytes	 Array of Byte	The bytes.
Percentage	 Float	Progress from 0 to 100.
Error Message	 String	Empty on success; otherwise why it failed.
Async Action	 Archive Bytes Async Action Object Reference	This node's operation. Call Cancel on it to stop the operation (no more events fire).

Create Archive

FUNCTION



Makes a ZIP archive of files and folders (folders with all their contents, under their own name if Include Folder Names, else their contents at the archive's root). Compression Level: 0 (store) to 9 (smallest). Use the Async version for big files.

Target: Runtime File Importer Subsystem

Async version: [Create Archive \(Async\)](#)

C++: `URuntimeFileImporterSubsystem::CreateArchive`

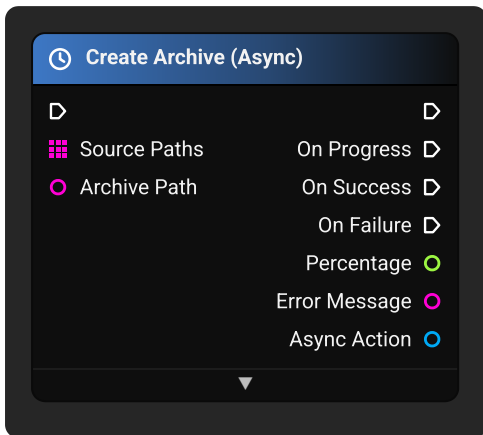
INPUTS

Pin	Type	Description
Target	 Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Source Paths	 Array of String	Files and folders to put in the archive. Folders are added with everything inside them.
Archive Path	 String	Full path of the ZIP file.
Compression Level	 Integer	0 (store, fastest) to 9 (smallest). 6 is a good balance. Default: 6
Include Folder Names	 Boolean	Keep the folder of each source path as the top folder in the archive. Default: <code>true</code>

OUTPUTS

Pin	Type	Description
Error Message	 String	Empty on success; otherwise why it failed.
Return Value	 Boolean	True on success. False if it failed (Error Message says why).

Create Archive (Async) ASYNC



Makes a ZIP archive of files and folders without blocking the game thread (see [Create Archive](#)).

Blocking version: [Create Archive](#)

C++: `UCreateArchiveAsyncAction::CreateArchiveAsync`

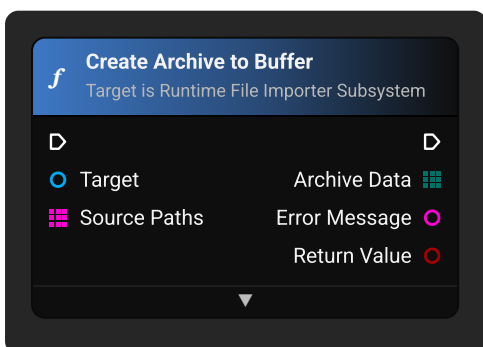
INPUTS

Pin	Type	Description
Source Paths	Array of String	Files and folders to put in the archive. Folders are added with everything inside them.
Archive Path	String	Full path of the ZIP file.
Compression Level	Integer	0 (store, fastest) to 9 (smallest). 6 is a good balance. Default: 6
Include Folder Names	Boolean	Keep the folder of each source path as the top folder in the archive. Default: <code>true</code>

OUTPUTS

Pin	Type	Description
On Progress	Exec	Fires as the operation advances (Percentage from 0 to 100). Other outputs are not set yet.
On Success	Exec	Fires when the operation has finished successfully.
On Failure	Exec	Fires if the operation failed or was cancelled; Error Message says why.
Percentage	Float	Progress from 0 to 100.
Error Message	String	Empty on success; otherwise why it failed.
Async Action	Create Archive Async Action Object Reference	This node's operation. Call Cancel on it to stop the operation (no more events fire).

Create Archive to Buffer FUNCTION



Makes a ZIP archive of files and folders in memory (e.g. to send over the network).

Target: Runtime File Importer Subsystem

Async version: [Create Archive To Buffer \(Async\)](#)

C++: `URuntimeFileImporterSubsystem::CreateArchiveToBuffer`

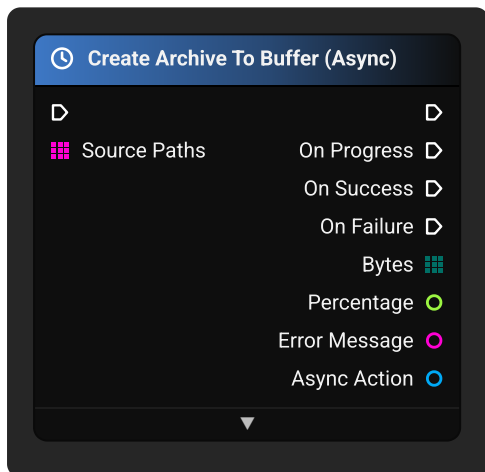
INPUTS

Pin	Type	Description
Target	 Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Source Paths	 Array of String	Files and folders to put in the archive. Folders are added with everything inside them.
Compression Level ADVANCED	 Integer	0 (store, fastest) to 9 (smallest). 6 is a good balance. Default: 6
Include Folder Names ADVANCED	 Boolean	Keep the folder of each source path as the top folder in the archive. Default: <code>true</code>

OUTPUTS

Pin	Type	Description
Archive Data	 Array of Byte	Bytes of a ZIP archive.
Error Message	 String	Empty on success; otherwise why it failed.
Return Value	 Boolean	True on success. False if it failed (Error Message says why).

Create Archive To Buffer (Async) ASYNC



Makes a ZIP archive of files and folders in memory without blocking the game thread.

Blocking version: [Create Archive to Buffer](#)

C++: `UArchiveBytesAsyncAction::CreateArchiveToBufferAsync`

INPUTS

Pin	Type	Description
Source Paths	 Array of String	Files and folders to put in the archive. Folders are added with everything inside them.
Compression Level ADVANCED	 Integer	0 (store, fastest) to 9 (smallest). 6 is a good balance. Default: 6
Include Folder Names ADVANCED	 Boolean	Keep the folder of each source path as the top folder in the archive. Default: <code>true</code>

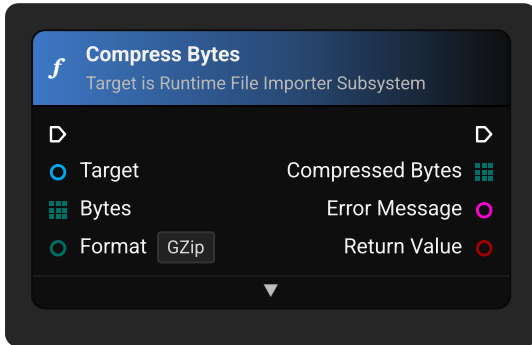
OUTPUTS

Pin	Type	Description
On Progress	 Exec	Fires as the operation advances (Percentage from 0 to 100). Other outputs are not set yet.
On Success	 Exec	Fires when the operation has finished successfully.
On Failure	 Exec	Fires if the operation failed or was cancelled; Error Message says why.
Bytes	 Array of Byte	The bytes.
Percentage	 Float	Progress from 0 to 100.

Pin	Type	Description
Error Message	 String	Empty on success; otherwise why it failed.
Async Action	 Archive Bytes Async Action Object Reference	This node's operation. Call Cancel on it to stop the operation (no more events fire).

Compress Bytes

FUNCTION



Compresses bytes (GZip like .gz files, Zlib, or raw Deflate), e.g. to save space or bandwidth. Level: 0 (store) to 9 (smallest).

Target: Runtime File Importer Subsystem

Async version: [Compress Bytes \(Async\)](#)

C++: `URuntimeFileImporterSubsystem::CompressBytes`

INPUTS

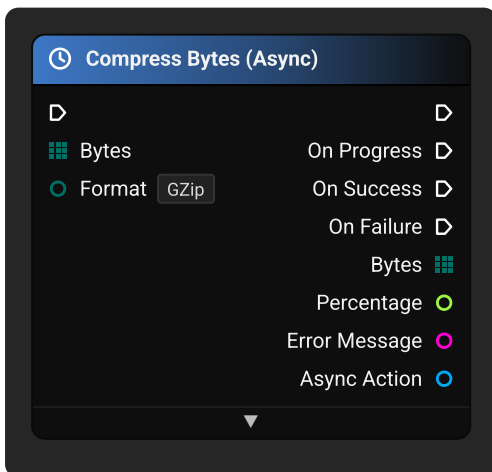
Pin	Type	Description
Target	 Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Bytes	 Array of Byte	The bytes.
Format	 ERuntime Compression Format	The compression format. Default: GZip
Level ADVANCED	 Integer	0 (fastest) to 9 (smallest). 6 is a good balance. Default: 6

OUTPUTS

Pin	Type	Description
Compressed Bytes	 Array of Byte	Compressed bytes (GZip, Zlib or raw Deflate; detected automatically).
Error Message	 String	Empty on success; otherwise why it failed.
Return Value	 Boolean	True on success. False if it failed (Error Message says why).

Compress Bytes (Async)

ASYNC



Compresses bytes without blocking the game thread.

Blocking version: [Compress Bytes](#)

C++: `UArchiveBytesAsyncAction::CompressBytesAsync`

INPUTS

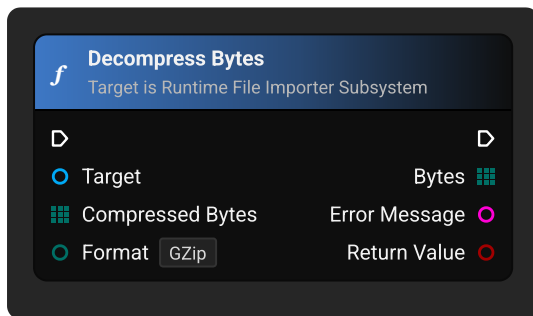
Pin	Type	Description
Bytes	 Array of Byte	The bytes.
Format	 ERuntime Compression Format	The compression format. Default: <code>GZip</code>
Level ADVANCED	 Integer	0 (fastest) to 9 (smallest). 6 is a good balance. Default: <code>6</code>

OUTPUTS

Pin	Type	Description
On Progress	 Exec	Fires as the operation advances (Percentage from 0 to 100). Other outputs are not set yet.
On Success	 Exec	Fires when the operation has finished successfully.
On Failure	 Exec	Fires if the operation failed or was cancelled; Error Message says why.
Bytes	 Array of Byte	The bytes.
Percentage	 Float	Progress from 0 to 100.
Error Message	 String	Empty on success; otherwise why it failed.
Async Action	 Archive Bytes Async Action Object Reference	This node's operation. Call Cancel on it to stop the operation (no more events fire).

Decompress Bytes

FUNCTION



Decompresses GZip (.gz), Zlib or Deflate bytes; GZip also detects the other two.

Target: Runtime File Importer Subsystem

Async version: [Decompress Bytes \(Async\)](#)

C++: `URuntimeFileImporterSubsystem::DecompressBytes`

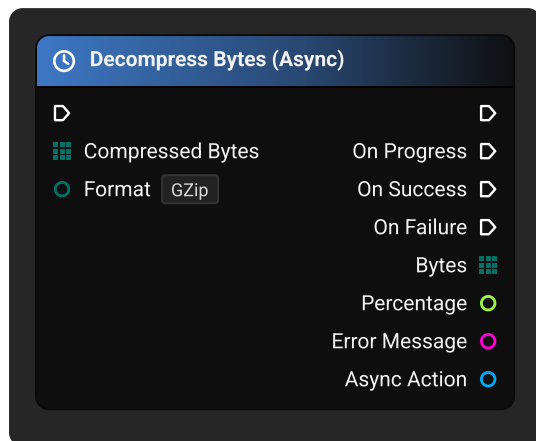
INPUTS

Pin	Type	Description
Target	 Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Compressed Bytes	 Array of Byte	Compressed bytes (GZip, Zlib or raw Deflate; detected automatically).
Format	 ERuntime Compression Format	The compression format. Default: <code>GZip</code>

OUTPUTS

Pin	Type	Description
Bytes	 Array of Byte	The bytes.
Error Message	 String	Empty on success; otherwise why it failed.
Return Value	 Boolean	True on success. False if it failed (Error Message says why).

Decompress Bytes (Async) ASYNC



Decompresses GZip, Zlib or Deflate bytes without blocking the game thread.

Blocking version: [Decompress Bytes](#)

C++: `UArchiveBytesAsyncAction::DecompressBytesAsync`

INPUTS

Pin	Type	Description
Compressed Bytes	Array of Byte	Compressed bytes (GZip, Zlib or raw Deflate; detected automatically).
Format	ERuntime Compression Format	The compression format. Default: GZip

OUTPUTS

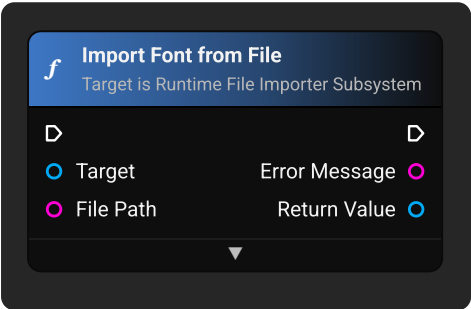
Pin	Type	Description
On Progress	Exec	Fires as the operation advances (Percentage from 0 to 100). Other outputs are not set yet.
On Success	Exec	Fires when the operation has finished successfully.
On Failure	Exec	Fires if the operation failed or was cancelled; Error Message says why.
Bytes	Array of Byte	The bytes.
Percentage	Float	Progress from 0 to 100.
Error Message	String	Empty on success; otherwise why it failed.
Async Action	Archive Bytes Async Action Object Reference	This node's operation. Call Cancel on it to stop the operation (no more events fire).

Font

Load TTF, OTF, TTC and WOFF fonts at runtime as Font assets for UMG text widgets, add typefaces (Bold, Italic, ...) to a font, and read font names and properties.

General 10 nodes

Import Font from File FUNCTION



Imports a font file (TTF, OTF, TTC, WOFF) as a Font for widgets: use it with Make Slate Font Info (Font Object) in a Text Block's Set Font, a Rich Text style, or the Runtime Document Viewer's Font. With Add To Font, the file becomes another typeface (e.g. "Bold") of that font instead, chosen with the font info's Typeface. Typeface Name empty: the file's style.

Target: Runtime File Importer Subsystem
 Async version: [Import Font From File \(Async\)](#)
 C++: URuntimeFileImporterSubsystem::ImportFontFromFile

INPUTS

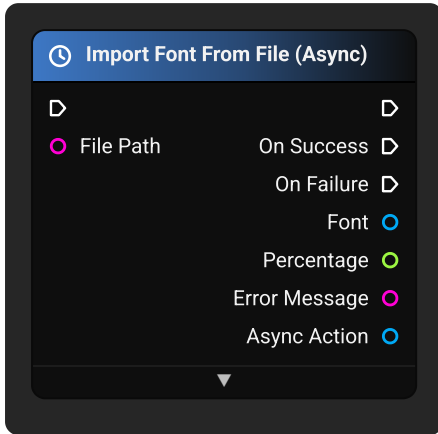
Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
File Path	String	Full path of the file on disk.
Add to Font ADVANCED	Font Object Reference	Optional. Add the typeface to this existing font (e.g. a Bold face to a Regular one) instead of making a new font. Default: <code>None</code>
Typeface Name ADVANCED	String	Name of the typeface, e.g. "Bold". Empty: the style name from the font file. Default: <code>(empty)</code>

OUTPUTS

Pin	Type	Description
Error Message	String	Empty on success; otherwise why it failed.
Return Value	Font Object Reference	The result, or None/empty if it failed (Error Message says why).

Import Font From File (Async)

ASYNC



Imports a font file as a Font without blocking the game thread (see Import Font From File).

Blocking version: [Import Font from File](#)

C++: `UImportFontAsyncAction::ImportFontFromFileAsync`

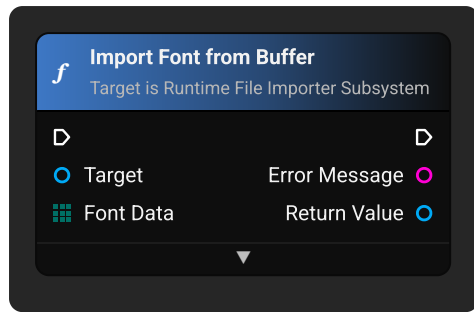
INPUTS

Pin	Type	Description
File Path	String	Full path of the file on disk.
Add to Font ADVANCED	Font Object Reference	Optional. Add the typeface to this existing font (e.g. a Bold face to a Regular one) instead of making a new font. Default: <code>None</code>
Typeface Name ADVANCED	String	Name of the typeface, e.g. "Bold". Empty: the style name from the font file. Default: (empty)

OUTPUTS

Pin	Type	Description
On Success	Exec	Fires when the operation has finished successfully.
On Failure	Exec	Fires if the operation failed or was cancelled; Error Message says why.
Font	Font Object Reference	The font.
Percentage	Float	Progress from 0 to 100.
Error Message	String	Empty on success; otherwise why it failed.
Async Action	Import Font Async Action Object Reference	This node's operation. Call Cancel on it to stop the operation (no more events fire).

Import Font from Buffer FUNCTION



Imports font file bytes in memory as a Font (see Import Font From File).

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::ImportFontFromBuffer`

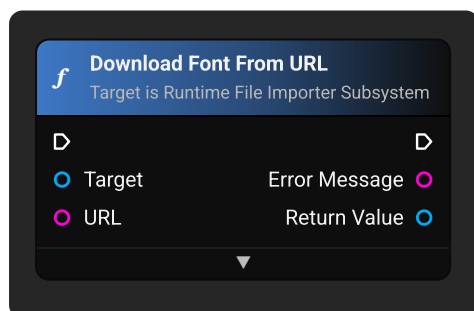
INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Font Data	Array of Byte	Bytes of a TTF, OTF, TTC or WOFF font file.
Add to Font ADVANCED	Font Object Reference	Optional. Add the typeface to this existing font (e.g. a Bold face to a Regular one) instead of making a new font. Default: <code>None</code>
Typeface Name ADVANCED	String	Name of the typeface, e.g. "Bold". Empty: the style name from the font file. Default: <code>(empty)</code>

OUTPUTS

Pin	Type	Description
Error Message	String	Empty on success; otherwise why it failed.
Return Value	Font Object Reference	The result, or None/empty if it failed (Error Message says why).

Download Font From URL FUNCTION



Downloads a font file and imports it. FREEZES the game until the download finishes; prefer the Async version.

Target: Runtime File Importer Subsystem

Async version: [Download Font From URL \(Async\)](#)

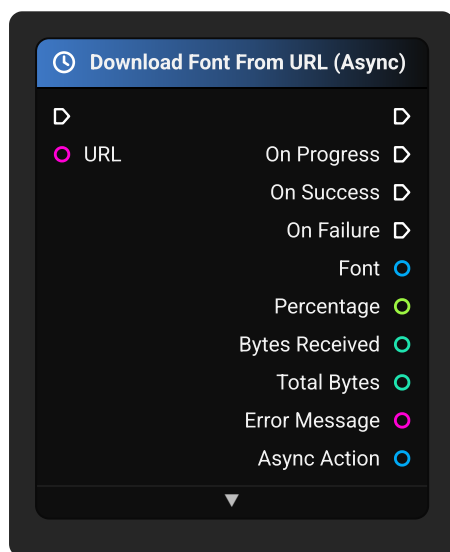
C++: `URuntimeFileImporterSubsystem::DownloadFontFromURL`

INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
URL	String	The http(s) address to download. Google Docs, Sheets, Slides and Drive share links are converted automatically.
Add to Font ADVANCED	Font Object Reference	Optional. Add the typeface to this existing font (e.g. a Bold face to a Regular one) instead of making a new font. Default: <code>None</code>
Typeface Name ADVANCED	String	Name of the typeface, e.g. "Bold". Empty: the style name from the font file. Default: <code>(empty)</code>

Pin	Type	Description
Save to File Path ADVANCED	String	Optional. Also writes the downloaded file there, unmodified. Default: (empty)
Timeout Seconds ADVANCED	Float	Gives up (and fails) if the download takes longer than this. Default: 30.0
OUTPUTS		
Pin	Type	Description
Error Message	String	Empty on success; otherwise why it failed.
Return Value	Font Object Reference	The result, or None/empty if it failed (Error Message says why).

Download Font From URL (Async) ASYNC



Downloads a font file and imports it as a Font.

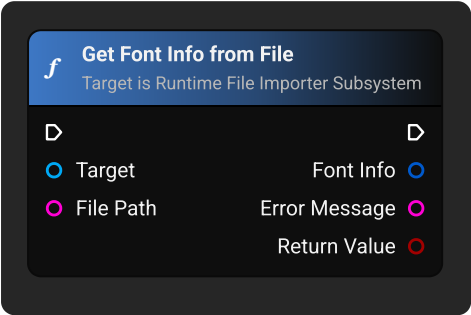
Blocking version: [Download Font From URL](#)

C++: `UDownloadFontAsyncAction::DownloadFontFromURLAsync`

INPUTS		
Pin	Type	Description
URL	String	The http(s) address to download. Google Docs, Sheets, Slides and Drive share links are converted automatically.
Add to Font ADVANCED	Font Object Reference	Optional. Add the typeface to this existing font (e.g. a Bold face to a Regular one) instead of making a new font. Default: None
Typeface Name ADVANCED	String	Name of the typeface, e.g. "Bold". Empty: the style name from the font file. Default: (empty)
Save to File Path ADVANCED	String	Optional. Also writes the downloaded file there, unmodified. Default: (empty)
Timeout Seconds ADVANCED	Float	Gives up (and fails) if the download takes longer than this. Default: 30.0
OUTPUTS		
Pin	Type	Description
On Progress	Exec	Fires as the operation advances (Percentage from 0 to 100). Other outputs are not set yet.
On Success	Exec	Fires when the operation has finished successfully.
On Failure	Exec	Fires if the operation failed or was cancelled; Error Message says why.
Font	Font Object Reference	The font.
Percentage	Float	Progress from 0 to 100.
Bytes Received	Integer	Bytes downloaded so far.

Pin	Type	Description
Total Bytes	 Integer	Size of the download in bytes, or 0 while the server hasn't said.
Error Message	 String	Empty on success; otherwise why it failed.
Async Action	 Download Font Async Action Object Reference	This node's operation. Call Cancel on it to stop the operation (no more events fire).

Get Font Info from File FUNCTION



A font file's family and style names, glyph count, and whether it's bold, italic or monospace.

Target: Runtime File Importer Subsystem

Async version: [Get Font Info From File \(Async\)](#)

C++: `URuntimeFileImporterSubsystem::GetFontInfoFromFile`

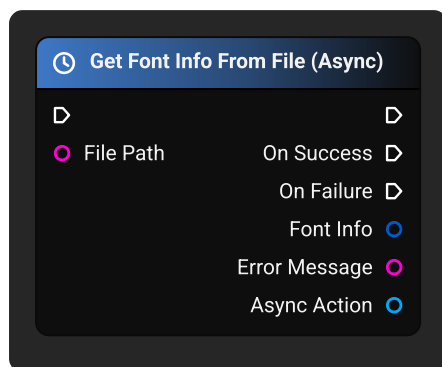
INPUTS

Pin	Type	Description
Target	 Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
File Path	 String	Full path of the file on disk.

OUTPUTS

Pin	Type	Description
Font Info	 Runtime Font Info	Names and properties of the font file.
Error Message	 String	Empty on success; otherwise why it failed.
Return Value	 Boolean	True on success. False if it failed (Error Message says why).

Get Font Info From File (Async) ASYNC



A font file's names and properties, without blocking the game thread.

Blocking version: [Get Font Info from File](#)

C++: `UGetFontInfoAsyncAction::GetFontInfoFromFileAsync`

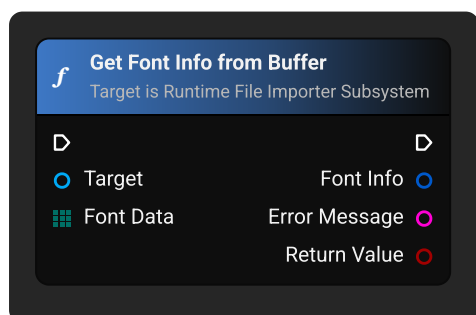
INPUTS

Pin	Type	Description
File Path	String	Full path of the file on disk.

OUTPUTS

Pin	Type	Description
On Success	Exec	Fires when the operation has finished successfully.
On Failure	Exec	Fires if the operation failed or was cancelled; Error Message says why.
Font Info	Runtime Font Info	Names and properties of the font file.
Error Message	String	Empty on success; otherwise why it failed.
Async Action	Get Font Info Async Action Object Reference	This node's operation. Call Cancel on it to stop the operation (no more events fire).

Get Font Info from Buffer FUNCTION



The info of font file bytes in memory.

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::GetFontInfoFromBuffer`

INPUTS

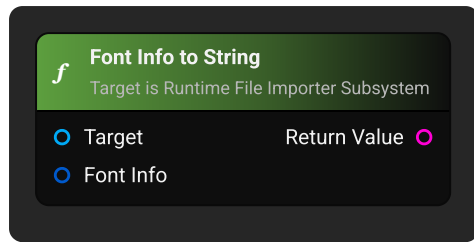
Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Font Data	Array of Byte	Bytes of a TTF, OTF, TTC or WOFF font file.

OUTPUTS

Pin	Type	Description
Font Info	Runtime Font Info	Names and properties of the font file.
Error Message	String	Empty on success; otherwise why it failed.
Return Value	Boolean	True on success. False if it failed (Error Message says why).

Font Info to String

PURE



A readable, multi-line description of font info.

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::FontInfoToString`

INPUTS

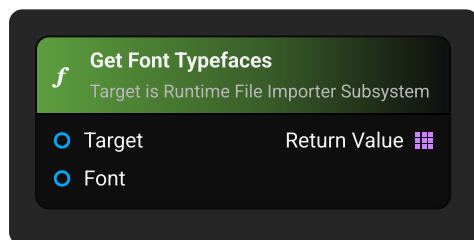
Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Font Info	Runtime Font Info	Names and properties of the font file.

OUTPUTS

Pin	Type	Description
Return Value	String	A readable, multi-line description of font info.

Get Font Typefaces

PURE



The typefaces of a Font ("Regular", "Bold", ...), for the font info's Typeface.

Target: Runtime File Importer Subsystem

C++: `URuntimeFileImporterSubsystem::GetFontTypefaces`

INPUTS

Pin	Type	Description
Target	Runtime File Importer Subsystem Object Reference	The Runtime File Importer Subsystem, from a Get Runtime File Importer Subsystem node.
Font	Font Object Reference	The font.

OUTPUTS

Pin	Type	Description
Return Value	Array of Name	The typefaces of a Font ("Regular", "Bold", ...), for the font info's Typeface.

Objects & widgets

The objects the nodes create or take, with their Blueprint variables and events, and the two UMG widgets: Runtime Document Viewer and Runtime File Drop Zone (in the widget Palette under Runtime File Importer).

Runtime Sound Wave

UTrackedSoundWaveProcedural · based on Sound Wave Procedural

A playable sound wave created at runtime by Runtime File Importer.

Unlike a plain SoundWaveProcedural, which consumes a one-shot queue, it keeps its decoded PCM data and streams it to the audio mixer from a playback cursor. That makes it seekable, pausable, loopable, re-playable and exportable. Audio can also be appended while it plays (streaming), in which case playback waits for more data instead of ending until streaming is finished. Playing one instance on several audio components at once shares that single cursor.

Its state is exposed to Blueprints as variables (Volume, Pitch, Looping, Paused, Playback Time, ...), and all playback events are broadcast on the game thread.

VARIABLES

Name	Type	Access	Description
Sample Rate	 Integer	Read only	
Num Channels	 Integer	Read only	
Metadata	 Runtime Audio Metadata	Read only	Tags read from the imported file (title, artist, album, ...).
Volume	 Float	Read / Write	Volume multiplier (0 or higher). Takes effect immediately, even while playing.
Pitch	 Float	Read / Write	Pitch multiplier (clamped to 0.125 - 4). Also changes playback speed. Takes effect immediately, even while playing.
Looping	 Boolean	Read / Write	Whether the sound starts over when it reaches its end. Takes effect immediately, even while playing. Ignored while streaming.
Paused	 Boolean	Read / Write	Pauses playback on every audio component playing this sound and fires On Playback Paused / Resumed. (Pausing an Audio Component directly also works, but doesn't fire these events.)
Playback Time	 Float	Read only	Current playback position, in seconds. Use Rewind Playback Time to change it.
Playback Percentage	 Float	Read only	Playback progress from 0 to 100.
Is Playback Finished	 Boolean	Read only	True once a non-looping sound has played to the end. Playing it again restarts from the beginning.
Current Amplitude	 Float	Read only	Loudness (RMS, 0-1) of the audio just played, e.g. for volume meters or simple lip sync.
Duration	 Float	Read only	Length of the audio in seconds (even when looping; for streaming sounds, what has been received so far).
Is Streaming	 Boolean	Read only	True while more audio can still be appended (until Finish Streaming Sound Wave).

Name	Type	Access	Description
Audio Properties	 Runtime Audio Properties	Read / Write	Every runtime-changeable property of the sound (volume, pitch, attenuation, sound class, submixes, ...). To change a few, get this, use "Set Members in Runtime Audio Properties", then set it back. Volume, pitch and looping apply immediately; the rest apply the next time the sound starts playing.
EVENTS			
Event	Outputs	Description	
On Playback Started	Sound Wave:  Runtime Sound Wave Object Reference	An audio component started playing this sound (also fires when replaying).	
On Playback Paused	Sound Wave:  Runtime Sound Wave Object Reference	Playback was paused by setting Paused.	
On Playback Resumed	Sound Wave:  Runtime Sound Wave Object Reference	Playback was resumed by clearing Paused.	
On Playback Looped	Sound Wave:  Runtime Sound Wave Object Reference	A looping sound reached its end and started over.	
On Playback Finished	Sound Wave:  Runtime Sound Wave Object Reference	A non-looping sound played to its end (for streaming sounds, only after streaming was finished). It is stopped automatically right after.	
On Playback Stopped	Sound Wave:  Runtime Sound Wave Object Reference	The sound stopped playing for any reason (finished, stopped by its audio component, destroyed, etc.).	

Runtime Mesh

URuntimeImportedMesh · based on Object

A 3D model imported at runtime by Runtime File Importer.

Static models get a Static Mesh: the whole model combined into one mesh, ready for a Static Mesh Component (Set Static Mesh) or Spawn Runtime Mesh Actor, which can also rebuild the model's node hierarchy. Models with bones, morph targets or animations get a Skeletal Mesh instead, with its Skeleton, Animations (Animation Sequences) and Physics Asset, ready for a Skeletal Mesh Component. It keeps the model's original data, so it can be exported again to any supported format.

VARIABLES

Name	Type	Access	Description
Static Mesh	 Static Mesh Object Reference	Read only	The whole model as one static mesh, with its node transforms applied. One material slot per material. Null for skeletal models.
Skeletal Mesh	 Skeletal Mesh Object Reference	Read only	The model as a skeletal mesh, for models with bones, morph targets or animations. Null for static models.
Skeleton	 Skeleton Object Reference	Read only	Skeleton of the Skeletal Mesh and Animations.
Animations	 Array of Animation Sequence Object Reference	Read only	The model's animations, playable on the Skeletal Mesh (Play Animation on its component).
Physics Asset	 Physics Asset Object Reference	Read only	Collision bodies of the Skeletal Mesh (also used for ragdolls).
Mesh Info	 Runtime Mesh Info	Read only	Vertex, triangle, material counts, bounds, file metadata, ...
Material Properties	 Array of Runtime Mesh Material Info	Read only	The model's materials as the file describes them (colors, textures, ...), one per material slot.

Name	Type	Access	Description
Textures	 Array of Texture 2D Object Reference	Read only	Every texture loaded from the model.
Source	 String	Read only	File path or URL the model was imported from (empty for buffers).
Materials	 Array of Material Interface Object Reference	Read / Write	Material of each slot: the scene's materials, then a default one if some triangles have none.

Runtime Mesh Actor

`ARuntimeImportedMeshActor` · based on `Actor`

Actor showing a Runtime Mesh. Static models: one Static Mesh Component with the combined model, or the model's node hierarchy rebuilt as scene components (named after the nodes) with a Static Mesh Component per mesh. Skeletal models: a Skeletal Mesh Component, which can play the model's animations. Spawn it with Spawn Runtime Mesh Actor, or place/spawn it yourself and call Set Runtime Mesh.

VARIABLES

Name	Type	Access	Description
Runtime Mesh	 Runtime Mesh Object Reference	Read only	The model shown.
Mesh Components	 Array of Static Mesh Component Object Reference	Read only	The Static Mesh Components showing the model.
Node Components	 Array of Scene Component Object Reference	Read only	Scene components of the model's nodes (empty when the hierarchy isn't kept), in the model's node order.
Skeletal Mesh Component	 Skeletal Mesh Component Object Reference	Read only	The component showing a skeletal model (null for static models).

FUNCTIONS

[Set Runtime Mesh](#)
[Find Node Component](#)
[Play Animation](#)
[Stop Animation](#)

Runtime Mesh Description

`URuntimeMeshDescription` · based on `Static Mesh Description`

Editable geometry of a mesh: vertices, vertex instances (a vertex's normal, UVs and color on one triangle), triangles and polygon groups (one per material slot). Made by Convert Mesh Object To Mesh Description from any mesh object.

Edit it with the engine's Mesh Description nodes (Get/Set Vertex Position, Set Vertex Instance UV, Create Vertex, Create Triangle, Delete Triangle, ...) or with the whole-array functions below, then turn it back into a Static or Skeletal Mesh with the Convert nodes (Convert Mesh Object To Static Mesh, ...), export it with Export Mesh Object, or rebuild an existing Static Mesh with its Build From Static Mesh Descriptions node.

Arrays of vertices and vertex instances are indexed by their ID; IDs of deleted elements keep their (unused) entries.

VARIABLES

Name	Type	Access	Description
Materials	 Array of Material Interface Object Reference	Read / Write	Material of each slot, in the order of Material Slot Names. Empty entries get the plugin's default material.
Material Slot Names	 Array of Name	Read / Write	Name of each material slot; polygon groups refer to slots by these names (Set Polygon Group Material Slot Name).

FUNCTIONS

- Get Vertex Positions
- Set Vertex Positions
- Get Vertex Instance Normals
- Set Vertex Instance Normals
- Get Vertex Instance Normal
- Set Vertex Instance Normal
- Get Vertex Instance UVs
- Set Vertex Instance UVs
- Get Vertex Instance Colors
- Set Vertex Instance Colors
- Get Vertex Instance Color
- Set Vertex Instance Color
- Get Num UVChannels
- Set Num UVChannels
- Get Polygon Group Material Slot Name
- Get Triangles
- Transform Mesh
- Recompute Normals and Tangents
- Flip Faces
- Generate UVs
- Get Bounds

Runtime Image

URuntimeImage · based on Object

Editable pixels in memory (no GPU texture): made by Import Image From File, Create Image or Convert Texture To Image, changed with the Image nodes (Resize, Crop, Adjust Colors, Draw Image, Set Image Pixel, ...), then turned into a texture (Convert Image To Texture, Update Texture From Image) or written to a file (Export Image To File). Pixels are 8-bit (sRGB colors, or linear data) or 32-bit float (HDR).

VARIABLES

Name	Type	Access	Description
Width	 Integer	Read only	Width in pixels.
Height	 Integer	Read only	Height in pixels.
Is HDR	 Boolean	Read only	True if the pixels are 32-bit float (HDR) rather than 8-bit.
Is sRGB	 Boolean	Read only	True if 8-bit pixels are sRGB colors; false for linear data (normal maps, masks).
Source	 String	Read only	File path or URL the image was imported from (empty for images made otherwise).

Runtime Video

URuntimeVideo · based on Object

A playing video, made by Import Video From File / URL / Buffer. Show its Texture anywhere a texture goes (a material, a UMG Image, Create Material From Texture), and control it with its variables (Time, Rate, Looping, Paused, Volume) or the matching subsystem functions. Files play with the plugin's own player where it can decode them (GIF and Motion JPEG AVI everywhere; any format Windows decodes, on Windows), otherwise and for URLs with the platform's player (Media Framework).

VARIABLES

Name	Type	Access	Description
Texture	 Texture Object Reference	Read only	What the video shows: Media Texture (platform player) or Texture 2D (plugin player).
Media Player	 Media Player Object Reference	Read only	The engine's player (platform player only; null for the plugin's player). Its own functions work too.
Time	 Float	Read / Write	Playback position in seconds. Setting it seeks.
Duration	 Float	Read only	Length in seconds (0 until opened, or for live streams).
Rate	 Float	Read / Write	Playback speed (1: normal). Platforms support different ranges; the plugin's player supports 0-8.
Looping	 Boolean	Read / Write	Starts over at the end.
Paused	 Boolean	Read / Write	Pauses or resumes playback.
Volume	 Float	Read / Write	Sound volume multiplier.
Is Playing	 Boolean	Read only	

Name	Type	Access	Description
Is Ready	 Boolean	Read only	True once the video is open (size, length and tracks known).
Is Finished	 Boolean	Read only	True once a non-looping video has played to the end.
Playback Percentage	 Float	Read only	Playback progress from 0 to 100.
Width	 Integer	Read only	
Height	 Integer	Read only	
Frame Rate	 Float	Read only	
Source	 String	Read only	File path or URL the video plays from.

EVENTS

Event	Outputs	Description
On Video Opened	Video:  Runtime Video Object Reference	Fires once the video is open and its size, length and tracks are known (it starts playing then if Play On Open).
On Video Open Failed	Video:  Runtime Video Object Reference Error Message:  String	Fires if the video couldn't be opened (unsupported format or codec, missing file, network error).
On Video End Reached	Video:  Runtime Video Object Reference	Fires when playback reaches the end (also each time a looping video starts over).
On Video Paused	Video:  Runtime Video Object Reference	
On Video Resumed	Video:  Runtime Video Object Reference	

Runtime Data

URuntimeData · based on Object

Structured data in memory, read from JSON, XML, CSV, TSV, INI or spreadsheet files (Import Data From File / String) or made with Create Data Object / Array or Convert Struct To Data. Read and change it with the Data nodes using paths ("player.items[2].name"): Get Data As String / Number / Boolean / Array, Set Data String / ..., Get Data Keys; turn it into a struct (Convert Data To Struct) or a Data Table, or write it as any data format (Export Data To File / String). A Runtime Data returned by Get Data Value is a view into the same data: changing one changes the other.

VARIABLES

Name	Type	Access	Description
Type	 ERuntimeData Type	Read only	What the value is: None (null), Boolean, Number, String, Array or Object. Same as Get Data Type with an empty path.
Count	 Integer	Read only	Items of an array, fields of an object, else 0. Same as Get Data Count with an empty path.
Source	 String	Read only	File path or URL the data was imported from (empty for data made otherwise).
Source Format	 ERuntimeData Format	Read only	Format the data was read from (Auto Detect for data made otherwise).

Runtime Document

URuntimeDocument · based on Object

A document with its formatting (headings, bold, italic, lists, tables, links) and pages: from Import Document From File / Buffer / String or Download Document From URL (Google Docs, Sheets, Slides and Drive links work). Show it with the Runtime Document Viewer widget, or put its Rich Text in a Rich Text Block (Create Document Text Style Set makes the styles).

VARIABLES

Name	Type	Access	Description
Title	 String	Read only	The document's title property, else its first heading. Same as Get Document Title.
Text	 String	Read only	All the text, as plain text (paragraphs separated by an empty line). Same as Get Document Text.
Rich Text	 String	Read only	All the text as rich text markup for a Rich Text Block (<h1>, , <i>, ... see Create Document Text Style Set). Same as Get Document Rich Text.
Num Pages	 Integer	Read only	Pages (PDF pages; 1 for other documents). Same as Get Document Num Pages.
Format	 ERuntime Text Format	Read only	Format the document was read from. Same as Get Document Format.
Metadata	 Map of String to String	Read only	Document properties: Title, Author, Subject, Keywords, Created, Modified, ... Same as Get Document Metadata.
Source	 String	Read only	File path or URL the document was imported from (empty for documents made from text).

Runtime Document Viewer

URuntimeDocumentViewer · based on Widget · UMG widget

Shows a document in a widget: a scrollable page per document page, with headings, bold and italic text, lists, tables and clickable links. Set Source to a file path or a URL (Google Docs, Sheets, Slides and Drive links shared with "Anyone with the link" work directly) and it loads when the widget is created, or call Load Document, Set Document or Show Text. Reads Word, OpenDocument, PDF, HTML, RTF, Markdown, spreadsheets, subtitles and text files.

VARIABLES

Name	Type	Access	Description
Source	 String	Read only	File path or URL loaded when the widget is created (local files also show in the designer).
Font	 Slate Font Info	Read only	Font of the text (empty: Roboto 16). Headings are larger and bold; code uses the monospaced font.
Text Color	 Slate Color	Read only	
Heading Color	 Slate Color	Read only	
Link Color	 Slate Color	Read only	
Code Color	 Slate Color	Read only	
Muted Color	 Slate Color	Read only	Color of quotes, page labels and table separators.
Padding	 Margin	Read only	Space around the document inside the scroll box.
Page Spacing	 Float	Read only	Space between pages (multi-page documents like PDFs).
Show Page Numbers	 Boolean	Read only	Shows "Page N" above each page of multi-page documents.
Line Height Percentage	 Float	Read only	Line height as a fraction of the font's.

Name	Type	Access	Description
Open Links in Browser	 Boolean	Read only	Opens clicked web links in the system's browser (On Link Clicked fires either way).
Show Scroll Bar	 Boolean	Read only	
EVENTS			
Event	Outputs		Description
On Document Loaded	Document:  Runtime Document Object Reference		A document finished loading (Load Document or Source) and is shown.
On Document Load Failed	Error Message:  String		Loading a document failed; the widget shows the error message.
On Link Clicked	URL:  String		A link in the document was clicked.
FUNCTIONS			
Load Document Set Document Show Text Get Document Get Num Pages Is Loading Scroll to Page Scroll to Top Scroll to End Get Scroll Offset Set Scroll Offset Set Font Set Colors			

Runtime File Drop Zone

URuntimeFileDropZone · based on Content Widget · UMG widget

An area files can be dragged onto from the system's file browser (e.g. Windows Explorer) while the game runs. Put any content inside it (an image, a "Drop files here" text, or a whole panel) and handle On Files Dropped, e.g. with the Import ... From File nodes. Only files the area accepts (Allowed File Type / Extensions) are passed on; others show a "can't drop" cursor. On Drag Entered / On Drag Left are for highlighting the area while files are over it.

VARIABLES

Name	Type	Access	Description
Allowed File Type	 ERuntime File Type	Read / Write	Kind of files accepted, by extension (Unknown: any kind).
Allowed Extensions	 Array of String	Read / Write	Extensions accepted, without dots (e.g. "png", "jpg"); empty: any extension. Combined with Allowed File Type.
Allow Folders	 Boolean	Read / Write	Accept folders too.
Allow Multiple Files	 Boolean	Read / Write	Accept several files at once (else only the first accepted one is passed on).

EVENTS

Event	Outputs	Description
On Files Dropped	File Paths:  Array of String	Files were dropped on the area: the accepted ones, as full paths.
On Drag Entered	File Paths:  Array of String	Files the area accepts were dragged over it (e.g. to highlight it).
On Drag Left	–	The files dragged over the area left it, or were dropped.

FUNCTIONS

Is Drag Over Filter Files

Runtime File Importer Subsystem

URuntimeFileImporterSubsystem · based on Game Instance Subsystem

Single Blueprint entry point for the Runtime File Importer plugin.

Get it with "Get Runtime File Importer Subsystem" (generated automatically by Unreal) and call its functions to import or export files at runtime. Every function here also has an async node (see RuntimeAudioAsyncActions.h) that doesn't block the game thread, reports progress and can be cancelled.

Imported sounds are Runtime Sound Waves: their Volume, Pitch, Looping, Paused, Playback Time, Audio Properties, Metadata, ... are variables on the sound wave itself, as are their playback events. The same settings can also be read and changed here, with functions that take the sound wave as an input. The On Audio ... events here fire for every import/export, whether started from a sync or an async node.

Imported 3D models are Runtime Meshes: a ready-to-use Static Mesh, or a Skeletal Mesh with its Skeleton, Animations and Physics Asset, plus the model's materials, textures and hierarchy (Spawn Runtime Mesh Actor shows it), which can be exported again in any supported format. Engine assets and actors can be exported too (Static/Skeletal Mesh, Animation, Physics Asset, Actor, or any of them with Export Mesh Object), and converted into one another (Convert ... nodes, Convert Mesh Object for any source and target).

Any mesh asset can be inspected (Get Static Mesh Details, Get Skeletal Mesh Details, ...: materials, sections, LODs, UVs, Nanite, collision, bones, ...) and changed at runtime (Editing nodes: materials, sections, LODs, sockets, collision), and its geometry edited with Mesh Descriptions (Convert Mesh Object To Mesh Description, then the Mesh Description nodes).

Images become textures (Import Texture From File/Buffer/URL) or editable Runtime Images (Import Image ..., then the Image nodes: Resize, Crop, Adjust Colors, Draw Image, ...). Any texture (Texture 2D, render target, cube, video texture) can be inspected (Get Texture Details), changed (Editing nodes), converted (Convert Texture Object ...) and exported to image files.

Videos play into textures (Import Video From File/URL/Buffer: Runtime Videos, controlled with their variables or the Playback nodes); video files can be inspected, converted, turned into frames, and frames, the screen or render targets written as videos (Export Frames To Video File, Start Video Recording).

Text files (TXT, Markdown, HTML, RTF, Word, OpenDocument, spreadsheets, any encoding) are read as text (Import Text ...) and written in any encoding. Data files (JSON, XML, CSV, TSV, INI, Excel, OpenDocument spreadsheets) become Runtime Data, read and changed with paths, converted to and from structs and Data Tables, and written in any data format. String Tables (localization) and subtitles (SRT, WebVTT, ASS) can be imported and exported too.

Documents (Word, OpenDocument, PDF, HTML, RTF, Markdown, spreadsheets, text; files, buffers or links, Google Docs / Sheets / Slides and Drive links included) become Runtime Documents with their formatting, shown by the Runtime Document Viewer widget.

Files: the system's Open / Save / folder dialogs, the Runtime File Drop Zone widget (files dragged onto the game), bytes loaded, saved or downloaded, and file info telling which nodes import a file. Archives: ZIP files listed, extracted and made, and bytes compressed (GZip, Zlib). Fonts (TTF, OTF, WOFF) become Fonts for widgets.

VARIABLES

Name	Type	Access	Description
Is Capturing	 Boolean	Read only	True while a microphone is being recorded.
Microphone Level	 Float	Read only	Loudness (RMS, 0-1) of the microphone input right now, e.g. for an input level meter. 0 when not capturing.
Is Recording Video	 Boolean	Read only	True while a video is being recorded.

EVENTS

Event	Outputs	Description
On Audio Imported	Sound Wave:  Runtime Sound Wave Object Reference Source:  String	Any audio import succeeded. Source is the file path or URL (empty for buffers).
On Audio Import Failed	Source:  String Error Message:  String	Any audio import (from a file, buffer or URL) failed. Source is the file path or URL (empty for buffers).
On Audio Exported	Destination:  String	Any audio export or conversion succeeded. Destination is the output file path (empty for buffers).

Event	Outputs	Description
On Audio Export Failed	Destination:  String Error Message:  String	Any audio export or conversion failed. Destination is the output file path (empty for buffers).
On Microphone Capture Started	Sound Wave:  Runtime Sound Wave Object Reference	Microphone capture started, recording into Sound Wave.
On Microphone Capture Stopped	Sound Wave:  Runtime Sound Wave Object Reference	Microphone capture stopped; Sound Wave now holds the complete recording.
On Mesh Imported	Runtime Mesh:  Runtime Mesh Object Reference Source:  String	Any mesh import succeeded. Source is the file path or URL (empty for buffers).
On Mesh Import Failed	Source:  String Error Message:  String	Any mesh import (from a file, buffer or URL) failed. Source is the file path or URL (empty for buffers).
On Mesh Exported	Destination:  String	Any mesh export or conversion succeeded. Destination is the output file path (empty for buffers).
On Mesh Export Failed	Destination:  String Error Message:  String	Any mesh export or conversion failed. Destination is the output file path (empty for buffers).
On Texture Imported	Texture:  Object Object Reference Source:  String	Any texture or image import succeeded. Texture is a Texture 2D or a Runtime Image; Source the file path or URL (empty for buffers).
On Texture Import Failed	Source:  String Error Message:  String	Any texture or image import failed.
On Texture Exported	Destination:  String	Any texture or image export (or image file conversion) succeeded. Destination is the output file path (empty for buffers).
On Texture Export Failed	Destination:  String Error Message:  String	Any texture or image export failed.
On Video Imported	Video:  Runtime Video Object Reference Source:  String	Any video opened (from a file, URL or buffer).
On Video Import Failed	Source:  String Error Message:  String	Any video failed to open.
On Video Exported	Destination:  String	Any video export, conversion or recording was written.

Event	Outputs	Description
On Video Export Failed	Destination:  String Error Message:  String	Any video export, conversion or recording failed.
On Video Recording Started	Output File Path:  String	A video recording started (it's written when stopped: On Video Exported / Failed).
On Text Imported	Object:  Object Object Reference Source:  String	Any text, data, Data Table, String Table or subtitle import succeeded. Object is the Runtime Data or Data Table made (none for text, lines, string tables and subtitles); Source the file path or URL (empty for buffers and strings).
On Text Import Failed	Source:  String Error Message:  String	Any text, data, Data Table, String Table or subtitle import failed.
On Text Exported	Destination:  String	Any text, data, Data Table, String Table or subtitle export (or file conversion) succeeded. Destination is the output file path (empty for buffers).
On Text Export Failed	Destination:  String Error Message:  String	Any text export or conversion failed.

FUNCTIONS

387 functions, listed in the node sections: [Audio](#), [Mesh](#), [Texture](#), [Video](#), [Text](#), [File](#), [Archive](#), [Font](#).

Structs

Options structs control how nodes read and write files; details and info structs describe a file or asset. Make or Break any of them with the engine's Make and Break nodes.

Runtime Audio Metadata

FRuntimeAudioMetadata • struct

Tags stored in an audio file (ID3, Vorbis comments, RIFF INFO, ...). Empty fields weren't present.

MEMBERS

Name	Type	Access	Description
Title	 String	Read only	
Artist	 String	Read only	
Album	 String	Read only	
Album Artist	 String	Read only	
Genre	 String	Read only	
Year	 String	Read only	Release year or date, as written in the file.
Track Number	 String	Read only	Track number, as written in the file (e.g. "3" or "3/12").
Comment	 String	Read only	

Runtime Audio Info

FRuntimeAudioInfo • struct

Information about an audio file, read without fully importing it.

MEMBERS

Name	Type	Access	Description
Format	 EAudio Format	Read only	
Duration	 Float	Read only	Length in seconds.
Sample Rate	 Integer	Read only	Sample rate the audio decodes to, in Hz.
Num Channels	 Integer	Read only	
Bitrate	 Integer	Read only	Average bitrate of the encoded data, in kilobits per second.
File Size	 Integer64	Read only	Size of the encoded data, in bytes.
Metadata	 Runtime Audio Metadata	Read only	

Runtime Microphone Device

FRuntimeMicrophoneDevice · struct

A microphone (or other audio input) that can be captured from.

MEMBERS

Name	Type	Access	Description
Device Index	 Integer	Read only	Pass this to Start Microphone Capture to record from this device.
Name	 String	Read only	
Id	 String	Read only	
Num Channels	 Integer	Read only	
Sample Rate	 Integer	Read only	
Is Default	 Boolean	Read only	True for the system's default input device.

Runtime Audio Properties

FRuntimeAudioProperties · struct

Every property of a Sound Wave that can be changed at runtime. Use "Get Audio Properties", change what you need, then "Set Audio Properties". Volume, pitch and looping apply immediately; the rest apply the next time the sound starts playing.

MEMBERS

Name	Type	Access	Description
Volume	 Float	Read / Write	Volume multiplier (0 or higher). Range: 0.0
Pitch	 Float	Read / Write	Pitch multiplier (0.125 - 4). Also changes playback speed. Range: 0.125 to 4.0
Looping	 Boolean	Read / Write	
Sound Class	 Sound Class Object Reference	Read / Write	Sound Class used for mixing (volume groups, ducking, etc.). Empty uses the project's default.
Attenuation Settings	 Sound Attenuation Object Reference	Read / Write	Distance attenuation and spatialization settings for 3D playback.
Priority	 Float	Read / Write	Used to decide which sounds keep playing when too many play at once. Range: 0.0 to 100.0
Virtualization Mode	 EVirtualization Mode	Read / Write	What happens when the sound is culled by concurrency, priority or voice limits.
Bypass Volume Scale for Priority	 Boolean	Read / Write	If true, volume doesn't lower this sound's priority.
Concurrency Set	 TSet<USoundConcurrency*>	Read / Write	Concurrency groups that limit how many instances of this sound play at once.
Override Concurrency	 Boolean	Read / Write	If true, Concurrency Overrides are used instead of Concurrency Set.
Concurrency Overrides	 Sound Concurrency Settings	Read / Write	
Source Effect Chain	 Sound Effect Source Preset Chain Object Reference	Read / Write	Effects applied to this sound before it is mixed.
Enable Base Submix	 Boolean	Read / Write	If false, the sound is only heard through its submix/bus sends.

Name	Type	Access	Description
Sound Submix	 Sound Submix Base Object Reference	Read / Write	Submix the sound is routed to. Empty uses the project's default.
Enable Submix Sends	 Boolean	Read / Write	
Submix Sends	 Array of Sound Submix Send Info	Read / Write	Additional submixes the sound is sent to.
Enable Bus Sends	 Boolean	Read / Write	
Pre Effect Bus Sends	 Array of Sound Source Bus Send Info	Read / Write	Source buses the sound is sent to before its source effects.
Post Effect Bus Sends	 Array of Sound Source Bus Send Info	Read / Write	Source buses the sound is sent to after its source effects.
Subtitles	 Array of Subtitle Cue	Read / Write	Subtitle lines, each shown at its time offset (in seconds) into the sound.
Subtitle Priority	 Float	Read / Write	When several sounds have subtitles, the highest priority is shown.
Mature	 Boolean	Read / Write	Marks the subtitles as mature content.
Manual Word Wrap	 Boolean	Read / Write	If true, subtitles keep their own line breaks instead of being wrapped automatically.
Single Line	 Boolean	Read / Write	If true, subtitles are shown on a single line.

Runtime File Info

FRuntimeFileInfo · struct

Size, dates and kind of a file or folder.

MEMBERS

Name	Type	Access	Description
File Path	 String	Read only	Full path.
File Name	 String	Read only	Name with extension, without the folder ("photo.png").
Extension	 String	Read only	Extension without the dot, lowercase ("png"; empty for folders).
Size	 Integer64	Read only	Size in bytes (0 for folders).
Is Directory	 Boolean	Read only	
Is Read Only	 Boolean	Read only	
Creation Time	 Date Time	Read only	When the file was created (UTC). Not every platform records it.
Modification Time	 Date Time	Read only	When the file was last changed (UTC).
File Type	 ERuntime File Type	Read only	Which nodes import the file, detected from its contents (and its extension).
Mime Type	 String	Read only	Media type, e.g. "image/png" ("application/octet-stream" if unknown).

Runtime Archive Entry

FRuntimeArchiveEntry · struct

A file or folder inside a ZIP archive.

MEMBERS

Name	Type	Access	Description
Path	 String	Read only	Path inside the archive, with / between folders ("textures/wood.png"; folders end with /).
Is Directory	 Boolean	Read only	
Size	 Integer64	Read only	Size in bytes once extracted.
Compressed Size	 Integer64	Read only	Size in bytes inside the archive.
Modification Time	 Date Time	Read only	When the file was last changed (as stored in the archive, local time).
Is Encrypted	 Boolean	Read only	True if the entry is password-protected (it can't be extracted).

Runtime Font Info

FRuntimeFontInfo · struct

Names and properties of a font file.

MEMBERS

Name	Type	Access	Description
Family Name	 String	Read only	Family name, e.g. "Open Sans".
Style Name	 String	Read only	Style within the family, e.g. "Bold Italic".
Full Name	 String	Read only	Full name, e.g. "Open Sans Bold Italic".
Version	 String	Read only	
Manufacturer	 String	Read only	Designer or foundry, as the file says.
Copyright	 String	Read only	Copyright notice.
Format	 String	Read only	"TrueType", "OpenType (CFF)", "TrueType Collection" or "WOFF".
Num Glyphs	 Integer	Read only	Number of characters' shapes (glyphs).
Num Faces	 Integer	Read only	Fonts in the file (more than 1 for collections, e.g. .ttc files).
Is Bold	 Boolean	Read only	
Is Italic	 Boolean	Read only	
Is Monospace	 Boolean	Read only	Every character has the same width (code fonts).

Runtime Mesh Material Slot Info

FRuntimeMeshMaterialSlotInfo · struct

A material slot of a static or skeletal mesh.

MEMBERS

Name	Type	Access	Description
Index	 Integer	Read only	Material index: what sections refer to, and what Set Material on a component takes as Element Index.
Slot Name	 Name	Read only	Name of the slot (Set Material By Name on components takes it).
Imported Slot Name	 Name	Read only	Name of the material in the file the mesh was imported from.
Material	 Material Interface Object Reference	Read only	
Material Name	 String	Read only	Name of the material asset (empty without one).
Material Path	 String	Read only	Object path of the material asset (empty without one).
Is Used	 Boolean	Read only	True when a section of any LOD draws with this slot.
UVChannel Densities	 Array of Float	Read only	Texel density of each UV channel for this material (used by texture streaming), when the engine computed it.

Runtime Mesh UVChannel Info

FRuntimeMeshUVChannelInfo · struct

A UV channel of a mesh LOD.

MEMBERS

Name	Type	Access	Description
Channel	 Integer	Read only	
Is Light Map Channel	 Boolean	Read only	True for the channel static lighting uses (Light Map Coordinate Index).
Has Bounds	 Boolean	Read only	True when the UVs could be read (meshes need "Allow CPU Access" in packaged games, except this plugin's) and the bounds are valid.
Min	 Vector 2D	Read only	Smallest and largest UV of the channel (0..1 when the UVs don't tile).
Max	 Vector 2D	Read only	

Runtime Mesh Section Info

FRuntimeMeshSectionInfo · struct

A section of a mesh LOD: triangles drawn with one material.

MEMBERS

Name	Type	Access	Description
Index	 Integer	Read only	
Material Index	 Integer	Read only	Material slot the section draws with.
Material Slot Name	 Name	Read only	
Num Triangles	 Integer	Read only	
Num Vertices	 Integer	Read only	
First Index	 Integer	Read only	First index of the section in the LOD's index buffer.
Cast Shadow	 Boolean	Read only	
Enable Collision	 Boolean	Read only	Static meshes: whether the section is part of the complex (per triangle) collision.
Visible in Ray Tracing	 Boolean	Read only	
Is Hidden	 Boolean	Read only	Skeletal meshes: hidden sections aren't drawn.
Num Bones	 Integer	Read only	Skeletal meshes: bones that move the section's vertices.
Max Bone Influences	 Integer	Read only	Skeletal meshes: most bones that move one vertex.

Runtime Mesh LODDetails

FRuntimeMeshLODDetails · struct

A level of detail of a static or skeletal mesh.

MEMBERS

Name	Type	Access	Description
Index	 Integer	Read only	
Num Vertices	 Integer	Read only	
Num Triangles	 Integer	Read only	
Screen Size	 Float	Read only	Screen size (fraction of the screen height the bounds cover) below which the next LOD is used.
Num UVChannels	 Integer	Read only	
UVChannels	 Array of Runtime Mesh UVChannel Info	Read only	
Has Vertex Colors	 Boolean	Read only	
Can Read Geometry	 Boolean	Read only	Whether the LOD's geometry can be read back (for export, conversion, UV bounds, mesh descriptions, complex collision).
Sections	 Array of Runtime Mesh Section Info	Read only	
Num Bones	 Integer	Read only	Skeletal meshes: bones the LOD uses.
Max Bone Influences	 Integer	Read only	Skeletal meshes: most bones that move one vertex.

Runtime Mesh Socket Info

FRuntimeMeshSocketInfo · struct

A socket of a static mesh, skeletal mesh or skeleton.

MEMBERS

Name	Type	Access	Description
Name	 Name	Read only	
Bone Name	 Name	Read only	Skeletal meshes and skeletons: the bone the socket is attached to.
Relative Transform	 Transform	Read only	Relative to the mesh (static meshes) or to its bone.
Tag	 String	Read only	Static meshes: the socket's tag.

Runtime Mesh Collision Info

FRuntimeMeshCollisionInfo · struct

Simple collision shapes of a static mesh or physics body.

MEMBERS

Name	Type	Access	Description
Has Collision	 Boolean	Read only	True when anything can collide with the mesh (simple shapes, or triangles used as collision).
Complexity	 ERuntimeCollisionComplexity	Read only	
Num Boxes	 Integer	Read only	
Num Spheres	 Integer	Read only	
Num Capsules	 Integer	Read only	
Num Tapered Capsules	 Integer	Read only	
Num Convex Hulls	 Integer	Read only	
Double Sided	 Boolean	Read only	Triangles collide from both sides.

Runtime Static Mesh Details

FRuntimeStaticMeshDetails · struct

Everything about a Static Mesh asset (Get Static Mesh Details).

MEMBERS

Name	Type	Access	Description
Name	 String	Read only	
Path	 String	Read only	Object path of the asset (a transient path for meshes made at runtime).
Is Runtime Mesh	 Boolean	Read only	True for meshes this plugin made (imports and conversions).
Num Vertices	 Integer	Read only	Vertices and triangles of LOD 0.
Num Triangles	 Integer	Read only	
Num LODs	 Integer	Read only	
Min LOD	 Integer	Read only	Lowest LOD index drawn (LODs below it are skipped).

Name	Type	Access	Description
LODs	 Array of Runtime Mesh LODDetails	Read only	
Materials	 Array of Runtime Mesh Material Slot Info	Read only	
Num UVChannels	 Integer	Read only	UV channels of LOD 0.
Has Vertex Colors	 Boolean	Read only	
Bounds	 Box Sphere Bounds	Read only	Local bounds, including the bounds extensions.
Size	 Vector	Read only	Size of the bounds, in centimeters.
Positive Bounds Extension	 Vector	Read only	
Negative Bounds Extension	 Vector	Read only	
Nanite Enabled	 Boolean	Read only	Nanite is enabled for the mesh (its settings in the editor; whether it has Nanite data in packaged games).
Has Nanite Data	 Boolean	Read only	The mesh has Nanite render data (Nanite meshes are built in the editor; meshes made at runtime have none).
Num Nanite Triangles	 Integer	Read only	
Num Nanite Vertices	 Integer	Read only	
Light Map Resolution	 Integer	Read only	
Light Map Coordinate Index	 Integer	Read only	UV channel used for static lighting.
Allow CPUAccess	 Boolean	Read only	Geometry stays readable on the CPU in packaged games (export, conversion, complex collision).
Support Ray Tracing	 Boolean	Read only	
Collision	 Runtime Mesh Collision Info	Read only	
Sockets	 Array of Runtime Mesh Socket Info	Read only	

Runtime Mesh Bone Info

FRuntimeMeshBoneInfo · struct

A bone of a skeletal mesh or skeleton.

MEMBERS

Name	Type	Access	Description
Index	 Integer	Read only	
Name	 Name	Read only	
Parent Index	 Integer	Read only	Index of the parent bone, or -1 for the root.
Parent Name	 Name	Read only	
Local Transform	 Transform	Read only	Reference pose, relative to the parent bone.
Component Transform	 Transform	Read only	Reference pose, relative to the mesh.

Runtime Skeletal Mesh Details

FRuntimeSkeletalMeshDetails · struct

Everything about a Skeletal Mesh asset (Get Skeletal Mesh Details).

MEMBERS

Name	Type	Access	Description
Name	 String	Read only	
Path	 String	Read only	
Is Runtime Mesh	 Boolean	Read only	True for meshes this plugin made (imports and conversions).
Num Vertices	 Integer	Read only	Vertices and triangles of LOD 0.
Num Triangles	 Integer	Read only	
Num LODs	 Integer	Read only	
Min LOD	 Integer	Read only	
LODs	 Array of Runtime Mesh LODDetails	Read only	
Materials	 Array of Runtime Mesh Material Slot Info	Read only	
Num UVChannels	 Integer	Read only	UV channels of LOD 0.
Has Vertex Colors	 Boolean	Read only	
Bounds	 Box Sphere Bounds	Read only	
Size	 Vector	Read only	
Positive Bounds Extension	 Vector	Read only	
Negative Bounds Extension	 Vector	Read only	
Skeleton	 Skeleton Object Reference	Read only	
Physics Asset	 Physics Asset Object Reference	Read only	
Shadow Physics Asset	 Physics Asset Object Reference	Read only	
Num Bones	 Integer	Read only	
Bones	 Array of Runtime Mesh Bone Info	Read only	Bones in hierarchy order (parents before children).
Morph Target Names	 Array of Name	Read only	Morph targets (blend shapes); a component plays one with Set Morph Target.
Sockets	 Array of Runtime Mesh Socket Info	Read only	The mesh's own sockets and its skeleton's.
Allow CPUAccess	 Boolean	Read only	Geometry stays readable on the CPU in packaged games (export, conversion, mesh descriptions).

Runtime Animation Notify Info

FRuntimeAnimationNotifyInfo · struct

A notify of an animation.

MEMBERS

Name	Type	Access	Description
Name	 Name	Read only	
Time	 Float	Read only	Seconds from the start.
Duration	 Float	Read only	Seconds (0 for a single moment).

Runtime Animation Details

FRuntimeAnimationDetails · struct

Everything about an Animation Sequence (Get Animation Details).

MEMBERS

Name	Type	Access	Description
Name	 String	Read only	
Path	 String	Read only	
Skeleton	 Skeleton Object Reference	Read only	
Length	 Float	Read only	Length in seconds (at a Rate Scale of 1).
Frame Rate	 Float	Read only	Keys per second.
Num Keys	 Integer	Read only	
Rate Scale	 Float	Read only	Playback speed multiplier.
Enable Root Motion	 Boolean	Read only	Root bone motion moves the actor instead of the mesh.
Is Additive	 Boolean	Read only	Additive animations add to a pose instead of replacing it.
Animated Bones	 Array of Name	Read only	Bones the animation moves.
Curve Names	 Array of Name	Read only	Float curves (morph target weights, material parameters, ...).
Notifies	 Array of Runtime Animation Notify Info	Read only	

Runtime Physics Body Info

FRuntimePhysicsBodyInfo · struct

A body of a physics asset.

MEMBERS

Name	Type	Access	Description
Bone Name	 Name	Read only	Bone the body follows.
Shapes	 Runtime Mesh Collision Info	Read only	
Simulated	 Boolean	Read only	Simulated bodies follow physics; kinematic ones follow the animation.
Collision Enabled	 ECollision Enabled	Read only	
Mass	 Float	Read only	Mass in kilograms.

Runtime Physics Constraint Info

FRuntimePhysicsConstraintInfo · struct

A constraint (joint) of a physics asset.

MEMBERS

Name	Type	Access	Description
Name	 String	Read only	
Bone 1	 String	Read only	Child bone.
Bone 2	 String	Read only	Parent bone.
Swing 1Limit	 Float	Read only	Angular limits in degrees (0: locked, 180: free).
Swing 2Limit	 Float	Read only	
Twist Limit	 Float	Read only	

Runtime Physics Asset Details

FRuntimePhysicsAssetDetails · struct

Everything about a Physics Asset (Get Physics Asset Details).

MEMBERS

Name	Type	Access	Description
Name	 String	Read only	
Path	 String	Read only	
Preview Mesh	 Skeletal Mesh Object Reference	Read only	Skeletal mesh the asset was made for (may be empty).
Bodies	 Array of Runtime Physics Body Info	Read only	
Constraints	 Array of Runtime Physics Constraint Info	Read only	

Runtime Skeleton Details

FRuntimeSkeletonDetails · struct

Everything about a Skeleton (Get Skeleton Details).

MEMBERS

Name	Type	Access	Description
Name	 String	Read only	
Path	 String	Read only	
Bones	 Array of Runtime Mesh Bone Info	Read only	
Sockets	 Array of Runtime Mesh Socket Info	Read only	
Virtual Bone Names	 Array of Name	Read only	
Curve Names	 Array of Name	Read only	Animation curve names known to the skeleton (morph targets, material parameters, ...).
Preview Mesh	 Skeletal Mesh Object Reference	Read only	Skeletal mesh shown when previewing the skeleton's assets (may be empty).

Runtime Physics Asset Options

FRuntimePhysicsAssetOptions · struct

How physics assets are generated for skeletal meshes.

MEMBERS

Name	Type	Access	Description
Body Shape	 ERuntime Physics Body Shape	Read / Write	
Min Bone Size	 Float	Read / Write	Bones whose vertices span less than this (in centimeters) get no body of their own. Range:  0
Create Constraints	 Boolean	Read / Write	Joins the bodies with constraints, so the mesh can ragdoll when simulated.
Angular Limit	 Float	Read / Write	Swing and twist limit of the constraints, in degrees (0: locked, 180: free). Range:  0 to 180

Runtime Mesh Import Options

FRuntimeMeshImportOptions · struct

How a 3D model file is read and turned into a Runtime Mesh.

MEMBERS

Name	Type	Access	Description
Up Axis	 EMesh Up Axis	Read / Write	Up axis of files that don't store one (OBJ, STL, PLY, Alembic). Auto: Y up for OBJ and Alembic, Z up for STL and PLY.
Units	 EMesh Unit	Read / Write	Unit of files that don't store one (OBJ, STL, PLY, Alembic). Auto: meters for OBJ and PLY, millimeters for STL, centimeters for Alembic.
Scale	 Float	Read / Write	Uniform scale applied after the unit conversion.
Transform	 Transform	Read / Write	Extra transform applied to the whole model (after Scale), e.g. to rotate it or move its origin.
Import Materials	 Boolean	Read / Write	
Import Textures	 Boolean	Read / Write	Loads the textures of the materials (embedded ones, or files next to the model).
Normal Map Convention	 ENormal Map Convention	Read / Write	Green channel convention of the file's normal maps. Most formats and tools use OpenGL.
Base Material	 Material Interface Object Reference	Read / Write	Parent material for every imported material (it should have the same parameters as the plugin's M_RuntimeMesh materials). Empty: the plugin's materials.
Recompute Normals	 Boolean	Read / Write	Ignore the file's normals and compute smooth ones (vertices shared by faces are smoothed).
Recompute Tangents	 Boolean	Read / Write	Ignore the file's tangents and compute them from the UVs.
Collision	 ERuntime Mesh Collision	Read / Write	
Animation Time	 Float	Read / Write	For animated files (Alembic, USD): the moment to import, in seconds from the start.
Mesh Type	 ERuntime Mesh Import Type	Read / Write	Static Mesh, Skeletal Mesh, or whichever fits the file.
Import Animations	 Boolean	Read / Write	Imports the file's animations (FBX takes, glTF animations) as Animation Sequences.

Name	Type	Access	Description
Import Morph Targets	 Boolean	Read / Write	Imports blend shapes as morph targets.
Animation Frame Rate	 Float	Read / Write	Frames per second animations are sampled at. Range: 1 to 240
Skeleton	 Skeleton Object Reference	Read / Write	Skeleton to use instead of creating one, e.g. to share animations with existing characters. Bones are matched by name; missing ones are added to it.
Bone Per Node	 Boolean	Read / Write	For skeletal meshes made from files without bones: a bone per node of the file (so parts can move separately), or a single root bone.
Create Physics Asset	 Boolean	Read / Write	Gives skeletal meshes a physics asset (collision, ragdoll).
Physics Asset Options	 Runtime Physics Asset Options	Read / Write	

Runtime Mesh Export Options

FRuntimeMeshExportOptions · struct

How a mesh is written to a 3D model file.

MEMBERS

Name	Type	Access	Description
Up Axis	 EMesh Up Axis	Read / Write	Up axis for formats that don't store one (OBJ, STL, PLY, Alembic). Auto uses each format's usual convention.
Units	 EMesh Unit	Read / Write	Unit for formats that don't store one (OBJ, STL, PLY, Alembic). Auto uses each format's usual convention.
Export Materials	 Boolean	Read / Write	
Export Textures	 Boolean	Read / Write	Writes textures too: embedded (GLB, FBX, USDZ, buffers) or next to the file (OBJ, glTF, USD, Collada).
Text Format	 Boolean	Read / Write	Writes the text variant of formats that have one (STL, PLY) instead of binary.
Export Skeleton	 Boolean	Read / Write	Writes skeletons, skin weights and morph targets (FBX, glTF). Off, or with other formats, skinned meshes are written as posed.
Export Animations	 Boolean	Read / Write	Writes animations (FBX, glTF).
Animation Frame Rate	 Float	Read / Write	Frames per second Animation Sequences are sampled at when exported (0: each animation's own rate). Range: 0 to 240
LODIndex	 Integer	Read / Write	Level of detail exported from Static and Skeletal Meshes (falls back to the last one). Range: 0

Runtime Mesh Convert Options

FRuntimeMeshConvertOptions · struct

How Convert nodes turn one kind of mesh object into another.

MEMBERS

Name	Type	Access	Description
Collision	 ERuntime Mesh Collision	Read / Write	Collision of created Static Meshes.
LODIndex	 Integer	Read / Write	Level of detail read from Static and Skeletal Meshes (falls back to the last one). Range: 0
Recompute Normals	 Boolean	Read / Write	Ignore the source's normals and compute smooth ones, e.g. after moving the vertices of a Mesh Description.
Recompute Tangents	 Boolean	Read / Write	Ignore the source's tangents and compute them from the UVs.
Pose Animation	 Anim Sequence Object Reference	Read / Write	Skeletal to Static: poses the mesh with this animation (empty: the reference pose, or a component's current pose).
Pose Time	 Float	Read / Write	Time of Pose Animation to use, in seconds. Range: 0
Skeletal Mesh	 Skeletal Mesh Object Reference	Read / Write	Skeletal mesh used with sources that have none: Animation Sequences (posed or exported on it) and Physics Assets (bodies placed on its bones). Empty: a loaded Skeletal Mesh using the asset.
Bone Per Node	 Boolean	Read / Write	Static to Skeletal: a bone per node of the source (actors and Runtime Meshes with a hierarchy), or a single root bone.
Skeleton	 Skeleton Object Reference	Read / Write	Skeleton for created Skeletal Meshes (bones matched by name, missing ones added). Empty: a new one.
Create Physics Asset	 Boolean	Read / Write	Gives created Skeletal Meshes a physics asset.
Physics Asset Options	 Runtime Physics Asset Options	Read / Write	
Animations	 Array of Anim Sequence Object Reference	Read / Write	Animations kept with the result (Runtime Meshes and exports), e.g. to export a character with its animations.
Animation Frame Rate	 Float	Read / Write	Frames per second Animations are sampled at (0: each animation's own rate). Range: 0 to 240

Runtime Mesh Animation Info

FRuntimeMeshAnimationInfo · struct

An animation of a model.

MEMBERS

Name	Type	Access	Description
Name	 String	Read only	
Duration	 Float	Read only	Length in seconds.
Frame Rate	 Float	Read only	
Num Frames	 Integer	Read only	

Runtime Mesh Info

FRuntimeMeshInfo · struct

Summary of a 3D model.

MEMBERS

Name	Type	Access	Description
Format	 EMesh Format	Read only	
Num Vertices	 Integer64	Read only	Vertices and triangles of the whole model (a mesh placed several times counts each time).
Num Triangles	 Integer64	Read only	
Num Meshes	 Integer	Read only	Distinct meshes in the model.
Num Nodes	 Integer	Read only	Nodes of the model's hierarchy.
Num Materials	 Integer	Read only	
Num Textures	 Integer	Read only	
Num UVChannels	 Integer	Read only	
Has Vertex Colors	 Boolean	Read only	
Bounds	 Box	Read only	Bounds in Unreal units (centimeters), after the import options were applied.
Size	 Vector	Read only	Size of the bounds, in centimeters.
File Size	 Integer64	Read only	Size of the source file in bytes (0 if unknown).
Metadata	 Map of String to String	Read only	Descriptive information from the file (Generator, Copyright, Application, ...).
Has Skeleton	 Boolean	Read only	True if the model has bones (it imports as a Skeletal Mesh with Mesh Type Auto).
Num Bones	 Integer	Read only	
Bone Names	 Array of String	Read only	
Morph Target Names	 Array of String	Read only	
Animations	 Array of Runtime Mesh Animation Info	Read only	
Material Names	 Array of String	Read only	
Node Names	 Array of String	Read only	Names of the model's nodes (its hierarchy), parents first.

Runtime Mesh Node Info

FRuntimeMeshNodeInfo · struct

A node of a Runtime Mesh's hierarchy.

MEMBERS

Name	Type	Access	Description
Name	 String	Read only	
Parent	 Integer	Read only	Index of the parent node, or -1 for a root.
Local Transform	 Transform	Read only	Relative to the parent.
Transform	 Transform	Read only	Relative to the model.
Meshes	 Array of Integer	Read only	Meshes the node draws (indices of the model's meshes).
Is Bone	 Boolean	Read only	True for bones.

Runtime Mesh Material Info

FRuntimeMeshMaterialInfo · struct

A material of an imported model, as described in the file.

MEMBERS

Name	Type	Access	Description
Name	 String	Read only	
Base Color	 Linear Color	Read only	Linear color; multiplied by Base Color Texture.
Metallic	 Float	Read only	
Roughness	 Float	Read only	
Emissive Color	 Linear Color	Read only	
Opacity	 Float	Read only	
Blend Mode	 ERuntime Mesh Blend Mode	Read only	
Alpha Cutoff	 Float	Read only	
Two Sided	 Boolean	Read only	
Base Color Texture	 Texture 2D Object Reference	Read only	
Normal Texture	 Texture 2D Object Reference	Read only	Already in Unreal's (DirectX) green channel convention.
Metallic Texture	 Texture 2D Object Reference	Read only	
Roughness Texture	 Texture 2D Object Reference	Read only	
Occlusion Texture	 Texture 2D Object Reference	Read only	
Emissive Texture	 Texture 2D Object Reference	Read only	
Opacity Texture	 Texture 2D Object Reference	Read only	

Runtime Text Import Options

FRuntimeTextImportOptions · struct

How text files are read.

MEMBERS

Name	Type	Access	Description
Encoding	 ERuntimeTextEncoding	Read / Write	Character encoding of the file. Auto Detect reads the byte order mark, else recognizes UTF-8, UTF-16, UTF-32 and Windows-1252.
Convert to Plain Text	 Boolean	Read / Write	HTML, RTF and Markdown files come as plain text (tags and formatting removed, paragraphs as lines); off: as written in the file. Word, OpenDocument and spreadsheet files always come as plain text (spreadsheets as tab-separated rows).
Line Ending	 ERuntimeLineEnding	Read / Write	Converts all line breaks to one style (Keep: as in the file).

Runtime Text Export Options

FRuntimeTextExportOptions · struct

How text is written.

MEMBERS

Name	Type	Access	Description
Encoding	 ERuntimeTextEncoding	Read / Write	Character encoding written (Auto Detect: UTF-8).
Write Byte Order Mark	 Boolean	Read / Write	Starts the file with a byte order mark (some Windows programs need it to recognize UTF-8). Not written when appending to a file.
Line Ending	 ERuntimeLineEnding	Read / Write	Converts all line breaks to one style (Keep: as in the text).
Append	 Boolean	Read / Write	File exports: adds to the end of the file (if it exists) instead of replacing it, e.g. for logs.

Runtime Text Info

FRuntimeTextInfo · struct

What a text file (or text) holds, from Get Text File Info.

MEMBERS

Name	Type	Access	Description
Format	 ERuntimeTextFormat	Read only	
Encoding	 ERuntimeTextEncoding	Read only	Character encoding of the file (of the document's text for Word, OpenDocument and spreadsheet files).
Has Byte Order Mark	 Boolean	Read only	True if the file starts with a byte order mark.
File Size	 Integer64	Read only	Size of the file in bytes (0 for text that isn't a file).
Num Characters	 Integer	Read only	Characters of the text (of the plain text for documents), line breaks included.
Num Words	 Integer	Read only	Words (runs of letters, digits and other non-space characters).
Num Lines	 Integer	Read only	Lines (0 for empty text).
Num Pages	 Integer	Read only	Pages of a PDF document (0 for other files).
Line Ending	 ERuntimeLineEnding	Read only	Line break style (Keep: none, or mixed).
Mixed Line Endings	 Boolean	Read only	True if the text mixes line break styles.
Metadata	 Map of String to String	Read only	Document properties when the file has them: Title, Author, Subject, Keywords, Description, Created, Modified, ...

Runtime Data Import Options

FRuntimeDataImportOptions · struct

How structured data files are read into Runtime Data.

MEMBERS

Name	Type	Access	Description
Encoding	 ERuntimeTextEncoding	Read / Write	Character encoding of the file (Auto Detect: from the byte order mark or the bytes).
First Row Is Header	 Boolean	Read / Write	CSV, TSV and spreadsheets: the first row names the columns (rows become objects with those keys); off: rows are arrays.
Detect Value Types	 Boolean	Read / Write	CSV, TSV, INI and XML: values that look like numbers or true/false become Numbers and Booleans; off: everything is a String.
Delimiter	 String	Read / Write	CSV: the character between values (empty: detected, comma or semicolon).
Sheet Name	 String	Read / Write	Spreadsheets: the sheet to read (empty: the first one).
All Sheets	 Boolean	Read / Write	Spreadsheets: reads every sheet, as an object with a field per sheet name (Sheet Name is then ignored).

Runtime Data Export Options

FRuntimeDataExportOptions · struct

How Runtime Data is written.

MEMBERS

Name	Type	Access	Description
Pretty Print	 Boolean	Read / Write	JSON and XML: one value per line, indented; off: everything on one line (smaller).
Write Header	 Boolean	Read / Write	CSV and TSV: writes a first row with the column names (the keys of row objects).
Delimiter	 String	Read / Write	CSV: the character between values (empty: comma).
Xml Root Name	 String	Read / Write	XML: name of the root element when the data isn't an object with a single field (which then names it).
Encoding	 ERuntime Text Encoding	Read / Write	Character encoding written (Auto Detect: UTF-8).
Write Byte Order Mark	 Boolean	Read / Write	Starts the file with a byte order mark (e.g. so Excel reads a UTF-8 CSV file's accents correctly).
Line Ending	 ERuntime Line Ending	Read / Write	Line break style (Keep: LF).

Runtime Subtitle Cue

FRuntimeSubtitleCue · struct

One subtitle: Text shown from Start Time to End Time (seconds).

MEMBERS

Name	Type	Access	Description
Start Time	 Float	Read / Write	
End Time	 Float	Read / Write	
Text	 String	Read / Write	The subtitle's text, lines separated by line breaks (formatting tags removed).
Identifier	 String	Read / Write	SRT: the cue number; VTT: the cue identifier; ASS: the style name. Optional.

Runtime Regex Match

FRuntimeRegexMatch · struct

A match of a regular expression in a text.

MEMBERS

Name	Type	Access	Description
Match	 String	Read only	The matched text.
Start	 Integer	Read only	Index of the match's first character in the text.
Length	 Integer	Read only	Number of characters matched.
Groups	 Array of String	Read only	The text of each capture group (the parts of the pattern in parentheses), in order.

Runtime Texture Import Options

FRuntimeTextureImportOptions · struct

How imported (and converted) images become textures.

MEMBERS

Name	Type	Access	Description
Pixel Format	 ERuntime Texture Pixel Format	Read / Write	GPU pixel format. Auto: 8-bit for ordinary images, half-float for HDR ones.
SRGB	 Boolean	Read / Write	True for colors (photos, base color, UI): the pixels are sRGB. Turn off for data textures (normal maps, roughness, masks, heights). Ignored for float pixels, which are always linear.
Generate Mipmaps	 Boolean	Read / Write	Smaller copies for distant surfaces (sharper and faster in 3D). Not needed for UI-only textures.
Compression Settings	 Texture Compression Settings	Read / Write	How materials treat the texture, as on texture assets: e.g. Normalmap for normal maps (sampled as Normal), Masks for linear data, User Interface 2D for UI. Runtime textures keep their pixel format either way.
Filter	 Texture Filter	Read / Write	Texture filtering (Default: the Texture Group's, normally trilinear/anisotropic; Nearest for pixel art).
Address X	 Texture Address	Read / Write	What happens outside the 0-1 UV range horizontally.
Address Y	 Texture Address	Read / Write	What happens outside the 0-1 UV range vertically.
LODGroup	 Texture Group	Read / Write	Texture group (World, UI, Character, ...): sets the default filtering and the LOD bias of the device profile.
Max Size	 Integer	Read / Write	Images larger than this (in pixels, on either side) are scaled down to fit, keeping their aspect ratio. 0: no limit. Range: 0
Flip Vertically	 Boolean	Read / Write	Flips the image upside down (for sources stored bottom-up).

Runtime Image Export Options

FRuntimeImageExportOptions · struct

How textures and images are written to image files.

MEMBERS

Name	Type	Access	Description
Quality	 Integer	Read / Write	JPEG quality, 1-100 (higher: larger and closer to the original). Range: 1 to 100
Include Alpha	 Boolean	Read / Write	Writes the alpha channel (PNG, BMP, TGA, EXR, GIF). Off: fully opaque pixels.
Dither	 Boolean	Read / Write	GIF: dithers the 256-color palette, which smooths gradients.
Exposure	 Float	Read / Write	HDR pixels written to 8-bit formats (PNG, JPEG, BMP, TGA, GIF) are clamped to 0-1; this scales them first (exposure, in stops).

Runtime Texture Convert Options

FRuntimeTextureConvertOptions · struct

How Convert Texture ... nodes make their result.

MEMBERS

Name	Type	Access	Description
Width	 Integer	Read / Write	Size of the result in pixels (0: the source's). If only one is set, the other keeps the aspect ratio. Range: 0
Height	 Integer	Read / Write	Range: 0
Resize Filter	 ERuntime Image Resize Filter	Read / Write	Filter used when the size changes.
Cube Face Size	 Integer	Read / Write	Size of each face of a Texture Cube result (0: a power of two near a quarter of the panorama's width). Range: 0
Pixel Format	 ERuntime Texture Pixel Format	Read / Write	Pixel format of the result. Auto keeps the source's precision (8-bit or float).
Generate Mipmaps	 Boolean	Read / Write	Mipmaps for Texture 2D and Texture Cube results (render targets generate their own when asked to).

Runtime Image Adjustments

FRuntimeImageAdjustments · struct

Color adjustments for Adjust Image Colors. The defaults change nothing.

MEMBERS

Name	Type	Access	Description
Brightness	 Float	Read / Write	Multiplies the colors (0: black, 1: unchanged, 2: twice as bright). Range: 0
Contrast	 Float	Read / Write	Spreads the colors away from (above 1) or toward (below 1) middle gray. Range: 0
Saturation	 Float	Read / Write	0: grayscale, 1: unchanged, above 1: more vivid. Range: 0
Hue Shift	 Float	Read / Write	Rotates the hues, in degrees (e.g. 180 turns red into cyan). Range: -360 to 360
Gamma	 Float	Read / Write	Above 1 brightens the midtones, below 1 darkens them. Range: 0.01
Exposure	 Float	Read / Write	Multiplies the colors by 2 to this power (photographic stops).
Tint	 Linear Color	Read / Write	Multiplies every pixel (alpha included) by this color.

Runtime Image Info

FRuntimeImageInfo · struct

Information about an image file, read without importing it.

MEMBERS

Name	Type	Access	Description
Format	 ERuntime Image Format	Read only	
Width	 Integer	Read only	
Height	 Integer	Read only	
Num Channels	 Integer	Read only	Channels stored in the file (1: grayscale, 3: RGB, 4: RGBA).
Bit Depth	 Integer	Read only	Bits per channel (8, 16, 32).
Is HDR	 Boolean	Read only	True for high dynamic range images (EXR, HDR, float TIFF): they import as float textures.
Has Alpha	 Boolean	Read only	True if the file can store transparency (it may still be fully opaque).
Num Frames	 Integer	Read only	Frames of an animated image (GIF); 1 for still images.
Duration	 Float	Read only	Length of an animated image, in seconds (0 for still images).
File Size	 Integer64	Read only	Size of the encoded data, in bytes.

Runtime Texture Details

FRuntimeTextureDetails · struct

Everything about a texture (any kind: 2D, cube, render target, video texture, ...).

MEMBERS

Name	Type	Access	Description
Name	 String	Read only	
Path	 String	Read only	Asset path (empty for textures made at runtime).
Type	 ERuntime Texture Object Type	Read only	
Is Runtime Texture	 Boolean	Read only	True for textures made at runtime (imported, converted, ...), false for assets.
Width	 Integer	Read only	Full size in pixels (of one face for cubes).
Height	 Integer	Read only	
Depth	 Integer	Read only	Depth of volume textures, slices of arrays, 6 for cubes, 1 otherwise.
Num Mips	 Integer	Read only	
Num Resident Mips	 Integer	Read only	Mips currently loaded on the GPU (streaming textures load the detailed ones only when needed).
Pixel Format	 String	Read only	GPU pixel format, e.g. "PF_DXT1", "PF_B8G8R8A8", "PF_FloatRGBA".
Is Compressed	 Boolean	Read only	True if the pixel format is block-compressed (DXT/BC, ASTC, ETC): smaller, but can't be changed directly.
Is HDR	 Boolean	Read only	True for float pixel formats (HDR).
Has Alpha Channel	 Boolean	Read only	True if the pixel format stores alpha.

Name	Type	Access	Description
SRGB	 Boolean	Read only	
Compression Settings	 Texture Compression Settings	Read only	
Filter	 Texture Filter	Read only	
Address X	 Texture Address	Read only	
Address Y	 Texture Address	Read only	
LODGroup	 Texture Group	Read only	
LODBias	 Integer	Read only	
Is Streamable	 Boolean	Read only	True if the texture streams its mips in and out as needed.
Is Virtual Texture	 Boolean	Read only	
Memory Size	 Integer64	Read only	GPU memory used now, in bytes.
Has CPUPixels	 Boolean	Read only	True if the exact pixels are in memory (runtime textures, and assets in the editor): reads are instant and lossless. Other textures are read back from the GPU (needs a renderer; not with -nullrhi).
Clear Color	 Linear Color	Read only	Render targets: the color they are cleared to.
Render Target Format	 String	Read only	Render targets: their format (RTF_...).
Media Player	 Media Player Object Reference	Read only	Video textures: the media player feeding them.

Runtime Video Open Options

FRuntimeVideoOpenOptions · struct

How videos are opened.

MEMBERS

Name	Type	Access	Description
Play on Open	 Boolean	Read / Write	Starts playing as soon as the video is open.
Loop	 Boolean	Read / Write	Starts over at the end.
Play Audio	 Boolean	Read / Write	Plays the video's sound (2D; Attach Video Sound makes it come from a place in the level).
Volume	 Float	Read / Write	Volume multiplier of the sound. Range: 0
Start Time	 Float	Read / Write	Where playback starts, in seconds. Range: 0
Player	 ERuntimeVideoPlayer	Read / Write	Which player plays the video.

Runtime Video Export Options

FRuntimeVideoExportOptions · struct

How videos are written.

MEMBERS

Name	Type	Access	Description
Frame Rate	 Float	Read / Write	Frames per second of the written video (frame sources), or 0 to keep the source's rate (conversions). Range: 0 to 240
Width	 Integer	Read / Write	Size of the written video in pixels (0: the first frame's / the source's). Frames of another size are scaled to fit. Range: 0
Height	 Integer	Read / Write	Range: 0
Quality	 Integer	Read / Write	1-100: JPEG quality of AVI frames, and how much of the automatic bitrate MP4 uses (higher: larger, better). Range: 1 to 100
Bitrate	 Integer	Read / Write	MP4 video bitrate in kilobits per second (0: automatic, from the size, frame rate and Quality). Range: 0
Audio	 Sound Wave Object Reference	Read / Write	Sound to put in the video (MP4 and AVI): a Runtime Sound Wave or another sound wave this plugin can read. Conversions keep the source's sound.
Include Audio	 Boolean	Read / Write	Conversions: writes the source's sound too (MP4 and AVI).
Loop Count	 Integer	Read / Write	GIF: how many times it plays (0: forever). Range: 0
Dither	 Boolean	Read / Write	GIF: dithers each frame's 256 colors (smoother gradients, larger files).
Image Sequence Format	 ERuntime Image Export Format	Read / Write	Image Sequence: the format of each frame's file.
Start Time	 Float	Read / Write	Conversions: the part of the source to write, in seconds (End Time 0: to the end). Range: 0
End Time	 Float	Read / Write	Range: 0

Runtime Video Info

FRuntimeVideoInfo · struct

Information about a video file, read without playing it.

MEMBERS

Name	Type	Access	Description
Format	 ERuntime Video Format	Read only	
Duration	 Float	Read only	Length in seconds (0 if unknown, e.g. live streams).
Width	 Integer	Read only	
Height	 Integer	Read only	
Frame Rate	 Float	Read only	Frames per second (average for variable rate videos).
Num Frames	 Integer	Read only	Number of frames (estimated from the length when the file doesn't say).
Video Codec	 String	Read only	Video compression, e.g. "H.264", "HEVC", "VP9", "Motion JPEG".
Has Audio	 Boolean	Read only	
Audio Codec	 String	Read only	Audio compression, e.g. "AAC", "Opus", "PCM".

Name	Type	Access	Description
Audio Sample Rate	 Integer	Read only	
Audio Channels	 Integer	Read only	
Bitrate	 Integer	Read only	Average bitrate of the whole file, in kilobits per second.
File Size	 Integer64	Read only	Size in bytes.
Can Read Frames	 Boolean	Read only	True if this plugin can read the frames on this platform (Get Video Thumbnail, Extract Video Frames, Convert Video File).

Runtime Video Details

FRuntimeVideoDetails · struct

Everything about a Runtime Video as it plays.

MEMBERS

Name	Type	Access	Description
Source	 String	Read only	File path or URL.
Player Name	 String	Read only	Which player plays it (e.g. "WmfMedia", "Plugin").
Is Ready	 Boolean	Read only	
Is Playing	 Boolean	Read only	
Is Paused	 Boolean	Read only	
Is Looping	 Boolean	Read only	
Supports Seeking	 Boolean	Read only	
Duration	 Float	Read only	
Time	 Float	Read only	
Rate	 Float	Read only	
Volume	 Float	Read only	
Width	 Integer	Read only	
Height	 Integer	Read only	
Frame Rate	 Float	Read only	
Video Format	 String	Read only	Video track format, e.g. "NV12", "BGRA".
Audio Channels	 Integer	Read only	
Audio Sample Rate	 Integer	Read only	
Video Tracks	 Array of String	Read only	Names of the tracks (with their language when known), per kind.
Audio Tracks	 Array of String	Read only	
Caption Tracks	 Array of String	Read only	
Selected Video Track	 Integer	Read only	Selected track of each kind (-1: none). Change with Select Video Track.
Selected Audio Track	 Integer	Read only	
Selected Caption Track	 Integer	Read only	

Enums

Formats, modes and kinds used by the nodes' enum pins.

EAudio Format

EAudioFormat • enum

Audio formats that can be imported.

VALUES

Value	Description
Auto Detect	
MP3	
WAV	
FLAC	
OGG Vorbis	
Uncompressed (Raw PCM)	
AIFF	
OGG Opus	

EAudio Export Format

EAudioExportFormat • enum

Audio formats that can be exported.

VALUES

Value	Description
WAV (16-bit PCM)	
OGG Vorbis	
FLAC (Lossless)	
Uncompressed (Raw PCM)	

ERuntimeFile Type

ERuntimeFileType · enum

What kind of file something is, i.e. which nodes import it.

VALUES

Value	Description
Unknown	Not a file type this plugin reads.
Audio	Import Audio ... nodes.
Mesh	3D models: Import Mesh ... nodes.
Image	Import Texture / Image ... nodes.
Video	Import Video ... nodes.
Text	Plain text, Markdown, HTML, RTF: Import Text / Document ... nodes.
Data	JSON, XML, CSV, TSV, INI, YAML, spreadsheets: Import Data ... nodes.
Document	Word, OpenDocument, PDF: Import Document / Text ... nodes.
Subtitles	SRT, WebVTT, ASS: Import Subtitles ... nodes.
Archive	ZIP archives: Archive nodes. GZip-compressed data: Decompress Bytes.
Font	TTF, OTF, TTC, WOFF: Import Font ... nodes.

ERuntimeCompressionFormat

ERuntimeCompressionFormat · enum

How Compress Bytes / Decompress Bytes store data.

VALUES

Value	Description
GZip	The .gz file format (header, compressed data, checksum). Decompressing: detected (GZip, then Zlib, then Deflate).
Zlib	Zlib stream (2-byte header, compressed data, checksum), e.g. for HTTP "deflate" or many file formats.
Deflate	Raw Deflate data, as inside ZIP archives.

ERuntimeCollisionComplexity

ERuntimeCollisionComplexity · enum

How a static mesh's simple and complex (per triangle) collision are used. Same as the mesh's Collision Complexity.

VALUES

Value	Description
Project Default	The project's default (simple shapes for physics, triangles for traces that ask for them).
Simple And Complex	Simple shapes for physics and simple traces, triangles for complex traces.
Use Simple Collision As Complex	Only the simple shapes, for everything.
Use Complex Collision As Simple	Only the triangles, for everything (they can't simulate physics).

ERuntimeUVProjection

ERuntimeUVProjection · enum

How Generate UVs projects the mesh onto a UV channel.

VALUES

Value	Description
Planar	Straight along the gizmo's X axis.
Box	From the six sides of a box (each triangle uses the side it faces most).
Cylindrical	Around the gizmo's Z axis.

ERuntimeMeshCollision

ERuntimeMeshCollision · enum

Collision generated for imported meshes.

VALUES

Value	Description
None	No collision: fastest to import.
Box	A box around the mesh: cheap, good for props that are only pushed around or stood near.
Complex	The triangles themselves (complex as simple): exact, for floors, walls and level geometry. Slower to import.
Convex	A convex hull around the mesh: cheap and closer than a box, for props that simulate physics.

ERuntimeMeshImportType

ERuntimeMeshImportType · enum

What an imported model becomes.

VALUES

Value	Description
Auto	A Skeletal Mesh when the file has bones or morph targets, otherwise a Static Mesh.
Static Mesh	Always a Static Mesh: skinned meshes are posed as in the file (bones, morph targets and animations are dropped).
Skeletal Mesh	Always a Skeletal Mesh: files without bones get a bone per node (Bone Per Node) or a single root bone.

ERuntimeMeshObjectType

ERuntimeMeshObjectType · enum

Kinds of objects the mesh nodes work with, as detected by Get Mesh Object Type.

VALUES

Value	Description
None	Not something with geometry, a skeleton or an animation.
Static Mesh	
Skeletal Mesh	
Skeleton	
Animation Sequence	
Physics Asset	
Runtime Mesh	
Actor	
Static Mesh Component	
Skeletal Mesh Component	
Mesh Description	A Mesh Description (Runtime Mesh Description or the engine's Static Mesh Description).

ERuntimeMeshConvertTarget

ERuntimeMeshConvertTarget · enum

What Convert Mesh Object turns an object into.

VALUES

Value	Description
Static Mesh	One Static Mesh (skeletal sources are posed: reference pose, or Pose Animation at Pose Time).
Skeletal Mesh	A Skeletal Mesh with its Skeleton (static sources get a root bone, or a bone per node).
Physics Asset	A Physics Asset with a body per bone (assigned to the Skeletal Mesh it is made for).
Runtime Mesh	A Runtime Mesh, keeping everything this plugin can export (hierarchy, skeleton, animations, materials).
Mesh Description	Editable geometry (a Runtime Mesh Description) with the materials: skeletal sources are posed, like for a Static Mesh.

ERuntimePhysicsBodyShape

ERuntimePhysicsBodyShape · enum

Shape of the bodies of generated physics assets.

VALUES

Value	Description
Capsule	
Box	
Sphere	

ERuntimeTextFormat

ERuntimeTextFormat · enum

Kinds of text files the Text nodes read. Documents and spreadsheets are read as their text.

VALUES

Value	Description
Auto Detect	Detected from the contents (falling back to the file extension).
Plain Text	
JSON	
XML	
CSV	
TSV	
INI	
Markdown	
HTML	
RTF	
SubRip Subtitles (SRT)	
WebVTT Subtitles (VTT)	
SubStation Alpha Subtitles (ASS/SSA)	
Word Document (DOCX)	
OpenDocument Text (ODT)	
Excel Spreadsheet (XLSX)	
OpenDocument Spreadsheet (ODS)	
YAML	
PDF Document	

ERuntimeTextEncoding

ERuntimeTextEncoding · enum

How text is stored as bytes.

VALUES

Value	Description
Auto Detect	Reading: from the byte order mark, else detected from the bytes. Writing: UTF-8.
UTF-8	
UTF-16 LE	
UTF-16 BE	
UTF-32 LE	
UTF-32 BE	
Windows-1252 (Latin-1)	Western European single-byte text (also ASCII). Characters it lacks are written as '?'. [1] https://en.cppreference.com/w/cpp/string/basic_string_view

ERuntimeLineEnding

ERuntimeLineEnding · enum

Line break style.

VALUES

Value	Description
Keep	Leave line breaks as they are (generated text uses LF). In text info: no line breaks, or mixed ones.
Unix (LF)	
Windows (CRLF)	
Classic Mac (CR)	

ERuntimeDataFormat

ERuntimeDataFormat · enum

Structured data files read into Runtime Data.

VALUES

Value	Description
Auto Detect	Detected from the contents (falling back to the file extension).
JSON	
XML	
CSV	
TSV	
INI	
Excel Spreadsheet (XLSX)	
OpenDocument Spreadsheet (ODS)	
YAML	

ERuntimeDataExportFormat

ERuntimeDataExportFormat · enum

Structured data files Runtime Data can be written as.

VALUES

Value	Description
JSON	
XML	
CSV	
TSV	
INI	
YAML	

ERuntimeData Type

ERuntimeDataType · enum

Kind of a Runtime Data value.

VALUES

Value	Description
None	Null, or no value at the path.
Boolean	
Number	
String	
Array	A list of values (Get Data As Array, Path "items[0]").
Object	Named fields (Get Data Keys, Path "player.name").

ERuntimeSubtitle Format

ERuntimeSubtitleFormat · enum

Subtitle file formats read.

VALUES

Value	Description
Auto Detect	
SubRip (SRT)	
WebVTT (VTT)	
SubStation Alpha (ASS/SSA)	

ERuntimeSubtitle Export Format

ERuntimeSubtitleExportFormat · enum

Subtitle file formats written.

VALUES

Value	Description
SubRip (SRT)	
WebVTT (VTT)	
SubStation Alpha (ASS)	

ERuntimeImageFormat

ERuntimeImageFormat · enum

Image file formats that can be imported.

VALUES

Value	Description
Auto Detect	Detect the format from the file's contents (falling back to its extension).
PNG	
JPEG	
BMP	
TGA	Truevision TARGA.
EXR	OpenEXR (HDR). Windows, Mac and Linux only.
HDR	Radiance HDR (RGBE).
TIFF	Windows, Mac and Linux only.
ICO	Windows icon.
ICNS	Mac icon. Mac only.
GIF	Still or animated GIF (the first frame as an image; every frame as a video).

ERuntimeImageExportFormat

ERuntimeImageExportFormat · enum

Image file formats that can be written.

VALUES

Value	Description
PNG	Lossless, with alpha. 8 bits per channel, or 16 for HDR images.
JPEG	Lossy, small, no alpha (Quality sets the size/quality balance).
BMP	Uncompressed, 24 or 32 bits (with alpha).
TGA	Truevision TARGA, run-length compressed, with alpha.
EXR	OpenEXR: half-float HDR with alpha. Windows, Mac and Linux only.
HDR	Radiance HDR (RGBE), no alpha.
GIF	256 colors, 1-bit alpha.

ERuntimeTexturePixelFormat

ERuntimeTexturePixelFormat · enum

How the pixels of a texture are stored on the GPU.

VALUES

Value	Description
Auto	8-bit color for ordinary images, half-float for HDR images (EXR, HDR, float pixels).
RGBA 8-bit	8 bits per channel, BGRA: 4 bytes per pixel.
Grayscale 8-bit	8-bit single channel (the image's luminance): 1 byte per pixel.
RGBA 16-bit Float	Half-float HDR: 8 bytes per pixel.
RGBA 32-bit Float	Full-float HDR: 16 bytes per pixel.

ERuntimeTextureObjectType

ERuntimeTextureObjectType · enum

Kinds of objects the texture nodes work with, as detected by Get Texture Object Type.

VALUES

Value	Description
None	Not a texture, image or video.
Texture 2D	
Texture Cube	
Texture 2D Array	
Volume Texture	
Render Target 2D	
Render Target Cube	
Media Texture	A video's texture (Media Texture).
Texture 2D Dynamic	Texture 2D Dynamic (e.g. from the engine's Download Image node).
Runtime Image	Editable pixels (Runtime Image).
Runtime Video	A Runtime Video (its current frame).
Other Texture	Another kind of texture (e.g. a Texture Render Target 2D Array); read as an image when possible.

ERuntimeTextureConvertTarget

ERuntimeTextureConvertTarget · enum

What Convert Texture Object turns an object into.

VALUES

Value	Description
Texture 2D	A Texture 2D. Cube textures become a panorama (equirectangular, twice as wide as high).
Render Target 2D	A Texture Render Target 2D, e.g. to draw on with Draw Material / Canvas nodes.
Texture Cube	A Texture Cube (e.g. for a sky or reflections) from a panorama (equirectangular) image.
Runtime Image	A Runtime Image: pixels to edit with the Image nodes.

ERuntimeImageResizeFilter

ERuntimeImageResizeFilter · enum

How Resize Image computes the new pixels.

VALUES

Value	Description
Nearest	Copies the closest pixel: blocky, keeps pixel art sharp.
Bilinear	Smooth.
Bicubic	Smooth and sharper (best for photos).

ERuntimeImageRotation

ERuntimeImageRotation · enum

Clockwise rotations.

VALUES

Value	Description
90 Degrees	
180 Degrees	
270 Degrees	

ERuntimeImageFlip

ERuntimeImageFlip · enum

VALUES

Value	Description
Horizontal	Mirror left to right.
Vertical	Upside down.

ERuntimeImageChannel

ERuntimeImageChannel · enum

A color channel of an image.

VALUES

Value	Description
Red	
Green	
Blue	
Alpha	

ERuntimeImageChannelSource

ERuntimeImageChannelSource · enum

Where Swizzle Image Channels takes a channel's value from.

VALUES

Value	Description
Red	
Green	
Blue	
Alpha	
Zero	0 everywhere.
One	1 everywhere.

ERuntimeImageBlendMode

ERuntimeImageBlendMode · enum

How Draw Image combines an image with the one it's drawn on.

VALUES

Value	Description
Alpha Blend	Normal layering; the drawn image's alpha decides how much it covers.
Additive	Adds the colors (brightens).
Multiply	Multiplies the colors (darkens).
Screen	Inverse multiply (brightens softly).
Replace	Replaces the pixels, alpha included.

ERuntimeVideoFormat

ERuntimeVideoFormat · enum

Video file formats (containers). Which ones play depends on the platform's video player; see Get Video Info.

VALUES

Value	Description
Auto Detect	Detect the format from the file's contents (falling back to its extension).
MP4	MPEG-4 (also .m4v).
MOV	QuickTime.
AVI	
MKV	Matroska.
WebM	
WMV	Windows Media (ASF).
GIF	Animated GIF.
MPEG-TS	MPEG transport stream (.ts, .m2ts).
MPEG	MPEG program stream (.mpg, .mpeg).

ERuntimeVideoExportFormat

ERuntimeVideoExportFormat · enum

Video file formats that can be written.

VALUES

Value	Description
MP4	H.264 video and AAC audio: plays everywhere, small files. Windows only (uses the system's encoder).
AVI	Motion JPEG video and PCM audio: every platform, larger files. Plays in this plugin (any platform), VLC, Windows players.
GIF	Animated GIF: every platform, 256 colors, no audio.
Image Sequence	One image file per frame (Image Sequence Format), numbered: Name_0000.png, Name_0001.png, ... No audio.

ERuntimeVideoPlayer

ERuntimeVideoPlayer · enum

How a video plays.

VALUES

Value	Description
Auto	Files the plugin can decode play with its own player (GIF and Motion JPEG AVI everywhere; on Windows, any format Windows decodes: MP4, MOV, WMV, MKV, ... with any renderer, D3D12 included); other files and URLs with the platform's player.
Platform Player	The platform's video player (Media Framework: Windows Media Foundation, Apple AVFoundation, Android MediaPlayer, or Electra if enabled). With D3D12 on Windows, the engine's WMF player needs Project Settings > WMF Media > Hardware Accelerated Video Decoding to show most videos.
Plugin Player	The plugin's own player (files only): GIF and Motion JPEG AVI on every platform, anything Windows decodes on Windows.

ERuntimeVideoTrackType

ERuntimeVideoTrackType · enum

Kinds of tracks in a video.

VALUES

Value	Description
Video	
Audio	
Caption	

EMesh Format

EMeshFormat · enum

3D model file formats that can be imported.

VALUES

Value	Description
Auto Detect	Detect the format from the file's contents, falling back to its extension.
FBX	
OBJ	
glTF (.gltf)	
glTF Binary (.glb)	
STL	
PLY	
USD (.usd, .usda, .usdc)	Universal Scene Description, text (.usda) or binary (.usdc); .usd can be either.
USDZ	
Alembic (.abc)	Alembic. Only available on Windows, Mac and Linux.
Collada (.dae)	

EMesh Export Format

EMeshExportFormat · enum

3D model file formats that can be exported.

VALUES

Value	Description
FBX (binary)	
OBJ	Wavefront OBJ, with a .mtl material file and textures next to it (when exporting to a file).
glTF (.gltf)	glTF 2.0 JSON, with a .bin buffer and textures next to it (embedded when exporting to a buffer).
glTF Binary (.glb)	
STL	
PLY	
USD (.usda)	Text USD, with textures next to it (when exporting to a file).
USDZ	USD packed with its textures into one file (for AR Quick Look, Reality Composer, ...).
Alembic (.abc)	Alembic. Only available on Windows, Mac and Linux.
Collada (.dae)	

EMesh Up Axis

EMeshUpAxis · enum

Which axis points up in a file.

VALUES

Value	Description
Auto	The file's own up axis, or the usual convention for formats that don't store one.
Y Up	
Z Up	

EMesh Unit

EMeshUnit · enum

Length unit of a file's coordinates.

VALUES

Value	Description
Auto	The file's own unit, or the usual convention for formats that don't store one.
Millimeters	
Centimeters	
Meters	
Inches	
Feet	

ERuntime Mesh Blend Mode

ERuntimeMeshBlendMode · enum

How a material's opacity is rendered.

VALUES

Value	Description
Opaque	
Masked	Fully opaque or fully transparent per pixel, cut at the Alpha Cutoff.
Translucent	

ENormal Map Convention

ENormalMapConvention · enum

Convention of the green channel in tangent-space normal maps.

VALUES

Value	Description
OpenGL (Y+)	Green points up (glTF, USD, Blender, Maya, ...). Converted to Unreal's convention on import.
DirectX (Y-)	Green points down, as Unreal expects (3ds Max, DirectX tools, ...).